

Effect of drying flow regimes on macroinvertebrates in headwater streams



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Frontispiece: Upstream reach on the Wungong Brook, February 2018 (photo by N. Carey)

Thesis Declaration

I Nicole Carey verify that in submitting this thesis;

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Statement of thesis structure:

This thesis has been designed to present the data collected during the degree of Doctor of Philosophy as a coherent piece of work incorporating a series of stand-alone manuscripts. Each manuscript was written according to the authorship guidelines of the journal that the manuscript has been accepted for publication, submitted for review, or will be submitted to in the future. A statement of author attribution precedes each manuscript. All chapters are presented in a similar format with continuous referencing and table and figure numbering to maintain the continuity of the document.

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Thesis Abstract

In many regions, global warming is causing drying in perennial streams and prolonged drying in intermittent streams. Flow regime strongly influences stream ecosystems, so changing regimes are predicted to alter stream biodiversity and function. This thesis investigated the effect of flow regime change (perennial to intermittent) on macroinvertebrate communities in headwater streams in south western Australia (SWA), a region that has already experienced substantial climatic drying. I used a multiple before-after, control-impact design where controls comprised stream reaches that remained perennial and impact reaches were perennial until 2000s and then became intermittent. Before-drying information came from Bunn's 1985 PhD thesis and I collected after-drying data from the same reaches in 2016-2017. Many species (including SWA endemics) were lost from streams that became intermittent, which also supported drying-adapted species, including new arrivals. Regardless of flow regime, high species turnover characterised both spatial and temporal beta-diversity in assemblages. Species traits did not adequately predict persistence under intermittency. Of the 13 formerly most-common species, seven were restricted to perennially flowing streams, and only four (two mayflies, one stonefly, one amphipod) adapted their life histories to intermittency. Bunn (1985) showed that leaf-litter breakdown was moderated by macroinvertebrates in these streams. By repeating his experiment, I showed that leaf-litter breakdown processes had remained similar in intermittent and perennial streams, despite the loss of most large macroinvertebrate shredders. To survive drying, invertebrates used a similar range of drought refuges identified in other studies. Altogether, these results demonstrate profound alterations to macroinvertebrate communities in streams forced into intermittency, particularly the loss of relictual and functionally important species. Given that current understanding of traits did not predict species persistence, life history research should focus on understanding trait

flexibility. Identification and protection of evolutionary refugia is urgent in regions undergoing climatic drying because substantial biodiversity loss is imminent there.

Chapter 1

General Introduction

Anthropogenic greenhouse gas emissions are causing shifts in the earth-climate system, including altered hydrology for much of the globe. Precipitation patterns are changing: the gradient between wet and dry regions is intensifying as rainfall diminishes in drier regions, and intensifies in others (Hoegh-Guldberg et al., 2018). Many of the world's regions will become increasingly water scarce as climate change progresses, exacerbating conflict between the water needs of humans and the natural environment (Gosling & Arnell, 2016; Dolan et al., 2021). The Mediterranean (hereafter med-) climate regions of the world are among those predicted to become drier as climate change progresses, due to reduced precipitation, higher temperatures and evaporation rates, leading to drying water regimes (Filipe et al., 2013; Polade et al., 2017; Zipper et al., 2021). In rivers and streams, the timing, duration and volume of flow, and the predictability of the flow regime is a fundamental abiotic control shaping freshwater ecosystems, interacting with organism life histories and physical tolerances to shape biotic assemblages and function (Boulton & Suter, 1986; Bunn & Arthington, 2002; Datry et al., 2017a). Therefore, shifts in flow regime are expected to profoundly impact aquatic faunal assemblages, particularly the transition towards drier states and less predictable flows.

The influence of flowing water on macroinvertebrates

Lotic environments are distinguished by unidirectional flow, and presence or absence of flowing water has profound implications for stream fauna, interacting with the physiology of individual organisms, as well as the physical and chemical attributes of their habitats at multiple scales (Resh et al., 1988; Boulton et al., 2014). Stream habitat is hierarchically structured across spatial and temporal scales, from the microhabitat (e.g. woody debris, patches of moss) to geological (e.g. climate, lithology) scales (Frissell et al., 1986).

Hierarchical structuring also governs the water balance within drainage networks formed by the geology, slope and climate of a particular landscape (Frissell et al., 1986). Within the nested hierarchy of the stream network, different habitats (such as riffles, runs, cascades or different types of pools) vary in the ratio of water depth and speed to the height of the roughness elements of the sediment, creating distinct flow environments with different habitat architecture that finally determine the range of microhabitats available to fauna (Frissell et al., 1986; Davis & Barmuta, 1989) in both perennial and non-perennial rivers.

In non-perennial rivers and streams, flow regimes differ based on the predictability and duration of the hydroperiod, and these two factors are well established as key influences on biotic processes (Boulton et al., 2014; Datry et al., 2017a). Broadly, seasonal flow regimes are predictably wet and dry according to season, whereas intermittent, episodic and ephemeral flow regimes occur on a gradient of decreasing predictability and duration (Boulton et al., 2014). The faunal assemblages of intermittent streams differ from those in perennially flowing streams because the processes of disconnection and drying alters the physical, chemical and biotic components of running waters (Boulton & Suter, 1986; Boulton & Lake, 1990; Leigh et al., 2016a). Patterns of drying differ between habitat patches at the reach and stream section scales, governed by hydrogeology at higher hierarchical scales (Stanley et al., 1997), which then determine the changes experienced by invertebrates at the

microhabitat scale. Consequently, biotic assemblages develop in the context of a particular flow regime and the habitat and microhabitat available, including the natural patterns of longitudinal and lateral connectivity usual for any particular river or stream (Bunn & Arthington, 2002).

Generally, species richness increases along a gradient towards increased flow permanence in rivers (Soria et al., 2017), but this is not always the case. Intermittent streams can harbour rich assemblages comparable to perennial streams (e.g. Boulton & Suter, 1986; Leigh et al., 2016b and references therein), and at larger spatial scales, the arrangement of intermittent and perennial reaches shapes biodiversity patterns along the river network. The spatial arrangement of differing hydroperiods along dendritic stream networks determines the location of dry season refuges and distributes resilient and resistant species in the catchment which, through their respective dispersal modes, shape recolonization patterns and community composition (Bogan et al., 2013; Cañedo-Argüelles et al., 2015; Tonkin et al., 2018; Gauthier et al., 2020). Headwater streams contribute disproportionately to biodiversity within catchments because of their high beta diversity and are particularly important when headwater reaches are perennial (Finn et al., 2011; Crabot et al., 2020). Drying in perennial headwater streams may thus disproportionately alter biodiversity patterns across the broader landscape, through loss of refuges and resilient colonist species.

Invertebrate adaptations to flow intermittency

Invertebrate adaptations to overcome intermittent drying are usually grouped into resistance and resilience traits (Robson et al., 2013; Bogan et al., 2017). Strachan et al. (2015) provide an extensive review of invertebrate strategies to survive drying (also see Bogan et al., 2017). Briefly, resistance traits are those that allow an organism to survive drying *in situ*, including the ability to aestivate (enter summer dormancy) known in some crustaceans, annelids, and insects including some Diptera, Odonata and Trichoptera (e.g. Chester & Robson, 2011;

Wickson et al., 2012; Strachan et al., 2014), or enter a desiccation-resistant life stage. The latter commonly comprises the eggs or cysts produced by many microcrustaceans, some dipterans, plecopterans and ephemeropterans (Boulton et al., 1992; Williams, 2006; Robson et al., 2011; Stubbington & Datry, 2013). Resilience traits enable organisms to escape drying pools by following environmental cues and facilitates recolonisation and gene flow within populations of species in catchments dominated by intermittent streams (Robson et al., 2011; Chester et al., 2015; Banegas-Medina et al., 2021). The onset of pool dry-down can prompt insects with an aerial adult stage to undergo eclosion and fly away. These insects spend varying amounts of time as adults that either move to other waterbodies, lay eggs into pools or on the dry streambed, or if longer lived, recolonise the stream when flows return (Bogan & Boersma, 2012; Strachan et al., 2015; Stubbington et al., 2017). A few freshwater animals can walk overland (for example a toebiter, *Abedus herberti* was documented walking at least 240 metres to reach a remnant pool in Sonoran desert stream – Boersma & Lytle, 2014) or crawl or swim to more permanent reaches upstream or downstream (Bogan et al., 2017). Some mayfly nymphs have also been shown to respond to dry-down cues by drifting downstream in search of more perennial waters (Chester et al., 2015 and references therein). Regardless of any resistance traits an organism may possess, its life history phenology must conform to the extant hydroregime in order to cope with flow intermittency. Concomitance between presence of surface water and phenological events including timing of hatching, length of aquatic development and timing of maturity, metamorphosis and emergence (in flying insects) will determine whether the species can persist under intermittent flow regimes (Robson et al., 2011; Strachan et al., 2015). Therefore pre-existing phenological adaptation to drying, or phenological plasticity, is also an important resilience trait in the context of drying flow regimes, where phenological mismatch has the potential to extirpate species.

Mediterranean climate streams

The med-climate regions of the world (the Mediterranean basin, southern South Africa, central Chile, central west coast of United States, and southwestern and southeastern parts of Australia) are biodiversity hotspots, characterized by high diversity and endemism, and the freshwater fauna is no exception (Myers et al., 2000; Bonada & Resh, 2013). Stream flow is typified by predictable seasonality between a winter peak in flow and summer low flows, and intermittency is common – a predictable disturbance that has occurred for hundreds or thousands of years in many streams (Resh et al., 1988; Hershkovitz & Gasith, 2013). Thus, resilience to drying is common across the med-stream fauna, which also tend to show seasonal variation in assemblage composition between winter flows (lotic assemblages) and summer pools (lentic assemblages) (Bogan & Lytle, 2007; Bonada et al., 2007). However, perennial streams tend to harbour the greatest number of species in med-stream networks, supporting both drying resilient and sensitive species, with progressively fewer species present with increasingly dry flow regimes (Arscott et al., 2010; Chester & Robson, 2011; Leigh et al., 2016b; Vander Vorste et al., 2021). Thus, assemblages of intermittent streams tend to be subsets of those present in nearby perennial streams, with recolonisation of intermittent reaches occurring from perennial refuges being the primary mechanism of community recovery (e.g. Chester & Robson, 2011; Chester et al. 2015).

Because of the dendritic structure of stream networks, the movement of many freshwater organisms across catchments is highly constrained (Frissel et al., 1986; Hughes et al., 2007; Bush & Hoskins, 2017). Thus, under increasingly dry climates, organism dispersal ability is predicted to become increasingly important in structuring metacommunities in med-regions (Cañedo-Argüelles et al., 2020), as permanent surface water diminishes and there are fewer opportunities to sustain dispersal of aquatic obligate, non-resistant and weaker overland dispersing organisms.

Med-region stream fauna share similarities in taxonomic composition, adaptive traits to drying, life histories and seasonal change in assemblages, due to the similarly predictable changes in temperature and flow regime (Bonada & Resh, 2013). Typically, invertebrates in rivers and streams in med-regions evolved long before the development of the Mediterranean climate, instead evolving in wetter, cooler climates; so med-region stream faunas are typically relictual (Bonada & Resh, 2013; Davies & Stewart, 2013). Because of their status as relicts, rather than sharing biogeographic history with desert adapted faunas, these species are considered particularly vulnerable to the impacts of climatic drying (Filipe et al., 2013; Kroll et al., 2018). For example, in southwestern Australia many relictual species were found to be restricted to higher rainfall areas of upland headwater catchments, whereas species occurring across the surrounding landscape tended to be evenly distributed generalists (Sutcliffe, 2003; see also Storey et al., 1990). Therefore, despite intermittent streams being common across med-regions, the loss of perennial streams is expected to reduce biodiversity significantly and would likely cause a shift in biogeographic affinities of the dominant species with loss of relictual species.

Expected impacts of flow regime change on macroinvertebrates

Understanding the changes to macroinvertebrate assemblages, life histories and functional organisation following hydroregime shifts is important for understanding the implications of climate-induced drying on the functioning of stream ecosystems. The few studies examining direct impacts of flow regime drying on macroinvertebrates have found that assemblages were greatly altered, including losses of species and cascading effects from loss of long-lived, poorly-dispersing predators and shredders (e.g. Bogan & Lytle, 2011; Pace et al., 2013; Crabot et al., 2021). Biotic assemblages have been observed to become more homogenized (Piano et al., 2020) or more spatially heterogenous (Bruno et al., 2019; Crabot et al., 2021) in response to drying. Therefore, the effects of flow regime change from perennial to

intermittent in SWA headwater streams may have resulted in the following: loss of species sensitive to drying (e.g. Bogan & Lytle, 2011; Crabot et al., 2021); restriction of sensitive species to perennially flowing refuges (e.g. Bogan et al. 2014; Robson et al. 2011; Chester et al. 2015); species with resilience traits occurring in refuge pools (Sheldon et al., 2010; Chester & Robson, 2011; Bogan et al., 2019; Bonada et al., 2020), and persistence through the dry season of taxa that are resistant to drying as dormant or desiccation-resistant life stages (Boulton et al., 1992; Steward et al., 2012). Upon resumption of flow, resilient taxa will begin recolonising stream reaches, whereas taxa with poorer powers of dispersal will be more limited in distribution and may become restricted to remnant perennial streams (Chester et al., 2015; Cañedo-Argüelles et al., 2015). However, given that drier climates have previously occurred in SWA (Bunn & Davies, 1990), the existing fauna may have resistance or resilience traits, and the ability to adapt their life histories to the new flow regime. It is likely that drought refuges will become important in sustaining resilient species in newly intermittent streams (Chester & Robson, 2011), whereas potential evolutionary refugia will become critical in sustaining sensitive relictual species (Keppel et al., 2012; Davies et al., 2013) which have very limited opportunity to move to higher latitudes or altitudes (Davies, 2010).

Studying the ecological impacts of global warming on streams

For invertebrates, and particularly insects, long term ecological datasets (i.e. repeated sampling over decades) are truly rare (Didham et al., 2020a; e.g. Hallman et al., 2020 and references therein). This is also true of freshwater insects (Jackson & Fureder, 2006), creating difficulties for the quantification of perceived insect declines (Didham et al., 2020b). Repeated sampling over a decade or more can clearly show the impacts of gradual changes, including the gradual onset of drier flow regimes in naturally intermittent streams (e.g. Pace et al., 2013). Sampling over shorter timeframes may elucidate impacts of threshold shifts,

such as sudden loss of perennial water and changed hydroregime (e.g. Bogan & Lytle, 2011). In the absence of repeated sampling, comparisons of historic and contemporary datasets in a before-after study design can show very clearly differences in faunal assemblages caused by either threshold or gradual changes, such as the impact of incremental warming over decades on tropical stream communities (Dudgeon et al., 2020). Where possible, multiple before-after, control-impact designs should be used to examine threshold or stepwise environmental changes, whereby the impacted systems are sampled over time alongside comparable unimpacted systems, because these designs isolate climate impacts from other changes that may have occurred during that time (Downes et al., 2002).

Didham et al., (2020b) discussed seven key challenges to the valid interpretation of insect declines in the present era of environmental change and provide a framework for developing robust designs within the limitations of a particular study. For example, the “false baseline effect” can occur when measurement begins after the perceived change is already happening, or, if the historical dataset was collected for an unrelated ecological question (or impact) and is thus biased. The “snapshot effect” can undermine time-point comparisons, particularly with paired observations of abundance (Didham et al., 2020b). Separating climate impacts from the multitude of other anthropogenic stressors acting on an ecosystem also poses challenges, especially in ecosystems where multiple anthropogenic stressors are the norm (unfortunately true for most freshwater ecosystems; Dudgeon et al., 2006; Omerod et al., 2010; Datry et al., 2018). Studies that separate potentially confounding impacts from climate change effects are valuable for predicting species responses to climate change in ecosystems experiencing multiple impacts. For example, isolating the differing (but potentially synergistic) effects of flow regulation from effects of intermittency caused by global warming (Belmar et al., 2018; Bruno et al., 2019, e.g. Chester et al., 2014). Space-for-time studies, or experimental manipulations of flow regime within mesocosms may also fill knowledge gaps

pertaining to the responses of species or communities on small scales, which can elucidate potential impacts at larger scales (e.g. Hille et al., 2014; Aspin et al., 2018). So, for any particular effect of global warming, a plurality of approaches will provide the most robust understanding of the range and type of impacts on a particular type of ecosystem. Studies from regions of the globe that are further along the trajectory of a particular climate change (in this case climatic drying) can provide a basis for predicting likely future outcomes in other regions that are currently less impacted. Such a ‘preview’ of the future may then enable ecosystem managers to put in place strategies for mitigating negative impacts, such as species losses. Therefore, the role of the research in this thesis is to quantify the effects of flow regime change on invertebrate communities in headwater streams in a region that has already experienced the effects of many decades of climate drying caused by global warming. It takes advantage of research conducted in headwater streams before drying caused flow regime change in most streams and it uses many of the same stream reaches in an mBACI design to separate the effects of flow regime change from other temporal changes. The prior research used (see below) was collected as a baseline, being the first exploration of invertebrate communities in streams in this region and is therefore fit for purpose. The streams lie entirely within forested protected water catchment, so human impacts other than climate change were minimal. Therefore, the results from this study provide a ‘preview’ of the effects of permanent flow regime change on headwater stream invertebrate communities subject to climate drying and complement conclusions from microcosm (e.g. Glasheen et al., 2017), mesocosm (e.g. Aspin et al., 2018) and field (e.g. Bogan et al., 2019) studies of the effects of drought and drying flow regimes on headwater stream ecosystems.

Research aims and thesis structure

This thesis aims to describe the *in situ* ecological impacts of flow regime change from historically perennial to seasonally intermittent flows on the macroinvertebrate fauna of

forested headwater streams. This research was conducted in southwestern Australia (SWA), a med-climate region that has undergone precipitation decline and profound changes to stream hydrology in recent decades (Petrone et al., 2010; Hughes et al., 2012; Andrys et al., 2017; MacFarlane et al., 2020). I examined changes to macroinvertebrate assemblage composition, patterns of biodiversity, life history phenology, contribution to leaf litter breakdown, and the role of dry season refuges in maintaining populations in recently intermittent streams (Figure 1.1). The temporal comparison in this study was possible due to the existence of a substantial body of work examining macroinvertebrate ecology of these streams prior to the onset of drying (Bunn's 1985 PhD thesis and associated publications: Bunn 1985; Bunn 1986; Bunn et al., 1986; Bunn 1988a; Bunn 1988b; Bunn 1988c). All but one of Bunn's study streams transitioned to intermittent flows in the early 2000s (see figures in Chapter 2: Study Area, Table 1.1). This provided an opportunity to gather empirical data on the direct effects of flow regime drying on macroinvertebrates, using a partial multiple before-after, control-impact study design (mBACI; see Downes et al., 2002). Importantly, one stream in the catchment and its tributaries have maintained perennial flows since records began, providing the essential 'control' reaches in the mBACI design, and allowing separation of the effects of drying from other potential impacts, such as warming. Such circumstances allowing direct comparison before and after a defined climate change impact are rare in stream ecology (*sensu* Jackson & Funderer, 2006), not least due to the requirement for quality "before impact" data coupled with climate change data (here, long term flow gauging data). Importantly, the study streams are in a forested water supply catchment, free from confounding environmental disturbances for at least four decades. Bunn conducted his pioneering research for the purpose of describing macroinvertebrate ecology, well prior to observed impacts of climate change on streamflow (thus avoiding the "false baseline" effect, *sensu* Didham et al., 2020a). There is a lexicon of definitions used to describe different non-

perennial flow regimes, encompassing differences in duration, predictability and cyclicity of flows (Costigan et al., 2017; Datry et al., 2017) however in this thesis, non-perennial streams are referred to as intermittent unless a more specific definition is required.

This research tested predictions of macroinvertebrate assemblage composition and biodiversity patterns (α , β) in response to flow regime drying (Chapters 3 and 4), assessed the validity of life history traits in predicting species responses (a key input into species distribution models; Chapter 5) and assessed the contribution of drought refuges to community persistence in newly intermittent streams (Chapter 7, Table 1.1, Figure 1.1). These comparisons enabled me to determine whether newly intermittent streams showed patterns expected in intermittent rivers and ephemeral streams (IRES, Figure 1.1). Because Bunn (1985; 1988) had conducted a leaf breakdown study, I was also able to examine whether there were differences in leaf litter decomposition in streams that were formerly perennial, but now intermittent. Leaf litter breakdown is a key ecosystem process moderated by macroinvertebrates that may be affected by flow regime drying (Figure 1.1). This was also an opportunity to examine impacts of flow regime drying on ecosystem function (Chapter 6). How this process responds to flow regime change has rarely been observed and is thus poorly understood. Impacts of climate change on ecosystem processes are of increasing interest as global warming progresses.

A stand-alone literature review chapter was not included in this thesis, to limit repetition of information provided in the introductions of the following chapters. Thesis chapters 3 – 7 are structured as journal manuscripts: two chapters are published and three are awaiting submission to journals (Table 1.1). Chapter 2 provides additional information on the study streams and catchment not included in the journal manuscript chapters. Chapter 8 synthesizes the conclusions of the research from chapters 3 – 7 (Table 1.1).

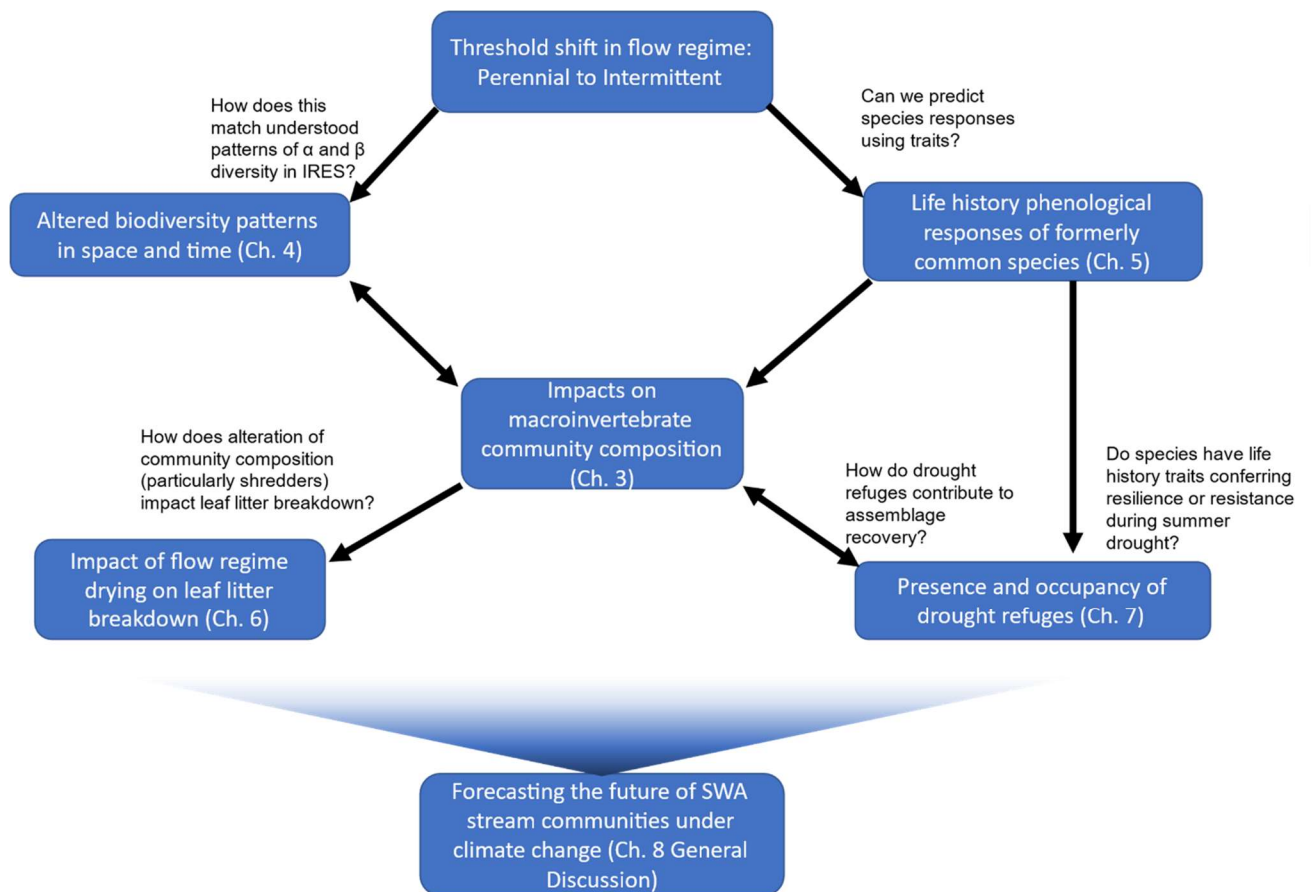


Figure 1.1: Schematic model of thesis chapters, including conceptual links. This thesis aimed to capture multiple elements of ecosystem response to flow regime change in NJF streams, as such there are links between each component that are not shown here.

Table 1.1: Summary of thesis structure, including key aims and publication status of research chapters.

Chapter title	Aims
Ch. 1 General Introduction	Introduction to the thesis and conceptual background
Ch. 2 Study sites	Provide information on the study streams, region and stream macroinvertebrate fauna
Ch. 3 “Loss of functionally important and regionally-endemic species from streams forced into intermittency by global warming” Target journal: <i>Global Change Biology</i> (in prep.)	Quantify and describe <i>in situ</i> impacts of flow regime drying on NJF stream communities at two temporal scales: 1) differences between contemporary assemblages and those historically present; and 2) seasonal patterns over one year between PER and INT streams.
Ch. 4 “Reduced alpha diversity and increased turnover (β -diversity) in formerly-perennial headwater streams affected by permanent drying” Target journal: <i>Oecologia</i> (in prep.)	To compare patterns of α (richness) and β diversity (turnover and nestedness) between PER and INT streams at two scales: spatial patterns within contemporary assemblages; and temporal patterns between 1981/82 (pre-intermittency) and contemporary assemblages at the same sites. This is a partial mBACI study design, PER = control reaches, INT = impact reaches.
Ch. 5 “Life-history traits are poor predictors of species responses to flow regime change in headwater streams” Published: Carey, N., Chester, E. T., Robson, B. R. (2021). <i>Global Change Biology</i> 27: 3547-3564 DOI: 10.1111/gcb.15673	To test predictions of life-history responses of 13 common macroinvertebrate species (12 insects and 1 crustacean) to <i>in situ</i> flow regime change, based on documented life histories and published traits (generally congeners or confamilials) as a predictor of future response. Size distributions, phenology, voltinism and synchrony were compared between a historic (1982/83) study and present (2016/17), between PER and INT reaches, thus comprising a partial mBACI study design.
Ch. 6 “Flow regime change alters shredder identity but not leaf litter decomposition in headwater streams affected by severe, permanent drying” Published: Carey, N., Chester, E. T. & Robson, B. J. 2021. <i>Freshwater Biology</i> 66(9): 1813-1830. DOI:10.1111/fw.b.13794	Examine changes to the decomposition process of jarrah leaves (key carbon source) and the role of macroinvertebrate shredders in streams impacted by flow regime drying, using a leaf bag experiment and two pre-planned comparisons (between contemporary PER and INT; temporal comparison between historic (1981/82) and contemporary breakdown patterns in PER and INT; an mBACI design).
Ch. 7	Identify drought refuges in INT streams and characterise the macroinvertebrate species using them. Conduct sediment rehydration experiments to identify taxa using desiccation resistant

Chapter title	Aims
“Drought refuges, refugia and flow-regime change in southern Australian streams”	life stages (e.g. eggs and aestivation). Determine the contribution of drought refuges to community recovery, compared to colonisation from the remaining PER stream.
Target journal: <i>Freshwater Biology</i> (in prep.)	
Ch. 8 General Discussion	Summarise key findings of Chapters 3-7 Synthesise information to project future changes to NJF stream communities under climate change

Chapter 2

Study area

The Wungong Catchment

The Wungong catchment is situated within the Northern Jarrah Forest (NJF), which extends along the Darling Scarp east of the Swan Coastal Plain (SCP) (centre of study area: S32°14.04'' E116°05.70''; Figure 2.1). The Darling Scarp is the western boundary of the Yilgarn Craton which covers much of the interior of southern Western Australia (Cassidy et al., 2006). The Precambrian Yilgarn Craton is a vast area of weathered, gently undulating landscapes with ancient drainage systems, salt lakes and few large rivers (Bettenay & Mulcahy, 1972). Only towards the western edge do valleys become steeper and short, created by east to west flowing rivers that flow over the Darling Scarp to the lowland SCP (Bettenay & Mulcahy, 1972). Jarrah forest covers the western edge of the plateau (up to the Scarp) however to the east, extensive clearing for agriculture has created a forested “island” which houses protected drinking water catchments, including the Wungong catchment (Figure 2.1).

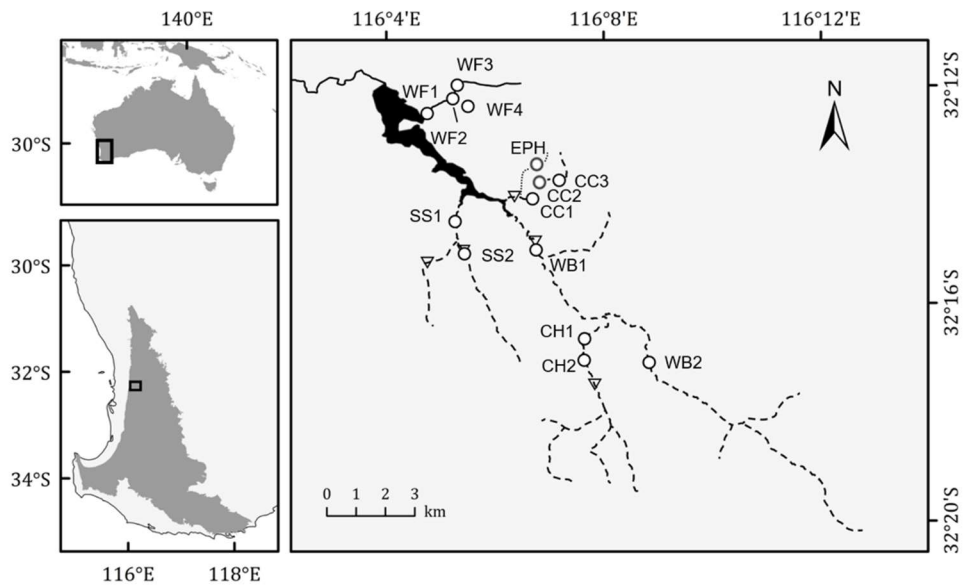


Figure 2.1: Study reaches in the Wungong Brook catchment of south-western Australia, within the Northern Jarrah Forest: perennial reaches (solid lines), formerly perennial, now intermittent reaches (dashed lines), and an ephemeral stream (EPH, dotted line) in panel A. Inverted triangles denote flow gauging stations (flow gauge at WF1 not visible). Sites used by Bunn (1988) were WF1, WF2, CC1, CC2, SS1, SS2, CH1 and CH2; additional sites sampled here were WF3, WF4, CC3, WB1, WB2 and EPH. The extent of the Northern and Southern Jarrah forest subregions shown (grey shading, middle left panel) however SWA has been subject to extensive clearing for agriculture and urban development; the true extent of forested area is shown in panel B (Google Earth imagery).

Most of the substantial creeks and rivers within the NJF have been dammed for water supply to Perth, surrounding towns and the Kalgoorlie goldfields, disconnecting upland headwaters

from lowland drainage. The other major land-uses within the NJF presently include conservation reserves and strip mining for bauxite (which is followed by revegetation using stored topsoil). However, the Wungong catchment has not been affected by mining for decades, and revegetation of the southern areas was completed in the 1980s. Historically, much of the jarrah forest (*Eucalyptus marginata* Wood) was been logged and the Wungong is not an exception – there is a legacy of logging and native forest regrowth since the 1870s (Ward, 2010). However, the Wungong has been a protected water supply catchment since 1909 so anthropogenic impacts on stream communities (aside from climate change) are negligible.

Climate, geology and streamflow generation

Southwestern Australia (SWA) experiences a mediterranean climate, with mild wet winters and prolonged summer drought. There is a distinct rainfall gradient across the NJF, average annual rainfall increases from around 700 mm on the eastern edge of the jarrah forest, to a maximum of around 1300 mm 10 to 12km east of the Scarp (Gentili 1989). Rainfall is unusually predictable by Australian standards (Bunn *et al.* 1986), however it has steadily declined since the 1970s (Silberstein *et al.*, 2012; Kinal & Stoneman, 2012; McFarlane *et al.*, 2020).

Rainfall totals measured at the study sites (Wungong Dam, Bureau of Meteorology station 009044) have steadily declined since records began in 1913 (Figure 2.2). Between 1913 and 1960, annual mean rainfall was 1386 mm dropping to 1011.4 mm between 1990 and 2017 – a 27% reduction. Unfortunately, there are no data for 1981/82 from station 009044, but at nearby Jarrahdale (Bureau of Meteorology station 009023) total annual rainfall in 1981 1222.8 mm and 1982 901.4 mm, compared with 1091.2 mm in 2016 and 1197.4 mm in 2017.

Therefore, neither this study nor Bunn's was conducted in a year of exceptionally high or low rainfall.

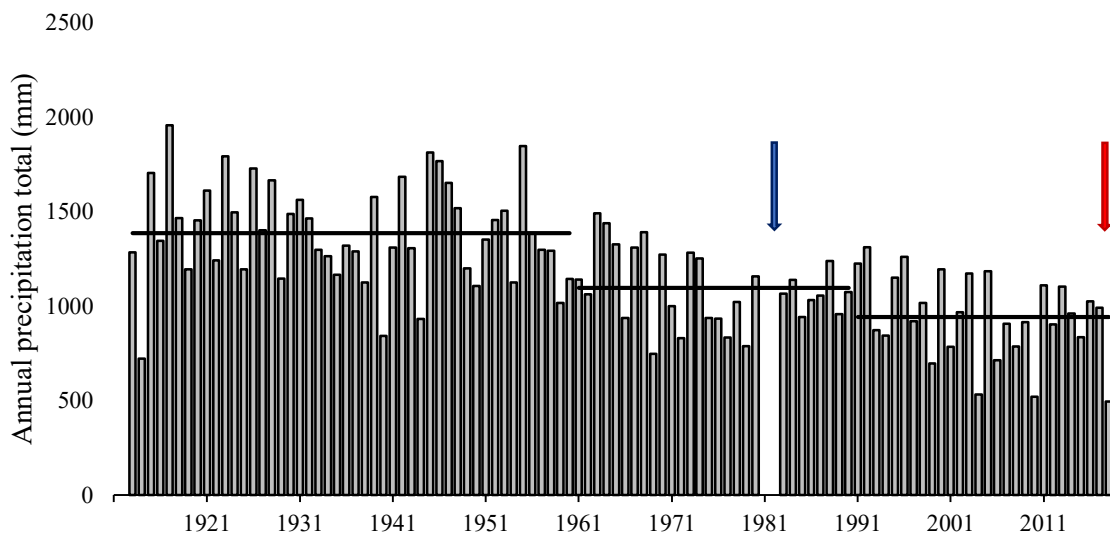


Figure 2.2: Annual rainfall total 1913 to 2018, recorded at Wungong Dam. Horizontal bar shows long term average rainfall at the site, 1913 to 1960; 1961 to 1990; and 1991 to 2018. Years 1981 and 1982 are missing data. Graph produced using rainfall data freely available from the Australian Bureau of Meteorology (station number 009044; bom.gov.au/climate/data). Blue arrow indicates timing of Bunn's study (1981/82), red arrow indicates timing of this study (2016/17).

The surface geology of the NJF consists of gravelly sands overlaying laterite duricrust in the uplands, interspersed with areas of outcropping granite (Churchwood & Dimmock, 1989). Alluvium collects in the lowlands, forming shallow swampy tracts on the Darling Plateau that gradually deepen westwards forming deep valleys along the Scarp (Churchwood & Dimmock, 1989). In the jarrah forest, this permeable weathered material is generally 20 to 70 metres deep over the weathering front (McArthur 2004). Winter rainfall infiltrates readily into the soil and may form perched aquifers on clay horizons, and then flows downslope to stream discharge zones. The soils have huge capacity to store water, which jarrah trees readily use over the dry summer months. Altogether, the combination of historically high

winter rainfall, soil water storage and high summer evapotranspiration rates have typified streamflow dynamics in the jarrah forest region (Schofield et al., 1989; Gentili, 1989).

This high soil water storage capacity has caused streamflows to be lower than might be expected from moderate (over 1300 mm at the study area) annual rainfall (Schofield et al., 1989), but these flows have historically been highly predictable and mostly perennial (Bunn 1985).

SWA has experienced progressive declines in annual precipitation since the 1970s, with less average winter rainfall, elimination of ‘wet years’, and more frequent anomalously dry years. Altogether, the post-1975 average rainfall is around 20% lower than the 1900-1975 average (Bates et al., 2008; McFarlane et al., 2020). With the onset of climatic drying, declines in rainfall (Figure 2.2), soil moisture and water tables have produced a disproportionately large decline in streamflow than expected from the 20% decline in precipitation (Silberstein et al., 2012; Ruprecht, 2018; Liu et al., 2019; Harper et al., 2020). The mechanism behind the up to 70% decline in streamflow caused by a 20% decline in rainfall is disconnection of streams from the groundwater inputs that supply baseflow in summer and autumn. In most Wungong catchment streams, this disconnection (causing flow cessation) manifested in the years 2000 to 2010 and caused most streams to cease flow in summer for the first time since records began (Petrone et al., 2010). Much of this streamflow decline has occurred since 2000 (flows in 2001-2012 were 36-52% less than in 1975-2000 across SWA streams; Ruprecht, 2018) which is independent of interannual rainfall variation (Kinal & Stoneman, 2012; Ruprecht 2018).

Since 2000, significantly higher winter rainfall volumes have been required to saturate stream discharge zones and form perched aquifers. This has reduced streamflow, causing later onset of flows, and has forced many streams to switch from perennial to intermittent flow regimes,

and made intermittent streams increasingly ephemeral (Petrone et al., 2010; Ruprecht 2018 and references therein). These threshold changes in flow regime are expected to be permanent and will intensify as rainfall continues to decline and catchments become drier (Reed et al., 2012; Andrys et al., 2017; Ruprecht 2018; McFarlane et al., 2020).

Catchments in the NJF have, until recent decades, supplied a large component of potable water to the city of Perth and surrounding regions (1890s to 1970s inclusive), but inflows to reservoirs have decreased in a stepwise fashion (Figure 2.3) to the extent that the catchments are no longer used for water supply (Petrone et al., 2010; Silberstein et al., 2012; Water Corporation 2020). Long term streamflow gauging data show clearly the transition of formerly perennial streams to intermittency (Figure 2.4).

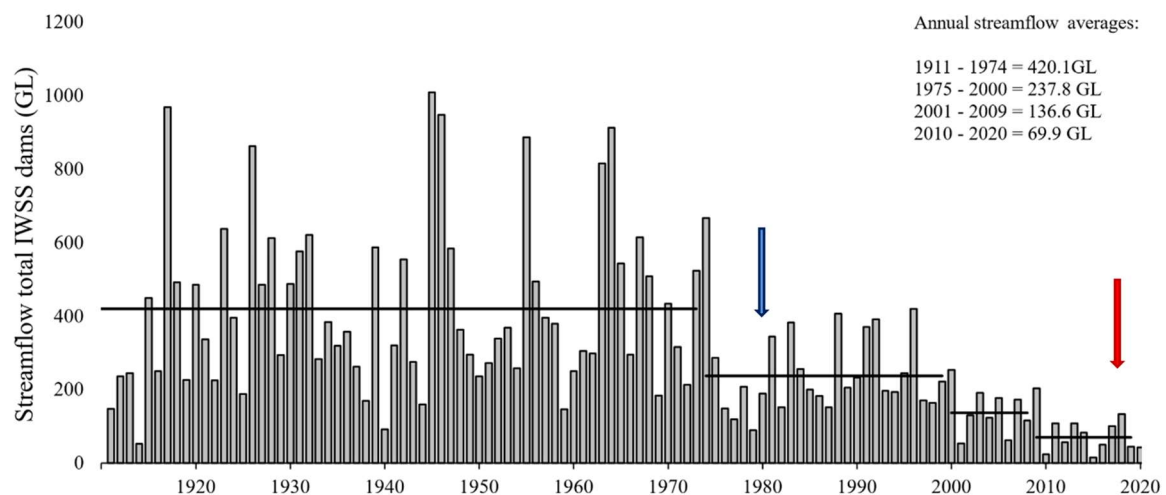


Figure 2.3: Total annual surface water inflows (GL/year) into water catchment dams on the Darling Scarp, formerly the sole source of potable water for Perth's Integrated Water Supply Scheme (IWSS). Streamflow totals have progressively dwindled to a fraction of the historical average, despite construction of new dams in the 1980s and 1990s. Data freely available from the Water Corporation (watercorporation.com.au). Blue arrow indicates timing of Bunn's study (1981/82), red arrow indicates timing of this study (2016/17).

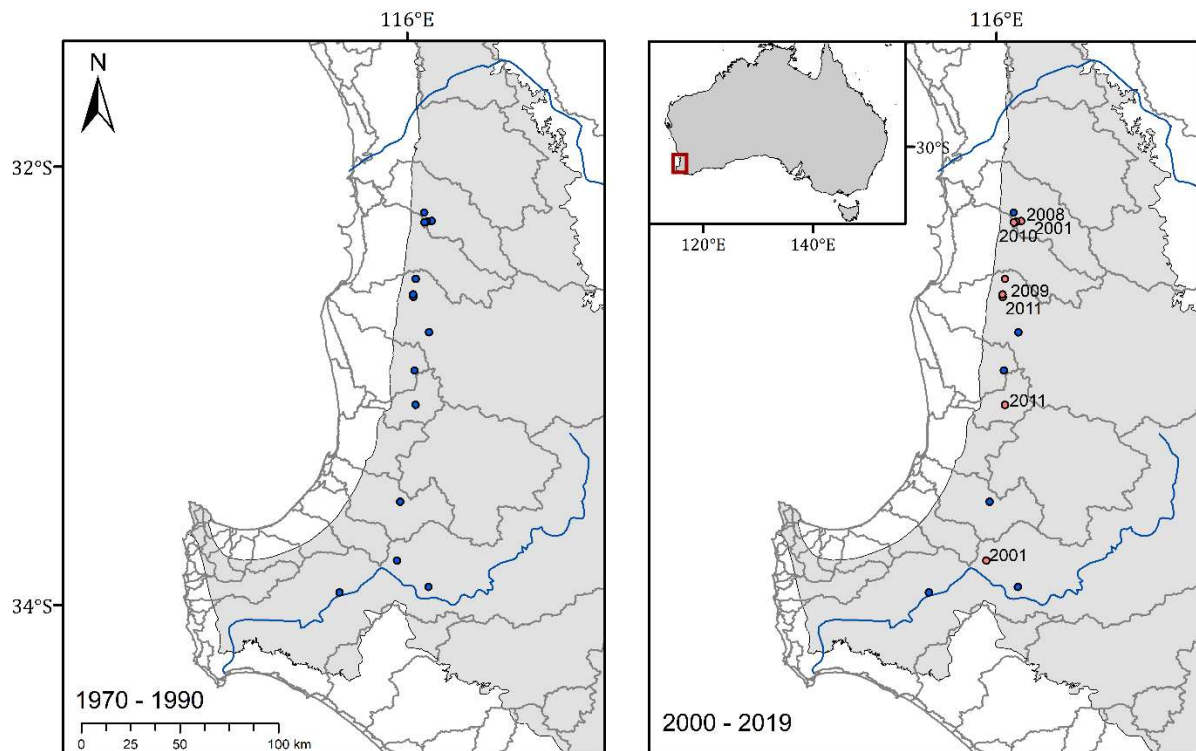


Figure 2.4: The Darling Scarp runs approximately between the Swan-Avon River (blue line North) and the Blackwood River (blue line South), bounded to the west by the Swan Coastal Plain. Grey shaded areas delineate the Northern and Southern Jarrah forests (approximately the extent of the Darling Plateau). Left panel shows low order streams with at least 50 years of gauging data, that were perennially flowing 1970 – 1990 (blue points mark gauging stations). Right panel shows current flow status at the same gauging stations, blue = perennial flow, orange = formerly perennial, now intermittent flows, with first year of drying shown. Note this is not an exhaustive list of perennial/ formerly perennial streams on the Darling Scarp, only those with complete flow gauging data (wir.wa.gov.au; stations see Table S2.1).

The SWA stream macroinvertebrate fauna

The stream fauna of SWA is primarily a Gondwanan relictual fauna, having evolved in cooler, wetter times (Davies & Stewart, 2013; Stewart et al., 2013). Whilst diversity is lower

in SWA than in southeastern Australia, endemism is very high (Bunn & Davies, 1990; Davies 2010). For example, all stream dwelling Leptophlebiidae mayflies and Gripopterygidae stoneflies are endemic, over 75% of the caddisflies, and a large proportion of dragonflies, which include several endemic genera many of Gondwanic origin (Bunn & Davies, 1990; Sutcliffe 2003; Hawking & Theischinger, 2004; Christidis 2006; Cartwright et al., 2013; Davies & Stewart, 2013). These attributes arise from a protracted period of geographic isolation, as the southwest is bounded by vast areas of land (Gascoyne, Midwest, Little Sandy Desert, Great Victoria Desert, Mallee, Nullarbor Plain) that have largely lacked permanent surface water for millennia (Bunn & Davies, 1990; Morton et al., 2011) and by the Southern and Indian Oceans.

Historically, the unusually predictable hydrological patterns in SWA streams was associated with predictable seasonal patterns in freshwater invertebrate communities with marked differences between winter high flows and summer baseflow (Bunn et al., 1988).

Considerable differences in species composition have been observed between adjacent catchments, depending on hydrology (perennial vs. intermittent) and the potentially limited ability of some fauna (e.g. amphipods) to disperse between upland catchments (Bunn et al., 1986; Storey et al., 1990). Species composition in stony upland headwater streams is also distinct from downstream lowland reaches flowing across sandplain (SCP) (Storey et al., 1990) and many species are restricted to forested streams in the Darling Scarp (e.g. Dean & Bunn, 1989; Sutcliffe 2003). Generally, perennial stream assemblages are more temporally persistent than those inhabiting seasonal streams, where species occurrences are more transient in space and time (Bunn & Davies, 2000). Because of their relictual lineage and geographic boundaries, many SWA stream fauna are thought to already be near their upper thermal maxima and at risk from further increases in water temperature (Davies 2010; Stewart et al., 2013). Indeed, streams running through cleared agricultural landscapes in

SWA are already too warm and many relictual species are no longer found there (Cook et al., 2013). So, although Gondwanan relict species in SWA have withstood drier times in the geological past (Bunn & Davies, 1990), they may be vulnerable to the impacts of climate drying because of increased water temperatures (decreased depths), prolonged periods of drying in historically and newly intermittent streams and substantial losses of hydrological (dams, dry sections of streambed, river salinisation) and terrestrial connectivity (land clearing for agriculture across SWA). Indeed, studies have already shown the negative effects of loss of hydrological connectivity on the distribution and gene flow of endemic freshwater species in SWA (e.g. mussels Klunzinger et al., 2015; frogs Cummins et al., 2019).

Flow regimes in Wungong catchment stream reaches

Flow in the Wungong Brook catchment has been gauged since the 1960s. Gauging data shows clearly the changes to stream flow regimes: decline in annual discharge across all streams (Figure 2.5), transition from perennial to seasonal flow in most reaches (see below), later commencement of flow (previously onset of increase from baseflow) and later peak flows in all streams (Figure 2.6). One stream, Waterfall Gully, remains perennial although annual discharge has declined in recent decades (Figure 2.5a). Furthermore, stream discharge now increases from baseflow in April where formerly it occurred in March, and peak flows now occur in September where formerly they occurred in August (Figure 2.6a). The remaining streams that were formerly perennial are now seasonally flowing. In the newly intermittent streams, flow commences in June instead of baseflow increasing in March-April, and peak flows occur in September instead of July-August (Figure 2.6b).

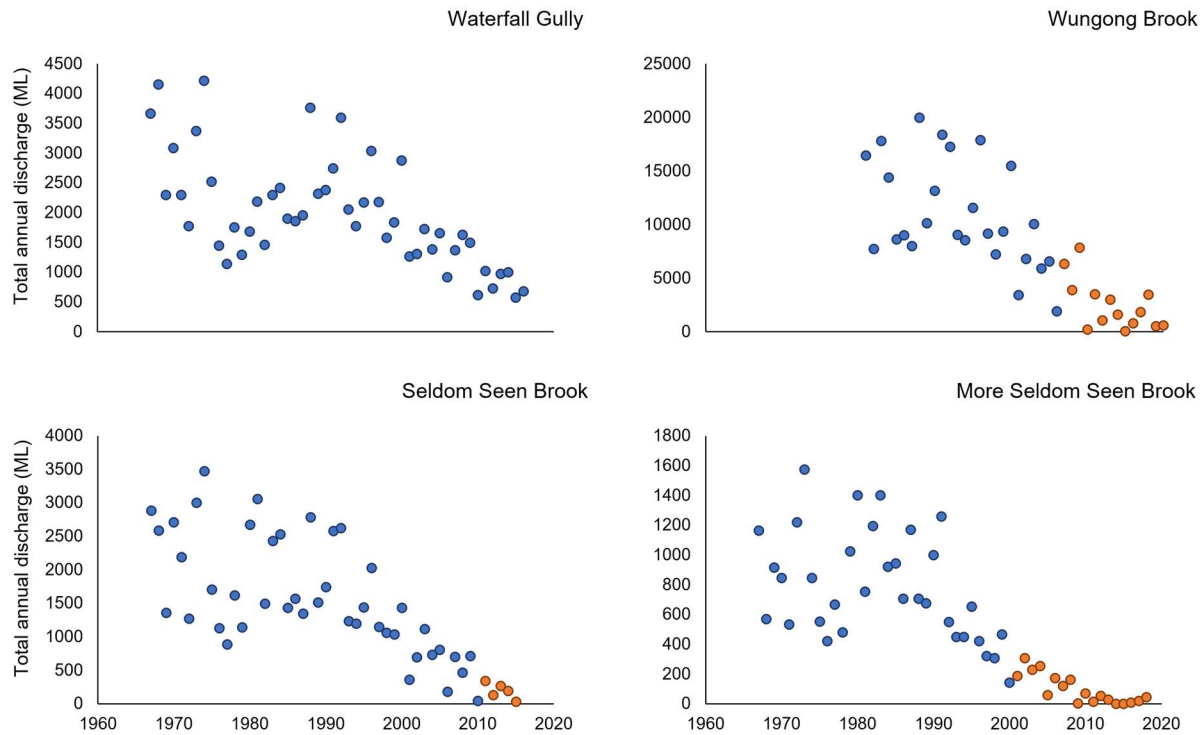


Figure 2.5: Total annual discharge (in ML; excluding years with extensive missing data), based on daily discharge measurements from four streams in the Wungong catchment, gauging began from 1982 (Wungong Brook, B) and 1967 (Waterfall Gully, A; Seldom Seen Creek, B; More Seldom Seen Creek, C) to most recent year available. Note differing scales on y-axes. Blue indicates years with perennial flow, orange indicates years post-flow regime drying (onset of intermittency). Data supplied by the WA Department of Water (www.wir.wa.gov.au; station information see Appendix 1).

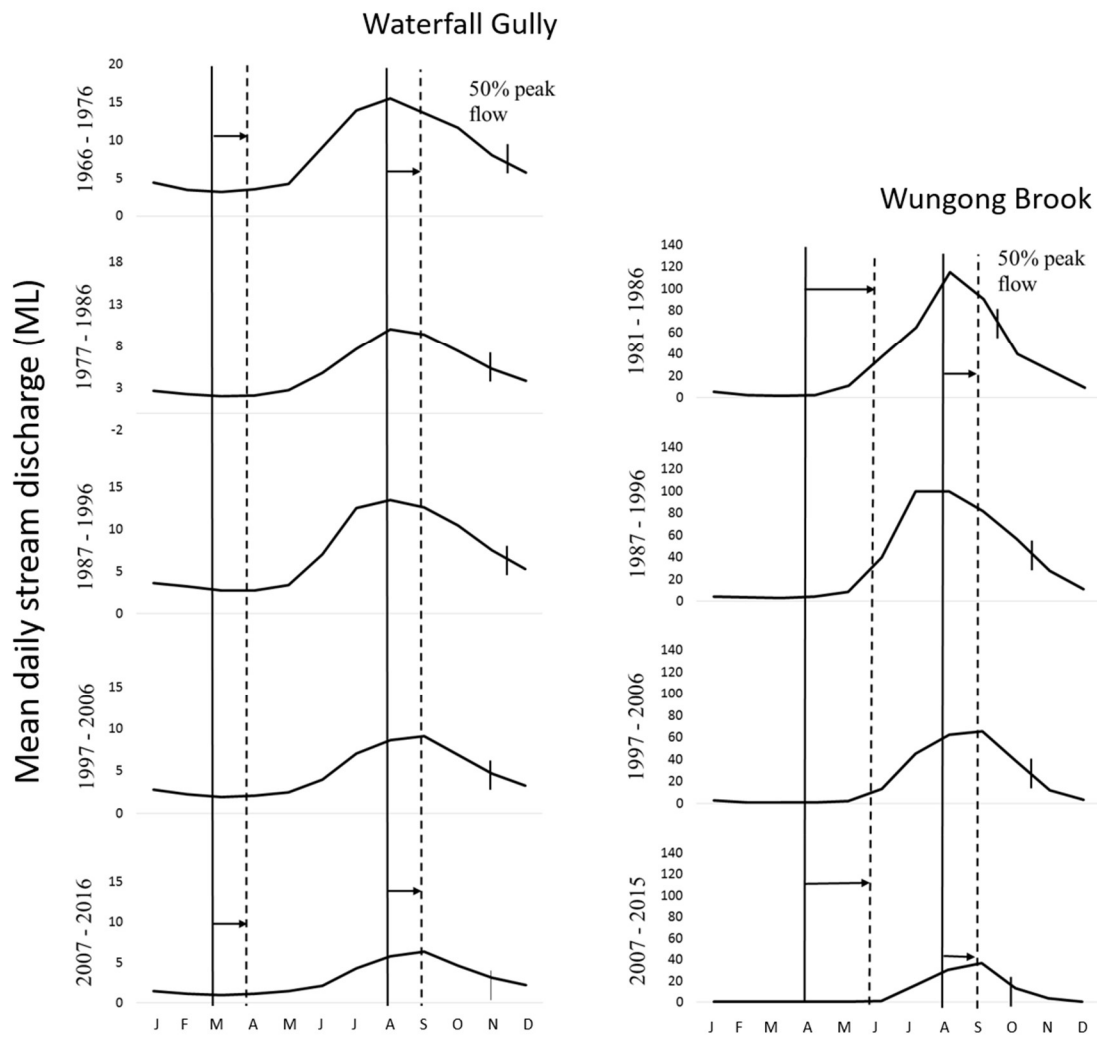


Figure 2.6: 10 year mean daily discharge at Waterfall Gully (left) and Wungong Brook (right). Solid vertical line indicates average onset of increased flows (left) and peak flows (right) in 1981-1990 (Wungong Brook) or 1967-1976 (Waterfall Gully), dashed vertical line indicates 2007-2016 timing. Date of 50% peak flow indicated by short vertical line, showing flows also taper off faster post-2006. Data freely available from the Western Australian Department of Water and Environmental Regulation (wir.water.wa.gov.au; gauging stations 616023 and 616041).

Study streams

Thirteen study sites were studied spanning the Wungong Brook and its four major tributaries, including eight study sites used by Bunn (1985) and an additional five sites, to capture the greater heterogeneity of flow regimes post-drying (Table 2.1; Figure 2.1). Below, each study stream and associated sites are described, and hydrological information presented.

Table 2.1: Historic (1981/82) and extant (2016/17) flow regimes at thirteen study reaches on five streams in the Wungong Brook catchment, including study reaches used by Bunn (1985, 1986, 1988) and Bunn et al., (1986), and five additional study reaches studied here. Historic flow regime information from Bunn (1985) or flow gauging data (wir.wa.gov.au); SR= study reach; FR = flow regime; superscript ‘A’ denotes additional reaches without historic flow information (assumed perennial). Number of sampling events where surface water present for sampling also given; (*) denotes dry season remnant pools located outside of, but nearby main study reach. Reaches on each stream were numbered from downstream to upstream.

	Site codes	Bunn SR	FR 1981/82	FR 2016/17	Sampling times
Waterfall Gully	WF1	Y	Perennial	Perennial	All
	WF2	Y	Perennial	Perennial	All
	WF3	N	Perennial	Perennial	All
	WF4	N	Perennial ^A	Perennial	All
‘Kunzia Creek’ (WF trib)					
Seldom Seen Brook	SS1	Y	Perennial	Intermittent	1,2,7
	SS2	Y	Perennial	Perennial	All
Curtis Creek	CC1	Y	Perennial	Intermittent	1,2,7
	CC2	Y	Perennial	Intermittent	1,2,3,6,7
	CC3	N	Perennial ^A	Intermittent	1,2,3,6,7
Wungong Brook	WB1	N	Perennial	Intermittent	1,2,3,4,5*,6*,7
	WB2	N	Perennial	Intermittent	1,2,3*,6,7
Chandler Brook	CH1	Y	Perennial	Intermittent	1,2,7
	CH2	Y	Perennial	Intermittent	1,2,7

Waterfall Gully

Waterfall Gully (WF sites; Figure 2.1) is the only stream in the study catchment that has not dried since records began (Figure 2.7; DWER gauge 616023). Three sites were located on Waterfall Gully: WF1 and WF2 are the same study sites used by Bunn in the early 1980s (sites 3 and 4 in his thesis); WF3 is situated in the upper reaches, and WF4 is situated on a tributary (Kunzia Creek). WF1 and WF2 were usually 30-50 cm deep, sediments were generally sandy/gravel with patches of cobbles and boulders (Figure 2.8), WF4 also had areas of outcropping granite bedrock. Dense riparian vegetation (teatree, *Agonis* sp.) at WF3

created a closed canopy, with deep organic substrates (up to 1 m deep in places). Kunzia Creek is a small, perennially flowing tributary, meeting Waterfall Gully just below WF2. Riparian vegetation (teatree and various sedges) created a closed canopy over small, braided channels seldom deeper than 20 cm, sediments comprised sand and organic matter.

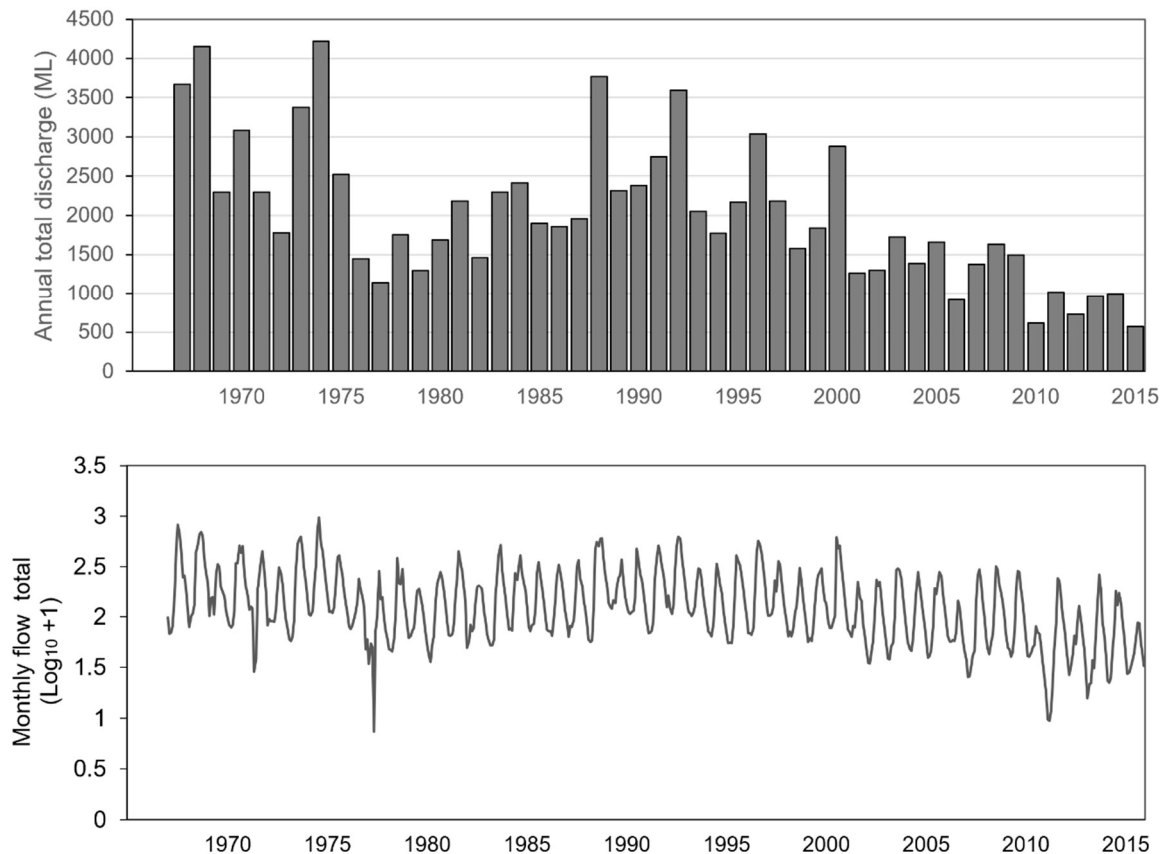


Figure 2.7: Annual total discharge (ML) at Waterfall Gully from 1967 to 2016 (top). Bottom: monthly streamflow totals at Waterfall Gully 1970 to 2016, log-transformed to clearly show there have been no months of zero flow since records began. All graphs produced using daily streamflow measurements, freely available from the Western Australian Department of Water and Environmental Regulation (gauge number 616023; wir.water.wa.gov.au). Note the apparent low-flow period in 1977 is actually due to missing data during that year.



Figure 2.8: Waterfall Gully is the only remaining perennial stream in the Wungong catchment (photos are of WF3). All reaches are densely vegetated, shallow (generally less than 50 cm) and had sand/gravel/organic sediments.

Seldom Seen Brook

Seldom Seen Brook dried for the first time in 2010, and whilst it has dried several times between 2011 and 2015 (Figure 2.9, DWER gauge 616021) in most years it retains a short, spring-fed perennial reach (at site SS2; Figure 2.1; Figure 2.10). The two sites on Seldom Seen Brook (SS1 and SS2) are both the same locations studied by Bunn (1985; sites 7 and 8). At the downstream end of this reach is a small weir pool and flow gauge, fortuitously positioned as summer refuge for aquatic species including a population of the mussel *Westralunio carteri* (listed as Vulnerable on the IUCN Redlist, IUCN 2021). Aside from this short reach, Seldom Seen Brook is ephemeral, with the downstream site SS1 retaining flows for only a short period in winter and spring. Both reaches were perennial when sampled by Bunn (1985).

The upper reaches are adjacent to a large granite inselberg, and the stream flows over bedrock and shallow regolith (Figure 2.10). SS1 is shallow (10-20 cm), with highly permeable sand/gravel/cobble sediments.

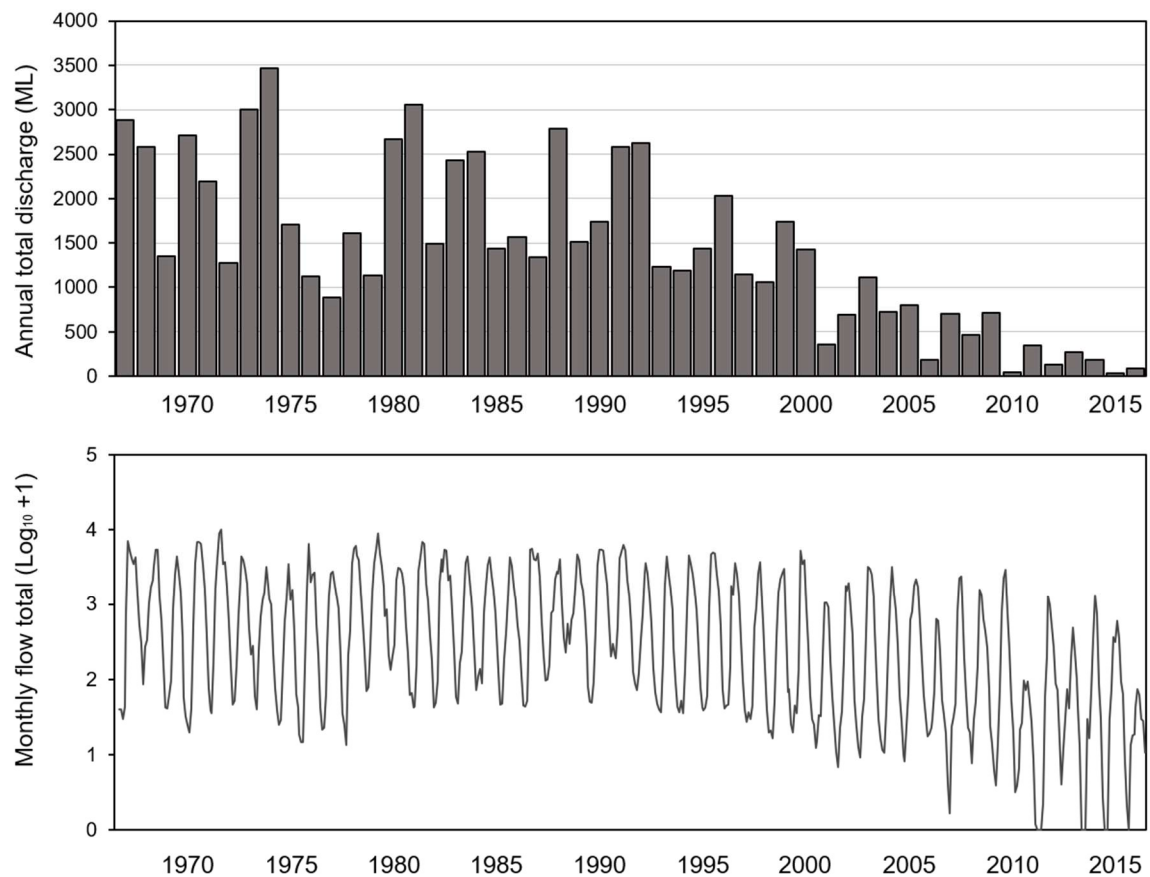


Figure 2.9: Annual total discharge (ML) at Seldom Seen Brook from 1967 to 2016 (top). Bottom: monthly streamflow totals at Seldom Seen Brook 1970 to 2016, log transformed so months of zero flow can be separated from those with very low flows. Seldom Seen Brook became intermittently flowing in 2011, however a spring on one reach maintained a trickle of perennial flow in some years since, including 2016-17. All graphs produced using daily streamflow measurements, freely available from the Western Australian Department of Water and Environmental Regulation (gauge number 616021; wir.water.wa.gov.au).



Figure 2.10: The spring-fed reach on Seldom Seen Brook is the only other perennially flowing water present in the Wungong catchment post-2010. The reach is shallow, following over bedrock, with small depositional areas (left). The flow gauging weir at the downstream end of this reach is a drought refuge in summer (top right). This section of Seldom Seen Brook runs along the northern edge of a low relief granite inselberg (bottom right).

Wungong Brook

The Wungong Brook is the largest waterway in the catchment and the only third order stream (WB sites, Figure 2.1). It has been gauged since the early 1980s, clearly showing the transition to intermittent flows in summer 2007 (Figure 2.11). The Wungong Brook does retain pools well into the dry season, some lasting until May (late autumn; Figure 2.12). The upper reaches (WB2) are wide and shallow, flowing over granite bedrock and interspersed with small cascades. The lower reaches (WB1) consist of deep pools broken by shallow riffle areas, which flow torrentially for a short period in mid-winter. Bunn (1985) did not have

sampling sites on the Wungong Brook (he erroneously named his study reaches on Chandler Brook as ‘Wungong Brook’), thus these two sites were additional in the 2016-2019 studies.

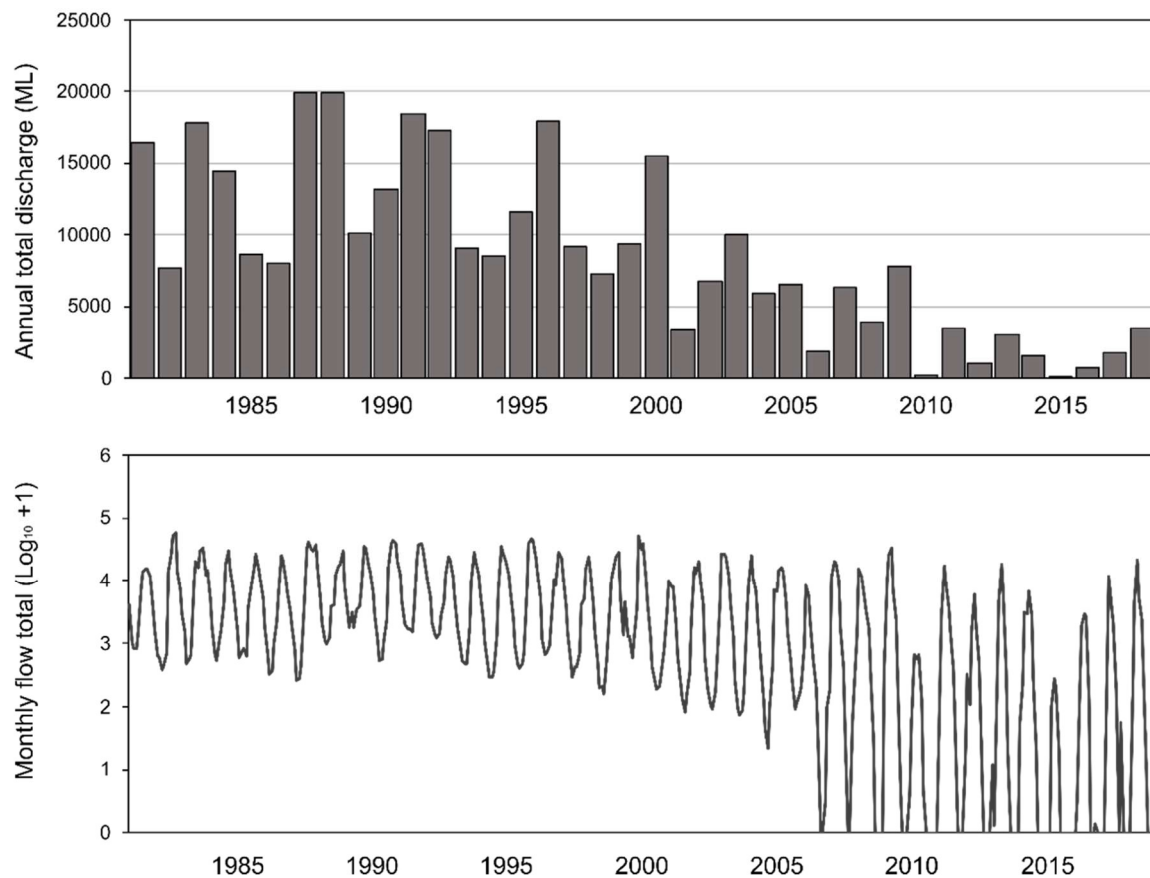


Figure 2.11: Annual total discharge (ML) at Wungong Brook from 1982 to 2016 (top). Bottom: monthly streamflow totals at Wungong Brook 1982 to 2016, log transformed so months of zero flow can be seen (separately from those with very low flows). Wungong Brook became intermittently flowing in 2007 and has flowed intermittently every year since (correct at 2020). All graphs produced using daily streamflow measurements, freely available from the Western Australian Department of Water and Environmental Regulation (gauge number 616041; wir.water.wa.gov.au).



Figure 2.12: Wungong Brook is the largest waterway in the catchment. WB1 (top left) consisted of deep pools and flashy riffle areas and retained pools well into the late dry season (middle), although most did eventually dry (right). Upstream reaches at WB2 (lower left) were wide, shallow pools and runs over granitic bedrock.

Curtis Creek

‘Curtis Creek’ is an officially unnamed tributary to the Wungong Brook (Figure 2.1), and whilst flow gauging data is incomplete (there are two gauges on this stream, but neither have more than a few years of data) Bunn had two sites on this stream while it was perennial (sites 1 and 2, corresponding to CC1 and CC2 in Figure 2.1). Curtis Creek arises from the base of a granite inselberg (Mt Curtis), where the headwaters comprise a small perennial spring-fed pool (above site CC3). There is a second perennial spring-fed pool further downstream at CC2 (Figure 2.13). Curtis Creek is otherwise intermittent along the rest of its length, CC1 has a short hydroperiod and was only sampled twice during this study; whereas the springs at CC2 and CC3 maintained trickling flows and small pools into the early dry season (Figure

2.13). Both CC1 and CC2 are bedrock dominated, whereas CC3 overlies deeper weathered material and has primarily sand/silt sediments.



Figure 2.13: The middle and downstream reaches on Curtis Creek are dominated by granitic bedrock, and flow swiftly in winter and dry quickly with the onset of summer (CC1 pictured right). The small spring on CC2 maintains trickling flows into early summer, before retracting into a small pool (left; spring in the background behind vegetation).

Chandler Brook

‘Chandler Brook’ is an unnamed tributary to the Wungong Brook, and the second largest waterway in the Wungong catchment. It is also the most strongly seasonal, barely retaining water once flow stops in December. Two sites on Chandler Brook were sampled by Bunn in the early 1980s when this stream was perennially flowing (sites 5 and 6, corresponding to CH1 and CH2; Figure 2.1); flow gauging data is incomplete (as for Curtis Creek). Note:

Bunn erroneously labelled two sites as being on ‘Wungong Brook’ in his study, but throughout this thesis these are referred to as Chandler Brook, including when referring to Bunn’s results. The downstream reach CH1 consists of large pools up to two metres deep, overlaying lateritic bedrock (Figure 2.14). The upstream reach has areas of deep sand overlaying laterite. During winter, Chandler Brook flows swiftly, by spring (October) flows had slowed, and by December, any remnant pools were well on the way to complete desiccation. Although the upper reaches of this stream were highly seasonal, crayfish burrows were common in the banks, indicating burrowing crayfish (likely gilgies *Cherax quinquecarinatus*, potentially koonacs *Cherax pressii*) were able to access damp refuges, potentially perched groundwater (but much below the streambed).



Figure 2.14: Chandler Brook transports a great deal of water during winter flows but dried very quickly in early summer. CH1 (left) has deep pools overlaying laterite; CH2 has deep sand with outcropping laterite (seen in foreground, right).

Coccinea Creek

‘Coccinea Creek’ is an unnamed, ephemeral stream arising from a granite inselberg spring, which flows for approximately six to eight weeks in late winter (EPH, Figure 2.1). This

stream was only sampled once (in August 2017) due to its short hydroperiod and was found to support a community of ephemeral stream specialists and stygophilic Amphipoda.

Chapter 3

Loss of functionally important and regionally endemic species from streams forced into intermittency by global warming.

The following chapter has been drafted in accordance with the journal *Global Change Biology*, which is its intended destination.

The manuscript is currently prepared for submission.

The following authors contributed to this manuscript as outlined below:

	Author order	Concept & study design	Fieldwork	Laboratory work	Data analysis	Writing manuscript	Review/edit paper
Nicole Carey	First	X	X	X	X	X	X
Edwin Chester	Second	X	X				X
Belinda Robson	Third	X			X		X

By signing this document, the Candidate and Principal Supervisor acknowledge that the above information is accurate and has been agreed to by all other authors.

Candidate

Principal Supervisor

Abstract

Climate change is altering hydrological cycles globally, and in Mediterranean (med-) climate regions is causing drying water regimes and loss of perennially flowing streams. Water regime exerts strong influence over stream assemblages, which have developed over geological timeframes with the extant flow regime, therefore sudden drying in formerly perennial streams is expected to have large, negative impacts on stream fauna. Contemporary (2016/17) macroinvertebrate assemblages of formerly perennial, recently intermittent streams in the southwestern Australian med-region were compared to a study conducted pre-drying (1981/82) in the same streams, using a multiple before-after, control-impact design. Recent intermittency had a profound effect on species composition, including the extirpation of nearly all Gondwanan relictual insect species from streams impacted by drying. New arrivals at intermittent streams tended to be widespread resilient species, with different biogeographic histories including desert adapted taxa. Intermittent streams also had very different species assemblages to each other, due in part to differences in the hydroperiod allowing establishment of distinct summer and winter assemblages in streams with longer-lived pools. Assemblage composition in the remaining perennial stream changed very little between the studies, and this stream is acting as an ark-type refuge for these ancient relictual species. The fauna of SWA upland streams is becoming more homogenised with that of the wider Western Australian region, as drought tolerant, widespread species replace locally endemic species. This study showed large *in situ* alterations to stream assemblage composition caused by climate change, and is a real-world demonstration of the existent threat posed to relictual med-region stream faunas by climate change.

Introduction

Global warming is altering hydrological regimes, profoundly affecting freshwater ecosystems, through changes to timing, duration and volumes of flows in rivers and streams worldwide (Döll & Schmeid 2012; Hoegh-Guldberg et al., 2018). Since flow regime is among the most important abiotic factors underpinning aquatic ecosystem structure and function in lotic systems (Poff, 1997; Gasith & Resh, 1999; Lytle & Poff 2004; Bernal et al., 2013), significant changes to flow regime caused by global warming are expected to have correspondingly significant impacts on aquatic community composition (Knouft & Ficklin 2017). However, opportunities to directly study such responses to climate change *in situ* have so far been rare, due in part to the scarcity of true “before” data (*sensu* Jackson & Füreder, 2006; Didham et al., 2020a,b) coupled with robust hydrological data. Here, we describe invertebrate community responses in headwater streams forced from perennial to intermittent flow regimes by global warming in a region where surface flows in streams are now only 20% of former long-term averages (McFarlane et al., 2020).

Aquatic macroinvertebrates underpin ecological function and trophic webs in freshwater ecosystems (Boulton et al., 2014). Macroinvertebrate assemblage composition often differs greatly between perennial and intermittent waterways, with the latter also exhibiting strong patterns of seasonal turnover in accordance with their abiotically dynamic environments (Williams & Hynes, 1976, 1977; Boulton & Lake, 1990, 1992a,b; Williams 2006; Bonada et al., 2007; Leigh et al., 2016b). Organisms inhabiting intermittent streams must possess drought resistance (e.g. desiccation resistant eggs) or resilience traits (e.g. long lived and/or highly mobile terrestrial adults; Strachan et al., 2015; Bogan et al., 2017); additionally, mounting evidence points towards the importance of drought refuges in landscapes that undergo seasonal (or supraseasonal) drying (Chester & Robson, 2011; Robson et al., 2013; Bogan et al., 2017; Bonada et al., 2020). Thus, stream intermittency is an environmental filter

which excludes species that do not possess life history adaptations to overcome the constraints of regular drying (Poff, 1997; Robson et al., 2011; Leigh & Datry, 2017), restricting those species to environments with permanent flow.

Many, perhaps most, streams in mediterranean climate regions (hereafter med-) are intermittent and seasonal dryness there is a natural disturbance that has occurred predictably in time and space for hundreds or thousands of years (Bonada & Resh, 2013). However, such regions also contain perennial streams which may not only support a distinct fauna but may also act as refuges for intermittent stream fauna in the surrounding catchment (e.g. Robson & Chester 2011; Chester et al. 2015). Med-climate regions are already experiencing drying climatic trends (Polade et al., 2017), that not only prolong drying in intermittent streams, but may cause perennial streams to dry out in dry years (Churchel & Batzer, 2006; Bêche et al., 2009; Skoulikidis et al., 2017) or to permanently transition from perennial to intermittent flow (Zipper et al., 2021)

In southwestern Australia (SWA), declining rainfall since 1975 has lowered groundwater tables and reduced winter runoff, causing disproportionately large reductions in streamflow of up to 70% (Petrone et al., 2010; Silberstein et al., 2012; McFarlane et al., 2020).

Historically, baseflows were generated through groundwater discharge into streams.

Following winter rains, saturated overland flow, shallow throughflow and formation of perched aquifers generated additional winter-spring flows (Gentili 1989). Subsequent disconnection of groundwater from streambeds (due to dropping groundwater tables), beginning around the years 2000 – 2001 in most catchments, continued the decline in streamflow independent of annual rainfall (Petrone et al. 2010; Hughes et al., 2012; Kinal & Stoneman, 2012). That is, most streams went from gaining to losing streams. For perennial streams, this usually caused a transition to intermittency that is understood to be permanent (based on climate modelling for SWA; Silberstein et al., 2012; Andrys et al., 2017). Thus

climate is a key driver of hydrological change across the region (McFarlane et al. 2020), as in other regions globally where streamflow declines are occurring.

Med-region aquatic biota tend to be highly endemic to each med-region, and many are relictual species that are often associated with perennial flows and cooler waters, particularly the southern hemisphere faunas of Gondwanan affinities (Bunn & Davies, 1990; Davies & Stewart, 2013; de Moor & Day, 2013; Stewart et al., 2013). Thus, the biodiversity of med-region streams is thought to be at particular risk from negative impacts of climatic drying, particularly species dependent on permanent flow (Filipe et al., 2013; Kroll et al., 2017). Furthermore, species losses can impact ecosystem functioning, particularly if biodiversity is low and those species are not functionally redundant (e.g. Jonsson & Malmqvist, 2000; 2003), as is the case in southwestern Australian med-stream fauna (Bunn & Davies, 1990).

The few studies investigating *in situ* impacts of water regime shifts from perennial to intermittent flow have found that the invertebrate assemblage does not recover. Cascading effects from loss of long-lived, poorly-dispersing but functionally important species have been observed (e.g. in desert stream pools, Bogan & Lytle, 2011), as well as the replacement of sensitive endemic species with widespread generalists (e.g., alpine rivers, Bruno et al., 2019), reduced alpha diversity and either assemblage homogenisation (e.g. alpine rivers, Piano et al., 2020) or marked increases in community turnover (e.g. humid continental streams, Crabot et al., 2021). Trends identified in mediterranean climate regions include reduced biodiversity of invertebrates (Leigh & Datry 2017), transitions in assemblage composition toward resilient taxa (Pace et al., 2013), and losses of sensitive species (Kroll et al., 2017; including freshwater fish Beatty et al., 2014; Jaric et al., 2019). It is widely anticipated that drying flow regimes would extirpate species that are not pre-adapted to intermittency, to be replaced by those with traits that allow survival (Kroll et al., 2017; Pyne & Poff, 2017).

Prior to the onset of widespread streamflow transition to intermittency, detailed ecological studies were conducted in SWA forested headwater streams, documenting macroinvertebrate assemblage composition and seasonal patterns (Bunn 1985; Bunn 1986; Bunn *et al.*, 1986), macroinvertebrate contribution to leaf litter processing (Bunn 1988b; Bunn 1988c) and the life history phenologies of 13 most-common species (Bunn 1988a). The existence of these ecological studies conducted over 20 years prior to the onset of flow regime change, and for the purpose of describing *in situ* community composition and characteristics (rather than measuring another impact or trend; *sensu* Didham et al., 2020b) provides a true baseline with which to compare change, a circumstance rare in stream ecology (*sensu* Jackson & Füreder, 2006; and insect ecology more broadly, Didham et al., 2020b) and thus an opportunity to directly study the impacts of climate driven flow regime drying in absence of other environmental changes.

We quantified the impacts of flow regime drying on med-climate stream communities of SWA, comparing differences in macroinvertebrate assemblages at two temporal scales: 1) differences between contemporary assemblages and those in an historical study when the same streams were perennial; and 2) seasonal patterns over one year between streams that remained perennial and those that have transitioned to intermittent flows. For the historical comparison, we used a multiple before-after, control-impact (mBACI) design, as these designs are optimal where there is a discrete, step change at the impact sites (Downes et al. 2002 p171), as there is with transition from perennial to intermittent flow regimes. To understand the impacts of climate change, we need information from both before and after change commences (Downes et al., 2002; Didham et al., 2020b). The difficulty of obtaining true ‘before’ data means that studies that can directly attribute a change in ecological processes to a state-change in environment are rare (i.e. the “false baseline” effect, *sensu* Didham et al., 2020b). We also compared assemblage composition changes over the course

of a year, because prior to flow regime change macroinvertebrate assemblages were strongly linked to seasonal changes in flow (i.e. increase from baseflow; Bunn 1988a). Sampling at regular intervals also avoids false inferences associated with “snapshot” effects (*sensu* Didham et al., 2020). Specifically, we hypothesised: (H₁) macroinvertebrate assemblage composition would differ between perennial reaches and those that have become intermittent, because assemblages in perennial reaches comprise a greater proportion of species without adaptations to intermittency than in intermittent reaches; (H_{2A}) perennial reaches would support a similar macroinvertebrate assemblage to that recorded by Bunn in the 1980s, because flow regime has not changed; whereas (H_{2B}) assemblages at reaches that have become intermittent will differ from those recorded by Bunn in the 1980s, because of the loss of species that require perennial flow; (H_{3A}) in perennial reaches, patterns of seasonal change will be similar in 2016/17 to those observed in 1981/82 because aquatic insect phenology has not changed; (H_{3B}) intermittent reaches will differ because they have more arid-zone adapted taxa so will have two (or more) seasonally distinct assemblages, as observed in intermittent streams elsewhere; and (H₄) perennial reaches would retain more relictual, regionally endemic species than intermittent reaches, which contain more widespread arid-adapted taxa, because drying displaces the former whilst providing habitat for the latter.

Methods

Study area

The mediterranean climate in SWA has been warming and drying for at least five decades (Bates et al., 2008), causing a widespread and permanent transition to intermittent streamflows in most catchments between 2000 to 2010, which is expected to occur in remaining perennial streams (Petrone et al. 2010; Hughes et al., 2012; Kinal & Stoneman 2012; Figure 2.3). Mediterranean climates typically show marked inter-annual variation in rainfall, but in this region, there has been a reduction of ~ 20% of mean precipitation since

1970 (Hope et al., 2010). Additionally, the “big wet” years that historically punctuated rainfall records no longer occur (the most recent was in 1974: 45 years ago), and totals that approach the long-term average (*circa* 830 mm p.a.) are now considered “wet” years (Ruprecht, 2018).

The study sites lie in the forested headwater streams of the Wungong Brook catchment, within the Northern Jarrah forest (NJF) of SWA (Figure 3.1). The area is protected water catchment, which means that no human disturbances are allowed and there has been no mining, logging or other disturbance in the catchment since Bunn’s studies. Some streams have small flow gauging weirs but none of the streams are regulated, although they all eventually terminate in the Wungong water supply reservoir. Consequently, the only changes to streamflow regimes have arisen from global warming. Thirteen reaches were selected spanning the Wungong Brook and each of its four major tributaries (Figure 3.1). Eight of these reaches were studied by Bunn (1986a; 1986b; 1988a) and Bunn et al., (1986) in the early 1980s. Because of the increased heterogeneity in flow duration caused by drying, five reaches were added to those sampled by Bunn to better represent this diversity (WF3, WF4 [a perennially flowing tributary of Waterfall Gully], Curtis Creek site 3 CC3, Wungong Brook site 1 WB1 and 2 WB2; Figure 3.1). At that time (and before), all study streams in this catchment were perennially flowing. Flow gauging data from this catchment from the 1960s onwards shows clearly the transition from perennial to seasonally intermittent flows in most of these streams (Figure 2.9; Figure 2.11). It also shows later commencement of flows in the now-intermittent streams; these flows originate from the formation of surface aquifers from rainfall and previously comprised the onset of wet-season increases from baseflow. Also, peak flows now occur later in all streams (Figure 2.6). In formerly perennial streams, flow now commences in June or July where formerly flows increased above baseflow in March-

April. These changes would be expected to alter seasonal patterns in stream invertebrate assemblages.

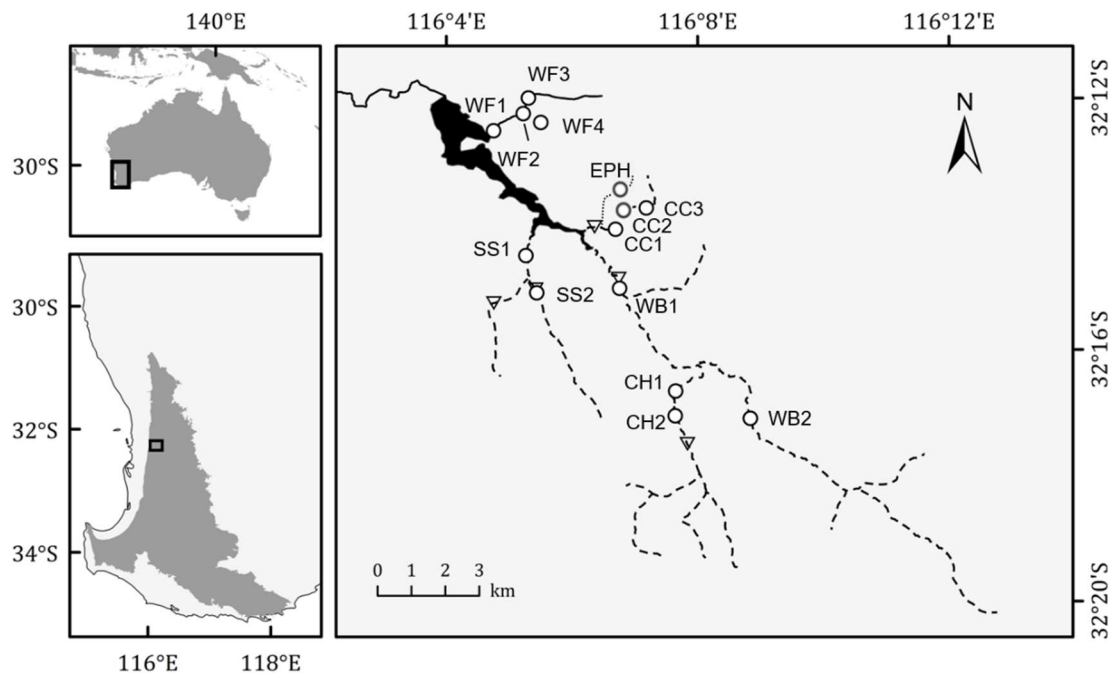


Figure 3.1: Study reaches in the Wungong Brook catchment of south-western Australia, perennial reaches (solid lines), formerly perennial, now intermittent reaches (dashed lines), and an ephemeral stream (dotted line). Inverted triangles denote flow gauging stations (flow gauge at WF1 not visible). Sites used by Bunn (1988) were WF1, WF2, CC1, CC2, SS1, SS2, CH1 and CH2; additional sites WF3, WF4, CC3, WB1, and WB2.

Only one stream in this catchment, Waterfall Gully (and its tributaries), remains perennial; flows there have not ceased since records began in 1966. One other stream, Seldom Seen Brook, has retained one short, spring-fed reach which is perennial in most years, slowing to a trickle in the summer months. Thus, at present, there are only two waterways that contain flowing water all year. Although discharge volumes have declined, annual flow patterns have changed little in these perennial reaches (Figure 2.6; Figure 2.7) and would not be expected to alter seasonal patterns in stream invertebrate assemblages.

Curtis Creek (CC), Chandler Brook (CH) and Wungong Brook (WB) are all formerly perennial streams. ‘Chandler Brook’ is an unnamed tributary to the Wungong Brook, and is the second largest waterway in the catchment after Wungong Brook. Flow gauging data there is incomplete, and records ceased after 2014. However, Bunn (1986) and Bunn et al. (1986) had sites in both Curtis Creek and Chandler Brook (erroneously named Wungong Brook by Bunn) when they were perennially flowing. Records show that Wungong Brook first dried out in 2007 and it is likely these two tributaries dried out around the same time (Figure 2.11). Currently, Chandler Brook is flashy, discharging large volumes of water over winter and spring, and quickly drying down in summer. During the dry phase in intermittent streams (February to June) streambeds were explored along most of their length, in search of remnant pools or seeps which may have acted as refuges for aquatic fauna. There were no permanent pools and no evidence of hyporheos in Chandler Brook (contains extensive areas of laterite bedrock), whereas Wungong Brook retained pools until at least May, although volumes were severely retracted. Flow gauging data at Curtis Creek is incomplete, however it was perennially flowing at the time of Bunn’s research (1981-1984) and has since switched to intermittent flows. The upper reaches are narrow, well vegetated, with deeper substrate overlaying granite and laterite bedrock, and the water was seldom deeper than 0.3 m at the study reaches. Curtis Creek dried completely by April 2017, however it retained two spring fed pools, at the base of a granite inselberg in the headwaters, and at site CC2.

NJF is dry sclerophyll forest, dominated by eucalypt trees: jarrah *Eucalyptus marginata* and marri *Corymbia calophylla*. This region has undergone a protracted period of isolation from the wetter environments of the continent’s south east (Davies & Stewart, 2013). So, despite the mediterranean climate, the stream invertebrate fauna are primarily Gondwanan relicts from wetter times, characterised by comparatively few Ephemeroptera, Plecoptera and Trichoptera (EPT) taxa and high endemism (Davies & Stewart, 2013).

Study design

Macroinvertebrate assemblage composition was analysed from each of the thirteen study reaches between October 2016 and August 2017 at approximately six-week intervals; seven sampling times in total. Sampling aimed to maximize variation in flow environments during the hydroperiod, commencing when all reaches were flowing, intermittent reaches then began to dry down and disconnect in November and December, and by autumn (April – July 2017) they had dried completely. Sampling therefore commenced in the austral spring 2016 and ended with austral late-winter in 2017, encompassing one year, similar to Bunn (1985).

As study reaches were representative of extant flow regimes across the Wungong catchment, patterns in assemblage composition due to flow regime, versus site specific differences, could be compared. This allowed distinction of spatial differences in macroinvertebrate communities in accordance with differing flow regimes. Then, as we sampled the same eight reaches as Bunn (1985, 1986, 1988c; Bunn et al., 1986), patterns in assemblage composition and functional organisation could be compared to determine temporal change between studies, at those same sites which either have, or have not undergone flow regime change. This comprises a multiple before-after control-impact (mBACI) design: there were multiple reaches, some of which remained perennial (control before vs after) and other which became intermittent (impact before vs after). These designs are ideal for identifying the effects of flow regime change because changes in the control (perennial) reaches comprise temperature increases and all other changes to which all streams were exposed, except for the flow regime change (Downes et al., 2002). The intermittent reaches, which were also exposed to flow regime change, can then be compared to the controls and the difference between them comprises the effect of flow regime change.

Bunn (1985; 1986; Bunn et al., 1986) sampled riffle zones only, taking two samples per reach per sampling occasion. Because of the increased heterogeneity in flow environment, and

different aims of this study (i.e. presenting empirical evidence of the impacts of climate change on aquatic communities, rather than characterising the macroinvertebrate community in previously little-studied streams), sampling effort in this study was up to four-fold greater per reach per sampling time. Where surface water permitted, four samples were taken from erosional habitats (riffles or runs) and four from depositional habitats (pools) giving a total of 8 samples per reach per sampling time. Heterogenous flow environments and associated microhabitat structure is known to have strong effects on macroinvertebrate community composition (Davis & Barmuta, 1989; Chester & Robson, 1999). Therefore, due to the reduced wetted area and increased temporal and spatial variation in flow environments associated with drying, increased sampling effort was necessary to fully capture community composition at a reach at a point in the hydroregime.

Invertebrate sampling

Bunn (1985) used a 0.1 m² Surber sampler (33 cm wide) with a 475 µm mesh net, which relied on flow velocity to wash invertebrates into the net and collect drift. However, the Surber technique is unsuitable during low-flow periods because material is not washed into the net and there are no drifting invertebrates. Instead, a dipnet sampling technique was adapted for either flowing or still water, and on any substratum type, using a 500 µm mesh, 30 cm wide net. In flowing reaches, the bottom edge of the net was pushed into the substrate (opening facing upstream), and a 30 x 30 cm area (0.09 m²) of substrate in front of the net gently disturbed by hand, dislodging invertebrates which were then washed into the net by flow. Larger stones within the 30 x 30 cm area were dislodged and brushed by hand, allowing any attached invertebrates to be washed into the net. This was timed for two minutes to capture drifting invertebrates and is much like the Surber technique used by Bunn (1985), who also used two-minute samples. This method was modified when flow had stopped in intermittent streams. A 30 x 30 cm area of substrate was vigorously disturbed by foot, and the

net quickly swept over for 30 to 60 seconds (dependent on depth), from the substrate to the surface to collect all disturbed invertebrates. Previous studies have shown that this method, where rocks are disturbed and the material beneath is collected into the net, collects more species than simply scrubbing the rocks alone (Giri et al., 2010). This method has been successfully used in other studies of intermittent streams, where flow is variable within and between sites at different times (e.g. Robson et al., 2005).

Samples were placed in plastic bags and preserved with 70-80% ethanol. In the laboratory, samples were sorted by emptying into a white tray and flooding with tap water, then carefully removing all invertebrates from substrates using a dissecting microscope mounted on a moveable arm, at 10x magnification. Invertebrates were identified to the lowest possible taxonomic level - species level for most groups excluding Diptera (Family level except Chironomidae, which were to genus), Ostracoda and *Uroctena* amphipods (genus level); Oligochaetes and other worms (Class level), and mites (Order).

Data analysis

To examine differences between contemporary flow regimes (H_1), patterns in macroinvertebrate assemblage composition were examined using PERMANOVA (Anderson et al., 2008). The model included three factors: sampling time (seven levels, fixed) stream (5 levels, fixed), and sites nested within streams (13 levels, random), and interactions between time and stream, and time and nested sites. Note that time is a fixed factor in mBACI designs (Downes et al., 2002) and had been chosen to coincide with those of Bunn, therefore time was fixed in all analyses. Sampling time, stream and sites were then tested for significant contribution to differences in dispersion of macroinvertebrate data (PERMDISP), as disturbances have the potential to either homogenise taxa or increase variability (Houseman et al., 2008).

To further examine H_1 , sample times were grouped into three hydroperiod stages: spring and early summer (October and December 2016) when all reaches were flowing; summer and autumn baseflows at perennial reaches, or disconnection and drying intermittent reaches (February, April and May 2017) and winter initial flows to peak flows (June and August 2017). Owing to the Mediterranean climate, seasonal changes in precipitation and temperature are very distinct in SWA and streamflows historically respond accordingly in a predictable manner. One-way analysis of similarity (ANOSIM) was used to test differences between the two flow regimes (factor flow regime: perennial, intermittent) within each hydroperiod stage (spring, summer, winter) separately across all reaches. Then, differences in macroinvertebrate assemblage composition between reaches at each hydroperiod stage were examined using one-way ANOSIM (factor hydroperiod stage) and visualized with non-metric ordination plots (nMDS). Similarity percentages (SIMPER) were calculated to identify key species contributing to dissimilarity between flow regimes, and then to determine species characterising each site and dissimilarity between sites (and identify the roles of relictual endemic and widespread generalist species) at each hydroperiod stage. The SIMPER analysis was also used to examine the contribution of endemic and relictual species to observed differences between reaches of differing flow regime (H_4).

To address H_2 macroinvertebrate assemblages were compared between historic (1981/82) and contemporary (2016/17) assemblages. Although the complete original data from Bunn's study was unobtainable for formal analysis, Bunn (1985, 1986) provided proportional dominance of species occupancy per site (pooled across time), only excluding very rare taxa (those that never comprised 5% or more of the total individuals at any site at any time). For comparison, macroinvertebrate data collected during 2016/17 from the same sites studied by Bunn in 1981/82 were standardised using this method (i.e. converted to proportional dominance per site and time, and very rare species removed). Due to differences in taxonomy

between times, some taxa were standardised between datasets to genus (some Trichoptera) or Family level (Tipulidae, Simuliidae). Thus, species assemblages excluding the rarest taxa were directly compared at each site sampled in 1981/82 and 2016/17 to test the mBACI design, noting that although sampling methods were similar, four times as many samples were taken in 2016/17 than 1981/82.

To address H₃, seasonal patterns in macroinvertebrate community composition were examined at each stream in 2016/17 and compared to patterns in 1981/82. By examining differences in macroinvertebrate assemblages between the seven sample times, seasonal patterns were analysed as a time series per stream (pooling reaches by stream). First it was necessary to determine whether spatial differences in assemblage composition occurred between reaches within each stream (and thus whether pooling reaches was appropriate). For each stream, a one-way ANOSIM (times pooled) was used to test for significant differences between its constituent reaches. The two reaches on Wungong Brook did not differ nor did those on Chandler Brook. SS2 was retained for this analysis, SS1 was excluded because only contained surface water on two occasions. Three reaches in Curtis Creek had small differences but only due to differing rates of drying in summer and autumn. Given that differences were small, these sites were pooled. Four reaches differed at Waterfall Gully—in particular a tributary (WF4) and the upper reach (WF3) differed greatly from the downstream reaches (WF1 and WF2, Bunn's 1981/82 sites). Habitats at these sites were very different, likely explaining differences in the fauna: WF3 was covered by dense tree cover and had deep (>50 cm) accumulations of flocculent organic matter covering the streambed; WF4 also had dense cover and organic substrates, but was shallow. By comparison, WF1 and WF2 had a mix of gravel/cobble riffle zones, sandy runs and depositions of organic material, and showed only small differences in assemblage composition. Therefore, WF1 and WF2 were pooled for subsequent analyses. Based on ANOSIM of macroinvertebrate assemblage

composition, pooled reaches used for analysis were Waterfall Gully (WF1, WF2), Seldom Seen Brook (SS2), Curtis Creek (CC1, CC2, CC3), Wungong Brook (WB1, WB2) and Chandler Brook (CH1, CH2). To conduct analysis of seasonal patterns, a one-way ANOSIM was conducted on pooled reaches testing for differences between times, and species contributing to seasonal differences were identified using Similarity Percentages (SIMPER).

Results

Assemblage composition differences between flow regimes (H_1)

Macroinvertebrate assemblage composition differed greatly across the thirteen study reaches. The PERMANOVA model explained only 38% of total variation, and all factors and interactions were significant. Reaches (nested within streams) explained the greatest proportion of variance (pseudo- $F_{8/324} = 3.8$, $P = 0.001$, 8.5%) together with the interaction between nested reaches and times (pseudo- $F_{30/324} = 1.7$, $P = 0.001$, 9.8%). Streams explained 7.3% (pseudo- $F_{4/324} = 2.5$, $P = 0.001$); sampling time explained 6.4% (pseudo- $F_{6/324} = 2.9$, $P = 0.001$), and the interaction between times and streams (pseudo- $F_{18/324} = 1.5$, $P = 0.001$) explained 5.7% of model variation.

Multivariate dispersion of macroinvertebrate assemblage composition varied throughout the year ($F_{6/324} = 5.7$, $P = 0.001$), being lowest in mid-spring flows (October mean = 60.1 ± 0.8) and highest in winter (August mean = 64.7 ± 1.1 ; June mean = 65.1 ± 0.8). Dispersion also differed between reaches within streams (i.e. reach level differences; $F_{12/324} = 6.7$, $P = 0.001$), with similar patterns to stream level dispersion (WF1 and WF2 lowest average dispersion, WB 1 and WB2 highest dispersion). Dispersion differed between streams ($F_{4/324} = 8.5$, $P = 0.001$); with perennial streams being the least variable (Waterfall Gully mean 60.5 ± 0.6 ; semi-perennial Seldom Seen Brook = 61.5 ± 1.0), and intermittent streams being more variable among reaches (Curtis Creek mean = 64.1 ± 0.7 ; Chandler Brook mean = 65.0 ± 1.1 ; and Wungong Brook mean = 65.9 ± 0.5). Thus, spatial variation among intermittent streams

decreased with length of time flowing, but was always higher than among perennial streams, which varied less through time.

Spring assemblages

During springtime, macroinvertebrate assemblages differed according to flow regime but differences were small (Global $R = 0.16$, $P = 0.001$), although average similarity of fauna was low among samples from both perennial (19.4%) and intermittent (11%) streams. Differences in frequency of occurrence of common species *Uroctena* sp. (Paramelitidae amphipods) and *Tanytarsus* sp. (Chironominae) were the greatest contributors to differences between flow regimes. However, mayflies were sensitive to flow regime: *Nousia* AV16 (Leptophlebiidae) was only found in intermittent reaches, whereas *Neboissephlebia occidentalis* (Leptophlebiidae) and *Offadens soror* (Baetidae) were only found at perennial reaches. Similarly, the dragonflies *Austroaeschna anacantha* (Telephlebiidae), *Austrosynthemis cyanitincta* (Synthemistidae), and caddisflies *Taschorema pallescens* (Hydrobiosidae) were only found at perennially flowing reaches, mostly restricted to Waterfall Gully (i.e. absent from SS2). Chironomidae assemblages also differed between flow regimes: *Tanytarsus* sp., *Cricotopus* sp. and *Polypedilum* sp. were more common at intermittent reaches, *Riethia* sp., *Stictoladius* sp., and *Stempellina* sp. were common at Waterfall Gully, and comparatively rarer at all other reaches. Therefore, many common species showed little consistent variation between times or flow regimes, but many less common species did.

Macroinvertebrate assemblages differed between reaches (Global $R = 0.5$, $P = 0.001$; Figure 3.2), apart from those on Curtis Creek and the intermittent site on Seldom Seen Brook (SS1) which did not differ to WB2 or CH1 (Table 3.1). Among-site variation was considerable but did not follow any particular pattern.

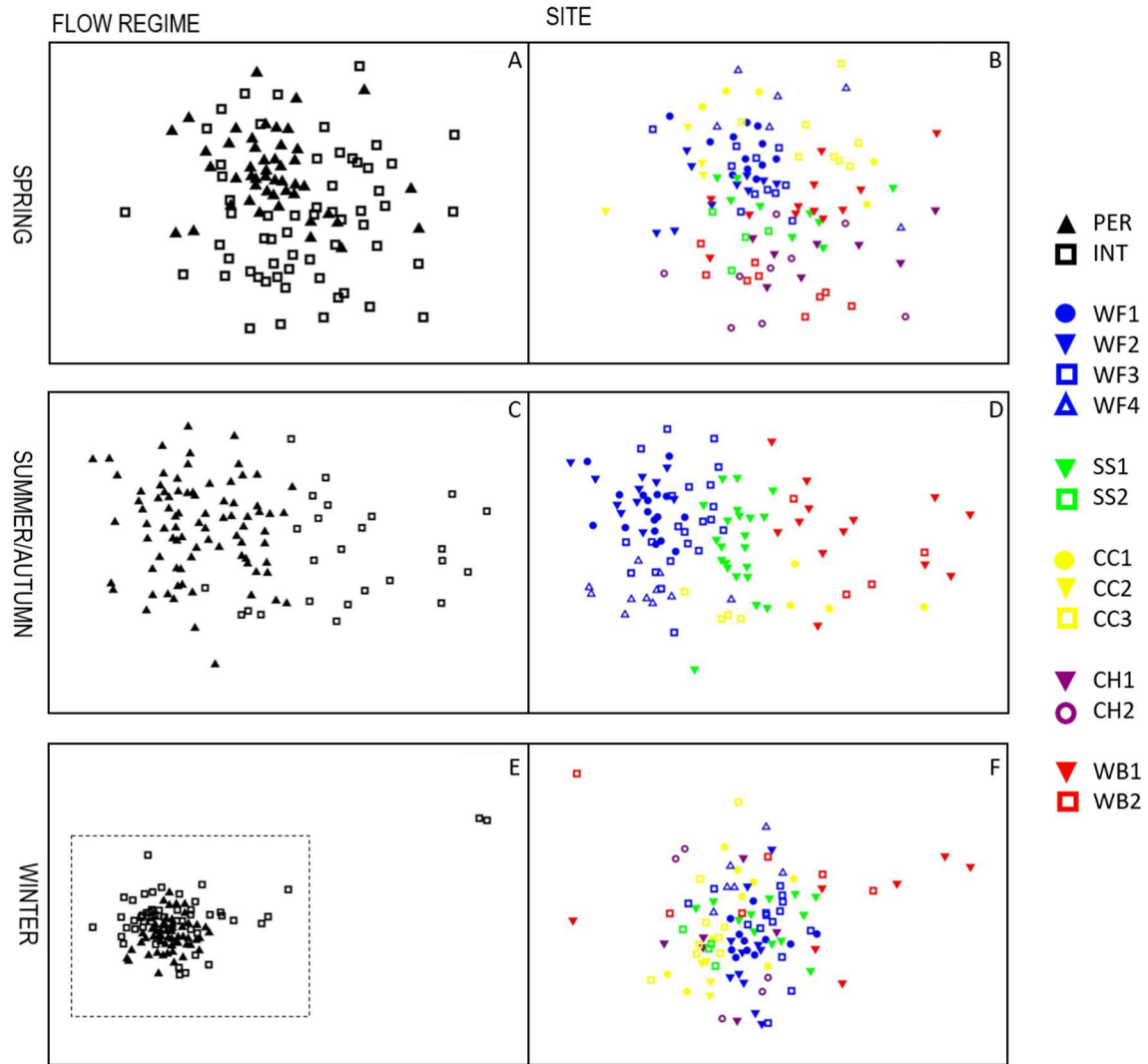


Figure 3.2: Ordination (nMDS) of macroinvertebrate assemblages present in perennial and formerly perennial, now intermittent streams (A, C, E) and thirteen sites (B, D, F) on five streams in the Wungong catchment (Waterfall Gully: WF1, WF2, WF3, tributary WF4; Seldom Seen Creek: SS1, SS2; Curtis Creek: CC1, CC2, CC3; Chandler Brook CH1, CH2; and Wungong Brook: WB1, WB2). Austral spring assemblages were collected in October and December 2016; Summer/Autumn in February, April and May 2017; and winter in June and August 2017. Panel F is an MDS subset of panel E, extent of subset shown in E.

Table 3.1: Assemblage composition in Austral spring (October and December 2016); summer/autumn (February, April and May 2017); and Winter (June and August 2017) at sites along five streams on a flow regime gradient from perennial to strongly seasonally intermittent. ASD = stream significantly different to all others at that sample time (one-way ANOSIM); site pairs or groups that did not differ during each hydroperiod stage given. Macroinvertebrate species contributing to 90% similarity per stream/time combination given (percentage contribution), bold indicates species not collected at any other stream. Note: SIMPER analyses between site level checked for differences in patterns of species contributions.

		Spring	Summer/Autumn	Winter
PERENNIAL FLOW	WF	ASD Chironomidae (6 sp., 48%); <i>Uroctena</i> sp.; <i>Nyungara bunni</i> ; <i>Cnephia tonnoiri</i> ; <i>Neboissophebia occidentalis</i> ; <i>Austrosynthemis cyantincta</i> .	ASD Chironomidae (6 spp., 60%); <i>N. bunni</i> ; <i>Lectrides parilis</i> ; <i>Uroctena</i> sp.; <i>Hudsonema aptus</i> ; <i>C. tonnoiri</i>	(WF3,WF4; -SS2 -CH2) (WF2, SS2; -CC1) (WF4,SS2) Chironomidae (5 spp., 50%); <i>Uroctena</i> sp.; <i>N. bunni</i> ; <i>C. tonnoiri</i> ; <i>L. parilis</i> ; Oligochaeta
	SS	SS2: ASD Chironomidae (5 spp., 70%) <i>Uroctena</i> sp.; <i>Perthia acutitelson</i> ; <i>Uroctena</i> sp.; <i>Bibulmena kadjina</i> (SS1); <i>Ilyodromus</i> sp.; <i>Nousia</i> AV16	SS2: ASD Chironomidae (5 spp., 62%); <i>B. kadjina</i> ; <i>Cherax quinquecarinatus</i> *; <i>P. acutitelson</i>	(SS2, WF3, WF4) (SS1,WF3, WF4) Chironomidae (5 spp., 41%); <i>B. kadjina</i> ; <i>C. quinquecarinatus</i> *; <i>Uroctena</i> sp.; <i>Leptoperla australica</i> ; <i>Archeosynthemis leachii</i>
DECREASING FLOW & SURFACE WATER PERMANENCE ↓	WB	(WB2,SS1) Chironomidae (5 spp., 80%); <i>Nousia</i> AV16; <i>Bibulmena kadjina</i> ; <i>C. tonnoiri</i>	(WB1,WB2) <i>Cloeon fluviatile</i> ; <i>Rhantus australis</i> ; <i>Cyclopoida</i> sp.; nr. <i>Cyperiscus</i> sp.; <i>Necterosoma darwini</i> ; <i>Procordulia affinis</i> ; Chironomidae (2 spp., 9%)	(WB1,WB2) <i>Perthia acutitelson</i> (47%); Chironomidae (4 spp., 27%); <i>N. darwini</i> ; <i>Nilobezzia</i> sp.; Anisoptera juv.
	CC	(CC1,CC2,CC3) Chironomidae (5 spp., 40%) <i>Uroctena</i> sp.; <i>Nousia</i> AV16; <i>B. kadjina</i> ; <i>Molophilus</i> sp., Oligochaeta	ASD Chironomidae (2 spp., 20%); <i>Culex</i> sp.; <i>R. australis</i> ; <i>B. kadjina</i>	(CC1,CC2,CC3) <i>Uroctena</i> sp.; <i>L. australica</i> ; <i>Simulium</i> sp.; Chironomidae (2 spp.,14%); Oligochaeta
	CH	(CH1,SS1) Chironomidae (4spp., 57%) <i>Nousia</i> AV16; <i>Ilyodromus</i> sp.	*dry	(CH1,CH2; -SS1; -CC3) Chironomidae (2 spp., 32%); <i>Uroctena</i> spp.; <i>Hydrobiosella michaelseni</i> ; Oligochaeta

Summer-autumn assemblages

In the summer and autumn months, perennially flowing reaches had receded to baseflow, and intermittent reaches underwent the process of disconnection, retraction and drying of most pools. Macroinvertebrates inhabiting these contrasting flow environments differed greatly (Global $R = 0.6$, $P = 0.001$), although average similarity remained low among samples from perennial (13.2%) and intermittent reaches (10%). Species that characterised perennially flowing reaches and differentiated them from intermittent reaches included the chironomids *Riethia* sp., *Stempelina* sp., and *Stictocladius* sp., mayfly *Nyungara bunni* (Leptophlebiidae), juvenile crayfish *Cherax quinquecarinatus* (Parastacidae) and shredding caddisfly *Lectrides parilis* (Leptoceridae). Several of these species were never collected from intermittent reaches during this study. At intermittent reaches, widespread and arid-zone adapted taxa including diving beetles *Rhantus suturalis* and *Necterosoma darwini* (Dytiscidae), mayflies *Cloeon fluviatile* (Baetidae) and *Bibulmena kadjina* (Leptophlebiidae), copepods (Cyclopoida), ostracods *Cypretta* sp., Notonectidae *Anisops hackeri* and dragonflies *Procordulia affinis*, and *Hemicordulia tau*. *Chironomus* sp. (Chironomidae) was abundant at intermittent reaches and SS2, and absent from Waterfall Gully. Except for *B. kadjina*, all insect taxa characterising intermittent reaches during the summer/autumn dry period were generalist species widely distributed across Australia including the arid zone.

Winter assemblages

By late June 2017, rainfall was sufficient to fill pools and commence flows at some intermittent reaches (with the exception of Chandler Brook sites and SS1); flows increased above baseflow at perennial reaches, so that by August all reaches were flowing.

Assemblages differed between flow regimes (Global $R = 0.178$, $P = 0.001$) and reaches (Global $R = 0.278$, $P = 0.001$); however, there was more overlap in fauna between reaches on different streams than there was in spring and summer. Notably, reaches in Waterfall Gully

differed from each other (except WF3 and WF4), but they did not differ to the perennial reach SS2, and some pairwise comparisons between WF sites and the most ephemeral sites (CH1, CH2, and SS1) also did not differ. These most ephemeral sites did not differ, and were dominated by amphipods *Uroctena* sp., Oligochaeta, and the chironomids *Austropelopia* sp. and Orthocladiinae DEC sp. 1. Anomalously, high densities of the otherwise rare SWA-endemic caddisfly *Hydrobiosella michaelsoni* occurred at CH1 during August.

Average similarity of faunal assemblages between intermittent reaches was very low (6.8%), as were those between perennial reaches (11%). Characterising species at intermittent reaches, that were rare/absent from perennial reaches, included the stonefly *Leptoperla australica* (particularly common at Curtis Creek), dytiscid beetle *Necterosoma darwini* and the dragonfly *Archaeosynthemis leachii*. The chironomid *Riethia* sp., mayfly *N. bunni*, caddisfly *L. parilis* and dragonflies *Austroaeschna anacantha* and *Austrosynthemis cyanitincta* typified perennial reaches. Tipulidae nr. *Limnophila* sp. and Simuliidae *Simulium* sp. were more common at intermittent reaches, whereas Tipulidae *Molophilus* sp. and *Tipula* sp., were more common at perennial reaches.

Long term differences in macroinvertebrate assemblages (H₂)

Six general patterns in distribution and dominance of common species (those comprising >95% total individuals at any site/time) occurred between the 1981/82 and 2016/17, pertaining either to losses of species due to stream flow drying, increases in relative proportions of some species to others, probable range expansions, or arrivals of new species (Table 3.2; Table 3.3).

A suite of species remained common at both perennial and formerly-perennial reaches (Table 3.3). This included the amphipod *Uroctena* sp., which showed reduced occurrence at some (not all) intermittent reaches (Table 3.3). Several widespread genera of Chironomidae were also among common taxa in both studies. Apparent fluctuations in their dominance in

2016/17 may be due to reduced numbers of other taxa, however an overall increase of these Chironomidae cannot be ruled out.

A second general outcome was absence or extreme rarity of species formerly common in 1981/82, across the entire catchment in 2016 (Table 3.2). For example, the predatory snail *Glacidorbis occidentalis* (Bunn) was formerly found in several streams (and was first described from this catchment, Bunn & Stoddart, 1983). The dytiscid beetle *Sternopriscus browni* was found twice at SS2 and WF3. The stonefly *Newmanoperla exigua* (Plecoptera) and caddisfly *Acritoptila* sp. (Hydroptilidae) were formerly common, but not captured during 2016/17 despite increased sampling effort.

A third group of species comprised those restricted to Waterfall Gully in 2016/17 (i.e. reaches that have never dried; Table 3.2) including a handful of species that were also present at SS2 but absent at intermittent reaches (e.g. chironomid *Rheotanytarsus* sp.). Two out of four most common mayflies and all three formerly common dragonflies were completely restricted to Waterfall Gully. Early instars of the dragonfly *Austroaeschna anacantha* were also present at the perennial reach on Seldom Seen Creek (SS2), but because this reach has dried out in most years since 2010 it is likely a sink population (wasted breeding effort). Trichoptera appear to be the worst affected group, as all seven formerly common and widespread species are completely restricted to Waterfall Gully, with reach SS2 a likely population sink for *Lectrides parilis* (Leptoceridae; also highly important shredders in NJF streams). Restriction to Waterfall Gully was the most frequent outcome. Formerly, all these species were widespread and abundant across SWA forested streams, and not considered remotely vulnerable or even specialised (Bunn 1985, 1986; Bunn et al., 1986; also see Sutcliffe 2003).

A fourth groups of species became proportionally more dominant in now-intermittent streams, including the ostracod *Ilyodromus* sp., and Simuliidae (*Cnephia tonnoiri*, and other

unidentified species). Simuliidae comprised over 20% of individuals in many instances, whereas previously (1981/82) they comprised <5% of the species at all sites. The amphipod *Perithia* sp. previously comprised a maximum of 5% of total individuals it comprised up to 20% of individuals at WB1.

A fifth group of species persisted at intermittent reaches but were much less common in 2016/17 than in 1981/82. These included mayflies *Bibulmena kadjina* and *Nyungara bunni*, chironomids *Riethia* sp., *Stempellina* sp., and caddisflies *Notalina* sp. With the exception of *B. kadjina*, these species remained proportionally dominant at Waterfall Gully.

A sixth group, that were previously not recorded by Bunn (1985, 1986), comprised several species that were very common in intermittent reaches, including two Australia-wide, generalist dragonflies, two Dytiscidae beetles, and hemipterans. The only hemipteran appearing on the 1981/82 species lists was the pygmy water boatman *Micronecta* sp. Each of the new species are highly mobile, arid adapted “pool” type predators, and were among the dominant species at Wungong Brook and Curtis Creek during drying phases (Chandler Brook did not hold any surface water long enough for their establishment). The ostracod *Cypretta* sp. was also numerically dominant in both reaches of Wungong Brook. During the winter and spring months a new mayfly, *Nousia* sp. AV16 (Leptoplebiidae), was common throughout intermittent reaches and absent from perennial reaches.

Lastly, a seventh guild of species was recorded as present, but not common, during 1981/82, and was prominent during disconnection and drying, particularly on the Wungong Brook. This guild included cyclopoid copepods, Dytiscidae beetles, a reportedly desiccation resistant dragonfly *Archeosynthemis leachii* (Theischinger 2006), and a widespread mayfly, *Cloeon fluviatile*. A generalist, pollution tolerant chironomid *Chironomus* sp. had a high proportional dominance across intermittent reaches, as did mosquito larvae.

Twelve Chironomidae species were among proportionally dominant species in 2016/17 that were either rare, absent or unidentified in 1981/82 (Table 3.3). Due to their difficult taxonomy, and limited access to reference material of the many unnamed species (necessitating identification to Genus level in most cases), changes in individual species' distributions in accordance with climate change are unclear. However, the formerly common endemic species *Botryocladus bibulmun* appears to have disappeared from the catchment, whereas others including *Paramerina* sp., *Thienemanniella* sp. and *Chironomus* sp. were apparently more common. Overall, there has been an increase in proportion of NJF stream fauna comprising Chironomidae between 1981/82 and contemporary assemblages.

Table 3.2: Relative composition (%) of the total invertebrate fauna sampled from NJF streams in 1981/82, compared to relative composition in 2016/17 (5 additional sites sampled only in 2016/17 in far right columns). Perennial reaches (blue) and formerly perennial, now- intermittent reaches (yellow). Taxa comprising more than 5% of total at any site on at least one occasion (either sampling period) are included. Oligochaeta spp., omitted as taxonomic resolution too broad to make inferences about occupancy change. Site abbreviations listed in methods. White = 0, light grey = 0.1-1%, medium grey = 1-5%, dark grey = 5-20%, black = >20% relative composition per site. Asterisk denotes species was present at that site in 2016/17 sampling, but rare (less than 5% of total at any sampling time) to differentiate from those that were absent in 2016/17. Where known, distribution given: SW = SWA endemic species; WA = Western Australian species (including arid zone); AUS = widespread across the Australian continent (and often beyond); UN = unknown, due to higher classification or lack of information; there are reports of *Bibulmena* spp. from Kakadu in the Northern Territory, this may be a congener of the SWA endemic *B. kadjina*. General response to drying also given (see text). Chironomidae *Diplocladius* now classified as *Stictocladus*.

		Distribution	Response	1981/82								2016/17								2016/17 addition					
				CC2	CC1	WF1	WF2	CH2	CH1	SS1	SS2	CC2	CC1	WF1	WF2	CH2	CH1	SS1	SS2	WF3	WF4	CC3	WB1	WB2	
Amphipoda	<i>Uroctena</i> sp	SW	1																						
	<i>Perthia</i> sp	SW	4									*													
Ostracoda	<i>Ilyodromus</i> sp.	W	4																						
Mollusca	<i>Glacidorbis occidentalis</i>	A																							
Acarina	<i>Hydracarina</i>	SW	2																						
Chironomidae	<i>Tanytarsus</i> sp	UN	2																						
	<i>Tanytarsus</i> sp	AU	1																						
	<i>Rheotanytarsus</i>	AU	3																						
	<i>Stempellina</i> sp	AU	4																						
	<i>Riethia</i> sp	AU	4																						
	<i>Botryocladus bibulmun</i>	SW	5																						
	<i>Botryocladus bibulmun</i>	SW	2																						
	<i>Cricotopus</i> sp	SW	2																						
	<i>Cricotopus</i> sp	AU	4																						
	<i>Polypedilum</i> sp	AU	3																						
	<i>Polypedilum</i> sp	AU	4																						
	<i>Ablabesmyia</i> sp	AU	?																						
<i>Paramerina</i>	SW	4																							
<i>?levidensis</i>	SW	4																							
<i>Stictocladus</i>	SW	5																							
Diptera	<i>Simuliidae</i> spp.	SW	5																						
	<i>Simuliidae</i> spp.	-	4																						
	<i>Cnephia</i> sp	SW	4																						
	Tipulidae sp	-	4																						
	Ceratopogonid.	-	1																						
<i>Empididae</i> spp.	-	1																							
Ephemeroptera	<i>Nyungara bunni</i>	SW	5																						
	<i>Nyungara bunni</i>	SW	5																						
	<i>Bibulmena kadjina</i>	SW	5									*													
	<i>Nebiossophlebia occidentalis</i>	SW	3																						
	<i>Offadens soror</i>	WA	3																						
Trichoptera	<i>Notalina</i> sp	SW	5																						
	<i>Hudsonema aptus</i>	SW	3																						
	<i>Hudsonema aptus</i>	SW	3																						
	<i>Acritoptila</i> sp	SW	2																						
	<i>Ecnomina</i> sp	WA	2									*	*					*	*						
	<i>Taschorema pallescens</i>	SW	3																						
	<i>Lectrides parilis</i>	SW	3																						
	<i>Oxethira retracta</i>	SW	2									*	*		*		*	*							
	<i>Smicrophylax australis</i>	SW	3																						
	<i>Hydrobiosella michaelsoni</i>	SW	4																						
	<i>Hydrobiosella michaelsoni</i>	SW	4																						
	<i>Maydenoptila</i>	SW	2										*								*				
Plecoptera	<i>Newmanoperla exigua</i>	SW	2																						
	<i>Leptoperla australica</i>	SW	4																						
	<i>Leptoperla australica</i>	SW	4																						
Odonata	<i>Austrosynthemis cyanitincta</i>	SW	3																						
	<i>Austrosynthemis cyanitincta</i>	SW	3																						
	<i>Austroaeschna anacantha</i>	SW	3									*	*				*	*	*						
	<i>Lathrocordulia metallica</i>	SW	3																						
Coleoptera	<i>Scirtidae</i> sp	-	5																						
	<i>Sternopriscus browni</i>	SW	2														*	*							

Table 3.3: Relative composition (%) of the total invertebrate fauna of NJF streams, in perennial reaches (blue) and formerly perennial, now- intermittent reaches (yellow). Taxa uncommon/ absent in 1981-81, but common (comprising more than 5% of total at any site on at least one occasion) during 2016/17 are included. Status in 1981/82 sampling given: Pres. = present in species list (Bunn 1985); ? = unknown, potentially explained by improved taxonomy, rather than absence in 1981/82; N.R. = not recorded in 1981/82. Where known, distribution given: SW = SWA endemic species; WA = Western Australian species (including arid zone); AUS = widespread across the Australian continent (and often beyond); UN = unknown, due to higher taxonomy or lack of information.

		1981/82	Dist.	CC1	CC2	WF1	WF2	CH1	CH2	SS1	SS2	WF3	WF4	CC3	WB1	WB2
Copepoda	Cyclopoida	Pres	-													
Ostracoda	<i>Cypretta</i> sp.	N.R.	WA													
Annelida	Hirudinidae sp.	Pres	-													
Chironomidae	<i>Austropelopia</i> sp.	?	WA													
	<i>Chironomus</i> sp.	Pres	AU													
	<i>Cladotanytarsus</i> sp.	?	AU													
	<i>Dicrotendipes</i> sp.	?	AU													
	Chironomini sp. "V15"	Pres	SW													
	<i>Stenochironomus</i> sp.	?	AU													
	Orthoclaadiinae "DEC1"	?	SW													
	<i>Paralimnophyes pullulus</i>	?	SW													
	<i>Paratanytarsus</i> sp.	?	AU													
	<i>Paratendipes</i> sp.	?	AU													
	<i>Rheocricotopus</i> sp.	?	AU													
	<i>Stictochironomus</i> sp.	?	AU													
	<i>Thienemanniella</i> sp.	Pres	AU													
Diptera	Culicidae sp.	Pres	-													
	Sciomyzidae sp.	?	-													
Ephemeroptera	<i>Cloeon fluviatile</i>	Pres	AU													
	<i>Nousia</i> sp. AV 16	N.R.	SW													
Trichoptera	<i>Hydrobiosella</i> sp. AV17	?	SW													
	<i>Notoperata</i> sp. AV4	?	SW													
	<i>Triplectides</i> sp. AV21	Pres	SW													
Odonata	<i>Aeschna brevistyla</i>	N.R.	AU													
	<i>Archeosynthemis leachii</i>	Pres	SW													
	<i>Hemicordulia tau</i>	N.R.	AU													
	<i>Zephyrogomphus lateralis</i>	Pres	SW													
Coleoptera	<i>Allodessus bistrigatus</i>	N.R.	AU													
	<i>Lancetes lanceolatus</i>	N.R.	AU													
	<i>Necterosoma regulare</i>	Pres	WA													
	<i>Platynectes decempunctatus</i>	Pres	AU													
	<i>Rhantus australis</i>	Pres	AU													
Hemiptera	<i>Anisops hackeri</i>	N.R.	AU													
	<i>Corixiodes juvenile</i>	-	-													
	<i>Siagara mullaka</i>	N.R.	AU													

Seasonal patterns in macroinvertebrate assemblages (H3) Waterfall Gully (always perennial):

Small differences in macroinvertebrate assemblage composition occurred throughout the year (Global $R = 0.26$, $P = 0.001$). Spring (October to December) assemblages were distinct, whereas summer and autumn assemblages overlapped, with different taxa appearing in August peak flows (Figure 3.3). Between October and December 2016, decreasing abundances of Leptophlebiidae mayflies were replaced by a 16-fold increase in baetids, which reversed in February. *Uroctena* sp. amphipods. also increased in average abundance likely reflecting decreased wetted area. The caddisfly *L. parilis* appeared between April and June 2017, whereas by August a large increase in the blackflies occurred in response to winter flows, driving 18% of average dissimilarity between times. Thus, whilst seasonal turnover did occur, much of the assemblage comprised permanent residents; this is similar to patterns reported in Bunn (1985, 1986) across the Wungong and North Dandalup catchments.

Seldom Seen Brook perennial reach (SS2: perennial most years and in 2016/17):

Assemblage composition at SS2 differed over the course of one year at a similar magnitude to Waterfall Gully (Global $R = 0.26$, $P = 0.001$). October (spring) assemblages were distinct from other times (Figure 3.3). *Uroctena* sp. were very common in October 2016 but had virtually disappeared by December, coinciding with increasing densities of *Chironomus* sp.: this drove 24% of average dissimilarity between times. During summer and autumn (trickling) baseflows, the assemblage overlapped between months, with differing frequencies of occurrence of *B. kadjina* mayflies, Chironomidae species and juvenile *Cherax quinquecarinatus* crayfish between times. By peak flows in August, nymphs of *B. kadjina* were absent, while there were increases or arrivals of a few taxa that were also common in Waterfall Gully.

Curtis Creek (perennial 1980s, now intermittent):

Stronger temporal patterns emerged at Curtis Creek (Global $R = 0.43$, $P = 0.001$), and four distinct assemblages emerged: springtime flows (October), summer drying (December to February), early flows (June, overlap with December) and winter flows (August; Figure 3.3). Declining flow between October and December was marked by losses of dipteran and mayfly taxa, presumably through eclosion. *Uroctena* sp. amphipods also disappeared, potentially retreating to subterranean refuges. Between December and February, with the onset of lentic conditions, there was a 30-fold increase in the average abundance of mosquito larvae as well as the arrival of Dytiscidae beetles. Species disappeared as pools dried, and *B. kadjina* persisted in spring-fed pools. August peak flows saw increases in *Uroctena* sp., and arrivals of *L. australica* stoneflies, and dipterans.

Wungong Brook (perennial 1980s, intermittent since 2008):

Assemblage composition differed through time (Global $R = 0.52$, $P = 0.001$), comprising distinct assemblages associated with the flow regime (Figure 3.3). Differences between times were driven primarily by turnover in Chironomidae taxa and arrivals of opportunistic pool specialists commonly found across Western Australia (including the arid zone; Appendix 2). Between October and December 2016 (decreasing flows), differences were largely in reduced average abundances of *B. kadjina* and four genera of Chironomidae, and an increased abundance of ostracods. In February, arrivals of *Cloeon fluviatile*, *Necterosoma darwini* and juvenile Notonectidae contributed most to differences. Average abundances of these species increased over autumn, concurrent with the arrival of additional Dytiscidae beetles, hemipterans and the cosmopolitan dragonfly *Hemicordulia tau* increased over autumn. With the onset of flows in June, much of this autumn assemblage was replaced by a winter assemblage, including the amphipod *Perithia acutitelson*, suspension-feeding chironomids, juvenile Leptoceridae caddisflies (likely *Triplectides* sp.) and the stonefly *L. australica*

Presumably, these taxa would have developed through the winter months; however torrential flows at Wungong Brook during August meant that low abundances of individuals were caught (due to dilution, and dangerous sampling conditions) so this could not be discriminated.

Chandler Brook (perennial 1980s, now intermittent):

Chandler Brook had surface water at only three sampling times, and each time harboured a distinct assemblage (Global $R = 0.46$, $P = 0.001$; Figure 3.3). In October the assemblage was dominated by the mayfly *Nousia* AV16, chironomids *Tanytarsus* sp. and *Paramerina levidensis*; these species had virtually disappeared by December, when receding pools were dominated by mosquitoes and *Chironomus* sp. Surface water was absent from Chandler Brook until August 2017; then the amphipod *Uroctena* sp., predatory chironomid *Austropelopia* sp. and Oligochaeta comprised the early winter assemblage.

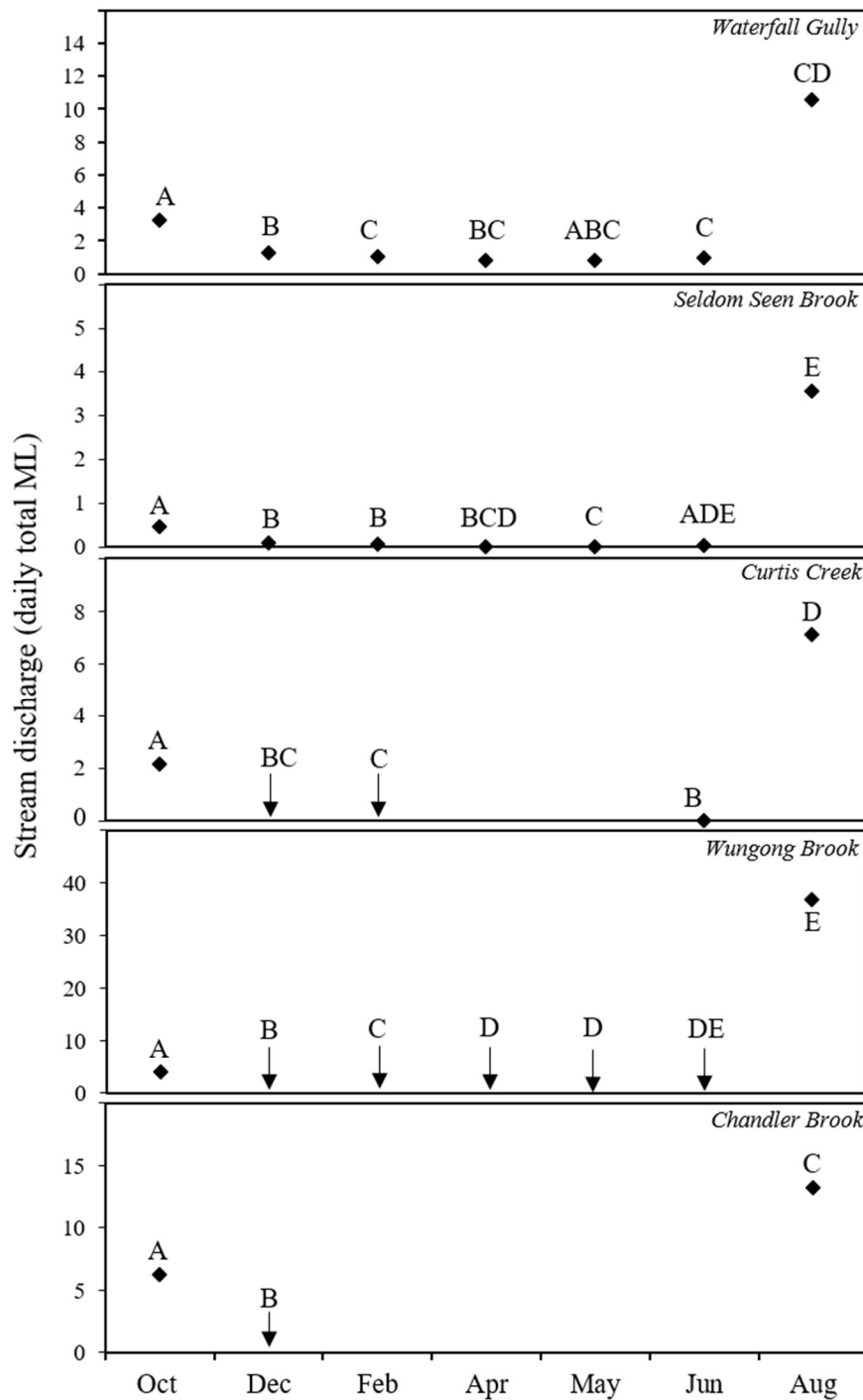


Figure 3.3: Temporal change in macroinvertebrate assemblages at 5 NJF streams over seven sampling events, approximate hydroperiod stage (daily streamflow at sampling) given, arrows indicate disconnected pools. Letters group assemblages that do not differ (one- way ANOSIM). Gauging data for Dept. Water and Environmental regulation (DWER), Waterfall Gully 616023; Seldom Seen Creek 616021; Curtis Creek 616143 (most recent data 2013), Wungong Brook 616041; Chandler Brook 616124 (most recent data 2013).

Discussion

Contemporary and long-term spatial differences in macroinvertebrate assemblages

The macroinvertebrate assemblages occupying formerly perennial, now intermittent streams differed greatly to those in reaches that had never dried. Whereas assemblages of med-region intermittent streams are typically shown to be a subset of that in perennial streams (e.g. Bonada et al., 2007; Chester & Robson 2011; Ruhí et al., 2017), we found little overlap in the common species occurring at perennial reaches and those that transitioned to intermittency. Essentially, streams that had undergone flow regime change harboured an entirely different fauna. Perennial reaches retained longer-lived species without desiccation resistance capabilities, whereas drought resistant species (Ostracoda, Copepoda and some Ephemeroptera) and overwhelmingly, drought resilient species (Hemiptera, Coleoptera, Diptera) dominated in intermittent streams. The fauna inhabiting intermittent reaches tended to have strong dispersal abilities, particularly the contingent of widespread lentic generalist taxa: beetles, hemipterans, dragonflies and dipterans that were common and are also found across the Australian continent. A few species likely use desiccation resistant immature stages to persist (e.g. stonefly *L. australica*, Ostracoda and Copepoda) or proliferate in now-intermittent streams (e.g. mayfly *Nousia* sp. AV16). Therefore, macroinvertebrate assemblage composition differed between perennial and intermittent reaches, because invertebrates in intermittent reaches had adaptations to survive intermittency (primarily resilience traits) whereas those restricted to perennial reaches evidently did not.

There were clear differences in distributions of formerly common, endemic, relictual species and widespread intermittent stream generalists, demonstrating that for many of the former, transition of flow regimes to intermittency comprises a threshold change causing extirpation in streams where they had previously been the most common species. This included monotypic genera that are ancient and unique to the region. Thus, the hypothesis (H₄) that

regionally endemic and relictual species would be particularly impacted by drying, which would create suitable habitat for widespread generalist species, was supported with few exceptions. The species commonly found at Waterfall Gully, and absent/rare in intermittent reaches, also comprised the most common species during 1981/82; thus, the hypothesis (H₂) that assemblages at perennial reaches would not have changed since Bunn (1986; et al., 1986), whereas those at reaches affected by flow regime change would differ, was also supported. Studies exploring regionwide changes in stream faunas in European med-regions predict transition from lotic adapted, cool water fauna, associated with perennial flows, to lentic adapted, generalist species and those with drought survival strategies (Pace et al., 2013; Floury et al., 2013; Alba-Tecedor et al., 2017), and sound a warning as to losses of regional biodiversity as climate-sensitive species become progressively displaced towards higher latitudes and altitudes (Kroll et al., 2017; Domisch et al., 2013; Sáinz-Bariáin et al., 2016). Similarly, Pyne & Poff (2017) predicted up to a 35% loss of species in the Californian med-region under conservative CO₂ emissions scenarios (also see Lawrence et al., 2010). Climatic trends are being repeated in med-regions globally, thus it is perhaps unsurprising the consequences predicted for Northern Hemisphere med-regions are, to an extent, playing out in SWA (Filipe et al., 2013; Bonada & Resh, 2013).

In some instances, turnover in species may offer taxonomic and/or functional redundancy, buffering communities and ecosystem function under climate drying scenarios (Rosenfeld, 2002; in desert streams, Boersma et al., 2014). For example, all three formerly most-common dragonflies reported by Bunn (*Austroaescha anacantha*, *Austrosynthemis cyanitincta* and *Lathrocordulia metallica*) were completely restricted to perennial reaches in 2016/17. These were replaced by widespread species adapted to shorter hydroperiods (fast larval development), or endemic species known to have drought resistance, for example larval *Archeosynthemis leachii* can reportedly withstand drying (Theischinger & Hawking, 2006).

Within mayflies, half of the formerly most common and widespread species are now completely restricted to Waterfall Gully, and two species (Leptophlebiidae *B. kadjina* and *N. bunni*) have apparently adapted to intermittency. Two new arrivals replaced those lost at intermittent reaches: a representative of the Gondwanan genus *Nousia* (see Christidis, 2006) was abundant in intermittent reaches during flow, and cosmopolitan species *Cloeon fluviatile* (Baetidae) occurred in receding summer pools, neither species were recorded in Bunn's study. Several suspension feeders were lost from intermittent streams, including caddisfly *Smicrophylax australis*; however, increases in suspension feeding Diptera supplemented this functional guild. Because turnover is occurring within some functional groups (shredders were a notable exception), this demonstrates that there may be some functional redundancy occurring within NJF stream communities experiencing transition to intermittency (*sensu* Boersma et al., 2014; Crabot et al., 2021; but also see Piano et al., 2020). This suggests that ecological roles historically performed by locally endemic species may be maintained, particularly within gathering collectors (replacement of mayfly species, increased abundance of Chironominae) and predators (some turnover in Odonata, arrival of new Coleoptera and Hemiptera). Within shredders, most large caddisfly species were extirpated whilst increases in abundance of the leaf-mining chironomid *Stenochironomus* sp. may be supplementing that role (as observed of this genus by de Mello Cione et al., 2021), however their efficacy as shredders in these streams is uncertain (but see Chapter 6). Whilst some taxonomic redundancy may also be occurring, our results suggest that more broadly, NJF stream communities may be becoming more homogenized with the wider SWA land division, as generalist taxa found across the continent replace specialist, locally endemic taxa.

Seasonality of macroinvertebrate assemblages

Bunn (1986; et al., 1986) noted that macroinvertebrate assemblages varied throughout the year, as organisms with life history adaptations to match the increases and decreases of flow

completed their life cycles (also see Bunn 1988a). There was some distinction between summer and winter assemblages, the former dominated by developing stonefly nymphs and some mayflies, and the latter by different mayflies and caddisflies. However, there was also a year-round assemblage of longer-lived or fully aquatic obligate taxa including dragonflies and caddisflies, amphipoda and isopoda. This general pattern was repeated at Waterfall Gully during this study. Surprisingly though, the general pattern of seasonal turnover also occurs at the short spring-fed reach on Seldom Seen Brook, suggesting that a trickle of flow is sufficient in these streams to support a year-around assemblage of permanent residents.

In contrast, at reaches that are now permanently intermittent, seasonally distinct assemblages occurred in accordance with the phase of flow, supporting H₃. Across intermittent streams, early colonisers were highly variable, as timing of the onset of flows was not uniform. The identities of early colonisers typified that reported from intermittent streams elsewhere, comprising Diptera (including Chironomidae) and stoneflies with desiccation resistant life stages (e.g. Boulton & Lake, 1992; Bogan et al., 2013). Resilient taxa colonised intermittent streams, homogenising the assemblages across intermittent reaches and streams through spring. Highly mobile opportunists arrived as streams began to dry. Some species present in remaining perennial streams, such as mayflies *Bibulmena kadjina* and *Nyungara bunni*, persisted in pools after longitudinal disconnection, however the predation pressure from arriving species probably puts significant pressure on their survival to adulthood. We emphasise the similarity between patterns observed in formerly perennial NJF streams and those widely reported from intermittent rivers in mediterranean and arid climate zones, where predictable (mediterranean) or episodic (arid) cyclic changes between lotic, lentic, and dry phases mediate temporally disparate faunal communities occupying the same space (Bogan & Lytle, 2007).

Chironomidae, an unexpected source of functional redundancy and community resilience?

The Chironomidae assemblages within the Wungong catchment streams exhibited as much complexity in flow regime preferences as the other faunal groups, and there were extirpations, incursions, and changes in relative abundance of Chironomidae between 1981/82 and the 2016/17 assemblages. For example, *Chironomus* sp. (Chironominae) and *Cricotopus* sp. (Orthoclaadiinae) are now ubiquitous in intermittent stream reaches during summer drying. Members of both these genera elsewhere have desiccation resistant life stages: *Chironomus* congeners can withstand losses of body water up to 84% (Frouz & Martena, 2015) and several genera were found to overwinter in dry temporary ponds as eggs or aestivating larvae, including *Cricotopus* sp., *Tanytarsus* sp., and *Polypedilum* sp. (Wiggins, 1980). A suite of Chironomidae were absent from, or rare in, intermittent stream reaches, including most of the species collected by Bunn (1986) including *Riethia* spp., *Stictocladius* sp. (formerly *Diplocladius uniserialis*).

The dynamics of Chironomidae in drying streams may be of particular importance, however this group are seldom accounted in most studies (usually beyond inclusion at the Family level, if at all) due to their difficult taxonomy, and status as a ‘pest’ organism, despite contributing a large proportion of species diversity in most systems (Raunio et al., 2011). Chironomidae have been demonstrated as becoming more dominant in streams impacted by climate change (e.g. warming in tropical streams, Dudgeon et al., 2020; simulated drying in mesocosms, Hille et al., 2014). Cañedo-Arguelles et al., (2016) propose that the Chironomidae are useful indicators of habitat condition and water scarcity, as their assemblages are closely tied to flow regime and possession of drought resistance traits. Functional organisation also differed according to hydrology: grazing collectors and shredders were more prominent in intermittent streams, and predators and filtering collectors were more common in perennial streams (Cañedo-Arguelles et al., 2016). Chironomidae are

often among the first colonisers in intermittent and ephemeral streams (Boulton et al., 1992), or in perennial streams following drought (Churchel & Batzer, 2006), and here the tanypod *Austropelopia prionoptera* and Orthocladinae sp. DEC1 were early colonisers of the driest streams. Ultimately, Chironomidae may offer some functional redundancy in invertebrate-moderated stream ecosystem processes, where the taxa previously associated with that process (e.g. breakdown of detritus by shredding macroinvertebrates) become rare or extinct. This is demonstrated in these study streams, as the replacement of key shredding leptocerid caddisflies *Notoperata tenax* and *Lectrides parilis* with leaf-mining *Stenochironomus* sp. midges, may fill the gap in detrital processes caused by their absence (Chapter 6). The role of Chironomidae, not only as indicators of drying flow regimes (*sensu* Cañedo-Argüelles et al., 2016), but also as ecological service providers in a changing world warrants more investigation, as does their elevation from ignored pest to important ecosystem component.

Drying flow regimes in med-region stream networks

Naturally intermittent streams can function as stepping stones for organisms dispersing between perennial streams (e.g. Robson et al., 2011), however as flow regimes become drier, intervals between flows are becoming longer and smaller streams are essentially disappearing. This essentially limits opportunities for organisms to move between and occupy aquatic habitats in space and time (e.g. Cañedo-Argüelles et al., 2020). Recent research suggests that under climate change, increasing streamflow intermittency in med-regions is increasing the importance of dispersal in determining community composition and metacommunity dynamics (Gauthier et al., 2020; Cañedo-Argüelles et al., 2020). That is, an organisms' ability to traverse increasingly dry landscapes is predicted to determine its distribution among the stream network. However, this assumes organisms with the ability to disperse are also successfully colonising newly intermittent streams (e.g. Chester et al., 2015; Li et al., 2020; Lancaster & Downes, 2017b; Downes & Lancaster, 2018). This does not

appear to be the case, as the suite of organisms extirpated by drying in NJF streams included many taxa with presumably strong powers of dispersal, including large dragonflies and caddisflies. These organisms are expected to be able to move between perennial and intermittent streams in this study (distances generally less than 3km) even if such dispersal resulted in wasted breeding effort due to the short hydroperiod of these streams. This implies other limitations to their recolonisation. Phenological mis-match is a potential barrier for some species, as the temporal constraints of seasonal drying may preclude oviposition in species that breed over summer and autumn (Bunn, 1988; Chapter 5). Species that undergo dormancy in the egg phase may also have specific requirements for hatching (although temperature has not changed substantially since Bunn's study; Chapter 5). Dispersal distances may be useful in predicting changing distribution patterns in larger landscapes, however at the scale of this small study catchment, dispersal distance did not appear to explain the absence of these species from intermittent streams.

Landscape level shifts in hydrology are expected to be a permanent change in SWA (Andrys et al., 2017; McFarlane et al., 2020), and are demonstrated here to have profound effects on stream assemblages. In SWA the capacity for species to contract their ranges has geographical limits, and inevitably extinction awaits many endemic species as these limits are crossed (Davies 2010). For example, species distribution modelling has shown that for some SWA endemic dragonflies, no suitable habitat will exist by 2055 (Bush et al., 2014). The results of this study support that prediction, and imply that protection of flows in perennial streams would be the only way to conserve the uniqueness of SWAs ancient fauna.

Furthermore, biodiversity in the southwest is naturally low, owing to protracted period of isolation from the southeast of Australia (Bunn & Davies, 1990), therefore losses of these species are likely to have ecosystem wide consequences. As climate change progresses, perennially flowing streams are predicted to all but disappear across SWA. A second region

wide threshold may be encountered as permanent pools and dams become drier, further depressing regional biodiversity. In particular, the outlook for SWAs stream-dwelling, endemic Gondwanan relictual fauna appears to be grim.

Chapter 4

Reduced alpha diversity and increased turnover (β -diversity) in
formerly-perennial headwater streams affected by permanent drying

The following chapter has been drafted in accordance with formatting for the journal *Oecologia* which is its intended destination.

The manuscript is currently prepared for submission.

The following authors contributed to this manuscript as outlined below:

	Author order	Concept & study design	Fieldwork	Laboratory work	Data analysis	Writing manuscript	Review/edit paper
Nicole Carey	First	X	X	X	X	X	X
Edwin Chester	Second	X	X				X
Belinda Robson	Third	X			X		X

By signing this document, the Candidate and Principal Supervisor acknowledge that the above information is accurate and has been agreed to by all other authors.

Candidate

Principal Supervisor

Abstract

Global warming is causing climatic drying, particularly in Mediterranean (med-) climate regions, leading perennially flowing streams to dry out and switch to intermittent flow regimes. Flow regime is a fundamental factor influencing stream ecosystems, therefore novel drying is expected to substantially alter biodiversity patterns. Examination of spatial and temporal patterns in beta diversity (turnover and nestedness) can elucidate broader changes to regional diversity caused by drying, which may also inform conservation efforts. Spatial and temporal patterns of macroinvertebrate alpha and beta diversity were examined in streams impacted by flow regime drying, compared to a historical “before drying” study in the same streams using a multiple before-after, control-impact (mBACI) study design. Alpha diversity was lower in reaches that switched to intermittent flows but was unchanged at the remaining perennial stream. Spatial beta diversity was dominated by turnover, species assemblages of intermittent reaches were not only distinct from that of perennial reaches, but also from each other. This is contrary to patterns understood to be common among med-region intermittent streams, where assemblages tend to be subsets of that occurring in perennial streams.

Temporal beta diversity was high at reaches that became intermittent, and also dominated by turnover. Some reaches also had partially nested faunas meaning that species lost through time were only partially replaced. Overall, flow regime drying altered biodiversity patterns in streams that became intermittent. High species turnover may be indicative of stream networks undergoing rapid drying due to climate change, shaped by lack of adaptations to drying among the perennial stream fauna, and the effects of localised environmental factors maintained by subsequent dispersal (or lack thereof).

Introduction

In stream ecosystems, the volume, timing, duration and predictability of flows constitutes a natural disturbance regime with which organisms have evolved (Gasith & Resh, 1999; Lake 2000). Global changes in precipitation patterns are altering the hydrological balance in many regions (Hoegh-Guldberg et al., 2018; IPCC 2021) thereby causing shifts in the most fundamental abiotic factor controlling aquatic community organisation (Resh et al. 1988; Lytle & Poff, 2004). In the mediterranean climate regions of the world (hereafter med-regions), hydrological change is manifesting as drying flow regimes: perennial rivers and streams are switching to intermittent flows, and intermittent rivers and streams are becoming drier (Polade, et al., 2017; Datry et al., 2018; Zipper et al., 2021). The ecological implications of hydrological change are gaining increased research attention, and anticipated impacts are becoming realized, such as species distribution shifts and range contractions, alteration of aquatic communities and extirpation of sensitive species (Bogan & Lytle, 2011; Filipe et al., 2013; Kroll et al., 2017; Pyne & Poff, 2017; Bruno et al., 2019; Piano et al., 2020). However, empirical studies before and after a defined climate change impact are still relatively rare (not just in stream ecology; *sensu* Jackson & Furdere, 2006; Urban et al., 2016; Didham et al., 2020a).

In the context of environmental change, there is growing interest in the role of beta (β) diversity in community organization and for predicting change in regional biodiversity (Soininen et al., 2018), including in aquatic ecosystems (e.g. Cañedo-Argüelles et al., 2015; Leigh & Datry, 2017; Sarremejane et al., 2017; Crabot et al., 2021). Beta diversity is the additive (or multiplicative, Baselga, 2010) diversity between local alpha diversities (α) within the regional species pool (gamma diversity γ) (Anderson et al., 2011), and in stream ecology may be defined at the scale of a connected stream network, or between rivers/streams in a drainage basin (e.g. Rolls et al., 2018). Appropriate assessment of β diversity may enable

prediction of changes to regional diversity from known changes in α diversity (*sensu* Socolar et al., 2016; Rolls et al., 2016). Some disturbances can spatially homogenize assemblages between sites (reducing β diversity); others may increase spatial heterogeneity (increasing β diversity) (Baselga, 2010; Socolar et al., 2016) or there may be no change at all (Horváth et al., 2019). Beta diversity can be increased either by increased turnover (species replacement between sites) or increased nestedness (species loss at a site without replacement). Scenarios where environmental change results in dominance of either turnover or nestedness have different conservation implications (Baselga, 2010; Socolar et al., 2016), for example in planning conservation reserves to protect a few highly diverse sites (high nestedness scenario) or where there is a requirement to protect many different habitats to maintain regional biodiversity (high turnover scenario). Measurement of temporal (as well as spatial) β diversity may point to associations between environmental factors and species sorting in space (Korhonen et al., 2010; Ruhí et al., 2017; Leigh et al., 2019; Jabot et al., 2020). This may also include identification of ark-type refuges (*sensu* Robson et al., 2013) that consistently harbour the bulk of the regional species pool through time.

Intermittent rivers and streams (those that predictably cease to flow for at least some part of the year; Datry et al., 2017a) are the most common type in med-regions. In these streams, drying is a natural disturbance that has occurred predictably in time and space for hundreds or thousands of years (Hershkovitz & Gasith, 2013). Intermittent streams in med-regions have an invertebrate fauna that possesses life history traits to cope with drying, including phenological traits, desiccation resistant life stages, or resilience traits, which are the ability to leave and recolonise intermittent reaches when more favourable conditions return (Strachan et al., 2015; Bogan et al., 2017). However, med-regions also contain perennial streams which often support species not adapted to drying, and which also act as refuges for resilient species during the dry season (e.g. Robson & Chester 2011; Soria et al., 2017). The

hydrology of intermittent streams can vary strongly both spatially and temporally. They may dry from the headwaters down, or from the lower reaches upstream, or in patches and they may retain spring-fed pools or flowing reaches (Bogan et al., 2017). This range of flow regimes and level of adaptation among stream fauna mean that med-regions typically have a high aquatic biodiversity, despite being dry biomes (Bonada et al., 2008; Bonada & Resh, 2013). Endemism also tends to be high, and in the Southern Hemisphere a suite of the fauna comprises relictual species originating prior to the breakup of the Gondwanan supercontinent (Figueroa et al., 2013; DeMoor & Day, 2013; Davies & Stewart, 2013; Bonada & Resh, 2013). These relictual faunas are from cooler, wetter times, and are one reason why med-regions are expected to be particularly vulnerable to extinctions caused by anthropogenic climate change (Filipe et al., 2013).

Arrangements of intermittent reaches along a river may determine the patterns of α and β diversity within the stream hierarchy, because of unidirectional flow (Stubbington et al., 2017; Tonkin et al., 2018). Alpha diversity typically decreases with increasing frequency or duration of flow intermittence (Datry et al., 2014; Soria et al., 2017), and when drying occurs in headwaters, α diversity is often reduced because recolonisation of aquatic obligate and short-range terrestrial dispersers is limited (e.g. Crabot et al., 2020). Conversely, stream networks with perennial headwaters often have higher α diversity, promoting downstream colonisation from headwaters to intermittent reaches via drift (e.g. Mackie et al., 2013; Sarremejane et al., 2017). Thus headwaters may be important dry season refuges (Chester and Robson 2011; Chester et al., 2015). Dynamics of drying acting upon dispersal processes and life history traits therefore shape β diversity, both between adjacent headwaters and downstream communities (Finn et al., 2011; Cañedo-Argüelles et al., 2015; Rolls et al., 2018; Gauthier et al., 2020). Consequently, spatial and temporal β diversity tend to be higher in intermittent than perennial streams, because of the marked difference in habitat heterogeneity

in space and time (Schriever et al., 2015; Rolls et al., 2016; Tonkin et al., 2017). Furthermore, because of the propensity for increasing dryness to eliminate species, and for perennial reaches to harbour taxa without drought survival strategies, macroinvertebrate communities within intermittent streams tend to be a subset of those found in proximal perennial reaches (Ascott, 2010; Chester & Robson, 2011; Datry et al., 2014; Datry et al., 2017; Vander Vorste et al., 2021), with varying contributions of intermittent-specialist taxa to regional biodiversity (e.g. Bogan et al., 2013).

Whilst nestedness between perennial and historically intermittent streams is often demonstrated in med-regions (e.g. Bonada et al., 2007; Chester & Robson 2011; Ruhí et al., 2017) there have been few opportunities to directly study the response of β diversity dynamics in streams exposed to threshold environmental change caused by global warming. Predicted reductions in flow permanence are expected to extirpate species not adapted to streamflow drying thereby altering assemblages (Filipe et al., 2013; Pyne & Poff, 2017; Kroll et al., 2017; Crabot et al., 2021); this has also been demonstrated in desert (Bogan & Lytle, 2011) and alpine streams (Piano et al., 2020; Bruno et al., 2019). It has been anticipated that increased drying would intensify nestedness patterns in med-stream communities, as species are progressively lost from reaches affected by drying and a smaller pool of resistant (or resilient) species remain (Datry et al., 2014; Vander Vorste et al., 2021). However, recent research examining newly intermittent rivers in humid-continental Europe found community change to comprise mostly turnover (Crabot et al., 2021). Potentially, different spatial and temporal dynamics occur in rivers that have transitioned from perennial to intermittent flow regimes than in rivers that have historically been intermittent, resulting in different patterns in β as well as α diversity.

The climate of southwestern Australia (hereafter SWA) has a mediterranean climate and has been experiencing an incremental decrease in rainfall yielding a 20% decline since the 1970s,

coupled with a 70% decline in streamflow. This has led to transition to intermittency in many (formerly) perennial streams, a change which is expected to be permanent (Andrys et al., 2017; McFarlane et al., 2020). Prior to the onset of intermittency, detailed ecological studies were conducted in SWA forested headwater streams, documenting macroinvertebrate assemblage composition and its seasonal patterns (Bunn 1985; Bunn 1986; Bunn *et al.*, 1986). The ecological studies conducted for the purpose of describing *in situ* community composition and characteristics (rather than measuring another impact or trend; *sensu* Didham et al., 2020b) provides a true baseline with which to compare change, a circumstance rare in stream ecology (*sensu* Jackson & Füreder, 2006; and insect ecology generally, Didham et al., 2020b). Furthermore, because the streams flow through protected, forested water catchment, drying has occurred in the absence of other (potentially confounding) environmental changes.

As climate change impacts have occurred earlier in SWA than in other med-regions (where similar changes are expected; Polade et al., 2017), these streams provide a particularly valuable opportunity to study actual outcomes of flow regime drying on stream biodiversity. We compared α and β diversity of invertebrates at two temporal scales: 1) contemporary differences between reaches that retained perennial flows and those that transitioned to intermittency; and 2) differences between contemporary assemblages and the historic study by Bunn (when those same stream reaches were perennial). For the historical comparison, we used a multiple before-after, control-impact (mBACI) design, as these designs are optimal where there is a discrete step-change at the impact sites (Downes et al. 2002 p171), as occurs with transitions from perennial to intermittent flow regimes. To understand the impacts of climate change, we need information from both before and after change commences, both from streams that have stayed perennial (controls) and from those that have become intermittent (impact) (Downes et al., 2002; Didham et al., 2020b). The difficulty of obtaining

true ‘before’ data is the reason that long-term studies of the effects of climate change on ecological processes are rare in most parts of the world (*sensu* Didham et al., 2020b).

We tested seven hypotheses, addressing two levels of spatial and temporal comparisons of α and β diversity. Firstly, we tested two hypotheses by making spatial comparisons of reaches under contemporary (2016/17) flow regimes, H₁: α diversity will be higher in reaches that remained perennial than in those that transitioned to intermittent flows because in med-regions, perennial reaches have been shown to harbour greater alpha diversity than reaches that dry (Chester & Robson 2011; Soria et al., 2017) and because the SWA stream fauna includes many Gondwanan relictual species likely to depend on perennial flows (Davies 2010; Davies & Stewart, 2013). H₂: assemblages in intermittent reaches will be a subset of those in remaining perennial reaches (i.e. high nestedness, low turnover in β diversity) as this is a widely reported pattern in med-regions (e.g. Chester & Robson, 2011) and because drying is expected to extirpate species that lack desiccation-resistance or resilience.

To examine patterns in α and β diversity between contemporary and historic assemblages in the same study reaches (Bunn 1985; 1986, all reaches perennial), we tested five hypotheses: H₃ α diversity will not differ between 1981/82 and 2016/17 in reaches that remained perennial because the flow regime has not changed. Note that this comparison is a key part of the mBACI design (before-control vs after-control) because it shows whether there have been temporal changes other than flow regime. H₄: α -diversity will be lower in 2016/17 in reaches that transitioned from perennial to intermittent flow than in the historic study (when same reaches were perennial) because drying will have extirpated susceptible taxa. Note that this comparison forms the before-impact vs after-impact contrast in the mBACI design. H₅: catchment level β diversity will be higher in 2016/17 because of increased heterogeneity of flow regimes (and therefore habitat types), than it was in 1981/82 when all reaches had perennial flow. H₆: within each reach, temporal β diversity will be lower at reaches that

remained perennial (in 2016/17 cf. 1981/82) than in reaches that have transitioned to intermittent flows (in 2016/17 cf. 1981/82). H₇: the 2016/17 assemblage at each intermittent reach will be a subset of that present in 1981/82 (when the reach was perennial) because drying will have extirpated species from those reaches.

Methods

Study area

The mediterranean climate in SWA has been warming and drying for at least five decades (Bates et al., 2008). Mediterranean climates typically show marked inter-annual variation in rainfall, but in this region, there has been a reduction of ~ 20% of mean precipitation since 1970 (Hope et al., 2010). Additionally, the “big wet” years that historically punctuated rainfall records no longer occur (the most recent was in 1974). The study streams are situated in the forested Wungong catchment, within the Northern Jarrah forest (NJF) of SWA (Figure 3.1). Rainfall totals measured at the study sites (Wungong Dam, Bureau of Meteorology station 009044) have steadily declined since records began in 1913 (Figure 2.2). Between 1913 and 1960, annual mean rainfall was 1386 mm dropping to 1011.4 mm between 1990 and 2017 – a 27% reduction in annual rainfall.

Historically, there were many perennial headwater streams in the NJF, their baseflows generated through groundwater discharge into the stream zone and formation of perched water tables following winter rains, generating throughflow and surface runoff (Gentili 1989). Declining water tables and eventual disconnection of groundwater from streambeds, beginning around the years 2000 – 2001 in most catchments, caused most streams to transition from gaining to perched streams, and for perennial streams caused a transition to intermittency (Petrone et al. 2010; Hughes et al., 2012; Kinal & Stoneman, 2012).

The NJF is dry sclerophyll forest, dominated by eucalypt trees, jarrah *Eucalyptus marginata* and marri *Corymbia calophylla*. The Wungong catchment is protected water catchment, which means that no human disturbances are allowed and there has been no mining or other disturbance in the catchment since Bunn's studies. Some streams have small flow gauging weirs but none of the streams are regulated, although they all eventually terminate in the Wungong water supply reservoir. Consequently, the only changes to streamflow regimes have been caused by the changing climate. Thirteen reaches were selected spanning the Wungong Brook and each of its four major tributaries (Figure 4.1; Table 2.1). Eight of these reaches were studied by Bunn (1986a; 1986b; 1988) and Bunn et al., (1986) in the early 1980s, when all streams were perennially flowing (Table 2.1; Figure 4.1). Flow gauging data from this catchment from the 1960s onwards shows clearly the transition from perennial to seasonally intermittent flows in most of these streams (Figure 2.5; Figure 2.9; Figure 2.11). Only one stream in this catchment, Waterfall Gully (and its tributaries), remains perennial; flows there have not ceased since records began in 1966. One other stream, Seldom Seen Brook, has retained one short, spring-fed reach which is perennial in most years, slowing to a trickle in the summer months (SS2). Thus, in 2016 and 2017, there were only two waterways that contain flowing water all year. Although discharge volumes have declined, intra-annual flow patterns have changed little in these perennial reaches (Figure 2.7) and would not be expected to alter seasonal patterns in stream invertebrate assemblages. The remaining streams (Wungong Brook, Curtis Creek and Chandler Brook) have switched to intermittent flows along their entire length and are now dry for 6 (or more) months per year.

Chandler Brook is the second largest waterway in the catchment after Wungong Brook, but flow gauging data there is incomplete, and records ceased after 2014. Bunn (1986) and Bunn et al. (1986) had sites in Chandler Brook and Curtis Creek when they were perennially flowing (*nb.* Bunn erroneously labelled Chandler Brook as Wungong Brook in his study).

Currently, Chandler Brook is flashy, discharging large volumes of water over winter and spring, and drying down quickly in summer. In 2016/17, during the dry phase in intermittent streams (February to June), streambeds were explored along most of their length, in search of remnant pools or seeps which may have acted as refuges for aquatic fauna. There were no permanent pools and no evidence of hyporheos in Chandler Brook (contains extensive areas of laterite bedrock), whereas Wungong Brook retained pools until at least May, although volumes were severely retracted.

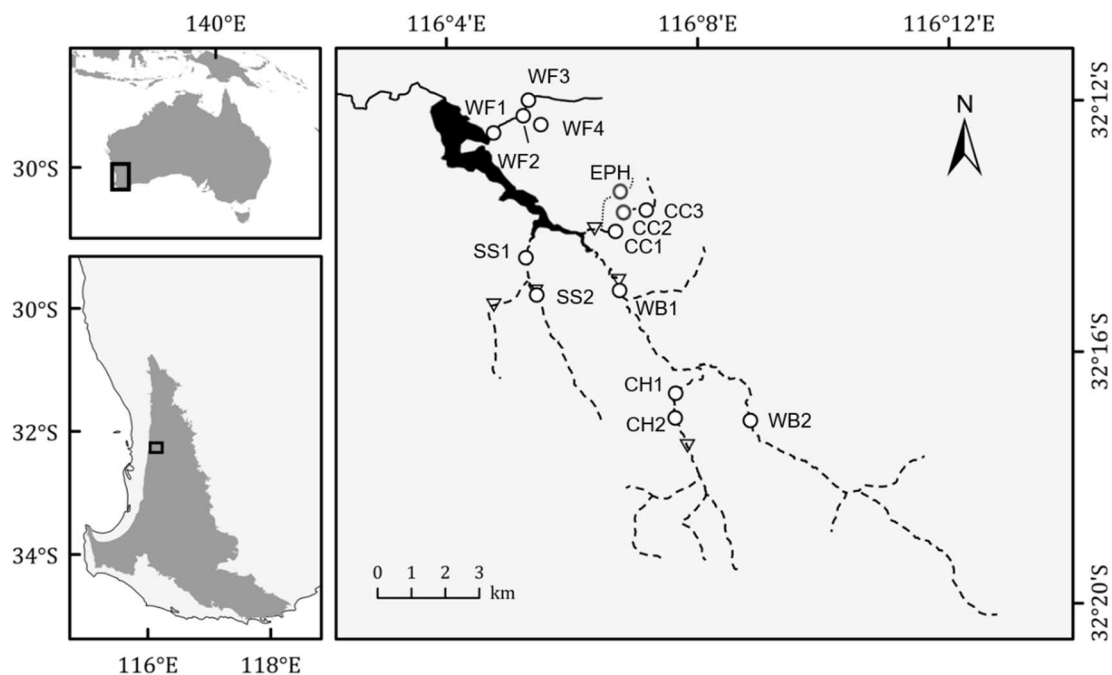


Figure 4.1: Study reaches in the Wungong Brook catchment of south-western Australia, perennial reaches (solid lines), formerly perennial, now intermittent reaches (dashed lines), and an ephemeral stream (dotted line). Inverted triangles denote flow gauging stations (flow gauge at WF1 not visible). Sites used by Bunn (1988) were WF1, WF2, CC1, CC2, SS1, SS2, CH1 and CH2; additional sites WF3, WF4, CC3, WB1, and WB2.

Flow gauging data at Curtis Creek is incomplete, however it was perennially flowing at the time of Bunn's research (1981–1984) and has since switched to intermittent flows. The upper

reaches are narrow and well vegetated, with deeper substrate overlaying granite and laterite bedrock, and the water was seldom deeper than 0.3 m at the study reaches. Curtis Creek dried completely by April 2017; however, it retained two spring fed pools at the base of a granite inselberg in the headwaters and at site CC2 (Figure 4.1). Because of the increased heterogeneity in flow duration caused by drying, five reaches were added to those sampled by Bunn to better represent this diversity: two sites in a perennially flowing tributary of Waterfall Gully (WF3, WF4), Curtis Creek site 3 CC3, and two sites on Wungong Brook (WB1, WB2; Figure 4.1; Table 4.1).

Table 4.3: Summary of flow regimes of Wungong catchment streams, including contemporary flow regime, distinguishing between intermittent streams that dry quickly, and those with persistent pools (Intermittent-short vs -long hydroperiod), and dry season surface water presence. Bunn SR = study reaches used by Bunn (1985, 1986, 1988) and Bunn et al., (1986). All study reaches were perennial until the early 2000s.

Stream name	Site codes	Bunn SR	Flow regime, hydroperiod	Dry season surface water
Waterfall Gully	WF1	Y	Remains perennial	Flowing
	WF2	Y	Remains perennial	Flowing
	WF3	N	Remains perennial	Flowing
	WF4	N	Remains perennial	Flowing
'Kunzia Creek' (WF tributary)				
Seldom Seen Brook	SS1	Y	Intermittent - short	Completely dry
	SS2	Y	Perennial reach most years	Trickling flow from spring
Curtis Creek	CC1	Y	Intermittent - short	Completely dry
	CC2	Y	Intermittent - protracted	Small springfed pool
	CC3	N	Intermittent - protracted	Small springfed pool (above)
Wungong Brook	WB1	N	Intermittent - protracted	Pools last until late summer
	WB2	N	Intermittent - protracted	Pools last until late Autumn
Chandler Brook	CH1	Y	Intermittent - short	Completely dry
	CH2	Y	Intermittent - short	Completely dry

In these streams, baseflow is supplied by groundwater, with higher flows in winter originating from the formation of temporary surface aquifers filled by rainfall (Gentili 1989; Petrone et al., 2010). Once groundwater tables dropped below the streambeds, streamflow resulted solely from those temporary surface aquifers (Petrone et al., 2010). Consequently, in formerly perennial streams, flow now commences in June or July where formerly perennial flows increased above baseflow in March-April (Figure 2.6). Thus as well as intermittency, these streams show a seasonal delay in flows in response to drying, as reported across forested streams in SWA (McFarlane et al., 2020).

All streams in the Wungong catchment were fresh and slightly acidic, and springtime water temperatures ranged between 14.3 and 17.8 °C (Table S4.1). Similarly, in 1982, Bunn found the streams to be fresh, slightly acidic and with temperatures from 10 – 19 °C throughout the year (Table S4.1). Conditions were consistent between the two studies in winter and spring (see Bunn 1985); the only marked differences were in summer. Seldom Seen Brook was ~8 °C warmer in summer 2016/17 than 1981/82 due to high ambient temperatures and trickling flows (Table S4.1), corresponding to the substantial decline in summer discharge (Figure 2.9). Drying pools in Wungong Brook also warmed up to 28.5°C in summer, much higher than any water temperatures measured in summer 1982 when the streams were flowing (Table S4.1, Bunn 1985). Summertime temperatures on Curtis Creek were moderated by groundwater inputs from the spring at reach CC2 and the disconnected spring-fed pools on Curtis Creek became more acidic in summer and into autumn (Figure 4.1, Table S4.1). Bunn did not record any pH levels below 5.45 in 1982 (Bunn 1985). Substantial amounts of CPOM that accumulated in the upper reaches of Curtis Creek likely contributed higher DOC concentrations, reducing pH. Stream temperatures in winter were also slightly lower in the 2016/17 study than recorded by Bunn, due to reduced groundwater inputs across the study area and thus less buffering capacity to cold air temperatures (frequently below freezing

overnight). Given that Bunn (1985) did not measure conductivity (he measured ionic composition and concentration in summer and winter (Na^+ , Mg^{2+} , Ca^{2+} and K^+)), comparisons are difficult. However, it appears that salinity has not changed in these streams (Table S4.1).

Study design

To test the hypotheses, we needed to sample intermittent and perennially flowing streams at different times throughout the year ($H_1 - H_2$) and also to compare contemporary (2016/17) invertebrate assemblages to those recorded by Bunn in 1981/82 ($H_3 - H_7$). Consequently, we collected macroinvertebrate species richness and abundance data from each of the thirteen study reaches (including the eight reaches sampled by Bunn) between October 2016 and August 2017 at approximately 6-week intervals, giving seven sampling times in total. Sampling at regular intervals avoids false inferences associated with “snapshot” effects (*sensu* Didham et al., 2020b) and the seasons we sampled matched those of Bunn (1985). Sampling targeted particular stages in the annual flow regime, commencing when all reaches were flowing (spring). Intermittent reaches began to dry and disconnect in November and December, and by autumn (April – July 2017) they had dried completely. Sampling therefore commenced in austral spring 2016 and ended with late winter 2017, encompassing one year, similar to Bunn (1985).

This design enabled us to use an mBACI design (Downes et al., 2002) whereby the assemblage composition recorded from perennial streams by Bunn comprised the before-controls and before-treatment (i.e. would become intermittent) and our samples comprised the after-controls (i.e. still perennial) and the after-treatment (i.e. now intermittent) samples. This design is ideal for identifying the effects of flow regime change because any changes detected in the control (perennial) reaches in 2016/17 comprise changes in temperature and any other changes to which all streams were exposed, except for flow regime change

(Downes et al., 2002). The intermittent reaches, which were also exposed to flow regime change can then be compared to the controls, and the difference between them comprises the effect of flow regime change. Two reaches on Waterfall Gully were sampled in both studies (WF1 and WF2) and have never stopped flowing, and thus were control sites under the mBACI design. As the spring-fed reach on Seldom Seen Brook (SS2) has dried completely in very dry years, it was considered a treatment reach along with the remaining five now-intermittent reaches (SS1, CC1, CC2, CH1 and CH2). Therefore, the mBACI design comprised two before-control/ after-control reaches, and six before-treatment/ after-treatment reaches, with a gradient of drying flow regimes (semi-perennial to strongly intermittent).

Bunn sampled riffle zones only, taking two samples per reach per sampling time. Because of the increased heterogeneity in flow environment, and different aims of this study (i.e. presenting empirical evidence of the impacts of climate change on aquatic communities, rather than characterising the macroinvertebrate community of previously little-studied streams), we increased sampling effort within each sampling time. Where surface water permitted, four samples were taken from erosional habitats (riffles or runs) and four from depositional habitats (pools) in each reach. Heterogeneous flow environments and associated microhabitat structure are known to strongly influence macroinvertebrate community composition (Davis & Barmuta, 1989; Robson & Chester, 1999). Climatic drying and intermittency reduce wetted area and increase temporal and spatial variation in flow environments. When streams dry, riffles dry out first and pools dry last (Stanley et al., 1997), so to characterize assemblages when streams were drying we needed to sample in pools as well as riffles. Therefore, increased sampling effort was necessary to fully capture community composition at reaches and times in the hydroregime.

Field sampling

Bunn (1985) used a Sürber sampler (0.1 m² area) with a 475 µm mesh net, which relied on flow velocity to wash invertebrates into the net and collect drift. However, the Sürber technique is unsuitable during low-flow periods because material is not washed into the net and there are no drifting invertebrates. Instead, a dipnet sampling technique was adapted for use in either flowing or still water and on any substratum type, using a 500 µm mesh 30 cm wide net. In flowing reaches, the bottom edge of the net was pushed into the substrate (opening facing upstream), and a 30 x 30 cm (0.09 m²) area of substrate in front of the net gently disturbed by hand, dislodging invertebrates which are then washed into the net by flow. Larger stones within the 30 x 30 cm area were dislodged and brushed by hand, allowing any attached invertebrates to be washed into the net. This was timed for two minutes to capture drifting invertebrates and is much alike the Surber technique used by Bunn (1985, also two minutes). This method was modified when flow had stopped in now-intermittent streams. A 30 x 30 cm area of substrate was vigorously disturbed by foot, and the net quickly swept over for 30 to 60 seconds (dependent on depth), from the substrate to the surface to collect all disturbed invertebrates. Previous studies have shown that this method, where rocks are disturbed and the material beneath is collected into the net, collects more species than simply scrubbing the rocks alone (Giri et al., 2010). This method has been successfully used in other studies of intermittent streams, where flow is variable within and between reaches at different times (e.g. Robson et al., 2005) and has been shown to deliver invertebrate species by abundance data not significantly different to data from a Sürber net (Robson et al., 2018). Samples were placed in plastic bags and preserved with 70% ethanol. In the laboratory, samples were sorted by emptying into a white tray and flooding with tap water, then carefully removing all invertebrates from substrates using a dissecting microscope mounted on a moveable arm, at 10x magnification. Invertebrates were identified to lowest possible

taxonomic level, being species level for most groups except Diptera, Ostracoda and *Uroctena* amphipods which were identified to genus or family level and Oligochaetes, other worms and mites which were identified to order.

Data analysis

Examination of aquatic invertebrate species richness across all thirteen reaches was only possible during flowing phases: spring (October 2016), early-summer (December 2016) and winter (August 2017). So, to test the hypothesis (H_{1A}) that α diversity will be higher in perennial reaches than in intermittent reaches when both are flowing, a mixed model ANOVA (SPSS v. 23) was used to test the effects of time (3 levels, fixed), flow regime (2 levels, fixed), reaches nested within flow regime (13 levels, random) and interaction between time and flow regime on the dependent variables species richness and abundance. Time was a fixed factor because we deliberately sampled at particular times to capture different phases of the flow regime and invertebrate life cycles, which may be constrained by flow regime in intermittent streams. That is, the different times were not included to increase representativeness (i.e. provide replication), so time could not be treated as a random factor (Quinn & Keough, 2002). Post-hoc Tukey's tests were used to compare pairwise differences between times. Assumptions of ANOVA were met for richness data, however abundance data required natural log transformation before it met the assumptions.

A chi-square test of association was used to examine differences in total species richness (α diversity) across four seasons: mid-spring (October), late-summer (February), early-winter (June), and winter (August), and between four streams: Waterfall Gully (WF1, WF2 only), Seldom Seen Brook (SS1, SS2), Curtis Creek (CC2, CC3) and Wungong Brook (WB1, WB2) (reaches pooled within streams). This tested the α -diversity relationship between streams and season, in the four streams which have complete seasonal data. Each has a different flow regime as a consequence of climate change since Bunn's study. At intermittent reaches: flow

commenced in early-winter (June), peak flows in winter (August); beginning of flow decline in mid-spring (October), dry-down in late-summer (February) and in perennial reaches: increase from baseflow in early-winter, peak flows in winter, beginning of flow decline in mid-spring and return to baseflow in late-summer. Seasons contained equal numbers of samples and therefore met the assumption of equal sampling effort for chi-square tests. Chandler Brook was excluded from this analysis because both reaches contained surface water on only three sampling occasions. Although Seldom Seek Brook has one perennial (SS2) and one intermittent (SS1) reach, species richness was pooled across the two reaches, as with the other three streams.

Patterns in contemporary β -diversity (H_2) were analysed using Baselga's (2010) framework for partitioning turnover and nestedness components. Species presence-absence data was pooled across times for each reach, and analysed using R package *beta.part* (Baselga & Orme, 2012), which allows calculation of total β -diversity (Sørensen-based; hereafter β_{SOR}), the turnover component (Simpson-based; hereafter β_{TUR}) and the nestedness component (hereafter β_{NES} ; note $\beta_{SOR} = \beta_{TUR} + \beta_{NES}$; *cf.* Baselga, 2010). Hypothesis H_2 : assemblages in intermittent reaches will be a subset of those in remaining perennial reaches was tested at two spatial scales. Firstly, H_{2A} : catchment level beta diversity primarily comprised nestedness; secondly, H_{2B} : reach scale β diversity within perennial and intermittent reaches will both comprise nestedness rather than turnover. To test H_{2A} , β_{SOR} was calculated across all reaches, to determine whether catchment level beta diversity primarily comprised turnover or nestedness. If less diverse reaches (expected to be the intermittent reaches) were a subset of more diverse reaches (expected to be the perennial reaches), then β_{SOR} would predominantly comprise β_{NES} . Conversely, if turnover between reaches was more important, then β_{SOR} would be dominated by β_{TUR} , meaning the species inhabiting less diverse reaches did not occur at more diverse reaches. To examine patterns in β -diversity (β_{SOR}) between flow regimes (H_{2B}),

reaches were analysed separately according to flow regime (perennial, five reaches and intermittent, eight reaches), and relative contributions of β_{TUR} and β_{NES} were determined for each.

The second aim of this study was to examine long-term changes in α and β diversity ($H_3 - H_7$). Unfortunately, Bunn's original data no longer exists, thus formal analysis of raw data from the two studies was not possible. Additionally, the ANOVA models used to test variables including species richness and density of individuals were not repeatable because Bunn (1985; 1986; et al., 1986) sampled three additional streams in the North Dandalup catchment which are no longer accessible (since flooded by large reservoir, and upper reaches under active bauxite mining). His sample data was divided into 2 seasons for analysis – summer and winter – this was not possible with the 2016/17 data because most streams were dry in summer. Fortunately, Bunn (1985, et al., 1986) provided proportional dominance of species occupancy per site (pooled across time), only excluding very rare taxa (those that never comprised 5% or more of the total individuals at any site at any time). So, to compare with this data, macroinvertebrate data collected during 2016/17 from the same eight sites studied by Bunn in 1981/82 was standardised using this method. That is, species x abundance data was converted to proportional dominance per site (pooled across times) and the very rare species removed (<5%). Due to differences in taxonomy between times, some taxa were standardised between datasets to genus (some Trichoptera) or Family level (Tipulidae, Simuliidae). This comprised the standardised dataset with which long-term comparisons of α -diversity and β -diversity were performed. Chi-square analysis of α -diversity was not possible using this reduced dataset, because sampling effort was not equal. I took 4 times the number of samples in 2016/17 than Bunn collected in 1981/82, so the comparison of proportional dominance data is based on a larger sampling effort in 2016/17. Therefore, total standardised

α diversity at each reach from both datasets was graphed for visual comparison to address hypotheses H_{3, 4}.

Next, β -diversity was examined between and within the two standardised datasets using Baselga's framework for presence absence data (Baselga 2010). To test H₅, β -diversity components (β_{SOR} , β_{TUR} , and β_{NES}) were compared between the 1981/82 and 2016/17 standardised datasets at the catchment level. We also compared β -diversity between the 2016/17 full (all taxa) and standardised datasets to interpret temporal differences in β -diversity without contributions of rare species (which had been included in the analysis of the 2016/17 data, above).

To address H_{6,7}, pairwise examinations of β -diversity at each reach between times (1981/82 vs 2016/17) were performed using the standardised datasets to examine spatial differences in temporal β -diversity in relation to flow regime change. This tested the mBACI design for β -diversity by comparing reaches that were perennial in both studies (before-control vs after-control) to reaches that were perennial in 1981/82 and intermittent in 2016/17 (before-impact vs after-impact).

Results

Alpha diversity and abundance between contemporary flow regimes (H₁)

When all five study streams were flowing (3 sampling times), α diversity differed significantly among times ($F_{2,156} = 63.5$, $P < 0.001$), flow regimes ($F_{1,11.4} = 9.1$, $P = 0.01$) and reaches nested within flow regime ($F_{11,156} = 2.8$, $P = 0.02$). There was no interaction between time and flow regime ($F_{2,156} = 2.7$, $P = 0.07$), therefore species richness was consistently higher in perennial streams (mean 9.8 ± 0.4 per sample) than intermittent streams (mean 6.9 ± 0.4 per sample) at all three times (Figure 4.2a). Each of the three times differed (all Tukey's post hoc pairwise tests < 0.01); mid-spring was the most species rich and winter the least

(Figure 4.2a). Among perennial reaches, the perennial reach in Seldom Seen Brook had the highest richness, followed by WF1 and WF2. Among intermittent reaches, the intermittent reach on Seldom Seen Brook had the highest species richness, followed by WB2 and the reaches on Curtis Creek (Figure 4.2b).

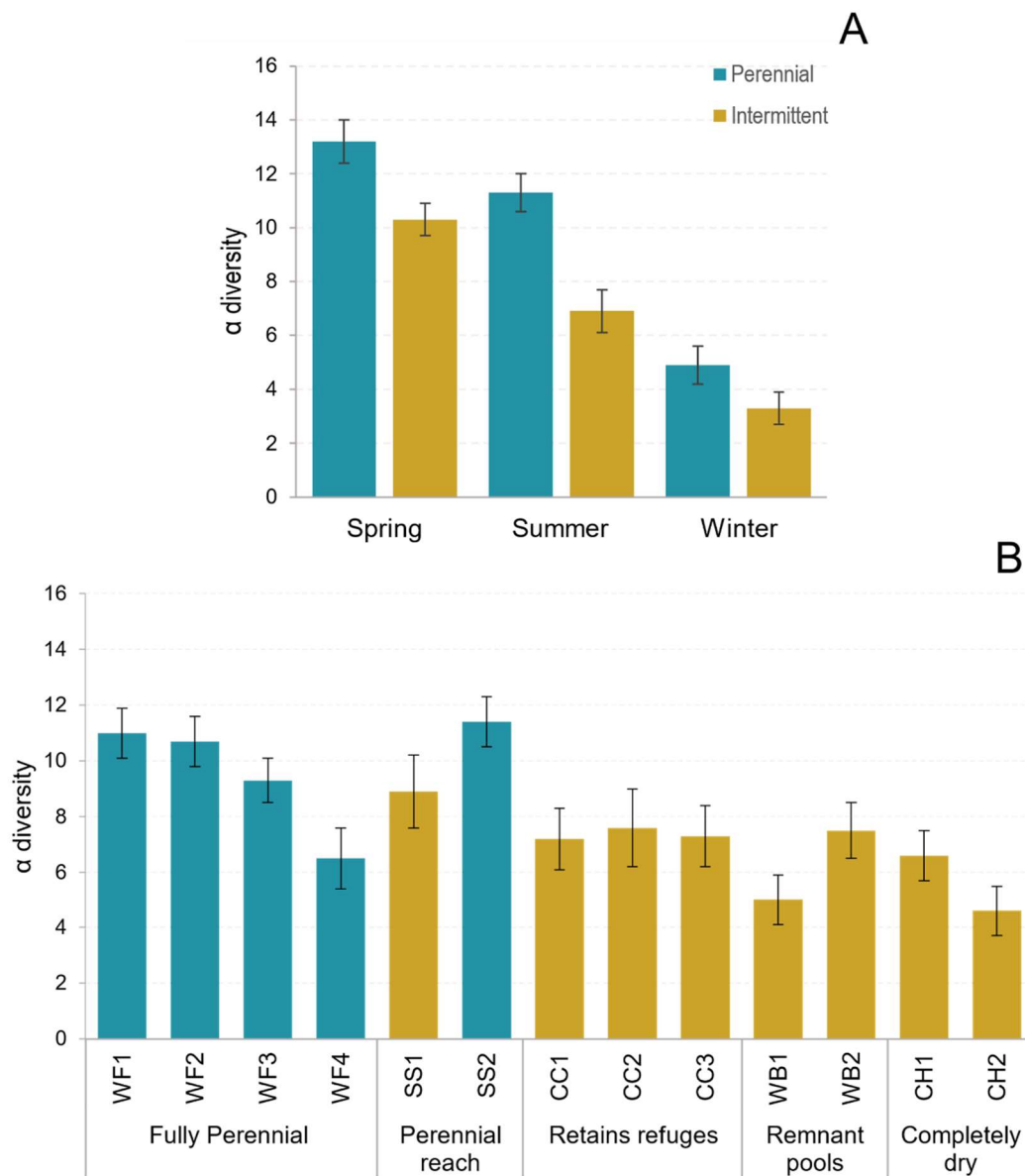


Figure 4.2: Least squares means ($\pm$ SE) of macroinvertebrate α diversity between perennial (blue) and now-intermittent (gold) flow regimes (A) and between reaches (B) averaged across 2016/17 flowing-phase sampling. In (A) α diversity was significantly different between flow regimes at all three times.

Abundance of macroinvertebrates during the flowing phase differed among sampling times ($F_{2,156} = 79.8$, $P < 0.001$) and reaches nested within flow regime ($F_{11,156} = 6.0$, $P < 0.001$), but not flow regimes ($F_{1,11.2} = 0.3$, $P = 0.6$; Figure 4.3). However, there was a significant interaction between time and flow regime ($F_{2,156} = 7.1$, $P = 0.001$) because abundances were higher in intermittent reaches than perennial reaches in spring but were higher in perennial reaches in early-summer and winter (Figure 4.3a). Intermittent flow regimes were not associated with consistently lower abundances overall (Figure 4.3b).

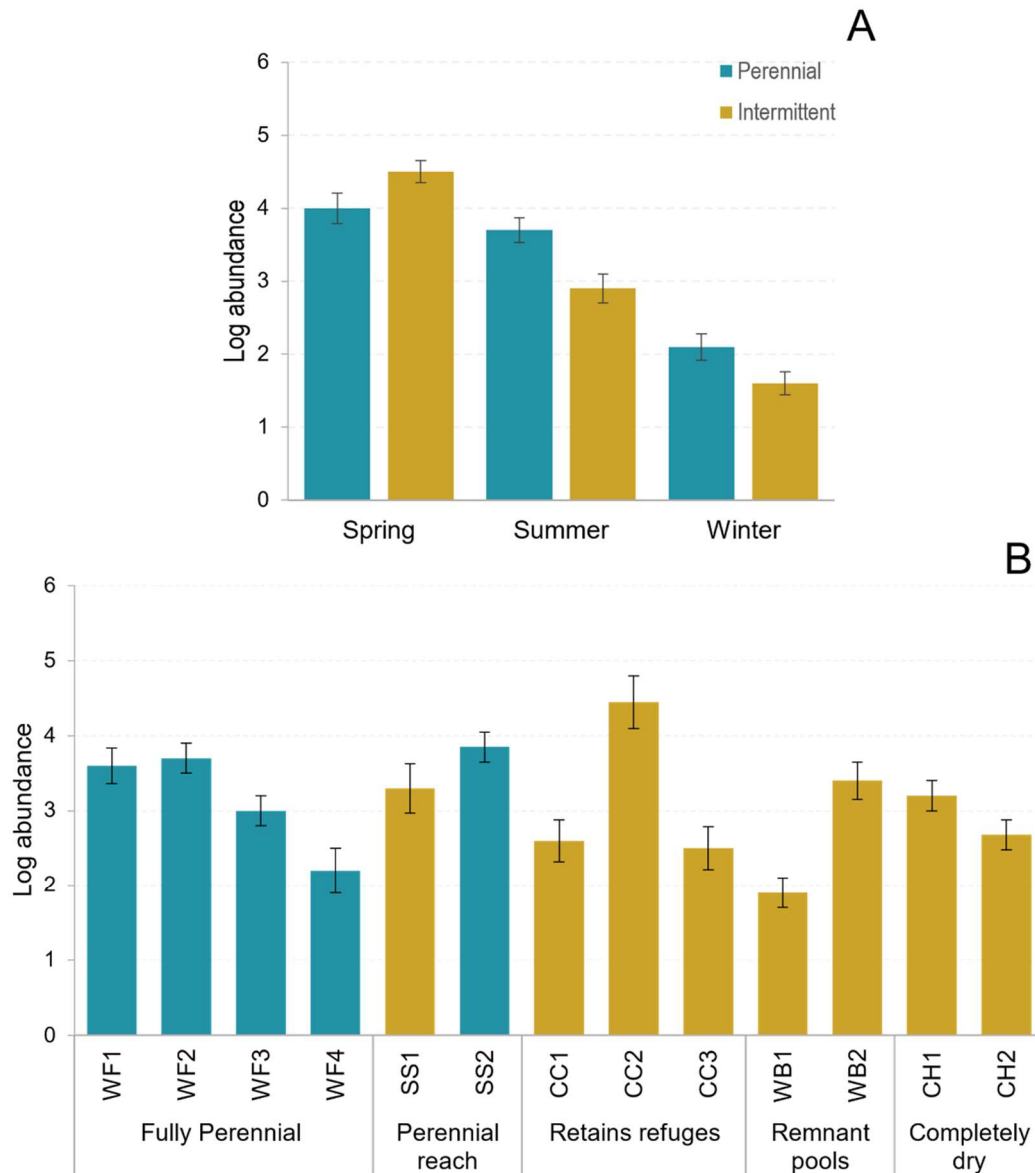


Figure 4.3: Least-squares means ($\pm$ SE) of log-transformed macroinvertebrate abundance between perennial (blue) and now-intermittent (gold) stream reaches (a) and between reaches (b) averaged across 2016/17 flowing-phase sampling.

Alpha diversity also differed among streams at different hydroperiod stages (H_{1B} : $\chi^2_9 = 450.0$, $P < 0.001$; Figure 4.4). Observed and expected diversity did not differ at Waterfall Gully, but observed richness was significantly higher than expected in spring at intermittent Wungong Brook and Curtis Creek (Figure 4.4). This is because diversity peaked in mid-spring at intermittent reaches and was consistently lower at other times of year. There was little

difference between observed and expected richness at Seldom Seen Brook, reflecting the perenniality of SS2, except in early winter, presumably associated with the arrival of new taxa colonizing winter flows.

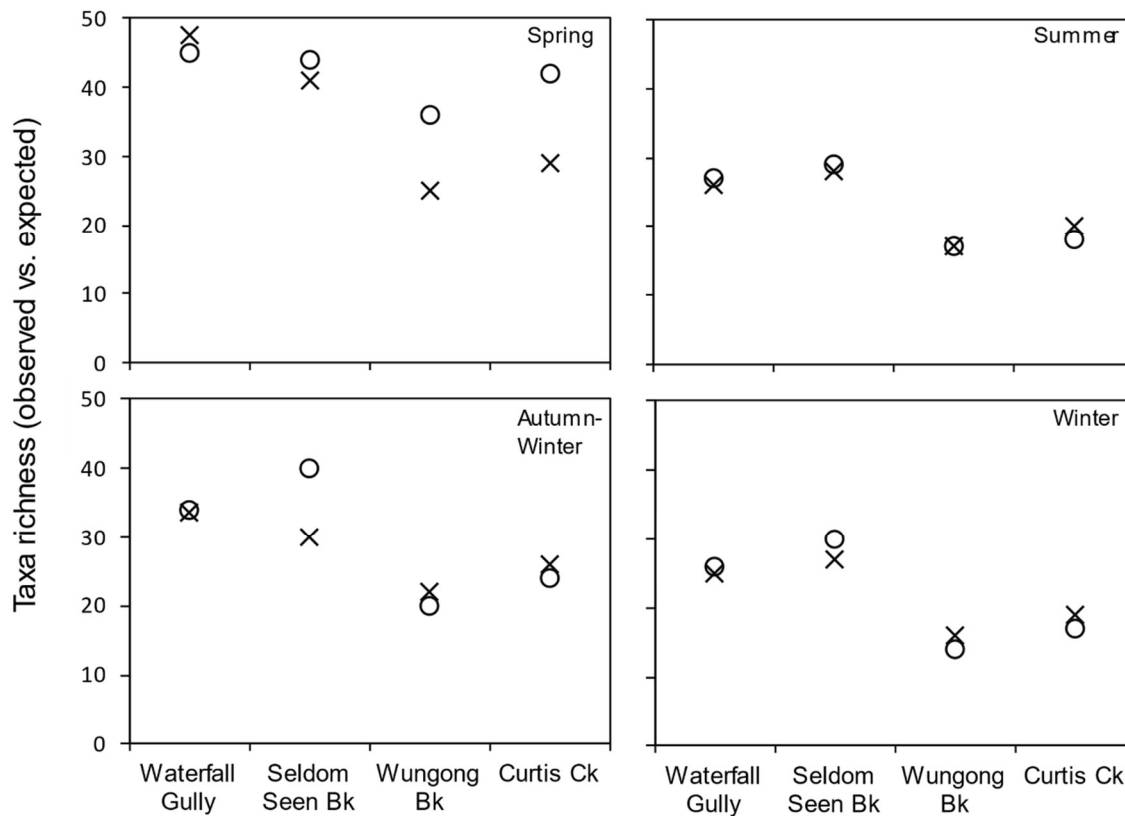


Figure 4.4: Observed (circles) vs expected (X) species richness in four streams (chi² test of association).

Beta diversity across contemporary flow regimes

Beta diversity across all study reaches (at the catchment scale; H_2) was high ($\beta_{\text{SOR}} = 0.82$), and turnover comprised more than 90% of total β -diversity ($\beta_{\text{TUR}} = 0.74$) whereas nestedness contributed less than 10% ($\beta_{\text{NES}} = 0.08$; Figure 4.5). Assemblages from intermittent reaches cannot then be a subset of those in perennial reaches (i.e. less diverse reaches are not nested within more diverse reaches), so hypothesis 2 was not supported. Instead, in the streams that

have become intermittent, different (although fewer) species have replaced those typical of perennial reaches. Within the reach types (H_3), β -diversity among intermittent reaches was highest ($\beta_{\text{SOR}} = 0.76$) and mostly comprised turnover ($\beta_{\text{TUR}} = 0.68/89.5\%$; $\beta_{\text{NES}} = 0.08/10.5\%$). Among perennial reaches, β -diversity was lower, but still substantial ($\beta_{\text{SOR}} = 0.55$), and also comprised turnover ($\beta_{\text{TUR}} = 0.48/87\%$; $\beta_{\text{NES}} = 0.07/13\%$, Figure 4.5). This means that macroinvertebrate assemblages were compositionally different among reaches of the same flow regime type, especially amongst intermittent reaches. However, it must be noted that four of the five perennial reaches are longitudinally connected. If perennial reaches had existed on different streams within this catchment, β -diversity may have been higher among perennial reaches.

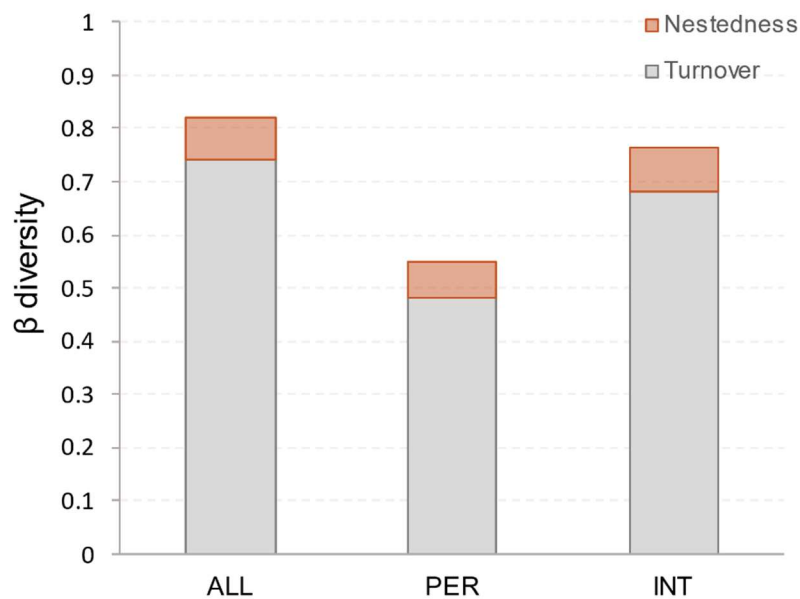


Figure 4.5: Sørensen's β diversity partitioned into turnover and nestedness components (cf. Baselga 2010), for all study reaches in the Wungong catchment (H_2), perennial reaches (5) and intermittent reaches (8) in 2016/17

Historical differences in alpha and beta diversity ($H_3 - H_7$)

Total α -diversity between the two standardised datasets (Bunn 1985, this study) showed little change in reaches that remained perennial (WF1, WF2, and SS2; Figure 4.6), supporting hypothesis 3. Reaches that transitioned to intermittent flows had between 26% (CH2) and 55% (CC2) fewer common taxa in 2016/17 than in 1981/82 (Figure 4.6), supporting hypothesis 4.

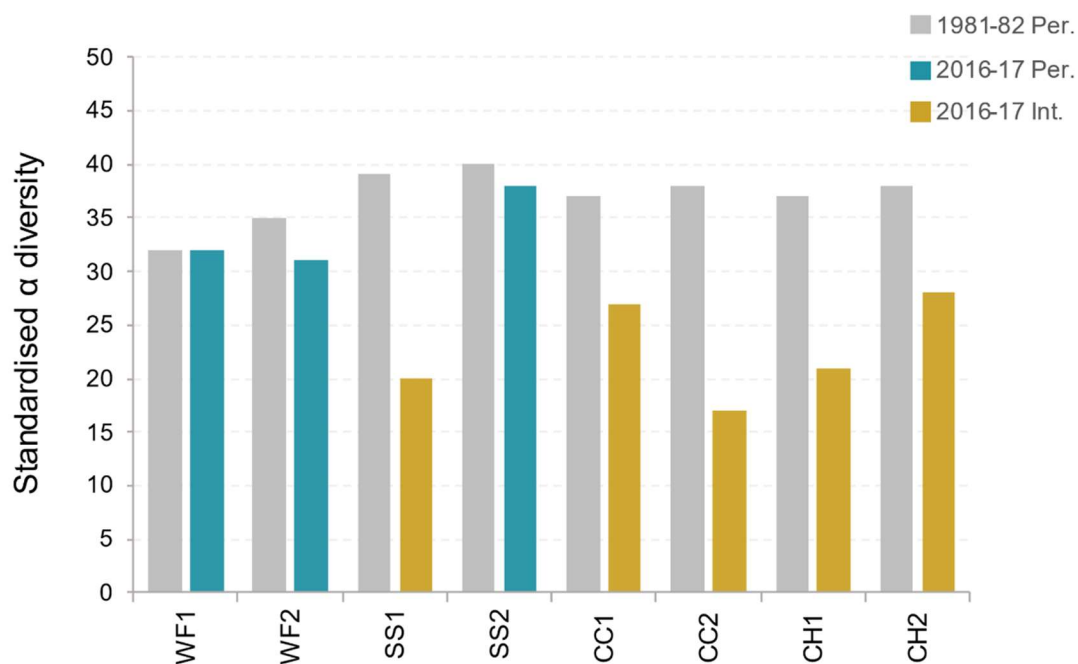


Figure 4.6: Comparison of standardised taxonomic richness (α diversity) measured at Wungong catchment reaches in 1981/82 (grey bars; Bunn 1985) and the same reaches in 2016/17, following flow regime change from perennial to intermittent at five reaches (blue bars) and remaining perennial flows at three reaches (gold bars).

Catchment level β diversity in 2016/17 was double that in 1981/82 (Table 4.2) supporting hypothesis 5. Beta diversity mostly comprised species turnover between reaches in 1981/82, consistent with Bunn's observations that macroinvertebrate communities were structured by flows and differing geologies between reaches; only 26% of taxa were nested within more species-rich reaches (Table 4.3). By 2016/17, turnover was an even larger proportion of β -

diversity and nestedness had declined to 16%. Therefore, the transition to intermittent flow regimes in most streams of the Wungong catchment has doubled spatial heterogeneity of the most common taxa and reduced nestedness among more diverse and less diverse reaches.

When comparing temporal change in β diversity between each reach sampled in 1981/82 and 2016/17, two patterns emerged. Total β diversity (β_{SOR}) was lowest at reaches that remained perennial (WF1 and WF2) supporting hypothesis 6 (Figure 4.7). Spring-fed SS2 retained perennial flows in most years but had higher β diversity (i.e. turnover between studies) than Waterfall Gully (Figure 4.7). The now-intermittent reach SS1 had higher β_{SOR} , and a comparatively greater contribution of β_{NES} , showing that a portion of the contemporary fauna is a subset of that recorded in 1981/82. At intermittent reaches, temporal β_{SOR} ranged between 0.56 and 0.69, and turnover was the largest component in most reaches (71% to 89%) except for CC1 where nestedness comprised 48% of temporal beta diversity. Consequently, hypothesis 7 was partially supported in some reaches (SS1, CC1, CH2) where the fauna was a partial subset of previous perennial flow regime fauna, but not in others (CC2, CH1) where the nestedness component was low (Figure 4.7).

Table 4.2: Beta diversity among reaches using the 1981/82 and 2016/17 standardised datasets, partitioned into turnover and nestedness components (sensu Baselga, 2010).

	β_{SOR}	β_{TUR} (%)	β_{NES} (%)
1981/82 standardised	0.31	0.23 (74%)	0.08 (26%)
2016/17 standardised	0.66	0.55 (83%)	0.11 (16%)

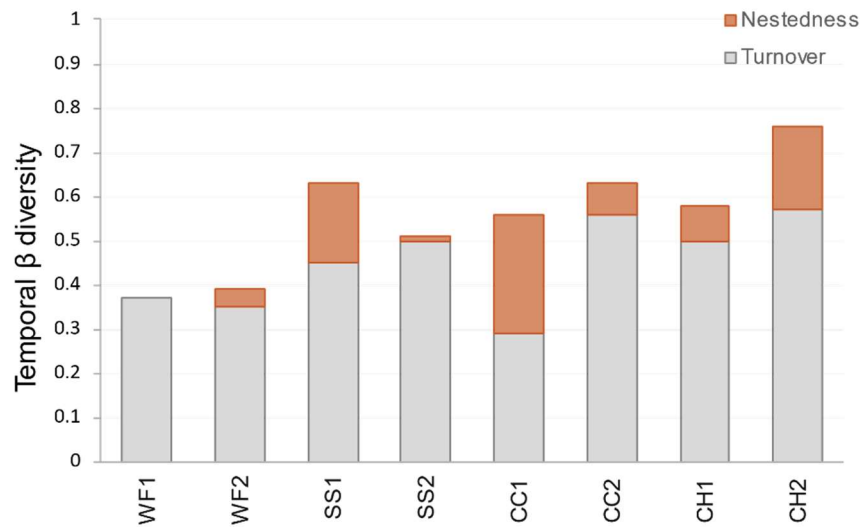


Figure 4.7: Temporal beta diversity (β SOR) between 1981/82 (Bunn 1985; 1986) and 2016/17 within each reach. Beta diversity partitioned into turnover (β TUR) and nestedness (β NES) components cf. Baselga (2010).

Altogether, the mBACI design showed that the onset of drying flow regimes in perennial streams caused a reduction of α diversity and caused turnover *in situ* between times. The reduction in taxa richness at reaches that dried also created a richness difference between times (i.e. temporal nestedness) which varied between sites, a pattern not observed at reaches that retained perennial flows. Transition to intermittency also increased the spatial heterogeneity of assemblage composition between stream reaches, such that contemporary assemblages of intermittent reaches were not subsets of those at perennial reaches, nor was there evident any homogenisation of the fauna due to drying. As expected, there was less temporal change at streams that retained perennial flows than those that transitioned to intermittent flows.

Discussion

Spatial and temporal patterns in alpha diversity

Alpha diversity was consistently lower in newly-intermittent reaches than in remaining perennial reaches, which the mBACI design showed was caused by reduced α diversity at impacted (intermittent) reaches between studies. Reaches that transitioned to intermittent flows had up to 55% fewer common taxa than recorded in 1981/82 (loss of between 10 – 21 spp.), differences much greater than observed at reaches that retained perennial flows (loss of 0 - 4 spp.). Thus, the transition to intermittency caused a substantial reduction in α diversity between studies, creating spatial differences in α diversity between contemporary flow regimes. Invertebrate abundances were not lower in intermittent reaches, showing that the lower α -diversity there was not simply a product of low abundance (i.e. the species-abundance relationship). Instead, a less diverse assemblage of temporarily abundant species occupied newly-intermittent reaches, also shown by distinct seasonal patterns with richness and abundance peaks in spring and declines over the summer months. Pronounced seasonal patterns are usually observed in intermittent streams, as organisms are constrained to timeframes imposed by regular drying and must adapt their life histories accordingly (Bogan & Lytle, 2007; Garcia-Rodger et al., 2011; Strachan et al., 2015). Altogether, differences in contemporary α diversity align with known patterns in intermittent streams across med-regions, in terms of lower α diversity and pronounced seasonal patterns in accordance with drying and rewetting (e.g. Leigh et al., 2016b; Soria et al., 2017).

In the streams that transitioned to intermittency, the presence of prolonged pools and potential drought refuges was associated with greater α diversity than reaches that lacked surface water during the dry season. Reaches on Curtis Creek had marginally higher α -diversity than reaches in Wungong Brook or Chandler Brook owing to the small, perennial, spring-fed refuge pools present in its channel. Curtis Creek has more than one granite

inselberg along its length and these inselbergs support aquifers that feed the springs (Centeno et al., 2010). The channel is also narrow and well shaded by vegetation, reducing evaporation and retaining more residual moisture in the streambed than in wider and less shaded streams (i.e. Wungong Brook, Chandler Brook). The importance of refuge pools in intermittent streams is well established (Chester & Robson 2011; Bogan et al., 2017; Bonada et al., 2020) and they certainly supported a population of one common mayfly species (*Bibulmena kadjina*) that otherwise would have become locally extinct (Carey et al., 2021 [Chapter 5]). It appears that not only do perennial refuge pools enable species to persist within catchments (e.g. Bogan et al., 2019), they at least partly sustain α diversity when perennial streams transition to intermittency.

Contemporary patterns in beta diversity

Beta diversity was high across the catchment in 2016/17 and comprised almost entirely turnover (species replacement) between sites. This was contrary to expectations because, based on current knowledge in med-regions, flow impermanence is understood to cause spatial nestedness of intermittent stream invertebrate assemblages within those of perennial streams (Cañedo-Argüelles et al., 2015, Leigh et al., 2016b; Stubbington et al., 2017). In recently intermittent NJF streams, high turnover between reach types was caused by replacement of some sensitive species with resistant/resilient taxa, presumably colonising from other localities. Between reaches with the same flow regime, β diversity (turnover) was highest among intermittent reaches, meaning that the assemblages in intermittent reaches were compositionally very different to each other, reflecting increased habitat heterogeneity in space and time in intermittent streams. Assemblage turnover was also substantial (albeit less pronounced) among perennial reaches, even though all except SS2 were hydrologically connected and so were expected to have similar assemblages. Regardless of their broad flow regime, each reach has different timing of high and low flows and this timing also differs

interannually with rainfall. These hydrological differences create substantial heterogeneity in the extent of aquatic habitat, its connectivity, and local flow conditions in both time and space, and these differences are reflected in invertebrate assemblage composition.

Temporal β -diversity patterns after 35 years of drying

Accompanying reduced α diversity at intermittent reaches between studies was an approximate doubling of catchment level β diversity when common species alone were considered. As increased catchment level β diversity was primarily an increase in turnover among reaches, this means the transition to intermittency in most streams caused assemblages to become more distinct, rather than resulting in a few highly diverse reaches with the assemblages at less diverse reaches nested within them. Overall, assemblage compositions in stream reaches in the Wungong catchment have become more distinct after flow regime change and continuing drying (see Chapter 3). As predicted, greater temporal β -diversity occurred at reaches that transitioned to intermittent flows, however, contemporary assemblages were not subsets of those present in 1981/82. This means that the suite of formerly common species extirpated by drying were only partially replaced by new arrivals or by formerly rare species that have now become common in intermittent streams.

Transitions to intermittency have been shown to reduce α diversity and alter patterns in β diversity (e.g. Crabot et al., 2021; Piano et al., 2020), and to extirpate sensitive, endemic and longer-lived aquatic obligate taxa (e.g. Bogan & Lytle, 2011). While groups reported as being particularly affected by seasonal flow cessation and drying typically include Ephemeroptera, Trichoptera and Plecoptera (e.g. Boulton 2003; Bonada et al., 2007; Stubbington et al., 2017; and in streams transitioning to intermittency; Lund et al., 2016; Belmar et al., 2019; Bruno et al., 2019) most Odonata and some endemic Chironomidae species were also lost from newly intermittent streams in this study (see Chapter 3). But since SWA has a range of desiccation tolerant species that replaced a portion of the extirpated drought-intolerant fauna, patterns

observed in intermittent reaches comprised turnover rather than nestedness. Had nestedness been the dominant pattern in intermittent reaches, α diversity would have been even lower.

Crabot et al. (2021) examined the impacts of widespread transition to intermittent flows in central European streams (where perennial streams were previously ubiquitous), comparing them to med-climate streams with a long history of intermittency. They found that spatial β diversity was high and dominated by turnover (93%) in the recently intermittent streams (central, temperate Europe) whereas in the historically intermittent streams (med-climate Europe), β diversity was lower and had a comparatively greater contribution from nestedness.

Pařil et al. (2019) also report high turnover among intermittent streams in the same temperate region (average 87% of β -diversity) and suggest lack of resistance to drying as a causative factor. Significantly increased turnover, rather than increased nestedness, was also demonstrated in experimental mesocosms testing the impacts of drought on temperate chalk-stream communities (Aspin et al., 2018). In contrast, Piano et al. (2020) report assemblage homogenisation in an alpine stream subject to recent drying, and no change in β diversity components compared to perennial reaches, although the authors advise caution in interpreting their results due to coarse taxonomic resolution. Flow intermittence has been demonstrated to reduce nestedness in med-stream diatom communities (Tornés & Ruhí, 2013), and to increase turnover in med-stream macroinvertebrates, particularly among aerial dispersers in networks with drying headwaters (Crabot et al., 2020). Given high turnover was also observed in newly intermittent streams in this study (in a med-region), it is possible that the disturbance experienced by stream communities exposed to novel intermittency is more important in determining β diversity patterns (nestedness vs. turnover) than climate type alone.

Change in stream biodiversity patterns caused by intermittency as a result of global warming

The sudden onset of regular drying was predicted to increase nestedness of assemblages between perennial and intermittent streams (e.g. Vander Vorste et al., 2021), because perennial reaches would act as refuges for biodiversity and supply colonists to intermittent reaches (Heshkovitz & Gasith, 2013; Robson et al., 2013). This could occur even if the short hydroperiod resulted in sink populations in intermittent streams. Hence it was expected that the transition to intermittency across much of the catchment would create patterns in biodiversity like those in intermittent river networks in other med-regions (e.g. Bonada et al., 2007; Clarke et al., 2010; Datry et al., 2014). Instead, novel drying eliminated much of the assemblage formerly present, and they were not replaced through dispersal from perennial reaches in succeeding years. Finer scale environmental differences (i.e. distinct hydrology) then sustained distinct invertebrate assemblages among reaches, including newly arrived species, creating a pattern of high turnover. Therefore, the imposition of an intermittent flow regime acted as an environmental filter that overrode other physical habitat characteristics, shaping diversity patterns in these assemblages (*sensu* Poff 1997; Lake et al., 2007), reflected in the far greater temporal β diversity observed at reaches that transitioned to intermittent flows.

High turnover is considered to be indicative of barriers to movement and lack of organism exchange (Baselga 2010; Anderson et al., 2011); in the Wungong streams, turnover may indicate that the onset of intermittency has created barriers to dispersal, even over small spatial scales. However, in this small catchment (*circa* 130 km²) distances between streams are between one and three kilometres and are therefore within range of moderate overland (flying) dispersers (e.g. Chester et al., 2015; Schriever et al., 2015). The catchment is forested with low to moderate relief, without obvious topographic barriers to flight (*cf.* Sabando et al., 2011; *sensu* Tonkin et al., 2018) and the assemblage in 1981/82 was dominated by flying

insects (i.e. likely moderate to strong overland dispersers across this catchment; Bunn et al., 1986; see also Bunn & Davies, 1990; Storey et al., 1990; Chapter 3; Chapter 5). Recent studies have shown that where appropriate resources are available, many invertebrates can successfully disperse to and colonise suitable reaches (Lancaster & Downes, 2017) but that limited oviposition success might more strongly influence detected occupancy patterns than dispersal limitation (Lancaster et al., 2010, 2020a, 202b). Many aquatic insects prefer to oviposit in riffles (Lancaster et al. 2020a), but these are the first habitats to dry out, potentially causing egg mortality; or oviposition opportunities may be lost altogether with the shift to a low-flow regime. In caddisflies, oviposition behaviours vary but may be fixed and inflexible (Lancaster et al. 2020b), constraining adaptation to changing habitat. In the Wungong streams, a dependence on flowing sites for oviposition would limit opportunities to Waterfall Gully. Certainly, the presence of refuges (pools and a flowing section) in Curtis Creek and Seldom Seen Brook did not increase α diversity in adjacent intermittent reaches. This suggests that those refuges either did not produce adult dispersers or that those adults were unable to lay eggs in other parts of the streams, possibly because any females emerging in late spring could not live long enough for flows to resume (7–8 months). Such constraints on adults suggests that drift may be relatively more important for local dispersal; but while drift alone may have a strong influence on occupancy patterns in rivers (Downes & Lancaster, 2018), intermittency clearly will reduce opportunities for this to happen. Overall, changes to dispersal driven by intermittency would, as observed, drive α diversity down and β diversity up.

After a disturbance, neither increased turnover nor increased nestedness between communities is necessarily a desirable or “better” outcome (Socolar et al., 2016), as both situations can result in losses of regional biodiversity through increased vulnerability to stochastic effects. Under other circumstances where assemblages of now-intermittent streams

are strongly nested within assemblages in the remaining perennial reaches, those perennial reaches would harbour the bulk of the regional species pool and so conservation of perennial flows would be essential for maintaining biodiversity of the entire catchment (e.g. Chester & Robson, 2011; *sensu* Robson et al., 2013; Ruhí et al., 2017). That is, loss of one intermittent stream to permanent drying or development would not necessarily result in catchment wide extinctions, as long as perennial flows were maintained in the perennial refuge streams. In contrast, this study identified high levels of spatial turnover between perennial and intermittent reaches, and *in situ* extirpations at now-intermittent reaches (temporal turnover). Even so, perennial reaches are still critical refuges, as many of the species harboured within them were absent from all intermittent streams in the Wungong catchment and are regionally endemic, Gondwanan relicts (Chapter 3, Chapter 5) that would go extinct should perennial flows cease — as is expected across SWA (Andrys et al., 2017; McFarlane et al., 2020). Furthermore, as intermittency increases assemblage distinctiveness between streams, stochastic loss of any stream could potentially cause catchment-wide extinctions of species. Therefore, flow regime drying has rendered all refuges within these stream networks important and worthy of conservation, because they are likely the source of most species within these spatiotemporally constrained communities.

Table S4.1: Summary of physicochemical variables measured in 2016/17 compared to measurements taken by Bunn (1985) in 1982 at the same sites and time of year (where available). Note differences in data collection and reporting: 1981/82 temperature and pH averaged over a six month period, whereas in 2016/17 ranges are given between reaches. ns = not sampled. Bunn (1985) measured ionic composition and concentration in summer and winter (Na^+ , Mg^{2+} , Ca^{2+} and K^+), reporting concentrations of salts as consistently low and dominated by Na^{2+} cations. Evapoconcentration in intermittent streams typically increases salinity during the drying phase (Gómez et al., 2017), evident during summer in disconnected pools (although conductivity remained low ~ maximum 400 $\mu\text{S}/\text{cm}$). However, in NJF streams groundwater baseflow is a significant source of salinity (Schofield et al., 1989), and the disconnection from groundwater has been demonstrated to reduce salinity reliably enough to be considered an indication of groundwater loss (e.g. Kinal & Stoneman, 2012). Therefore, it is likely that salinity has not changed markedly in these streams (or may have even declined), however this cannot be confirmed due to non-comparable data.

	Waterfall Gully	Seldom Seen Brook	Curtis Creek	Wungong Brook	Chandler Brook
Temperature (°C)					
<u>Spring</u>					
Water temperature range 2016	14.3-14.8	14.5-14.6	13.4-15.6	14.5-17.8	14.5-16.0
Mean water temperature 1982	16	14	16	ns	14.5
<u>Summer</u>					
Water temperature range 2016	17.9-20.0	24.6	18	22.0-28.5	Dry
Mean water temperature 1982	19	16.5	22	ns	17
<u>Autumn</u>					
Water temperature range 2016	16.0-16.9	15.2	15	14.7-17.6	Dry
Mean water temperature 1982	15	14.5	16	ns	14
<u>Winter</u>					
Water temperature range 2016	10-10.5	9.5	8-11	8.5-10	8-12
Mean water temperature 1982	13	10.5	11	ns	10
pH					
Mean summer pH 1982	6.5	6.3	6.4	ns	6.4
Mean winter pH 1982	6.25	6.25	6.25	ns	6.25
2016 – spring	5.9-6.3	5.9	5.85-6.1	5.6-6.2	5.8-5.9
2017 - summer	5.3-5.7	7.3	4.28	6.8-7.6	Dry
2017 - autumn	5.2-6.2	5.3	4.69	5.5-6.2	Dry
Conductivity ($\mu\text{S}/\text{cm}$)					
Spring 2016	150-230	160-200	170-230	280-310	200
Summer 2017	150-200	230	130	200-400	Dry
Winter 2017	160-250	240	210	100	Dry

Chapter 5

Life history traits are poor predictors of species responses to flow regime change in headwater streams.

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The following authors contributed to this manuscript as outlined below:

	Author order	Concept & study design	Fieldwork	Laboratory work	Data analysis	Writing manuscript	Review/edit paper
Nicole Carey	First	X	X	X	X	X	X
Edwin Chester	Second	X	X				X
Belinda Robson	Third	X			X		X

By signing this document, the Candidate and Principal Supervisor acknowledge that the above information is accurate and has been agreed to by all other authors.

Candidate

Principal Supervisor

Abstract

Recent climate change is altering the timing, duration and volume of river and stream flows globally, and in many regions perennially flowing rivers and streams are drying and switching to intermittent flows. Profound impacts on aquatic biota are becoming apparent, due in part to the strong influence of flow regime on the evolution of life history. We made predictions of life history responses for thirteen common aquatic invertebrate species (four caddisflies, five mayflies, two stoneflies, a dragonfly and an amphipod), to recent flow regime change in Australian mediterranean climate streams, based on historic studies in the same streams. Size distributions, phenology, voltinism and synchrony were compared, revealing five main responses. More than half of the species were restricted to perennially flowing streams and were absent from those that had switched to intermittent flows (including all four caddisfly species). These formerly common species are at risk of extinction as climate change progresses. Two mayfly species had divergent responses in voltinism and synchrony, and one relied on drought micro-refuges to persist. One stonefly species changed development timing to suit the new flow regime, and the amphipod species retreated to subterranean refuges. Two formerly common species were not detected at all during 2016-17. In addition, a new mayfly species and a caddisfly species proliferated under new flow regimes, because they had life histories suited to brief hydroperiods. Importantly, previous life history rarely predicted species' actual responses to climate-driven flow regime change, raising doubts about the veracity of predictions based on species traits. This is because a species' potential for flexible phenology or growth rate is not necessarily indicated by life history traits.

Keywords

Body size, climate change, ephemeral streams, extirpation, intermittency, invertebrates, mBACI, phenology, rivers, species traits

Introduction

One key negative outcome of climate change is the likely mismatch between the timing of organisms' life history events and the environmental conditions with which they have evolved (Senseth & Mysterud, 2002). Phenological mismatch has already been documented across ecosystems (such as forests, e.g. Ibáñez et al., 2010), species interactions (such as plant-pollinator relationships, Forrest 2015), and food webs (trophic asynchrony, Thackeray et al., 2010). The phenology of freshwater species has evolved in accordance with extant water regimes (Bunn & Arthington 2002; Lytle & Poff 2004), governed by precipitation patterns which are rapidly changing under global warming (Döll & Zhang, 2010; IPCC 2014; Hoegh-Guldberg et al., 2018), making them particularly vulnerable to climate change. Sudden shifts in the presence or absence of water can lead to losses of entire cohorts: a few cases have already been documented (e.g. Lund, Wissinger & Peckarsky, 2010; Bogan & Lytle, 2011), but such losses are almost certainly more widespread. Invariably, losses of species and declines in biodiversity will change assemblage structure, energy flows and the essential processes which these organisms regulate.

In stream ecosystems, flow regime (the timing, duration and velocity of flow) is the most important abiotic factor governing ecosystem processes and organism life histories, including invertebrates, which comprise the greatest component of freshwater biodiversity (Gasith & Resh, 1999; Bernal, von Schiller, Sabater & Martí, 2013; Clarke, Mac Nally, Bond & Lake, 2008). Many regions are now experiencing climatic warming and drying, causing declines in surface water flows and groundwater inputs to streams (as water tables decline). In regions where this has been occurring for decades, such as southwestern Australia (SWA), the result has been the transition of perennially flowing rivers and streams to intermittent flow regimes (Petrone, Hughes, Van Niel & Silberstein, 2010; Döll & Schmeid, 2012; Datry et al., 2017b). Whilst intermittent rivers and ephemeral streams (IRES) are widespread in many climate

regions (Datry, Bonada & Boulton, 2017a; Datry et al., 2018), increased frequency and duration of dry phases and loss of perennial surface water are expected to adversely impact many stream invertebrate faunas because they have evolved in catchments with reliable perennial flows. In mediterranean-climate regions, the stream invertebrate fauna consists primarily of endemic relictual species from cooler wetter times, rather than arid-adapted taxa (Bonada & Resh, 2013; Davies & Stewart 2013), making them particularly vulnerable to flow regime change. Furthermore, perennially flowing streams act as dry season refuges for nearby intermittently flowing streams (Chester & Robson, 2011), so the loss of perennial flows from catchments removes refuges and thus reduces biodiversity across the catchment, not just in the affected stream.

A considerable knowledge gap exists regarding the responses by species with different types of life history to the impact of environmental changes caused by climate change (Filipe, Lawrence & Bonada, 2013). Many studies have predicted species' responses to changes in particular climate variables, based on their life history characteristics, without being able to test whether the predictions are accurate because the predicted climatic changes have not yet occurred in real landscapes (e.g. Robson, Chester and Austin, 2011; Urban et al., 2016; Serra-Diaz and Franklin, 2019). Some studies have used life histories to predict population responses (e.g. Chester, Miller, Valenzuela, Wickson & Robson, 2015), with mixed success. There is an urgent need to determine whether species life histories do in fact predict their responses to current and future climate change. Because it is such a profound change in a key driving variable, with strong links to invertebrate life histories, the transition from perennial to intermittent flow regimes in rivers provides an ideal climate change impact to examine how species responses are influenced by their life histories.

Aquatic organism life histories play a pivotal role in persistence in intermittent streams, as life cycles are constrained by the process of regular drying, which eliminates individuals that

are not capable of withstanding drying (Gasith & Resh, 1999; Boulton, 2003; Robson, et al., 2011). There are examples of phenological plasticity (e.g. development time and voltinism) in insects inhabiting both intermittent and perennial streams (e.g. Lehmkuhl, 1973; Storey & Quinn, 2011), however the extent of such adaptability among other species cannot be assumed and is likely dependent on the organism's evolutionary history (Bonada et al., 2017). In addition, freshwater macroinvertebrates may possess drought resistance and/or resilience traits. Resistance traits enable organisms to persist in situ, commonly through desiccation resistant life stages (Robson et al., 2011; Strachan, Chester & Robson, 2015). Resilience traits enable organisms to escape drying waterbodies, and then rapidly recolonise upon resumption of flows. Drying cues may trigger the emergence of larval insects, or retreat to perennial refuges by aquatic obligate organisms (Strachan et al., 2015). When perennial rivers and streams transition to seasonal flows, species persistence depends on whether they possess resistance or resilience traits, and their phenological flexibility. However, species' ability to adapt their life history phenology to suit new flow regimes has rarely been studied, in part due to the rarity of detailed life history records coupled with reliable flow regime data.

This research tested predictions of life history responses by invertebrates to in situ flow regime change in SWA headwater streams. This was possible because of detailed life history studies carried out prior to changes in flow regime (Bunn 1988), and the existence of long-term flow regime data for the study streams, which documents their transition from perennial to seasonal flows. Bunn (1988) documented the life histories of the 13 most common benthic macroinvertebrate species in headwater streams in the northern jarrah forest in SWA, in the early 1980s when perennially flowing headwater streams were common in the region. These 13 species (12 insects and 1 crustacean) are endemic to SWA, and are Gondwanan relicts from cooler wetter times, that have persisted into contemporary mediterranean climates (Davies & Stewart, 2013). In common with faunas of south-eastern temperate climate regions

of Australia (e.g. Marchant et al., 1984; Campbell, Parnorng & Treadwell, 1998), the species in southwestern streams tended to have univoltine life cycles (single generation per year), often with slow larval growth and no diapause phase (mayflies and caddisflies), poorly synchronised development and extended emergence, and with evidence of cohort splitting into one and two year life cycles (e.g. dragonflies). The larval development time of most of these species in the 1980s is longer than the current (2010s onward) average hydroperiod duration in these streams. Therefore, we predicted that species that have not adapted to the seasonal flow regime (via phenology or desiccation resistance/ resilience) would likely be extirpated from these streams. Other studies have noted that species with longer development times, or that are univoltine (rather than multivoltine) are potentially vulnerable to local extinction during drought (Lake 2003; Bogan & Lytle, 2011; Lawrence et al. 2010; Chessman 2015; temperature regime, Rivers-Moore et al. 2013; hydrological alteration and temperature change, De Walt, Favret & Webb, 2005). We used the existing data on species life histories from Bunn (1988) and other studies describing responses to hydrological change (e.g. Bogan & Lytle, 2011; Lawrence et al. 2010; Chessman 2015; Chester et al., 2015; Strachan et al., 2015) to predict the responses by these 13 species to intermittency. Generally, species with larval development time overlapping with summer or autumn (when intermittent streams were dry) were regarded as needing to adapt. Relevant life history traits (e.g. synchrony, cohort-splitting, voltinism, according to Bunn, 1988) and potential to possess adaptations to desiccation (Strachan et al. 2015) influenced predictions.

The aims of this study were to: 1) determine the efficacy of documented life histories as a predictor of future response to environmental change caused by climate change – in this case, drying flow regimes, and 2) identify life histories or taxonomic groups which may have greater extinction risk or less ability to adapt. To do this, life histories were grouped into types as the basis for hypotheses predicting particular responses to flow regime change,

which were then tested using life history data for the same species in the same stream reaches from before and after the transition from perennial to seasonal flow regimes. To isolate the effects of flow-regime change from other factors that changed through time, a multiple before-after, control-impact (mBACI) design was used.

Methods:

Climate change in the study area

The mediterranean climate in SWA has been warming and drying for at least five decades (Bates et al., 2008). Mediterranean climates typically show marked inter-annual variation in rainfall, but in SWA it has declined by 20% of long-term mean precipitation since 1970 (Hope, Timbal & Fawcett, 2010). Additionally, the “big wet” years that historically punctuated rainfall records no longer occur (the most recent was in 1974 – 45 years ago), and totals that approach the long-term average (i.e. *circa* 830 mm p.a.) are now considered “wet” years. Within the study area, rainfall totals (measured at Wungong Dam, Bureau of Meteorology station 009044) have steadily declined since records began in 1913, and the 1990-2017 mean was 27% below that between 1913-1990. This pattern is congruent with that of the wider SWA region (McFarlane et al., 2020).

As rainfall declined, a disproportionately larger decline in streamflow occurred, of around 70% (Petrone et al., 2010; Silberstein et al., 2012). This was caused by declining groundwater tables; disconnection of groundwater from streambeds occurred around the years 2000 – 2001 in most catchments, resulting in continued decline in streamflow independent of annual rainfall (Petrone et al. 2010; Hughes, Petrone & Silberstein, 2012; Kinal & Stoneman, 2012). That is, most streams went from gaining to losing streams, causing further losses in surface flow and the transition of perennial streams to intermittency. In some SWA catchments, interactions between climatic drying and historical management of forests (overstocking in some regrowth areas), and widespread water stress of forest vegetation, added further

downward pressure on water tables that were already declining due to climate change (Batini, 2007; Liu, Harper, Smettem, Dell & Liu, 2019; Harper, Smettem, Ruprecht, Dell & Liu, 2019). Nevertheless, climate remains the key driver of hydrological change across the region (McFarlane et al., 2020), as in other regions globally where streamflow declines are occurring.

Study design

mBACI designs are optimal for detecting the effects of environmental change through time because they enable separation of the variable of interest (here flow regime) from other temporal changes (Downes et al., 2002). The mBACI design has multiple control reaches that have remained perennial and were sampled by Bunn (1988) (the before-control samples) and again by us (the after-control samples). It has multiple impact reaches that have transitioned to intermittent flows and were sampled by Bunn (1988) when they were perennial (before-impact reaches) and by us now that they are intermittent (after-impact reaches). If flow regime is the sole cause of changes in life histories, then the life histories of species in the control reaches would be the same in the before-control and after-control data, but the before-impact life histories would differ from the after-impact life histories. If other variables had altered life histories across the catchment, then the after-control data would show different life histories to the before-control data. If flow regime change had not altered life histories, there would be no difference between the before-impact and after-impact results.

This research was conducted in the forested headwater streams of the Wungong Brook catchment, within the northern jarrah forest (NJF) of SWA (Figure 5.1). This catchment is entirely forested and has been a protected water supply catchment since 1909, ensuring that only climate-related variables have affected stream ecosystems. Hence no additional confounding factors needed to be controlled. NJF is dry sclerophyll forest, dominated by eucalypt trees: jarrah *Eucalyptus marginata* and marri *Corymbia calophylla*. This region has

undergone a protracted period of isolation from the wetter environments of the continent's south east (Davies & Stewart, 2013). So, despite the mediterranean climate, the stream invertebrate fauna are primarily Gondwanan relicts from wetter times, characterised by comparatively few Ephemeroptera, Plecoptera and Trichoptera (EPT) taxa and high endemism (Davies & Stewart, 2013).

Thirteen reaches were selected spanning Wungong Brook and each of its four major tributaries. Eight of these reaches were studied by Bunn in 1983-84 (1986; 1988) and Bunn, Edward & Loneragan, (1986) (Figure 5.1). In the period of Bunn's studies (and before), most streams in this catchment were perennially flowing (i.e. before-impact and before-control in the mBACI design). Flow gauging data from this catchment from the 1960s onwards shows clearly the transition from perennial to seasonal flows in most of these streams (Figures 5.2, 5.3). It also shows later commencement of flows in the now seasonal streams (this was previously the onset of wet season increases from baseflow) and later peak flows in all streams (Figure S5.1 [2.6]). One stream, Waterfall Gully (and its tributaries), remains perennial (reaches labelled WF, Figure 5.1); flows there have not ceased since records began in 1966. All other formerly perennial watercourses in the Wungong catchment have transitioned to seasonal flows. Only one other stream, Seldom Seen Brook, has retained one short, spring-fed reach (SS2 Figure 5.1) which is perennial in most years, slowing to a trickle in the summer months (reach SS1 is intermittent). Thus, at present, there are only two waterways that contain flowing water all year (hence they represent the before-control/after-control comparison in the mBACI design). In the intermittent streams (i.e. representing the before-impact/after impact comparison), flow now commences in June or July, whereas formerly perennial flows increased above baseflow in March-April (Figure 2.6). Since disconnection from near-surface groundwater aquifers, streamflow is entirely dependent on surface run-off from rainfall, causing the later commencement of flows. Because of the

increased heterogeneity in flow duration caused by drying, five additional reaches were added to the study to better represent this diversity (Figure 5.1).

Curtis Creek (CC, Figure 5.1) and Chandler Brook (CH, Figure 5.1) are intermittent, but both were formerly perennial streams. Bunn (1986) and Bunn et al. (1986) had sites in both these streams when they were perennially flowing (Bunn erroneously labelled Chandler Brook as Wungong Brook in his study). Chandler Brook is the second largest waterway in the catchment after Wungong Brook, but flow gauging data is incomplete. Records show that Wungong Brook first dried out in 2008 and Chandler Brook likely dried at the same time (Figure 5.2). Currently, Chandler Brook is flashy, discharging large volumes of water over winter and spring, and quickly drying in summer. During the dry phase in intermittent streams (February to June) streambeds were explored along most of their length, in search of remnant pools or seeps which may have acted as refuges for aquatic fauna. There were no permanent pools and there was no evidence of hyporheos in Chandler Brook (much of which is underlain by bedrock). The upper reaches of Curtis Creek are narrow and well vegetated, with deeper substrate overlaying bedrock; water was seldom deeper than 0.3 m at the study sites. It contains perennial pools sustained by groundwater seeps. Lastly, Coccinea Creek (EPH, Figure 5.1) arises from a granite outcrop spring and may always have had intermittent flow (ungauged); it is currently ephemeral, flowing from July-September. This stream was only sampled once in August 2017 (winter) due to its short hydroperiod. While this was not formally part of mBACI comparisons, sampling here did allow inferences to be made about responses at the low end of the flow spectrum in this system.

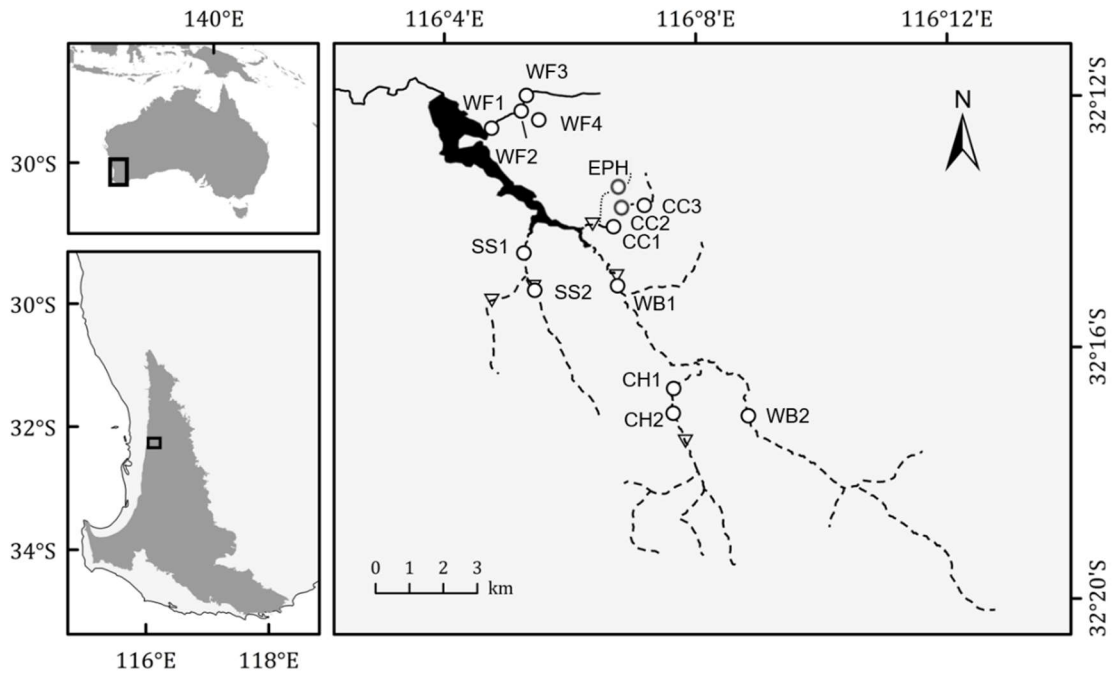


Figure 5.1: Study reaches in the Wungong Brook catchment of south-western Australia, perennial reaches (solid lines), formerly perennial, now intermittent reaches (dashed lines), and an ephemeral stream (dotted line). Inverted triangles denote flow gauging stations (flow gauge at WF1 not visible). Sites used by Bunn (1988) were WF1, WF2, CC1, CC2, SS1, SS2, CH1 and CH2; additional sites WF3, WF4, CC3, WB1, WB2 and EPH.

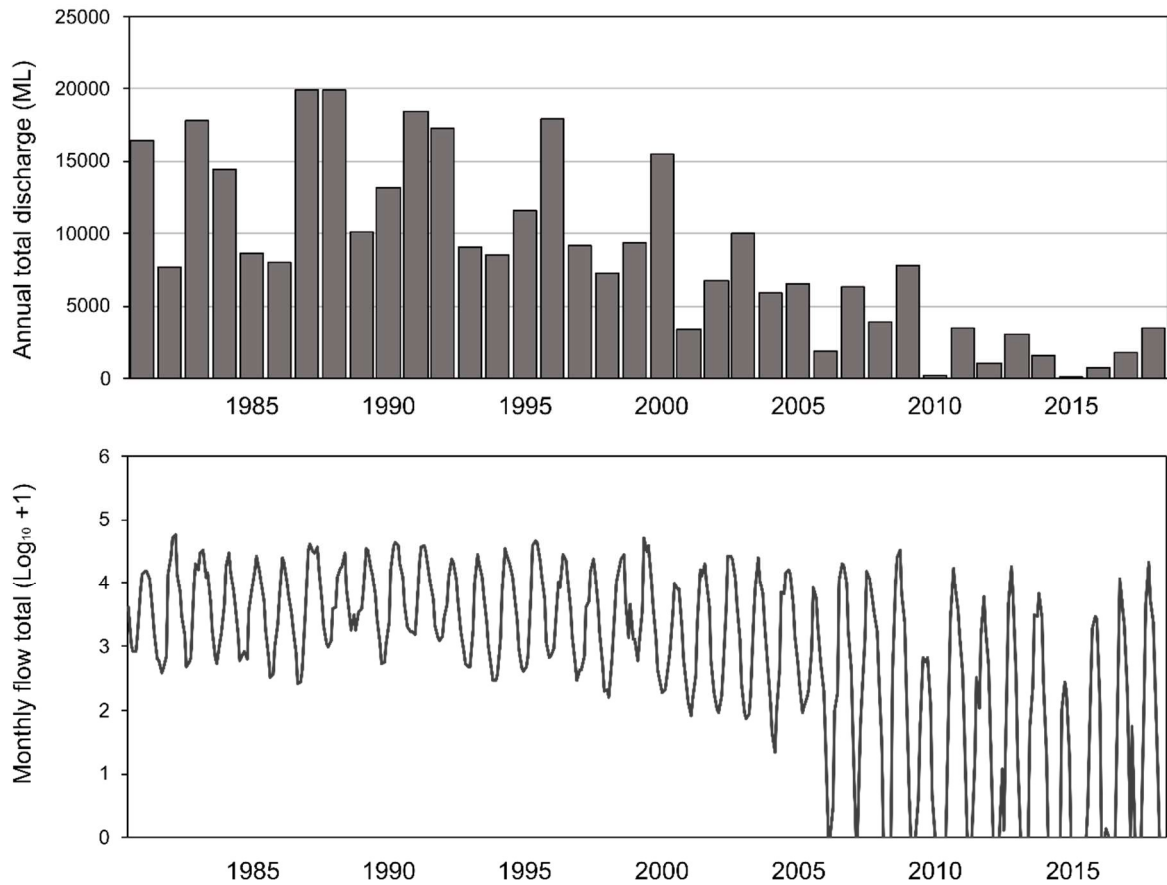


Figure 5.2: Annual total discharge (ML) at Wungong Brook from 1982 to 2016 (top). Bottom: monthly streamflow totals at Wungong Brook 1982 to 2016, log transformed so months of zero flow can be seen (separately from those with very low flows). Wungong Brook became intermittently flowing in 2007 and has flowed intermittently every year since (correct at 2020). All charts produced using daily streamflow measurements, freely available from the Western Australian Department of Water and Environmental Regulation (gauge number 616041; wir.water.wa.gov.au).

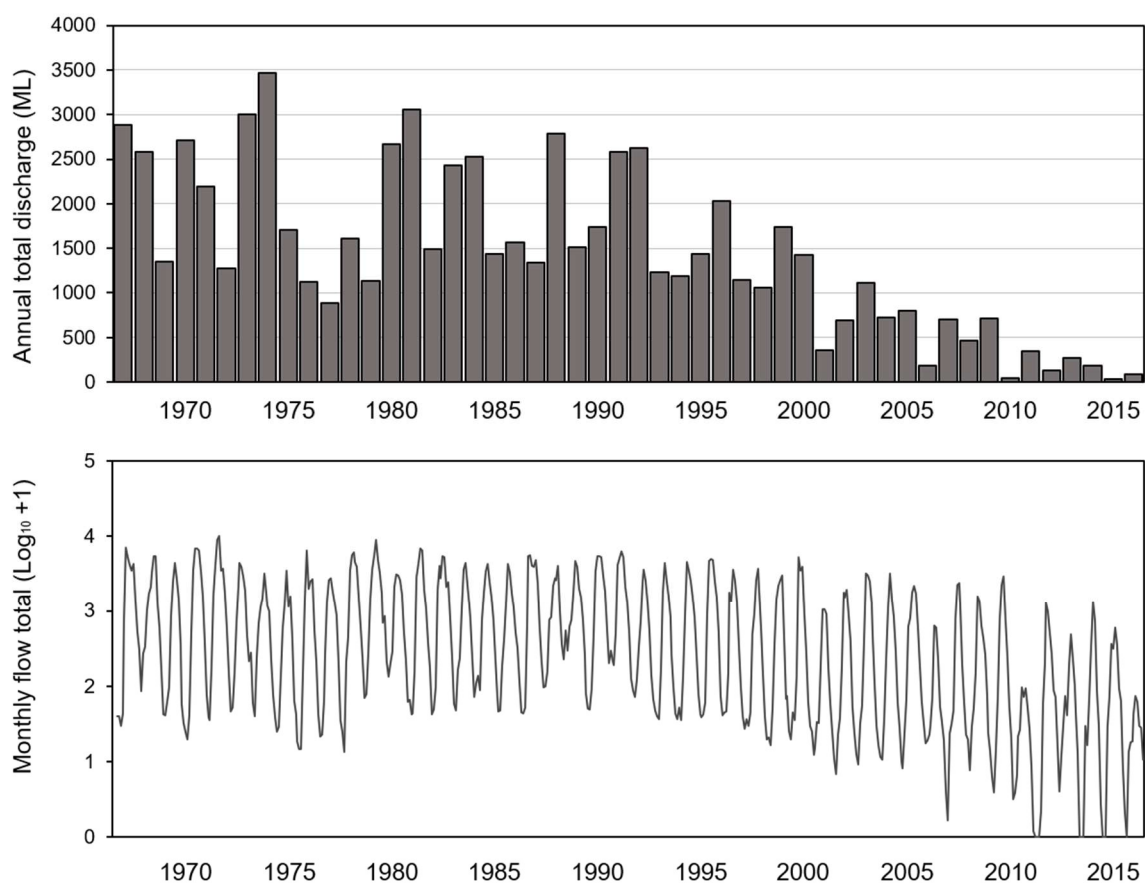


Figure 5.3: Annual total discharge (ML) at Seldom Seen Brook from 1967 to 2016 (top). Bottom: monthly streamflow totals at Seldom Seen Brook 1970 to 2016, log transformed so months of zero flow can be seen (separately from those with very low flows). Seldom Seen Brook became intermittently flowing in 2011, however a spring on one reach maintained a trickle of perennial flow in some years since, including 2016-17. All charts produced using daily streamflow measurements, freely available from the Western Australian Department of Water and Environmental Regulation (gauge number 616021; wir.water.wa.gov.au).

Temperature measurements made by Bunn (1985) using maximum-minimum thermometers, and by the authors using temperature loggers (Thermodata™ iB Tags, Thermodata Pty. Ltd.), show patterns that reflect the change in flow regime perhaps more strongly than rising air temperatures (Figure S5.2). Bunn’s recorded means show that all the streams in the early 1980s had quite constant temperatures of around 15 °C in both summer and winter; indicating

the groundwater origin of summer baseflow (constant temperature cooler than air temperature) (Figure S5.2). Waterfall Gully has remained perennial and still shows this consistent temperature regime, but in Seldom Seen Brook (where only a trickle of perennial flow remained) temperatures were $> 20^{\circ}\text{C}$ in December and peaked at 25°C in February. Temperatures in summer flows in Curtis Creek also exceeded 20°C in December, but disconnected pools were cooler, with temperatures similar to those recorded by Bunn (1985). As these pools were located on groundwater seeps, this likely determined their temperature. In contrast, pools in Chandler Brook that were not groundwater-fed reached temperatures $> 35^{\circ}\text{C}$ before drying out, far exceeding any water temperatures recorded by Bunn (1985). Interestingly, winter temperatures recorded in our study were consistently slightly cooler than those recorded by Bunn (1985) (Figure S1). Bunn (1985) rarely recorded temperatures below 10°C , yet we recorded them often in winter, indicating that cooler air temperatures more strongly influence water temperatures in all streams now that groundwater inputs are reduced or absent. Overall, water temperatures in these streams rarely exceeded 20°C (with the exception of disconnected, drying pools), which is the temperature threshold for sensitive stream invertebrates in SWA (Stewart, Close, Cook & Davies, 2013). Changes in habitat temperature are therefore an attribute of flow regime change in each stream, rather than being an independent factor arising from climate change — the impact of the slight rise in average annual air temperature in SWA since 1983-84 is tiny compared to the threshold effect resulting in loss of groundwater. In any case, the use of the mBACI design (see below) separates changes in life history due to flow regime change from the effects of temperature or other variables that may have acted across the study area.

Size class distributions of the thirteen most-common macroinvertebrate species at the study streams were recorded by Bunn (1988) in the early 1980s, providing detailed descriptions of species phenology (i.e. “before” data) with which to compare after 35 years of streamflow

drying (i.e. “after” data). These were: four Trichoptera: *Notoperata tenax* Neboiss and *Lectrides parilis* Neboiss (Leptoceridae), *Taschorema pallescens* (Banks) (Hydrobiosidae), and *Smicrophylax australis* (Ulmer) (Hydropsychidae); five Ephemeroptera: *Bibulmena kadjina* Dean, *Neboissophlebia occidentalis* Dean, and *Nyungara bunni* Dean (Leptophlebiidae), *Tasmanocoenis tillyardi* Lestage (Caenidae), and *Offadens soror* (Ulmer) (Baetidae; formerly *Baetis soror*); two Plecoptera (Gripopterygidae): *Leptoperla australica* (Enderlein) and *Newmanoperla exigua* (Kimmins); one Odonata: *Austroaeschna anacantha* Tillyard (Telephlebiidae; formerly in Aeshnidae), and one Amphipoda: *Uroctena* sp. Nicholls (Paramelitidae).

We predicted responses to drying for each species based on previously known reproductive and phenological traits (Table 5.1). Univoltine and synchronous species, or those with larval development time exceeding one year where growth occurred during times when intermittent streams were dry, were predicted to be extirpated from intermittent streams (*B. kadjina*, *T. tillyardi*, *N. tenax*, Table 5.1). Stoneflies are known to use desiccation-resistant eggs (Strachan et al. 2015) so although they were univoltine, synchronous and developed during summer and autumn, we predicted that they would adapt to synchronous development during winter-spring (Table 5.1). Univoltine species with > 1 cohort p.a. (*N. occidentalis*, *N. bunni*, *T. pallescens*, *S. australis*, *L. parilis*, *Uroctena* spp.) and asynchronous, multivoltine species (*O. soror*) were predicted to adapt to growth during the flow period in intermittent streams or to be absent from them (if they had no desiccation-resistant stage). The dragonfly *A. anacantha* and the caddisfly *L. parilis* showed cohort-splitting (Bunn, 1988) indicating the capacity to expedite growth, so we predicted that these species would persist in intermittent reaches. We sampled the same stream reaches as Bunn with the exception of Foster Brook (which is in a different catchment and is not now accessible), and with the addition of the sites described above, for the same 13 taxa in 2016-17. The size frequency distributions of

two additional species, common in 2016-17, were also measured: the mayfly *Nousia* AV16 (Leptophlebiidae) and the suspension-feeding caddisfly *Hydrobiosella michaelsoni* (Ulmer) (Philopotamidae). *Nousia* AV16 was not recorded by Bunn (1986; et al. 1986; 1988) but was found in large numbers in all formerly perennial, now-intermittent streams. *Hydrobiosella michaelsoni* was present in the Wungong catchment in the early 1980s, but generally in low numbers. It was present in high numbers in winter 2016 in Chandler Brook.

Each sample reach was visited seven times between October 2016 and August 2017, at approximately six-week intervals and macroinvertebrates were collected. In mid-spring, all reaches were flowing, intermittent reaches then began to dry and disconnect in early summer (November and December), and by autumn (April – July 2017) they had dried completely. The sampling therefore commenced in the austral spring 2016 and ended with austral late-winter to early spring in 2017, encompassing one year.

Invertebrate sampling

Bunn (1988) used a Sürber sampler (250 µm mesh) to collect macroinvertebrates, but this method requires sufficient flow to wash material into the net and is therefore ineffective during low flow and disconnected phases in seasonal streams. Robson et al. (2018) compared Sürber and dip net samples from 12 headwater streams and showed that the invertebrate assemblage (species x abundance) collected did not differ. Thus, at perennial reaches and during the flowing phase at intermittent reaches, a 500 µm mesh dip net was used in the same way as a Sürber sampler. The opening was faced into the oncoming flow, and a 30 cm x 30 cm area directly in front of the net gently disturbed by hand to dislodge invertebrates, which were then washed into the net by the flow. A 500 µm mesh size was judged sufficient to capture all individuals for the study species: notably, both Bunn and this study recorded a minimum head width of around 0.1 mm (albeit measured slightly differently) for *Uroctena* sp., that were captured efficiently regardless of differences in the net. Where available, four

samples were taken in erosive habitat (riffles and runs), and four from depositional habitats (pools). When flows were insufficient, a 30 cm x 30 cm area of substrate was disturbed by foot, and the net vigorously swept through the water column to collect any dislodged and suspended invertebrates, collecting four samples per reach where surface water was present.

There was also increased sampling effort in this study in comparison to Bunn (1988), who collected two samples from each study reach. We collected between four and eight samples (depending on surface water availability) per reach per time. The rationale for increasing sampling effort was to strengthen inferences about change in reproductive and life history phenology due to climatic drying, with increased heterogeneity of flow environments, rather than baseline characterisation of phenological traits (already completed by Bunn 1988).

Determination of size classes and analysis of life history patterns

Invertebrate sizes were measured in the same way as Bunn (1988). Head capsule widths (or lengths, for the amphipod *Uroctena* sp.) were measured to the nearest 2 μm using a dissecting microscope fitted with an eyepiece graticule. Final instars (Trichoptera) or mature nymphs (Ephemeroptera and Plecoptera – identified by presence of blackened wing buds and/or presence of adult compound eyes) were identified to determine emergence timing and size at emergence for each species. Head capsule length of the amphipod *Uroctena* sp. was measured from the base of the antennule to the posterior edge of the head capsule, rather than the first thoracic segment as in Bunn (1988). This measure was used because some specimens were curled over, affecting measurement accuracy, and the head capsule measurement was thus more accurate; it is also widely used in studies of stream amphipods (e.g. Vadher et al., 2015). Presence of gravid females was also recorded.

Bunn (1988) subjectively assigned size classes to head capsule width-frequency data, using peaks in size frequencies and presented the data as graphs; unfortunately, his original data is no longer available. Therefore, we redrew Bunn's figures (with permission from the original

publisher) as raw size frequency data, as this provided the best comparison between datasets. Although we did not compare the number of size classes per species between the two studies, we did compare patterns in relative size frequency, because body size increases with age for both insects and amphipods.

Table 5.1: Reproductive cycle and phenology of thirteen formerly most-common species in Northern Jarrah Forest perennial streams recorded in 1983-84 (Bunn, 1988), the hypothesized response of each species to flow regime drying, and their observed responses (# corresponds to response type number given in Results text).

Species	Reproductive cycle 1982-83	Phenology 1983-84	Hypothesised life history response	Observed life history response	#
Ephemeroptera					
Leptophlebiidae					
<i>Bibulmena kadjina</i>	Univoltine: 1 synchronised cohort	Spring → Autumn	Displacement to perennial streams, unless phenology changes	Reduced synchrony, occupancy of autumn drought refuges	3
<i>Neboissophlebia occidentalis</i>	Univoltine, overlapping cohorts	Autumn/winter → Summer/autumn	Increased synchrony, development during flow period in intermittent streams	Restricted to perennial streams Did not adapt	1
<i>Nyungara bunni</i>	Univoltine, overlapping cohorts, little synchrony	Autumn/winter → Summer/autumn	Increased synchrony, development during flow period in intermittent streams	Reduced development time, increased synchrony to adapt to intermittency	2
Caenidae					
<i>Tasmanocoenis tillyardi</i>	Univoltine, synchrony	Late summer → early summer	Displacement, unless development expedited	Not found 2016-17	-
Baetidae					
<i>Offadens soror</i>	Unclear	All year	Adaption to new flow regime. Hatching → eclosion during flow period	Restricted to perennial streams Did not adapt	1
Plecoptera					
Gripopterygidae					
<i>Leptoperla australica</i>	Univoltine, synchrony	Summer & Autumn	Delayed hatching if able, development to suit new regime	Development timing began later in intermittent streams, expedited larval growth	4
<i>Newmanoperla exigua</i>	Univoltine, synchrony	Summer & Autumn	Delayed hatching if able, development to suit new regime	Not found 2016-17	-
Trichoptera					
Hydrobiosidae					
<i>Taschorema pallescens</i>	Bivoltine, synchronised cohorts	Winter → Summer, Summer → Spring	Switch to a single cohort in intermittent streams, recolonize from perennial streams	Restricted to perennial streams Did not adapt	1
Hydropsychidae					
<i>Smicrophylax australis</i>	Bivoltine, loosely synchronous, variable	Autumn/Winter → Spring	Become univoltine in intermittent	Restricted to perennial streams	1

			streams develop during flow period	Did not adapt	
Leptoceridae					
<i>Lectrides parilis</i>	Univoltine, cohort splitting	Autumn → Spring Autumn → Early summer	Alter hatching time to suit flow regime. Expedite development (shown by ability to split cohort)	Restricted to perennial streams/reaches Did not adapt	1
<i>Notoperata tenax</i>	Univoltine, synchrony	Winter → Autumn	Persist in intermittent streams, if eclosion before Autumn drying	Restricted to perennial reaches Did not adapt	1
Odonata					
Telephlebiidae					
<i>Austroaeschna anacantha</i>	Cohort splitting, 1 and 2 year development	All year	Persistence in intermittent streams, if larval development is expedited (potential shown by ability to split cohort)	Restricted to perennial streams/reaches Did not adapt	1
Amphipoda					
Paramelitidae					
<i>Uroctena spp.</i>	Univoltine	Variable	No known desiccation resistance – loss from streams without refuges	Retreat to groundwater refuges, hyporheos or granite inselbergs	5

Results

Two of the thirteen species recorded by Bunn (1988), the stonefly *N. exigua* and the mayfly *T. tillyardi* were not collected from these streams during this study. Of the eleven species present in 2016-17, our predicted responses to intermittency, based on species life history, were wrong in nearly every case (Table 5.1). This is because species with the same reproductive cycles and phenologies in 1983-84 displayed different responses to intermittency in 2016-17. Some species which, in theory, were predicted to be well placed to adapt to the new flow regime did not adapt their phenology (*N. occidentalis*, *O. soror*, *N. exigua*, *T. pallescens*, *S. australis*, *L. parilis*, *N. tenax*, *A. anacantha*, Table 5.1). Rather, five general life history responses to intermittency were identified: (1) retreat to perennially flowing reaches, (2) increased population synchrony, (3) decreased synchrony and occupancy of drought refuges, (4) alter phenology to suit new flow regime, and (5) retreat to groundwater refuges.

Response 1: Retreat to perennially flowing reaches

Seven of the 13 species studied by Bunn are now only found in perennially flowing reaches in Waterfall Gully, making this the most common response. Of these, two species, the caddisfly *L. parilis* and dragonfly *A. anacantha* were also collected in small numbers at the semi-perennial reach at Seldom Seen Brook. At Waterfall Gully, *A. anacantha* appeared to have two cohorts present in October and again in July 2017. This is suggestive of the cohort splitting observed by Bunn (1988), but too few mature nymphs were collected to be certain (Figure S3). A trickle of perennial flow between 2016 and 2017 was sufficient for *L. parilis* (Figure S4) and *A. anacantha* larvae to persist in Seldom Seen Brook; however, this reach often dried out post-2010 so these populations are likely sinks. Bunn found *L. parilis* had two partially overlapping cohorts in Waterfall Gully, with mature nymphs present continually between spring (October) and autumn (May) (Figure S4). We also found *L. parilis* present for most of the year, but mature nymphs were only collected in summer (December – February) and winter (July).

We found the remaining caddisfly species studied by Bunn (*T. pallescens*, Figure S5; *S. australis*, Figure S6; and *N. tenax*, insufficient numbers to describe phenology; Figure S7) and two mayflies (*O. soror*, Figure S14 & S15; and *N. occidentalis*, Figure S16 & S17) only in Waterfall Gully, which had remained perennial. The phenology and life history of these six species had not changed in Waterfall Gully since 1983-84. In 1983-84, *Taschorema pallescens* was widespread and displayed a determinate (i.e. inflexible) life history phenology across streams in the Wungong catchment (Bunn 1988). In contrast, *S. australis* displayed different phenology and degree of synchrony in response to differing variability in flow velocity between streams (Bunn 1988). *Notoperata tenax* development was synchronised across streams, growing throughout the winter months and emerging in early autumn – timing that could fit the intermittent stream flow regime. The two mayflies had multiple over lapping

cohorts with potentially several generations per year (particularly *O. soror*, Bunn 1988), yet did not colonise intermittent streams during the flowing phase.

Response 2: Increased population synchrony

In 1983-84, the mayfly *Nyungara bunni* was found to have several apparently overlapping cohorts with little synchrony in development (Bunn 1988; Figure 5.4a); this remained unchanged at Waterfall Gully in 2016-17 (Figure 5.4b). At Seldom Seen Brook, although flow did not cease during the study period, *N. bunni* expedited its development time as a single synchronised cohort (Figure 5.4c). Nymphs were not found in this reach by the summer months, potentially in response to higher temperatures (Figure S5.1), or to the slowly trickling flow. Large nymphs of *N. bunni* were also found at Curtis Creek in early summer (December) 2016, when the stream was drying (Figure 5.4d), indicative of emergence just prior to drying. Nymphs were not collected there at any other time, indicating the presence of a single synchronous cohort.

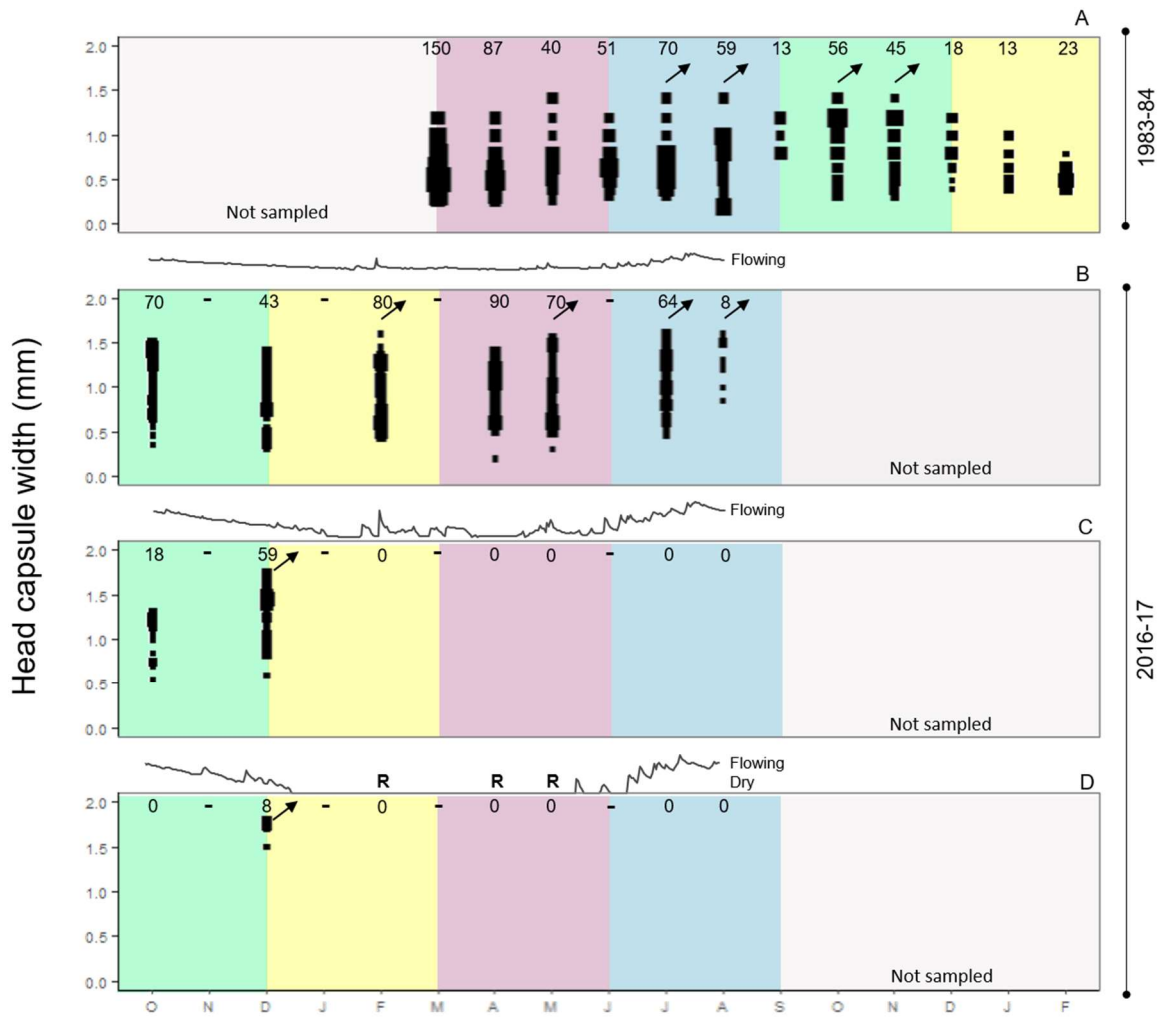


Figure 5.4: Size distribution of mayfly nymphs *Nyungara bunni* (Leptophlebiidae) collected from Waterfall Gully in 1983-84 (A, perennial), and in 2016-17 from Waterfall Gully (B, perennial), Seldom Seen Brook (C, semi-perennial) and Curtis Creek (D, intermittent), during Austral spring (green), summer (yellow), autumn (purple) and winter (blue). 1983-84 data (A) redrawn from Bunn (1988) with permission from CSIRO publishing. Number of individuals collected given at top of each panel (0 means sampling occurred but no individuals were collected), solid arrow indicates final instars with blackened wing buds (ready for emergence). Size of squares indicate proportion of individuals of that size. Presented above each 2016-17 panel is log-transformed flow data from Waterfall Gully (B), Seldom Seen Brook perennial reach (C), coinciding with the sampling period (21/10/2016 to 11/08/2017; DWER stations 616023 and 616022).; and Curtis Creek (C) 2015 hydroperiod (data not available for sampling period; DWER station 616144). R = only drought refuges remaining.

Response 3: Decreased synchrony and occupancy of drought refuges

In 1983-84, the mayfly *B. kadjina* had a strongly synchronised univoltine life history, hatching in spring, nymphs grew throughout summer and autumn, emerging as adults over the winter months (Figure 5.5a). In contrast, development in 2016-17 was unsynchronised, with at least two loosely overlapping cohorts at Seldom Seen Brook (Figure 5.5b) and Curtis Creek (Figure 5.5c).

Curtis Creek dried by April, except for small, groundwater-fed refuge pools sparsely distributed in the streambed (Figure 5.6a,b). If the nymphs occupying these refuges survived, they may have supplied the eggs for hatching in the following winter (i.e. for the winter cohort). Notably, nymphs collected from drought refuge pools dwarfed ($> 2000\ \mu\text{m}$) the maximum size class reported for *B. kadjina* ($1211\ \mu\text{m}$) by Bunn (1988; Figure 5.7). Nymphs collected in autumn (April) 2017 were also bigger than those ready to emerge in summer (December) 2016, despite not having reached their final instar (Figure 5.7).

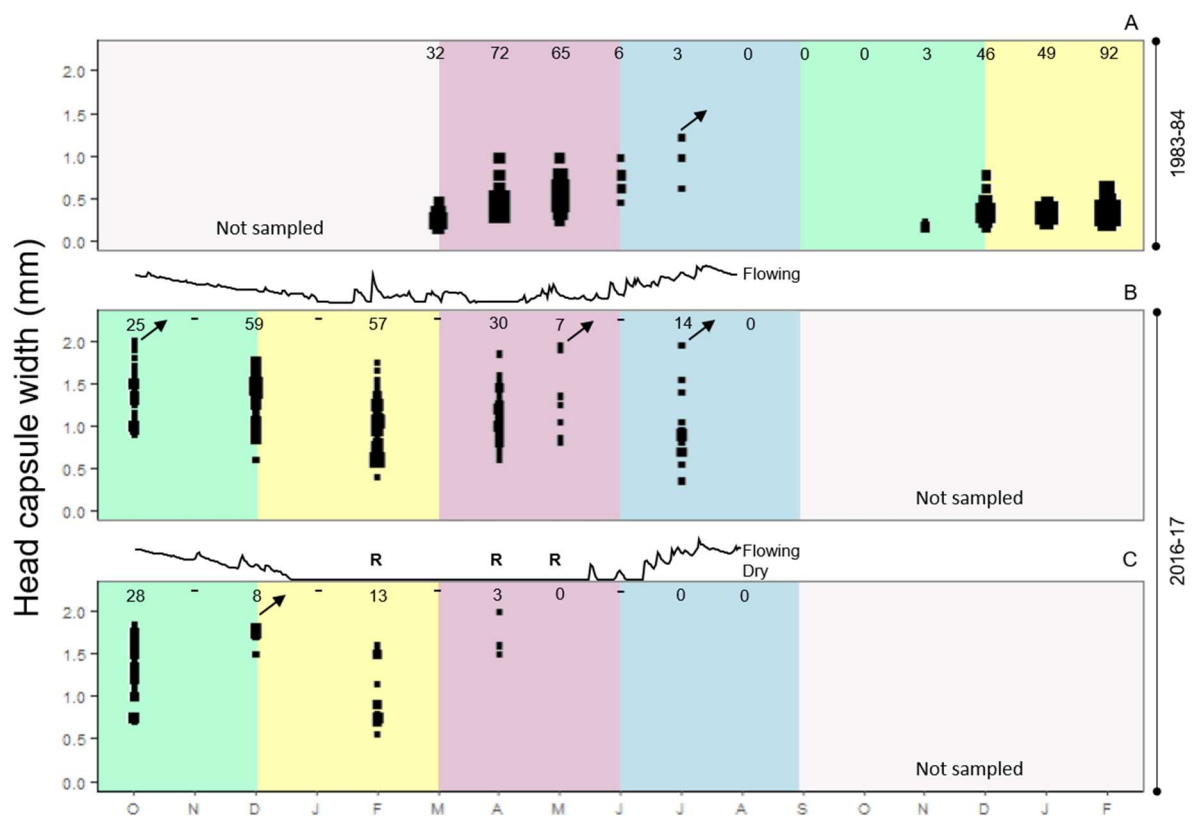


Figure 5.5: Size distributions of mayfly nymphs *Bibulmena kadjina* (Leptophlebiidae) nymphs collected from Seldom Seen Brook in 1983-84 (A), and in 2016-17 from Seldom Seen Brook (B) and Curtis Creek (C). 1983-84 data (A) redrawn from Bunn (1988) with permission from CSIRO publishing. Caption as for Figure 5.4.

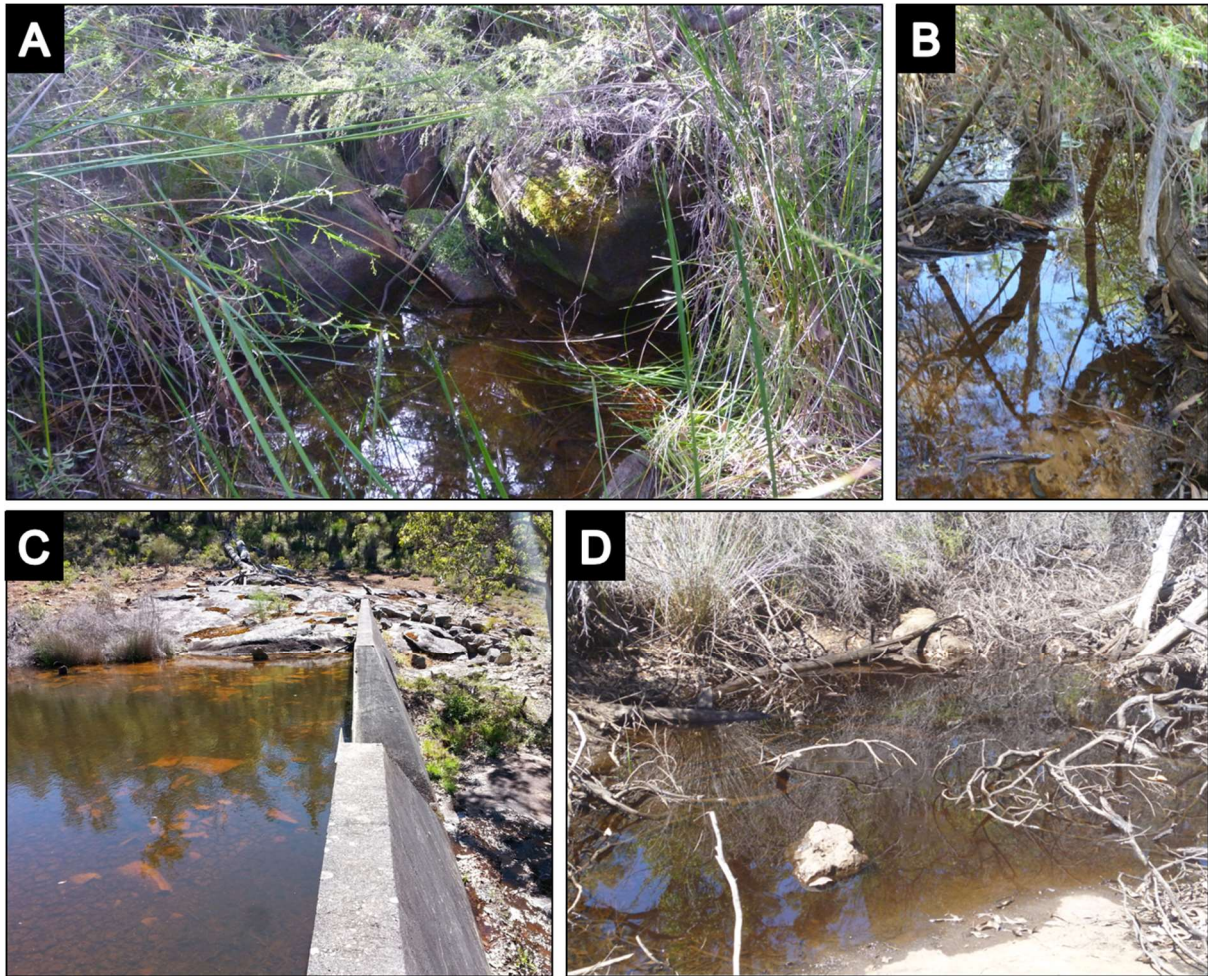


Figure 5.6: Examples of drought refuge pools in streambeds in the Wungong catchment in April 2017, from which *Bibulmena kadjina* and other macroinvertebrate species were collected: A = small springs; B = perennial groundwater seeps at the base of granite inselbergs (here at the headwaters of Curtis Creek); C = weir pools (here a flow gauging weir on Seldom Seen Brook). For comparison, an example of a receding (intermittent, non-refuge) pool (D) on the Wungong Brook during summer and autumn.

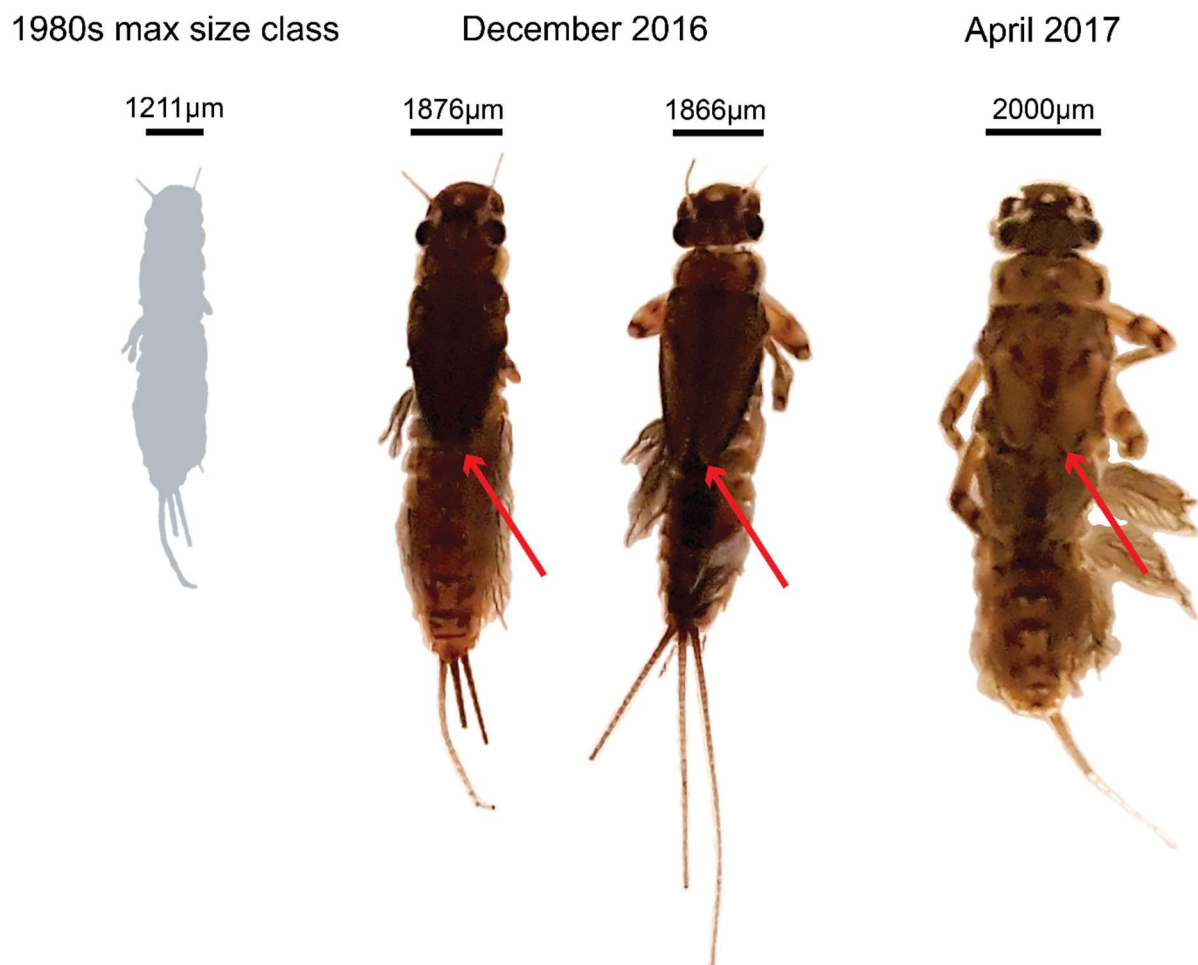


Figure 5.7: *Bibulmena kadjina* nymphs collected from Curtis Creek in December 2016 (centre), and from a drought refuge in April 2017 (right; drought refuge A in Figure 6). In 1983-84, the head capsule width of final instar *B. kadjina* was 1211µm (left); compared to much larger nymphs in 2016-17. Arrows point to the mature wing buds on the December specimens, compared to the developing wing buds on the April specimen indicating that it had not yet reached full size.

Response 4: Alter phenology to suit new flow regime

The stonefly *L. australica* was reported as an autumn breeder in 1983-84 (Figure 5.8a), but autumn is now the driest period of the year, with no surface water present in most streams, making this phenology untenable. In 2016-17, small nymphs were collected in early winter (June), soon after flow resumed (Figure 5.8b-d), and medium-sized and mature nymphs were collected in late winter (August). At Wungong Brook, hatching occurred even later in the year: hatchlings were collected in August 2017, and mature nymphs were found in mid-

spring (October) (Figure 5.8d). Therefore, this species successfully shifted its larval phenology to the new flow period.

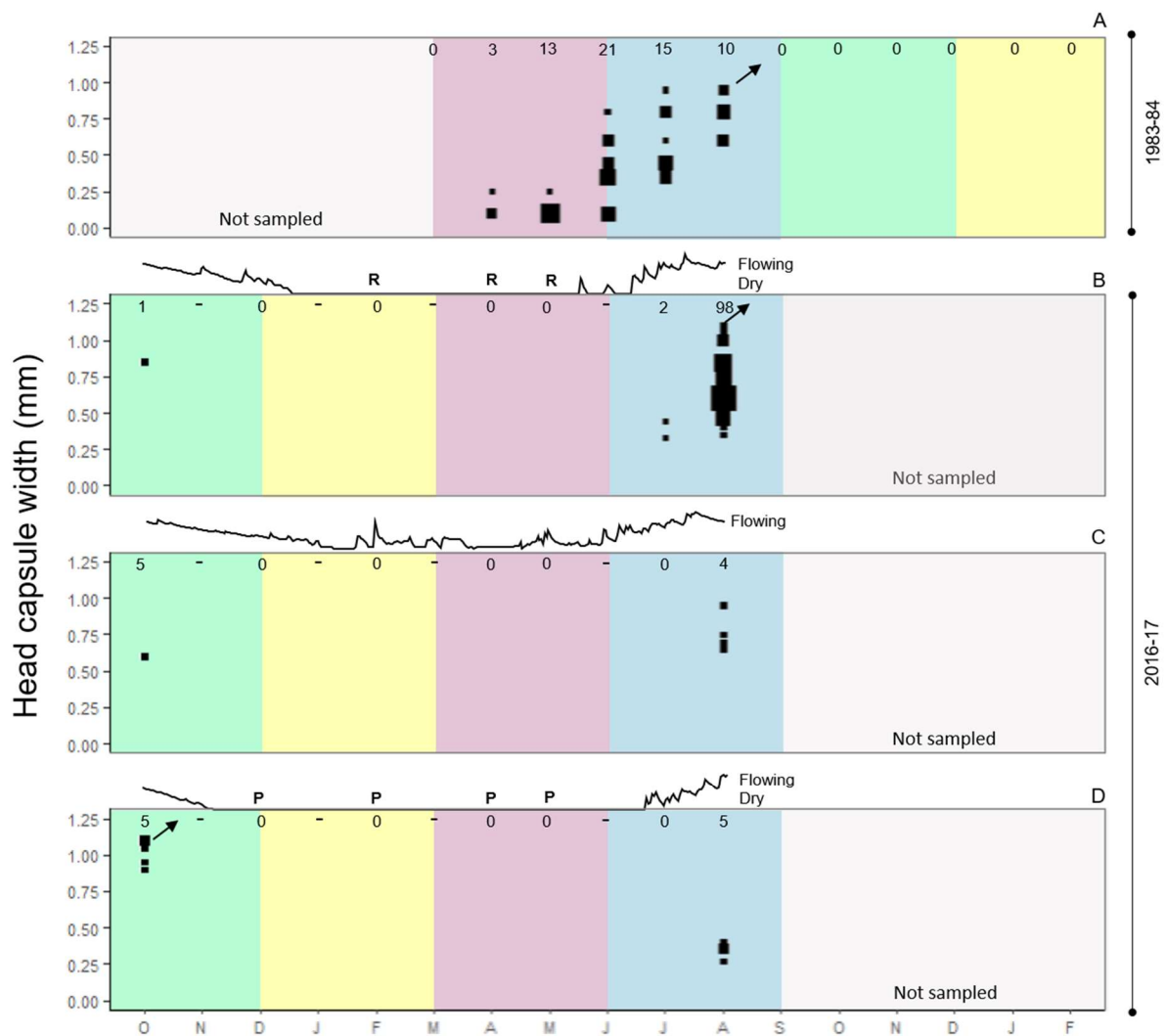


Figure 5.8: Size distributions of stonefly nymphs *Leptoperla australica* (Gripopterygidae) collected from perennial Foster Brook in 1983-84 (A, similar pattern observed in perennial Chandler Brook in 1983-84), and in 2016-17 from Curtis Creek (B), Seldom Seen Brook (C) and Wungong Brook (D). 1983-84 data (A) redrawn from Bunn (1988) with permission from CSIRO publishing. Caption as for Figure 4, except P = receding (non-refuge) pools.

Response 5: Retreat to groundwater refuges

In the early 1980s, adults of the amphipod *Uroctena* sp. were found in most size classes all year round in Waterfall Gully, Chandler Brook and Seldom Seen Brook (Figure 5.9a). In

2016-17, *Uroctena* sp. were still found at all sizes throughout the year in Waterfall Gully (Figure 5.9b), showing that their phenology has not changed there. However, at Curtis Creek, abundances in all size classes peaked in spring then showed large declines as the stream dried (Figure 5.9d). By late summer, Curtis Creek had dried completely, but a range of sizes of *Uroctena* sp. were found in the groundwater-fed drought refuge pools (Figure 5.9d). When flows resumed, a few individuals (including adults) were present, ready to repopulate Curtis Creek.

Uroctena sp. were also observed emerging from cracks atop granite inselbergs in winter 2017 and in 2018, associated with groundwater that begins to seep to the surface between winter and early spring. One inselberg had sufficient outflow to form a gently flowing ephemeral stream (Coccinea Creek) in winter 2017, in which high abundances of *Uroctena* sp. of all sizes were found. These occurrences confirmed that these paramelitid amphipods are able to switch from being a surface dweller to stygofauna during dry times (and *vice versa*; Figure 5.9e).

In Waterfall Gully, gravid females were not detected year-round as they had been by Bunn in 1983-84 (Figure 5.9b). However, Bunn (1988) attributed that occurrence to unusually elevated spring-fed flows, as in Chandler Brook and Curtis Creek they had occurred only from early winter to spring. Any change in phenology in the perennial Waterfall Gully was difficult to detect, because breeding females only occurred in occasional samples of 100+ individuals in 2016-2017; but their appearance in spring and autumn appeared consistent with the historical pattern. *Uroctena*'s appearance in the now intermittent and ephemeral streams was sporadic (Figure 5.9c-e).

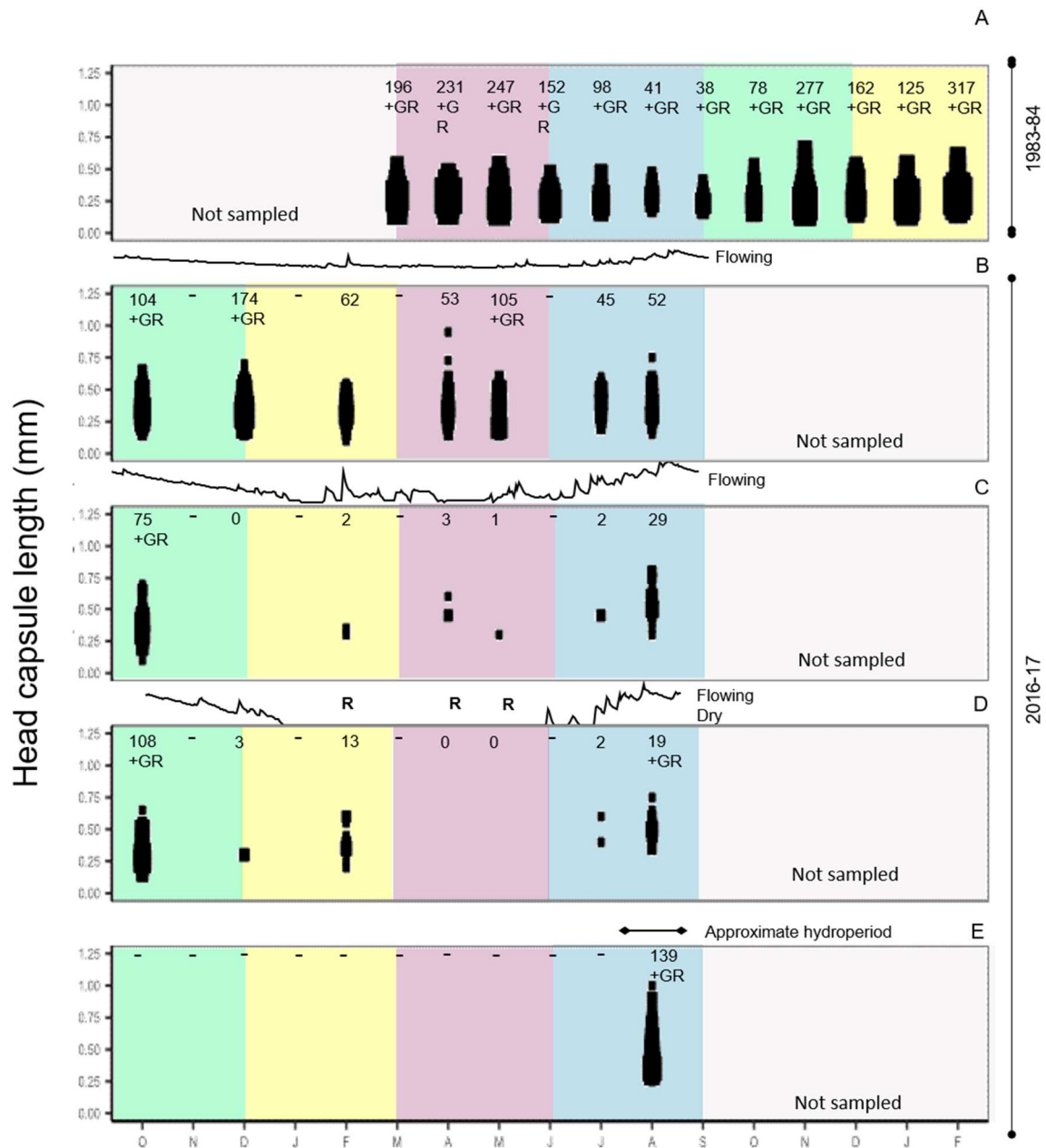


Figure 5.9: Size distributions of the amphipod *Uroctena* sp. (Paramelitidae) collected from perennial Waterfall Gully in 1983-84 (A, similar pattern observed in perennial Chandler Brook in 1983-84), and in 2016-17 from Waterfall Gully (B), Seldom Seen Brook (C), Curtis Creek (D), and Coccinea Creek (E). 1983-84 data (A) redrawn from Bunn (1988) with permission from CSIRO publishing. GR = presence of gravid females. Note phenology and flow regime both unchanged in Waterfall Gully. Hydrographs, rest of caption as for Figure 5.4.

Newly arrived species: fast univoltine life cycle

The leptophlebiid mayfly *Nousia* AV16 was not recorded by Bunn (1988) but was present in high numbers in all formerly perennial streams in spring (October) 2016. It is likely hatching occurred in mid-winter and eclosion occurred before sampling in December (Figure 5.10).

Nousia was found in the greatest numbers at Chandler Brook (Figure 5.10a), which has the shortest hydroperiod, yet was absent from Waterfall Gully. The caddisfly *H. michaelsoni* was only found in Chandler Brook in winter; but the size data suggested two cohorts (Figure .11).

It appears to have very rapid larval development, going from hatching to pupation between July and October.

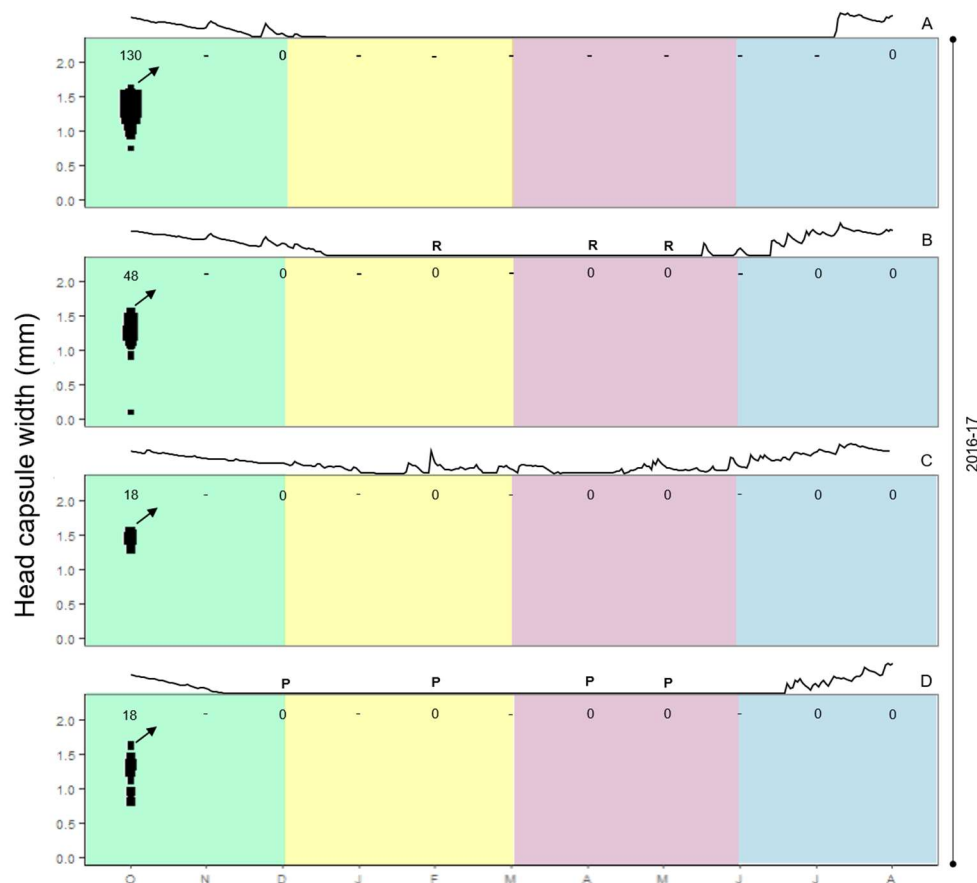


Figure 5.10: Size distributions of mayfly nymphs *Nousia* sp. AV16 (Leptophlebiidae) at Chandler Brook (A), Curtis Creek (B), Seldom Seen Brook (C) and Wungong Brook (D) in 2016-17. Caption as for Figure 4, except 2013 hydroperiod data presented for (A) as more recent data is not available (DWER station 616124).

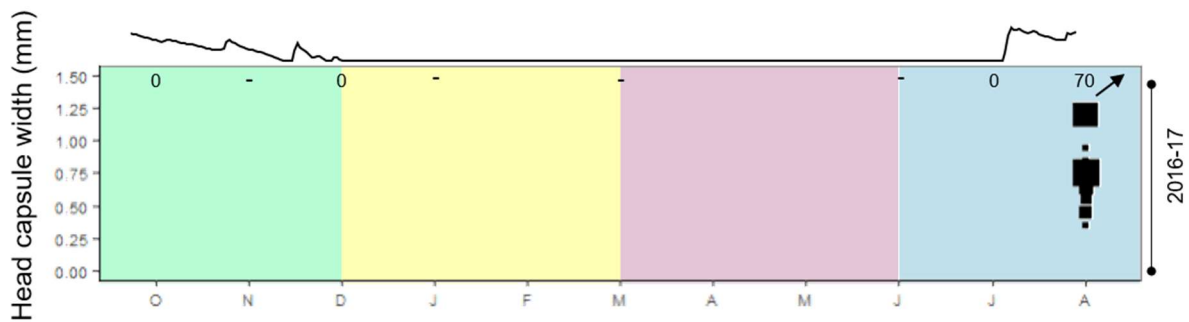


Figure 5.11: Size distributions of larval caddisflies *Hydrobiosella michaelsoni* (Philopotamidae) at Chandler Brook in 2016-17. Arrow indicates final instars present (prior to pupation). Hydrograph caption as for Figure 10A.

Discussion

For more than half of the species studied, life history phenology did not change following transition from perennial to intermittent flow regimes, restricting those species to perennially flowing waters. Furthermore, detailed information on species' antecedent life history rarely predicted their actual responses to climate-driven flow regime change, raising doubts about the veracity of predictions based on species traits. This is because species' potential for flexible phenology or growth rate is not necessarily indicated by life history traits. Species with the same life history may show quite different responses to intermittency, so there was no clear type of life history or taxon that showed a greater extinction risk or lesser ability to adapt. All of the seven species we report as restricted to perennial reaches were regarded as abundant, tolerant and widely distributed species by Bunn (1988). Many studies have predicted extirpation or even extinction of rare species as a result of climate change, but our results show clearly that common, widespread species will also be vulnerable to the effects of global warming. The species extirpated from streams by intermittency included all four

Trichoptera, one leptophlebiid (*N. occidentalis*) and one baetid mayfly (*O. soror*), and the dragonfly *A. anacantha*.

Predicted and observed changes to freshwater species distributions caused by climate change include retreat into higher altitude headwaters and range contractions to higher latitudes (e.g. Heino et al., 2009; Domisch et al., 2013; Markovic et al., 2014), however there are invariably limits to the environment available for range contractions (especially in SWA, Davies 2010). Perennial streams in mediterranean-climate regions support a higher proportion of taxa without traits adaptive for drying (Bonada, Rieradevall & Prat, 2007; Bogan et al., 2017). Species now restricted to perennial reaches in the Wungong Catchment will likely contract their ranges southward under climate change, provided suitable habitat remains within SWA; species with narrow ranges within regions that may in future have no perennial water face the additional challenge of movement given that most rivers flow east-west, not north-south (Davies 2010).

Larval development timing may not be the only critical life history event disrupted by flow regime drying. Many freshwater insects have very specific requirements for oviposition sites, such as stones protruding from the water in either fast or slow flows (Lancaster, Downes & Reich, 2003; Reich & Downes, 2003; Reich, Hale, Downes & Lancaster, 2011). So, the supply of suitable oviposition sites can limit larval populations (Lancaster, Downes & Arnold, 2010). If streambeds are dry when females are ready to oviposit or there is less wetted area, then oviposition opportunities may limit larval population sizes and species persistence in drying streams. The oviposition habits of Western Australian freshwater insect species have never been studied, nor have egg mass characteristics been described for any species. However, this often-ignored life history stage may in fact explain why some species which were predicted to have some capacity to adapt to flow regime change (according to

larval development phenology e.g. *Smicrophylax australis*, *N. tenax* and *N. occidentalis*) did not persist in now-intermittent streams.

Resistance, resilience and flexible life history phenology

Four species were found to have adapted their life history phenology to the new flow regime, through changes in timing and duration of development and the use of resistance or resilience strategies to survive drying. The stonefly *Leptoperla australica* changed its phenology from autumn to winter breeding to coincide with the onset of flows in now-intermittent streams. Stoneflies may have plastic life histories in accordance with environmental conditions, including flow regime (e.g. Hynes & Hynes, 1975; Gripopterygidae, Storey & Quinn, 2011; Winterbourn 2010). Desiccation-resistant eggs are known in some stoneflies (Chester & Robson, 2011; Strachan et al., 2015) and it is likely that *L. australica* uses them to persist in now-intermittent streams. Storey & Quinn (2011) also suggested that some stonefly species may over-summer as long lived adults; but in SWA adults probably cannot live through the high temperature/low humidity conditions which have been shown to reduce adult life span (Collier & Smith, 2000).

Another stonefly, *Newmanoperla exigua* had also been an autumn breeder in 1983-84, so we expected that it would respond to drying in a similar way to *L. australica*. However, *N. exigua* was not found in 2016-17. If the presence of nymphs was confined to the months between August and October, our sampling may have missed them. Misalignment of environmental cues may offer another explanation for the absence of *N. exigua* during the present study. Autumn water temperatures in 1983-84 were around 20°C (Bunn 1988), whereas now, when flow commences in intermittent streams in mid-winter, the water temperature ranges between 9 and 11°C. If *N. exigua* hatching is constrained to the higher autumn temperatures — and temperature can cue critical life history events in stoneflies (e.g.

Brittain 1991) — they simply would not hatch until after winter, limiting time available for development to adulthood and resulting in extirpation.

Two Leptophlebiidae mayflies (*N. bunni*, *B. kadjina*) adapted their life histories to suit intermittent flow regimes but displayed opposite patterns in synchrony. *Nyungara* has multiple overlapping cohorts in perennial flows, but at now-intermittent reaches it was strongly synchronous. This pattern has been observed in other species/genera of Leptophlebiidae mayflies (Storey & Quinn 2011). *Nyungara* may persist in intermittent streams through desiccation-resistant eggs, or (more likely) by recolonisation from perennial Waterfall Gully, (via adult flight, a resilience trait; Chester & Robson 2011; Chester et al., 2015) because nymphs do not show up until many months after flow commences. *Nyungara* nymphs have an estimated upper thermal tolerance of 21.9°C (Stewart et al., 2013) and when eclosion occurred in December in intermittent streams, water temperatures were 21°C (Seldom Seen Brook) and 23°C (Curtis Creek); temperature may cue synchronised emergence by this species. Water temperature may also explain the absence of the baetid mayfly *O. soror* from now-intermittent streams, as it has an upper thermal tolerance of 20.5°C (Stewart et al., 2013) which was exceeded in all streams except Waterfall Gully.

Bibulmena kadjina underwent several changes to life history phenology following flow regime change. This mayfly switched from a tightly synchronised univoltine life cycle (Bunn, 1988) to either an unsynchronised bivoltine or a univoltine life cycle with a cohort split in early summer. Additionally, nymphs occupying drought refuges grew to sizes much larger than those from Bunn's study. If these nymphs survived to eclosion, they would become larger, presumably fitter and more fecund adults. Unlike drying pools, drought refuge pools were apparently predator-free, and as few other invertebrate species were found there, interspecific competition for food would also be low. The ability to use drought refuge pools is a resistance trait that enables some aquatic species to persist during supra-seasonal

droughts (Chester & Robson, 2011; Bogan et al. 2015). Cohort-splitting has been predicted as a way for species to survive desiccation because it enables bet-hedging (Strachan et al., 2015), in this case via the use of perennial drought refuge pools. Relatively few of these very large nymphs may be needed to repopulate streams, if their fecundity is higher than that of smaller individuals.

Amphipoda are known to be able to retreat to the hyporheos to withstand temporary stream drying (Stubbington 2012; Vadher et al., 2018). However, although the amphipod *Uroctena* sp. occurred in all streams, the streams primarily overlay granite bedrock, so there is no hyporheos. Instead, *Uroctena* sp. were observed emerging from springs and seeps in granite inselbergs which flow once winter rainfall has been sufficient for them to overflow into streams. Amphipods may also retreat into groundwater below streambeds through cracks in the granite.

Uroctena sp. was reported to have a year-long life cycle, with females reaching maturity at the end of their first year of life, and brooding occurring either year around, or during winter and spring (depending on stream discharge characteristics, Bunn, 1988). In 2016-17, hydroperiods in most streams were much shorter than one year. Coccinea Creek had the shortest hydroperiod (~ 6-8 weeks), yet gravid female *Uroctena* were observed moving downstream in spring-flows from the granite outcrop in mid-winter. It is unclear whether breeding is now constrained to shortened periods of surface flow, or whether it breeds in the subterranean environment. The latter is possible, because Paramelitidae are primarily hypogean taxa, and stygal Paramelitidae are highly diverse in the Australian arid zone (Humphreys 2001). They are hypothesised to be descended from fauna originating in the more humid distant past that became progressively restricted to groundwaters through continental aridification (Humphreys 2001; Väinölä et al., 2008). It may be that *Uroctena*

spp. are in fact surface water vagrants and that drying flow regimes due to climate change are continuing a trajectory towards becoming true stygofauna.

Range expansion of species with life histories suited to flow intermittency

Following environmental change, replacement of species restricted to perennial flows with arid-zone adapted generalists may occur. The arrival of the mayfly *Nousia* sp. AV16, which had a short, synchronous life cycle in intermittent streams and was absent from perennial flowing reaches is one example. Mayflies of the genus *Nousia* are widespread across the Australian continent and have Gondwanan ties to South America (Christidis 2006); AV16 is endemic to SWA. Some Leptophlebiidae mayflies are known to have desiccation resistant cysts (Lemhkuhl, 1973; Boulton, Stanley, Fisher & Lake, 1992), a strategy almost certainly used by *Nousia* sp. AV16 to colonise intermittent streams where the previous mayfly assemblage has been displaced by drying.

The free-living caddisfly *Hydrobiosella michaelsoni* (Philopotamidae) was distributed historically, and in 2016-17, in low densities across the Wungong catchment (Bunn 1986; Bunn et al., 1986). However, in winter 2017, it was found in high abundance at Chandler Brook which is markedly ephemeral. *Hydrobiosella* is an omnivorous suspension feeder associated with small, fast flowing streams, and favours intermittency (Dean & Bunn, 1989). Dean & Bunn (1989) report a synchronised univoltine life cycle in intermittent streams, with pupation occurring in spring: they caught adult females in autumn (May) and they likely oviposited as flows began. Presumably, the adults (females at least) are sufficiently long-lived, strong dispersers able to populate Chandler Brook when flows began in mid-late winter. Thus, *H. michaelsoni* and *Nousia* sp. AV16 are examples of endemic relictual species that have responded positively to shifts in flow regime brought about by climate change. The presence of *H. michaelsoni* is especially important as Trichoptera generally appear to be among the taxa most negatively impacted.

Predicting changes to life histories and distributions of macroinvertebrates

Bunn's detailed descriptions of species' life histories, as they were expressed in populations prior to onset of climate drying, were not sufficient to predict subsequent adaptation. For more than half of the species studied, the mismatch between their life history phenology and a greatly changed flow regime was critical; thus, they are now restricted to perennially flowing waters. Species that became restricted to perennially flowing waters will likely show southward range contraction as drying progresses and perennial streams cease to exist in NJF catchments. Thus far, global warming ($> 1^{\circ}\text{C}$ in SWA) and drying has altered the abiotic conditions aquatic invertebrates respond to, notably hydroperiod, but also water temperatures in intermittent streams, persistence of drought refuge pools, wetted area and dispersal pathways (Lake 2011; Chester et al., 2015; Bogan et al., 2017; Datry et al., 2017b). Recent predictions of warming of up to $3\text{--}4^{\circ}\text{C}$ by 2090 (IPCC 2014) means that these trends will continue to accelerate until at least the end of the century. Few invertebrates appear on conservation listings in SWA (a common problem worldwide; Chichorro, Juhlén & Cardoso, 2019), however, recently some SWA stream fauna have been identified as candidates for conservation listing (Penniford et al., 2017; Penniford, 2018). But we contend that the numbers of SWA stream-dwelling species under threat of range contraction and eventual extinction due to climate drying will be far greater than identified under current predictions because many, perhaps most, common species will be affected.

Recent advances in species distribution modelling (SDM) attempt to address gaps in the capacity to predict species and community change beyond simple range shifts (Zurell et al., 2016; Serra-Diaz & Franklin, 2019). Authors widely decry a lack of knowledge of species life history characteristics such as phenology, dispersal ability and demography as barriers to robust prediction under climate change (Zurell et al., 2016; Urban et al., 2016; Moor 2017; Serra-Diaz & Franklin, 2019). Yet, for aquatic macroinvertebrates, many life history

characteristics are simply not known and a lack of new natural history research remains a problem in advancing predictive capacity based on traits (Kefford et al., 2020), even if current patterns could reliably predict those in the future. Furthermore, life history traits may be mis-assigned to species, based on those known for congeners and con-familials (Hamilton et al., 2020). A prime example in this study is the assumption that Gripopterygidae stoneflies and Leptophlebiidae mayflies possess desiccation-resistant embryonic eggs (Boulton et al., 1992; Strachan et al., 2015). The persistence of *L. australica* in now-intermittent streams is almost certainly due to a desiccation resistant egg (or diapausing larvae, also known in some stoneflies; Strachan et al., 2015), whereas, con-familial *N. exigua* appears extirpated across the catchment. Likewise, the leptophlebiid *N. occidentalis* was completely restricted to perennial streams, whereas con-familial *Nousia* sp. AV16 had expanded its range into all streams that had transitioned to intermittency. If *N. occidentalis* had a desiccation resistant life stage, we expect it would have been present in intermittent reaches.

Predicting species responses to flow regime change based on what con-familial species are known to do is unlikely to be successful; while detailed information on species' antecedent life history may poorly predict their actual responses to climate-driven flow regime change largely because measurements of extant life history traits do not necessarily reveal species' potential for flexibility in phenology or growth rate. A better understanding of the role of life history in each species' chance of adapting under climate change is needed, especially for those in which potential range shifts are constrained by geography. Experiments that, for example, look for an adaptive response when an environmental challenge is presented will be useful. We also urge a greater emphasis on comparative studies with past data in places where climate change has already effected significant environmental change. Valuable lessons can be learnt by challenging naïve (as it turns out) models for species survival with threshold changes to habitat driven by global warming.

Supplementary material

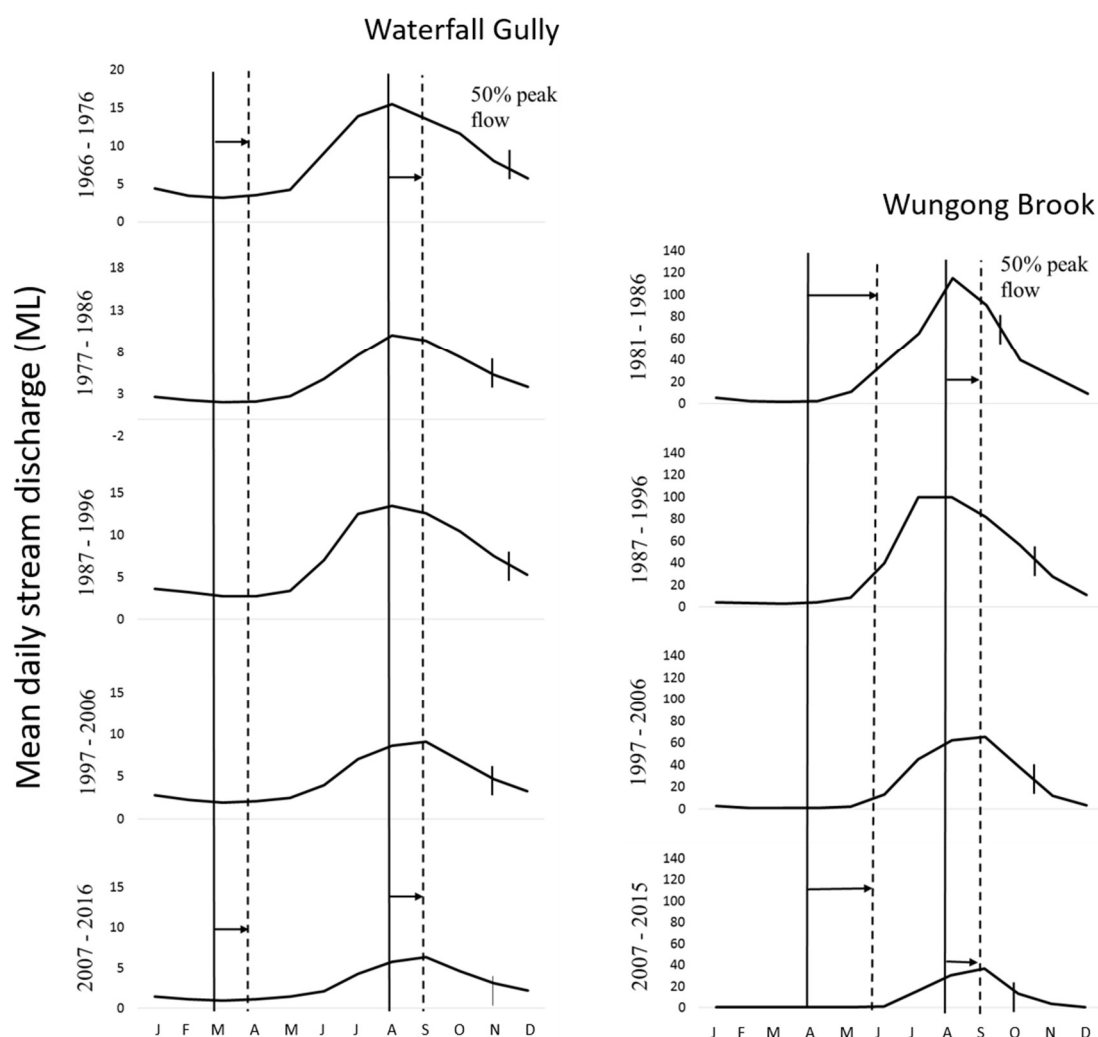


Figure S5.1 [Figure 2.6]: 10 year mean daily discharge at Wungong Brook (left) and Waterfall Gully (right). Solid line indicates 1981-1990 (Wungong Brook) or 1967-1976 (Waterfall Gully) average onset of increased flows (left) and peak flows (right), dashed line indicates 2007-2016 timing. Data freely available from the Western Australian Department of Water and Environmental Regulation (wir.water.wa.gov.au; gauging station 616041 and 616023).

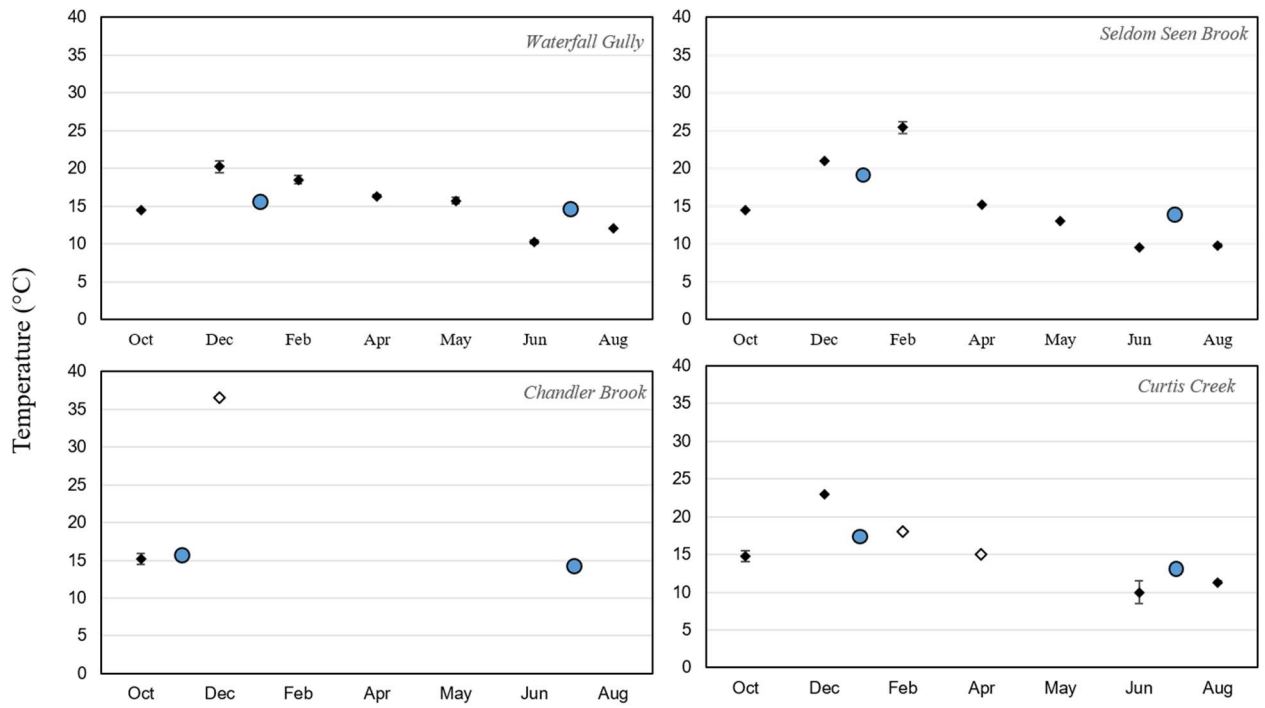


Figure S5.2: Mean temperature per stream (where >1 site was available for sampling), for sites with flow (solid symbols) and those reduced to standing pools (open symbols). Blue circles represent mean summer and winter temperatures recorded by Bunn (1985).

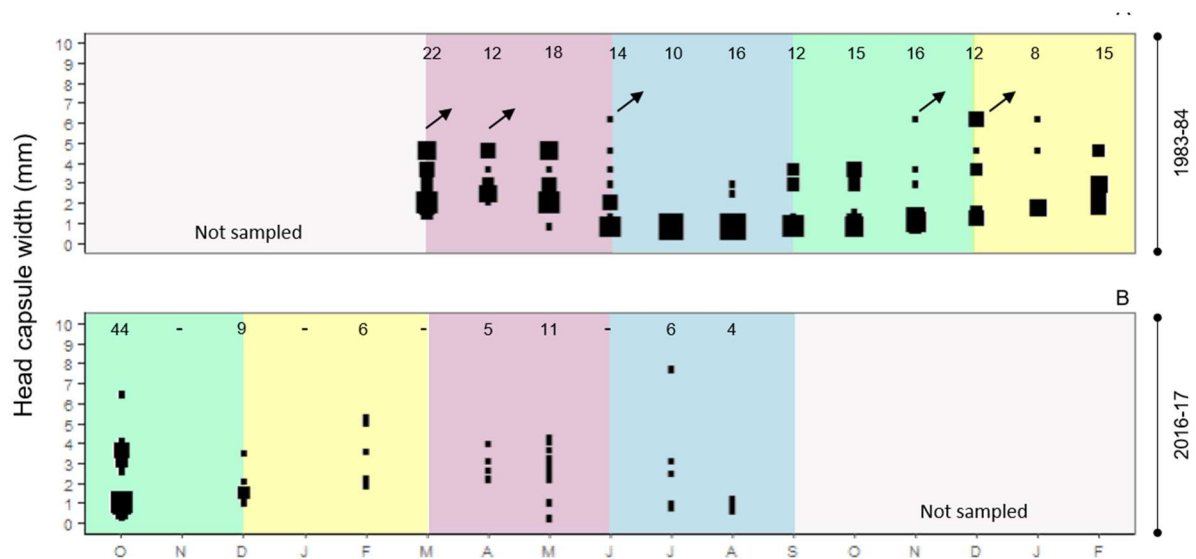


Figure S5.3: Size distribution of dragonfly nymphs *Austroaeschna anacantha* (Telephlebiidae) collected from perennial Waterfall Gully in 1983-84 (A) and in 2016-17 (B), during Austral spring (green), summer (yellow), autumn (purple) and winter (blue). 1983-84 data (A) redrawn from Bunn (1988) with permission from CSIRO publishing. Number of individuals collected given, solid arrow indicates final instars. Size of squares indicate proportion of individuals of that size. Note phenology and flow regime both unchanged.

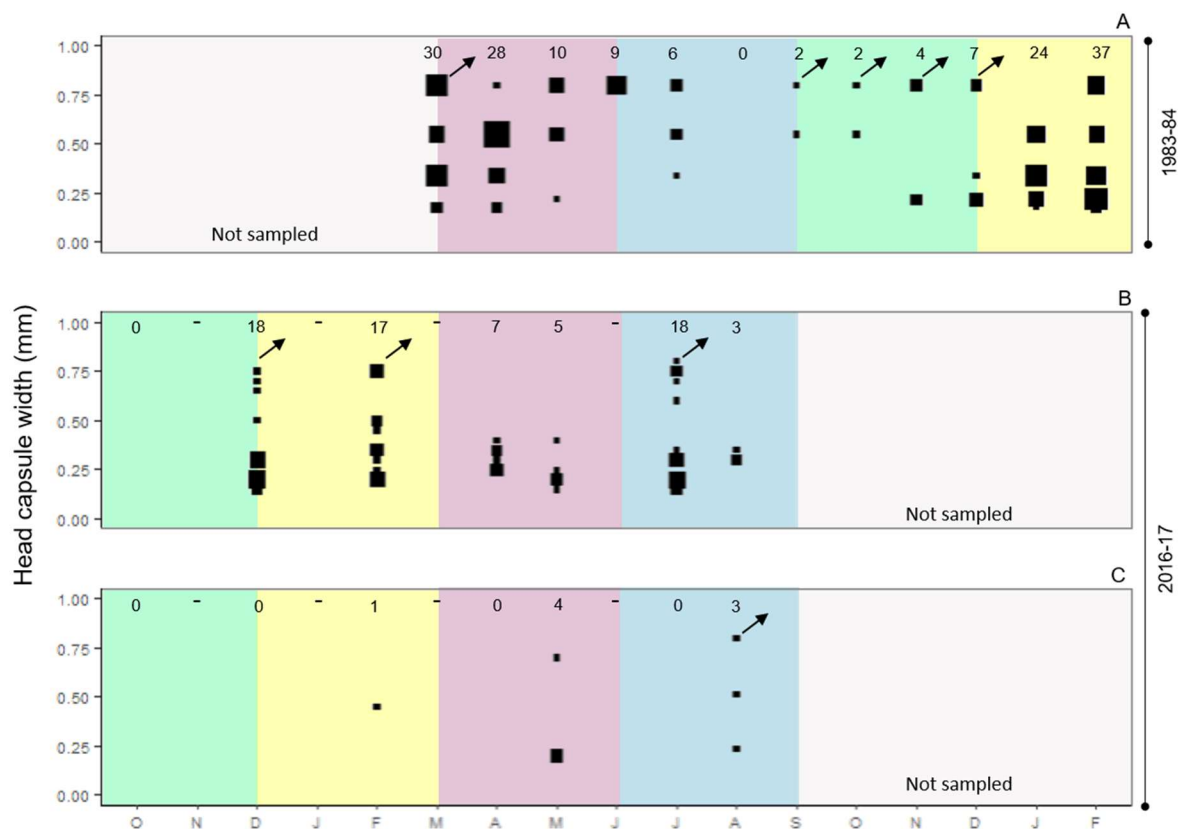


Figure S5.4: Size distribution of larval caddisflies *Lectrides parilis* (Leptoceridae) collected from Waterfall Gully in 1983-84 (A), and in 2016-17 from Waterfall Gully (B) and Seldom Seen Brook (C). Arrows indicate final instars pre-pupation. Note that fewer individuals were recorded in 2016-17 with higher sampling effort although these streams remained perennially flowing. Note phenology and flow regime both unchanged.

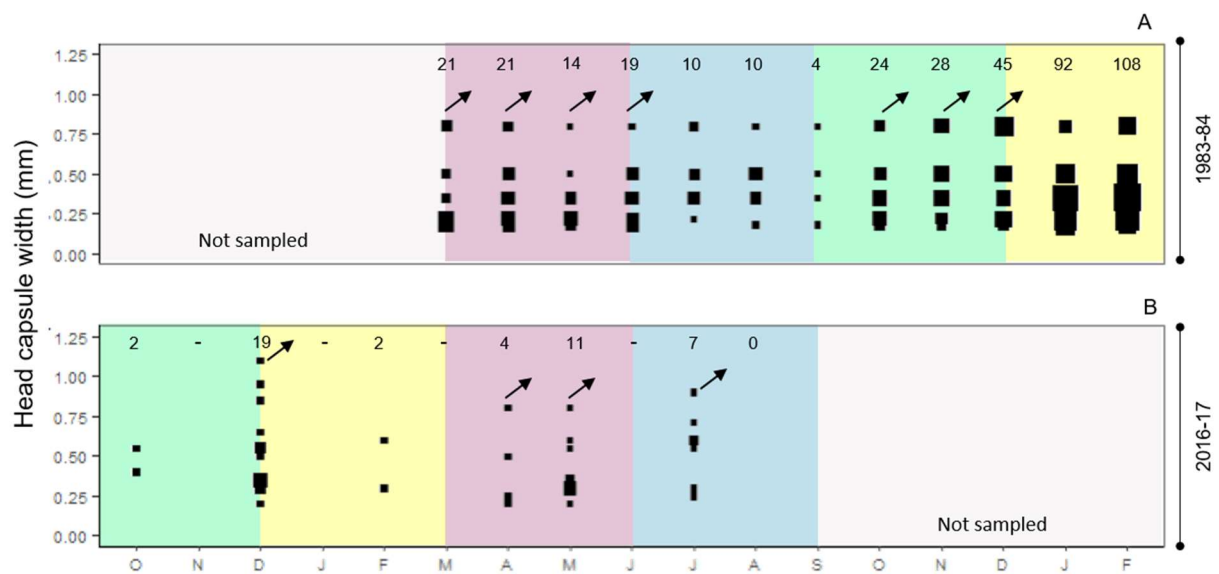


Figure S5.5: Size distribution of larval caddisflies *Taschorema pallescens* (Hydrobiosidae) collected from Waterfall Gully in 1983-84 (A), and in 2016-17 (B). Arrows indicate final instars pre-pupation. Note that fewer individuals were recorded in summer and autumn 2016-17 with higher sampling effort although this stream remained perennially flowing. Note phenology and flow regime both unchanged

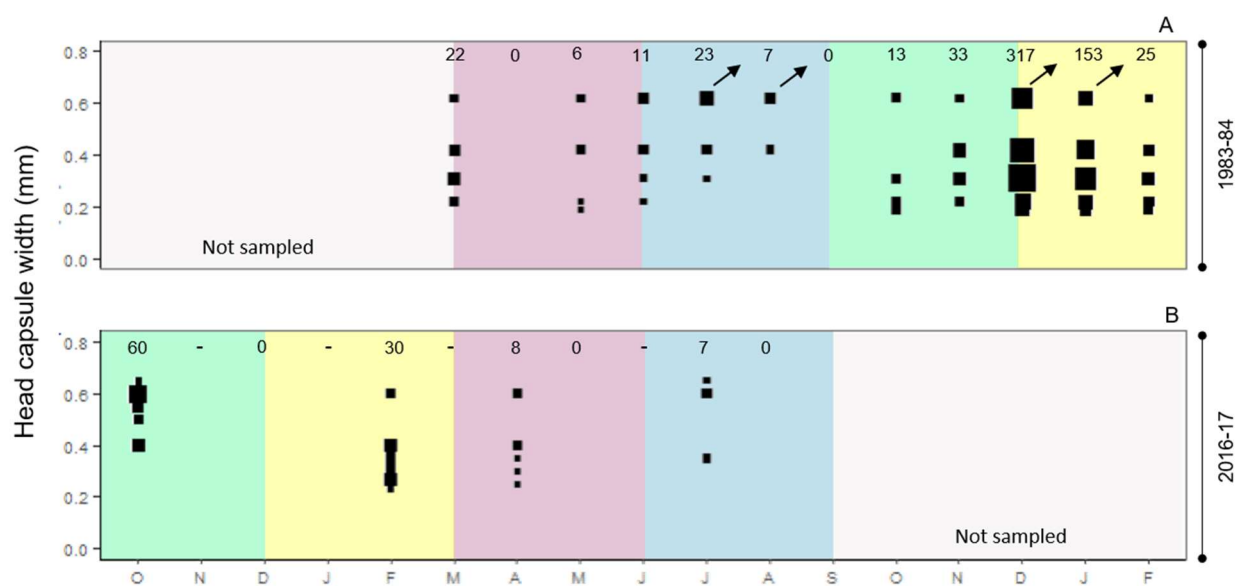


Figure S5.6: Size distribution of larval caddisflies *Smicrophylax australis* (Hydropsychidae) collected from Waterfall Gully in 1983-84 (A), and in 2016-17 (B). Arrows indicate final instars pre-pupation. Note that fewer individuals were recorded in 2016-17 with higher sampling effort although this stream remained perennially flowing. Note phenology and flow regime both unchanged.

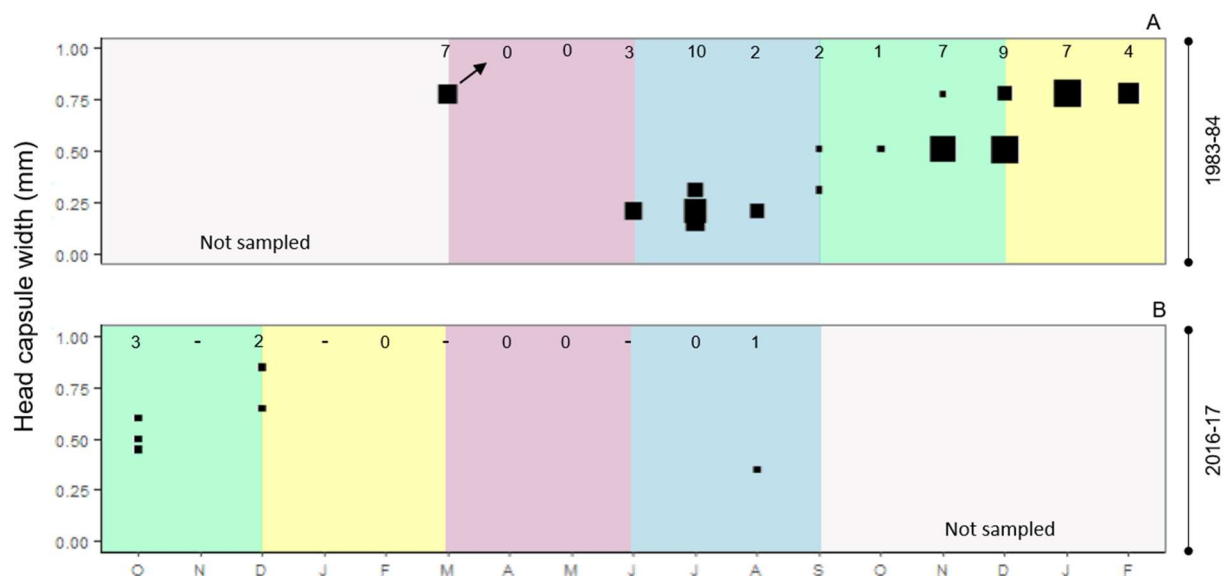


Figure S5.7: Size distribution of larval caddisflies *Notoperata tenax* (Leptoceridae) collected from Waterfall Gully in 1983-84 (A), and in 2016-17 (B). Arrows indicate final instars pre-pupation. Note that fewer individuals were recorded in 2016-17 with higher sampling effort although this stream remained perennially flowing.

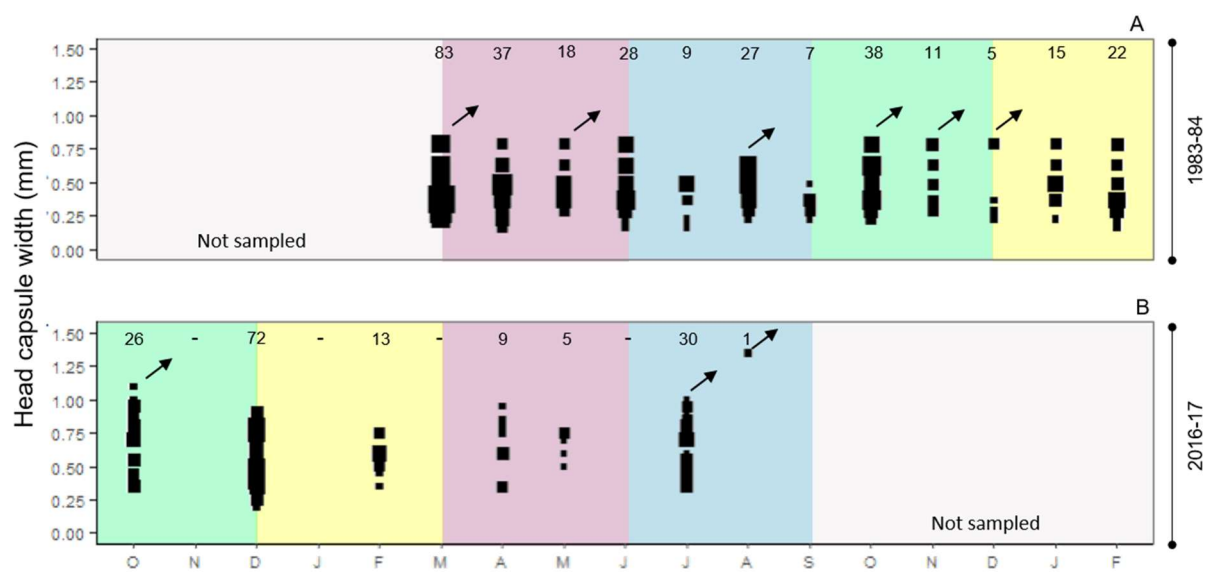


Figure S5.8: Size distribution of mayfly nymphs *Offadens soror* (Baetidae) collected from Waterfall Gully in 1983-84 (A), and in 2016-17 (B). Arrows indicate final instars with blackened wing buds. Note that fewer individuals were recorded in autumn 2016-17 with higher sampling effort although this stream remained perennially flowing. Note phenology and flow regime both unchanged.

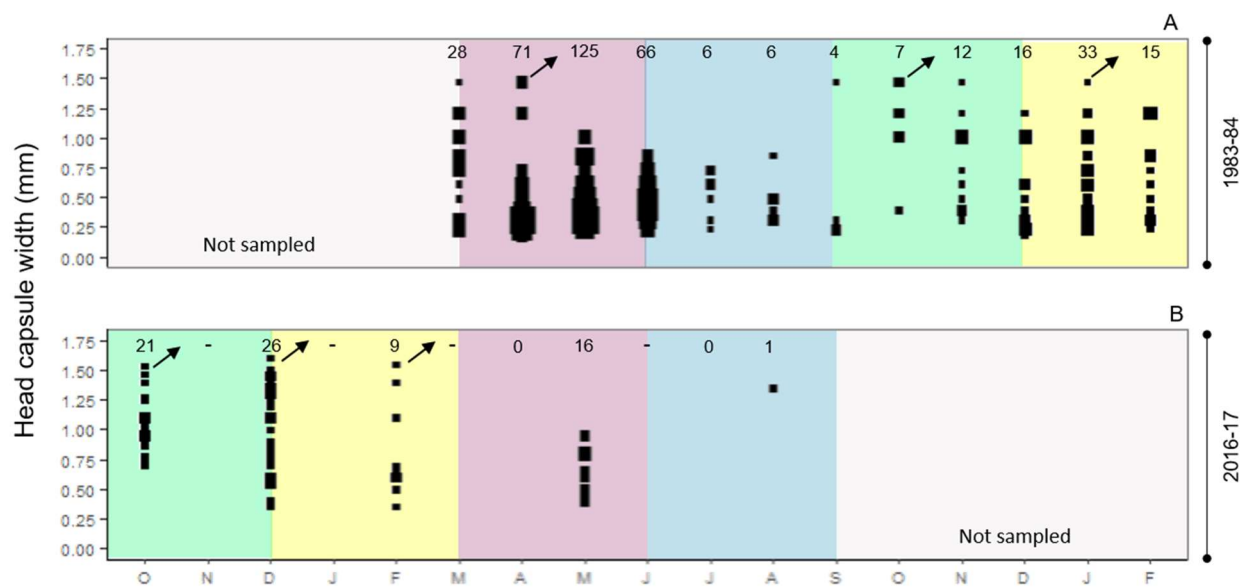


Figure S5.9: Size distribution of mayfly nymphs *Neboissophlebia occidentalis* (Leptophlebiidae) collected from Waterfall Gully in 1983-84 (A), and in 2016-17 (B). Arrows indicate final instars with blackened wing buds. Note that fewer individuals were recorded in autumn, winter 2016-17 with higher sampling effort although this stream remained perennially flowing. Note phenology and flow regime both unchanged.

Chapter 6

Flow regime change alters shredder identity but not leaf litter decomposition in headwater streams affected by severe, permanent drying.

The following chapter has been published in the journal *Freshwater Biology*.

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Note: Several figures and tables appearing as supplementary information in the published version have been incorporated into the main text of this chapter, for ease of examination.

The following authors contributed to this manuscript as outlined below:

	Author order	Concept & study design	Fieldwork	Laboratory work	Data analysis	Writing manuscript	Review/edit paper
Nicole Carey	First	X	X	X	X	X	X
Edwin Chester	Second	X	X				X
Belinda Robson	Third	X			X		X

By signing this document, the Candidate and Principal Supervisor acknowledge that the above information is accurate and has been agreed to by all other authors.

Candidate

Principal Supervisor

Abstract

1. Climate change is altering hydrologic regimes globally. In the mediterranean climate region of south-western Australia (SWA), climate drying has caused many perennial streams to switch to intermittent flow regimes. Shifts in flow regime are expected to alter physical and biological processes in streams, including litter decomposition, which is the basis of detrital food webs.
2. Decomposition of jarrah (*Eucalyptus marginata*) leaves and associated macroinvertebrates, were measured over 320 days in 2018-19 using leaf bags in four headwater streams in SWA. Two streams retained perennial reaches and two were formerly-perennial streams that are now intermittent. Pre-planned comparisons that formed a partial multiple before-after, control-impact design (mBACI) were used to compare the results to an experiment conducted in 1982-83 in some of the same streams when all were perennially flowing. Both experiments used coarse and fine mesh bags containing 10 g of dry leaves.
3. In one perennial stream, coarse bags lost more weight than fine bags at the last sampling time only, when shredding caddisflies arrived on the leaf bags. In the other perennial stream, leaf-mining chironomids entered fine mesh bags and accelerated decomposition so that they lost more weight than the coarse mesh bags. There was no difference in weight loss between fine and coarse mesh leaf bags in the two intermittent streams.
4. In 2018-19, decomposition was slower in dry reaches of intermittent streams than in perennial reaches. Leaf weight loss increased with the resumption of flow in intermittent streams, so that by the end of the experiment, similar amounts of leaf weight had been lost in intermittent and perennial reaches. Thus, although the

temporal pattern of decomposition differed between intermittent and perennial reaches, after 320 days, they had reached a similar endpoint.

5. Over similar experimental duration, mean leaf weight remaining in perennial reaches at the end of the experiment did not differ between the 1982-83 study and 2018-19, showing that leaf decomposition had not changed in reaches that retained perennial flow. As mean leaf weight remaining also did not differ between intermittent and perennial streams in 2018-19, leaf decomposition was robust to flow regime change. However, since 1982-83, these streams have lost populations of shredding caddisflies and stoneflies, which were replaced by other shredders (e.g. leaf mining chironomids *Stenochironomus* sp.) showing that there was some redundancy amongst shredder species.
6. As climate change progresses, drying flow regimes will become commonplace in mediterranean (and other) climate regions globally. This study indicates that litter decomposition may be maintained as streams transition to intermittency although shredder species may change. However, the impact of shredding species on leaf decomposition varies amongst studies, so effects of the loss of shredder species sensitive to drying may also vary, and in some cases their loss may substantially alter ecosystem processes.

Keywords

Flow regime, climate change, leaf litter breakdown, shredding macroinvertebrates, drying

Introduction

Globally, anthropogenic climate change is altering precipitation patterns, impacting stream ecosystems and processes. In particular, mediterranean climate regions are undergoing warming and drying trends, with reduced precipitation, increasing frequency and intensity of drought conditions and changing seasonality of rainfall (Döll & Zhang, 2010; Filipe et al., 2013; Hoegh-Gulberg et al., 2018). Among the impacts of a drying climate is the reduction in surface water flows and groundwater depths, and as these progress perennially flowing rivers and streams transition to intermittent flow regimes (Datry et al., 2017). For example, the mediterranean climate region of south-western Australia (SWA) has undergone significant rainfall declines, causing many perennial headwater streams to switch to intermittent flows (Petrone et al., 2010; Silberstein et al., 2012). As flow regime is one of the most important factors governing communities and processes in mediterranean climate streams (Gasith & Resh, 1999; Bernal et al., 2013), a rapid transition to intermittent flow would be expected to substantially alter stream ecosystem function. However, to date there have been few opportunities to directly study changes in stream ecosystem processes *in situ* following flow regime change due to global warming, not least because of limited historical data with which to compare present conditions. Fortunately, prior to the onset of streamflow drying, the ecology of macroinvertebrates inhabiting SWA headwater streams was intensively studied (Bunn, 1985, 1986; Bunn et al., 1986; Bunn 1988a), including experiments examining in-stream processing of detritus from the dominant forest tree, the jarrah *Eucalyptus marginata* (Bunn, 1988b; Bunn, 1988c). These experiments provide historical baseline information on organic matter decomposition which is rare in stream ecology (and ecology generally). Given the enormous need for knowledge of how climate change actually affects ecological processes (as opposed to predictions of effects on processes), Bunn's research represents a rare opportunity for comparison of a fundamental ecological process in stream ecosystems.

In forested headwater catchments, detritus from riparian tree species is the primary carbon source, and transformation of coarse material into smaller fractions is the main pathway by which this detritus becomes available to consumers in these ecosystems. Once leaf litter arrives in-stream, physical processes of leaching and abrasion commence (Gessner et al., 1999; dos Santos Fonseca et al., 2013). Then, the decomposition process is regulated by the chemical composition of detritus (Ferreira et al., 2016; Zhang et al., 2018), and extrinsic factors including water temperature, dissolved oxygen availability and nutrients (Tank et al., 2010) and the microbial decomposer and detritivore communities (Boyero et al., 2007; Navel et al., 2011; but see Heard & Richardson, 1995; Usio et al., 2001). In developing this model for leaf litter decomposition in streams, many studies have used coarse and fine mesh leaf bags exposed in-stream. Coarse mesh is designed to allow access to invertebrate shredders that turn coarse organic matter to fine organic matter, whereas fine mesh is designed to exclude shredders and estimate the contribution of microbes to decomposition (Tiegs et al., 2009; Tank et al., 2010).

Flow permanence is likely an important limiting factor on decomposition rate (Corti et al., 2011; Datry et al., 2011). Naturally intermittent rivers and streams provide a mosaic, in time and space, of expanding and contracting inundated, flowing, drying and dry habitat; and ecosystem processes such as leaf litter decomposition vary across this habitat mosaic (Abril et al., 2016; Von Schiller et al., 2017). In perennially flowing water, volume and velocity of flows, temperature, and nutrients influence decomposition (Tank et al., 2010; dos Santos Fonseca et al., 2013); whereas in intermittent streams, periodicity of flows (drying-rewetting cycles) and duration of drying pools become important (Boulton, 2003; Dieter et al., 2013). Once the streambed is dry, photodegradation and colonisation by terrestrial microbial and invertebrate detritivores slowly continue the decomposition of organic matter (von Schiller et al., 2017; Datry et al., 2018a). These processes pre-condition organic material so that when

flows resume, dissolved and particulate matter are readily released, often as a substantial downstream “pulse” (Corti & Datry, 2012; Shumilova et al., 2019). These observations of differences in organic matter decomposition processes between intermittent and perennial streams indicate that when a stream transitions from perennial to intermittent flows, substantial changes in organic matter decomposition are likely. Such changes may include slower decomposition (while streams are dry), a more pulsed release of dissolved and particulate organic matter and changes in shredder species identity (to species tolerant of drying).

Aquatic organisms adapted to flow intermittency possess a suite of traits to cope with impermanence, through both their physiology and life history phenology. These traits include desiccation resistance in one or more life stages, rapid larval development (particularly flying insects), and the ability to retreat and recolonise from d refuges, either *in situ* (e.g. hyporheos), or from nearby permanent water sources (Chester & Robson, 2011; Strachan et al., 2015; Bogan et al., 2017). As flow regimes change from perennial to intermittent, macroinvertebrate assemblages would be expected to change to become dominated by organisms that possess these traits (Robson et al., 2011). Effects on detritus decomposition will depend in part upon the ability of shredding macroinvertebrate species to persist through dry periods (e.g. Wickson et al., 2012).

This study examined aspects of the leaf litter decomposition process in streams where flow regimes have changed from perennial to intermittent due to climate change using a leaf bag experiment with a partial multiple before-after, control-impact (mBACI) design with two pre-planned comparisons. The two pre-planned comparisons were: 1) contemporary comparison of leaf decomposition between perennial and intermittent (formerly perennial) stream reaches; 2) comparison between contemporary litter decomposition in perennial streams and those from perennial streams 35 years ago in the same catchment, before the transition in

flow regime (Bunn, 1985; 1988c). We hypothesised (H₁) that contemporary decomposition of *Eucalyptus marginata* leaves (weight lost from coarse and fine mesh leaf bags) would only differ between perennial and intermittent streams during the dry phase, so that on the resumption of flows, rapid leaching would occur, so that decomposition would be similar between streams. We hypothesised (H₂) that contemporary decomposition (weight lost from coarse and fine mesh leaf bags) in perennial stream reaches would not differ from that recorded by Bunn (1988c) in perennial streams in 1982-83. We hypothesised (H_{3a}) that shredding macroinvertebrates would process a similar amount of leaf material in intermittent and perennial stream reaches (i.e. similar weight lost in coarse mesh leaf bags), and consequently, that (H_{3b}) leaf bags which excluded invertebrates (fine mesh bags) would break down more slowly than those which allowed invertebrate access (coarse mesh bags).

Methods

Climate change in South-western Australia

The climate in SWA has been warming and drying for at least five decades, with declines in annual rainfall as well as higher temperatures (Timbal, Arblaster, & Power, 2006; Bates, Hope, Ryan, Smith, & Charles, 2008). Whilst rainfall in this region historically exhibits marked inter-annual variation, there has been a reduction of around 20% of mean precipitation since 1970 (Hope, Timbal, & Fawcett, 2009; Andrys, Kala & Lyons, 2017). Additionally, the “big wet” years that historically punctuated rainfall records no longer occur (the most recent was in 1974 – 45 years ago). Within the study area, rainfall totals (Wungong Dam, Bureau of Meteorology station 009044; Figure S6.1[2.2]) have steadily declined since records began. The 1913 to 2018 mean rainfall total was 1188 mm; between 1913 and 1960, the mean was 1386 mm, declining to 1095 mm during the period 1961 to 1990. Between 1991 and 2018 the mean had dropped further to 943 mm – a 32% reduction from the 1913 to 1960 mean, congruent with that of the wider SWA region.

As rainfall declined, streamflow underwent a disproportionately larger decline of around 70% (Petrone et al., 2010; Silberstein et al., 2012). This was caused by declining groundwater tables, and disconnection from streambeds began around the year 2000 – 2001 in the forested catchments, resulting in continued decline in streamflow independent of annual rainfall (Petrone et al., 2010; Hughes et al., 2012; Kinal & Stoneman, 2012). Perennial streams with groundwater-fed baseflow became intermittent ‘losing’ streams. In some SWA catchments, additive interactions between climatic drying and historical management of forests (overstocking in some regrowth areas), and widespread water stress, added further downward pressure on water tables that were already declining due to climate change (Batini, 2007; Liu et al., 2019; Harper et al., 2019). Nevertheless, climate remains the key driver of hydrological change across the region, as in other regions globally also experiencing streamflow declines.

Study streams

The study was located in the Wungong Brook catchment, in the Northern Jarrah Forest (NJF) of SWA (Figure 6.1). This catchment is covered by native forest and lies upstream of an urban water-supply dam; consequently, the area is designated as protected water catchment and access is restricted. Since the late 1980s, no bauxite mining, commercial logging or other disturbances have been permitted at the study sites. Stream water quality, substratum and fringing vegetation composition remain unchanged. Climate change is the only human impact on these streams. Four streams in the Wungong catchment (Waterfall Gully, Seldom Seen Brook, Curtis Creek, Chandler Brook) were chosen based on flow regime, the availability of long-term flow gauging data and to best replicate the experimental sites used by Bunn (1985; 1988c). Stream substrata comprise gravel and sand with areas of concreted laterite or granite bedrock. NJF is dry sclerophyll forest, dominated by jarrah *E. marginata* and marri *Corymbia callophylla* (Bunn 1985). The stream invertebrate fauna is characterised by few

Ephemeroptera, Plecoptera and Trichoptera (EPT) taxa and high endemism (Davies & Stewart, 2013).

In the Wungong catchment, climatic drying has caused transition from perennial to intermittent flow regimes in most streams, including later commencement of winter flows and later peak flows in all streams (Figures S6.2, S6.3, S6.4). Seldom Seen Brook dried out for the first time in 2011 but contains one spring-fed reach that has perennial base flow in all but the driest years (Figure S6.3). Only one stream, Waterfall Gully, has remained perennial although annual discharge has declined by 70% (Figure S6.2). In the other streams, streamflow is entirely dependent on surface run-off from rainfall, so flow now commences in June or July instead of discharge increasing above baseflow in April-May (e.g. Figure 2.6). Curtis Creek and Chandler Brook became intermittent in the early 2000s. Currently, Chandler Brook is flashy, discharging large volumes of water in winter and spring, but drying down quickly in summer. Comparison of water temperatures measured by Bunn (1985) using maximum-minimum thermometers with our temperature measurements using temperature loggers (ThermodataTM iB Tags, Thermodata Pty. Ltd.) showed that temperatures increased dramatically in drying pools (Figure S5.2), but that otherwise they were not dissimilar.

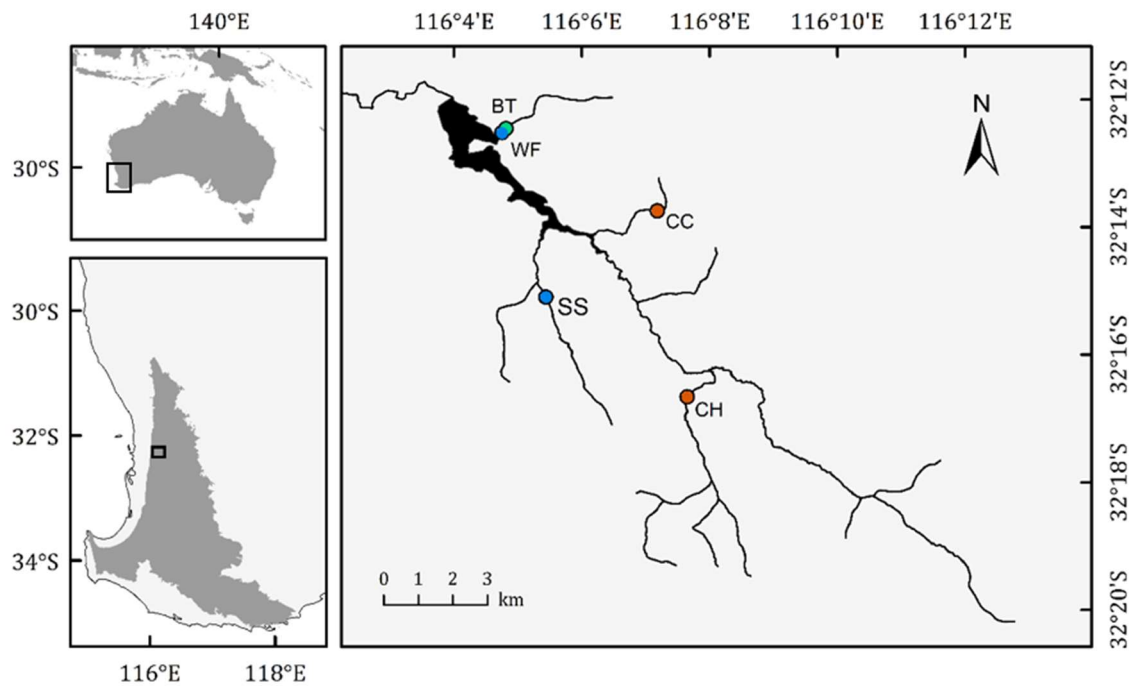


Figure 6.1: The study area is within the Wungong catchment, forested low order streams in the Northern Jarrah Forest of Western Australia. Extent of the Northern and Southern jarrah forests shaded (bottom left). Leaf bags were anchored in perennial (blue WF; Waterfall Gully; SS, Seldom Seen Brook) and formerly perennial, now- intermittent (orange CC, Curtis Creek; CH, Chandler Brook) reaches. The procedural control (brick tether treatment) was located at Waterfall Gully (green BT). SS is the same location used by Bunn (1988c).

Study design

Leaf bags were attached in a single layer on wire sheets that were anchored to the streambed in random positions using tent pegs. Six leaf bags were attached to each sheet (a total of ten sheets per stream reach) in random order. Bags were collected five times (without replacement) over a period of 320 days, with the time intervals matching those used by Bunn (1988c) as closely as possible.

mBACI designs are optimal where there is a discrete, step change at impact sites (Downes et al. 2002 p171), as occurs with transition from perennial to intermittent flow regimes (Figures

S6.3, S6.4). One strength of mBACI designs is that they enable us to quantify temporal change through comparing controls for the before and after period (Downes et al., 2002). In the present study, comparing perennial reaches from both time periods, thirty years apart, would provide controls for any changes not attributable to the transition to intermittency. We assumed that the temporal comparison was realistic, because we had no evidence that either 1982-83 or 2018-19 were atypical over a five-year mean (in terms of rainfall, stream discharge, temperature). Since Bunn used methods similar to contemporary litter bag experiments, and there is no reason to believe that his results are flawed or faulty, they could be used in a partial mBACI design to understand changes in leaf decomposition as a result of climatic drying forcing intermittency onto perennial streams.

A full mBACI design comprises four pre-planned comparisons: (1) between the control and impact sites before the impact occurs (ideally the sites are similar); (2) between the control sites (where no impact has occurred), sampled before and after the impact at the other sites, to quantify the effects of temporal changes unrelated to the impact; (3) between the impact sites, sampled before and after the impact to quantify the impact; (4) between the after-control and after-impact sites, to separate the effects of the impact from other temporal changes. Bunn sampled a reach in Seldom Seen Brook in 1982-83 and since this stream is still perennial it could be used as a before-control to compare with samples taken in 2018-9 to provide after-control data. The other before-control site was a perennial stream sampled by Bunn, Foster Brook; this stream is now beneath a large reservoir so it was replaced as the other after-control site by Waterfall Gully, which has also remained perennial since 1982-83. This enabled us to make the second pre-planned comparison listed above. We have no before-impact streams, because Bunn's leaf bag experiment was only carried out in two streams, meaning that we could not carry out the first or third pre-planned comparisons listed above. In 2018-2019 we used two additional reaches, in Chandler Brook and Curtis Creek; both

streams that had been perennial in 1982-83 but transitioned to intermittency in the early 2000s. These comprised the after-impact streams and enabled us to make the fourth pre-planned comparison listed above. Therefore, our study design is a partial mBACI design that enables us to quantify any temporal changes aside from flow regime change by comparing leaf decomposition in the perennial streams between 1982-83 and 2018-19, and to quantify the impact of flow regime change by comparing leaf decomposition in perennial and intermittent reaches in 2018-19. We do not know what litter decomposition was like in Chandler Brook or Curtis Creek in 1982-83 when they were perennial, but their substrata, environmental conditions, flows and fauna were similar to those in Waterfall Gully in 1982-83 (Bunn 1985), so it is likely that leaf decomposition was also similar. This means that we must interpret our long-term comparisons with caution.

Comparing contemporary results to those from past studies is not necessarily easy or straightforward, yet the importance of using before data in assessing environmental impacts has been made clear (Downes et al. 2002; Didham et al., 2020b). Although the complete original data from Bunn (1988c) was unobtainable for formal analysis, we obtained mean leaf weights remaining at the end of Bunn's experiment from figures in Bunn (1985) and tested these against weights remaining at the end of our experiment to make the before-after-control comparison of the mBACI design. Invertebrate data was also taken from Bunn (1985), as presence-absences per experimental reach, to compare with contemporary leaf bag assemblages.

There were other challenges in comparing our contemporary results to those of Bunn. While we chose to carry out this study in a single catchment to minimize confounding with other landscape changes, one of the two streams Bunn studied, Foster Brook, lies in a different (although nearby) catchment. Nevertheless, Bunn's historical results from that stream could be compared to those from our contemporary substitute, Waterfall Gully, because these two

streams had similar invertebrate fauna, flow and substrate characteristics in 1982-83 (Bunn 1985). The other stream that Bunn (1985) studied, Seldom Seen Brook, had bauxite mining in its catchment that caused some sedimentation in the study reach and he attributed the slower leaf bag decomposition there to sedimentation (Bunn 1985). This means that as a perennial control reach, Seldom Seen Brook may underestimate the leaf weights lost by the end of the experiment in 1983. Bunn (1988c) tethered leaf bags in-stream to house bricks, while we enclosed leaves inside plastic mesh (Gutterguard™) cages (“flat packs”) that lay directly on the bed. Our method was logistically much easier and avoided confounding our results: it is now known that bricks could potentially alter near-bed flows around leaf bags (Robson, Chester, & Davis, 1999 and references therein), thereby altering macroinvertebrate assemblages and the decomposition process. This difference between the two studies could be addressed through the use of a procedural control experiment that directly compared the brick tethering method to our ‘flat pack’ design under perennial flow conditions in Waterfall Gully. For the brick tether treatment, leaf bags (both coarse and fine mesh) were attached to house bricks using cable ties and distributed haphazardly in the reach with the flat packs. Leaf weight lost and invertebrate assemblage composition were assessed at the same dates as the main experiment and were then compared to determine whether the brick-tether method delivered different results to the flat packs under the same in-stream conditions. In all other respects, Bunn’s experiment meets contemporary standards for leaf bag experiments.

Another challenge in studying the effects of climate change is that it may alter variables other than flow regime change. In this case, the composition of Jarrah (*E. marginata*) leaves may have changed as a result of global warming (increased temperatures and CO₂ concentrations, reduced soil moisture, duration of winter-spring rains, increased evaporation) between the early 1980s and now. Both this study and that of Bunn (1988c) used naturally abscised leaves; *Eucalyptus* trees withdraw the majority of nutrients and other valuable compounds

prior to leaf fall, so the nutritional content of these leaves is minimal. In the 1980s, a study of the terrestrial decomposition of *E. marginata* leaves showed that among-site variation in leaf litter nutrient content and decomposition rates was small, even between burnt and unburnt sites (O'Connell & Menagé, 1983). There is evidence that short-term intense drought reduced the nitrogen content of jarrah leaves in the Wungong catchment, which in turn may affect decomposition (Qiu et al., 2013). However, even if leaves have become less nutritious because of climate change, leaves entering both perennial and intermittent streams will be affected, and therefore this variation will be partitioned into change in the controls between 1982/83 and 2018/19 in the mBACI design, so it can be separated from the effect of flow regime change using our study design.

Leaf bag preparation and placement

Emulating Bunn (1985, 1988c), freshly fallen, abscised *E. marginata* leaves were collected from stands of forest within the Wungong catchment over several days in January 2018, coinciding with peak litter fall of *Eucalyptus* species in these forests. Leaves were dried for 24 h at 50 degrees Celsius, to halt biological processes. Once dried, the petioles were removed, leaves were weighed accurately to 10.0 grams and placed inside leaf bags.

Coarse mesh bags, made from plastic GutterGuard™ had a maximum aperture of 5 mm. For the fine mesh bags, nylon curtain fabric with a maximum aperture of 200 µm was sewn into 20 x 15 cm bags, and then placed inside a gutter guard case for protection. Bunn (1985, 1988c) used 100 µm mesh for his fine litter bags and 3.36 mm mesh for his coarse bags.

Collection and processing of leaf bags and invertebrates

Leaf bags were anchored in streams on the 14th of February 2018; reaches at Waterfall Gully and Seldom Seen Brook had steady baseflow, whereas Curtis Creek and Chandler Brook were completely dry. Flow commenced in Wungong Brook in winter on the 11th of June 2018, it is assumed flows commenced at Curtis Creek and Chandler Brook at a similar time

(i.e. approximately 117 days after leaf bag deployment). Leaf bags were sampled in May (79 days, autumn), July (128 days, mid-winter), September (199 days, early spring), November (255 days, spring) and January 2019 (320 days, mid-summer). At each time, 5 randomly chosen bags of each mesh type were collected using a 500 μm mesh net. Leaf bags were transferred to a plastic bag and placed on ice for transport back to the laboratory as per Bunn (1985, 1988c).

In the laboratory, leaf bag contents were each emptied into a white tray and flooded with water. Fine mesh bags were initially rinsed before being cut open, to remove any invertebrates attached to the outside. Organic material, sand and lodged invertebrates were gently brushed off the leaves, which were then inspected for leaf-mining Chironomidae. Trays were then carefully inspected for invertebrates, which were preserved in 70% ethanol for later identification to the lowest taxonomic level possible. At 79 days, Curtis Creek and Chandler Brook were dry and almost no invertebrates were present on the leaf bags. Consequently, macroinvertebrate assemblages were only compared between inundated leaf bags. Leaves were dried in an oven at 50 °C overnight, or until completely dry (up to 24 h). Dried leaves were then accurately weighed to determine loss in each bag over the time in-stream. These are the same methods used by Bunn (1985, 1988c).

Data analyses

In our design, different leaf bags were sampled randomly at each time without replacement to estimate decomposition (weight loss), ensuring that times were sampled independently; therefore this was not a repeated measures design. Time is a fixed factor in mBACI designs (Downes et al., 2002, p168) and also in leaf decomposition experiments where there are hypotheses about particular temporal patterns. That is, we expected leaf weight to decline and invertebrate abundance and richness to increase with time. We also matched our sampling times to be as close as possible to those of Bunn (1985, 1988c). So, there were clear *a priori*

expectations about patterns of leaf weight loss and invertebrate assemblage composition at each level (sampling date) of the time factor and we wished to make inference about those particular times, so time was treated as a fixed factor in all analyses (Quinn & Keough, 2002, p 176). Due to torrential flows in Chandler Brook, we were unable to collect the leaf bags in September, creating a missing cell in the design that caused a variable number of reaches among times.

We tested the hypotheses for leaf decomposition (weight loss) using the two pre-planned comparisons, and the procedural control experiment added a second set of leaf weights from the reach at Waterfall Gully. Initially, we used a three factor ANOVA — time: 5 levels (79, 128, 199, 255 and 320 days), fixed; stream reaches: 4 or 5 levels, fixed; and mesh size: 2 levels, fixed — to analyse leaf weights from the 2018-19 experiment. The main effect of time and the three-way interaction were highly significant because of the differences between sampling times expected *a priori* as a result of decomposition (results not presented). So, to more closely examine the effects of flow regime and mesh size (hypotheses 1, 3_{a,b}), we analysed leaf weights at each time separately using a two factor ANOVA (stream reaches: 4 or 5 levels, fixed; and mesh size: 2 levels, fixed). This made it easier to deal with the missing reach (Chandler Brook) in September and to include the brick-tether leaf bags as another ‘reach’, although they were actually placed in the same reach as the flat packs in Waterfall Gully. Tukey’s HSD tests were used to test differences between particular stream reaches. Assumptions were checked and log₁₀ transformation used where necessary.

Macroinvertebrate assemblages (species x abundance per leaf bag) in 2018-19 were first analysed using a four-factor PERMANOVA (Primer-E; Anderson, Gorley, & Clarke, 2008) with the factors: time, 5 levels, fixed; reach type, 2 levels (perennial or intermittent), fixed; mesh size, 2 levels (coarse or fine), fixed; stream nested within reach type, 2 levels, random. To further examine the effect of reach type, assemblages were analysed at each time

separately using a three-factor PERMANOVA with the same factors (reach type, mesh size, and reach type nested within stream). A crossed two-factor ANOSIM design testing mesh size (2 levels) and reach (4 levels) was used to determine the effectiveness of fine mesh bags at excluding invertebrates in each stream reach at each time and to quantify effect sizes using the *R*-value. Similarly, changes in assemblage composition over time were examined in coarse mesh bags only using a crossed two-factor ANOSIM (stream reach 4 levels, time 5 levels). SIMPER was used to identify key taxa contributing to assemblage change over time at each stream separately.

To further investigate invertebrate assemblages on leaf bags, taxa were grouped into three functional feeding groups: shredders, predators, and (gathering and filtering) collectors (no scraping or piercing species were found). The leaf mining chironomid *Stenochironomus* sp. was included in the shredder guild, and often occurred in high numbers in both coarse and fine mesh bags (Figure 6.2). *Stenochironomus* sp. were able to enter through holes in fine mesh bags as tiny first instar larvae, potentially from eggs laid onto the bags. Total number of individuals of each functional feeding group was compared between streams and times using a two-factor (streams, 4 levels, fixed; times 5 levels, fixed) ANOVA of data from coarse mesh bags only. The assumptions of normality and equal variance were satisfied in each of these analyses except for the collector guild, where they were met after log-transformation.

Lastly, the role of shredder density in leaf weight loss was tested using single-factor ANCOVAs for each stream reach separately: leaf dry weight remaining was the dependent variable and the independent variable was time (5 levels, fixed) with shredder density as the covariate. The assumptions of normality and equal variance were satisfied in each of these analyses except for Curtis Creek; transformation of the weight remaining met these assumptions.

Differences in Eucalyptus leaf litter decomposition following 35 years of drying

The results from the procedural control experiment were determined using Tukeys HSD tests from the two factor ANOVA of leaf weights described above. For macroinvertebrate assemblages, times were pooled and a crossed two factor (mesh size: 2 levels; method: 2 levels) ANOSIM based on a Bray- Curtis similarity matrix was used (Primer -E; Clarke, Gorley, Somerfield, & Warwick, 2014).

Given there were multiple control and impact locations the statistical model for the partial mBACI design could be simplified to a series of t-tests (Downes et al., 2002). Each of which tested one of the two pre-planned comparisons. Estimates from each reach represent the combined effects of leaf decomposition processes over an equivalent duration in the two studies (1982-83 327 days, 2018-19 320 days). The first pre-planned comparison showed whether, in the perennial streams, the mean weight of material remaining from the ten grams of *E. marginata* leaves in coarse or fine leaf bags (analysed separately) differed between 1982-83 (Bunn 1985, 1988c) and 2018-19 (the before-control vs after-control pre-planned comparison). If this showed no difference in mean leaf weight remaining, the control-impact comparison using the ‘after’ data could be used to test the effect of the transition to intermittency (2nd pre-planned comparison). Again, two t-tests tested the second pre-planned comparison: coarse and fine mesh separately. Since variances were unequal for all t-tests, the formula for unequal variance was used.



Figure 6.2: Remnants of *Eucalyptus marginata* leaves, showing shredding activity of *Notoperata tenax* (Trichoptera; A) and *Stenochironomus* sp. (Chironomidae; B). Leaves were collected in January 2019 (320 days) from coarse mesh bags in Waterfall Gully (A) and Seldom Seen Brook (B).

Results

Patterns in leaf litter decomposition between perennial and formerly-perennial streams in 2018-19 (Hypotheses 1 and 3)

Leaf bags in perennial reaches in Waterfall Gully and Seldom Seen Brook were inundated from the start of the experiment, whereas those in intermittent Curtis Creek and Chandler Brook were on dry stream bed for up to 117 days. This difference in the duration of immersion is reflected in the patterns of initial weight loss, where leaf bags in perennial reaches lost significantly more weight than those in intermittent reaches (stream reach: $F_{4,40} = 82.7$, $P < 0.001$; Table 6.1, Figure 6.3). Flows commenced in the intermittent reaches in early winter (mid-June, Figure 5.3c,d); weight loss in those streams then accelerated, but did not catch up to either perennial stream until 255 days (spring, November) when only leaf weights at intermittent Chandler Brook (mean weight 7.07 g) were higher than at perennial sites (mean weight range 6.06 – 6.27 g; Figure 6.3). In mid-winter (July), mean leaf weights in the two perennial reaches (Waterfall Gully 6.7 g, Seldom Seen Brook 6.8 g) did not differ, but were lower than the two intermittent reaches (stream reach $F_{4,41} = 22.78$, $P < 0.001$; mean leaf weights Chandler Brook 7.3 g, Curtis Creek 7.7 g). By spring (September), intermittent Curtis Creek still had a higher mean leaf weight (6.9 g) than the perennial stream reaches (Waterfall Gully 6.16 g, Seldom Seen Brook 5.81 g). By the end of the experiment in summer (January 320 d), the divisions between the perennial and intermittent streams were less clear (although there were still differences $F_{4,62} = 27.7$, $P < 0.001$). Intermittent Curtis Creek had a significantly higher mean leaf weight remaining (6.5 g) than the perennial streams, but Chandler Brook (6.38 g) did not; it did not differ from either Curtis Creek or perennial Seldom Seen Brook (5.85 g). Perennial Waterfall Gully had significantly lower mean leaf weight remaining than all other streams (4.85 g). Therefore, the delay in immersion experienced by leaf bags in the intermittent streams ultimately did not slow decomposition as

they had caught up before the next period of intermittency (Figure 6.3), supporting hypothesis 1.

Table 6.1: Mean weight lost from leaf bags at the first (79 d) and final (320 d) collection time in 2017.

At 79 d mean weights at Waterfall Gully and Seldom Seen Brook did not differ (Tukeys HSD $P = 0.99$); Curtis Creek and Chandler Brook did not differ (Tukeys HSD $P = 0.34$); but these pairs differed from each other (Tukeys HSD all $P < 0.001$).

Flow regime	Stream	Mesh type	Weight lost (g) 79 d	Standard error (g)	Weight lost (g) 320 d	Standard error (g)
Perennial	Waterfall Gully	Coarse	2.9	0.22	6.1	0.19
		Fine	3.5	0.22	4.3	0.19
Perennial	Seldom Seen Brook	Coarse	2.8	0.24	3.9	0.22
		Fine	3.4	0.24	4.4	0.18
Intermittent	Curtis Creek	Coarse	1.6	0.24	3.6	0.19
		Fine	0.9	0.24	3.4	0.19
Intermittent	Chandler Brook	Coarse	1.1	0.17	3.8	0.19
		Fine	0.8	0.17	3.5	0.16

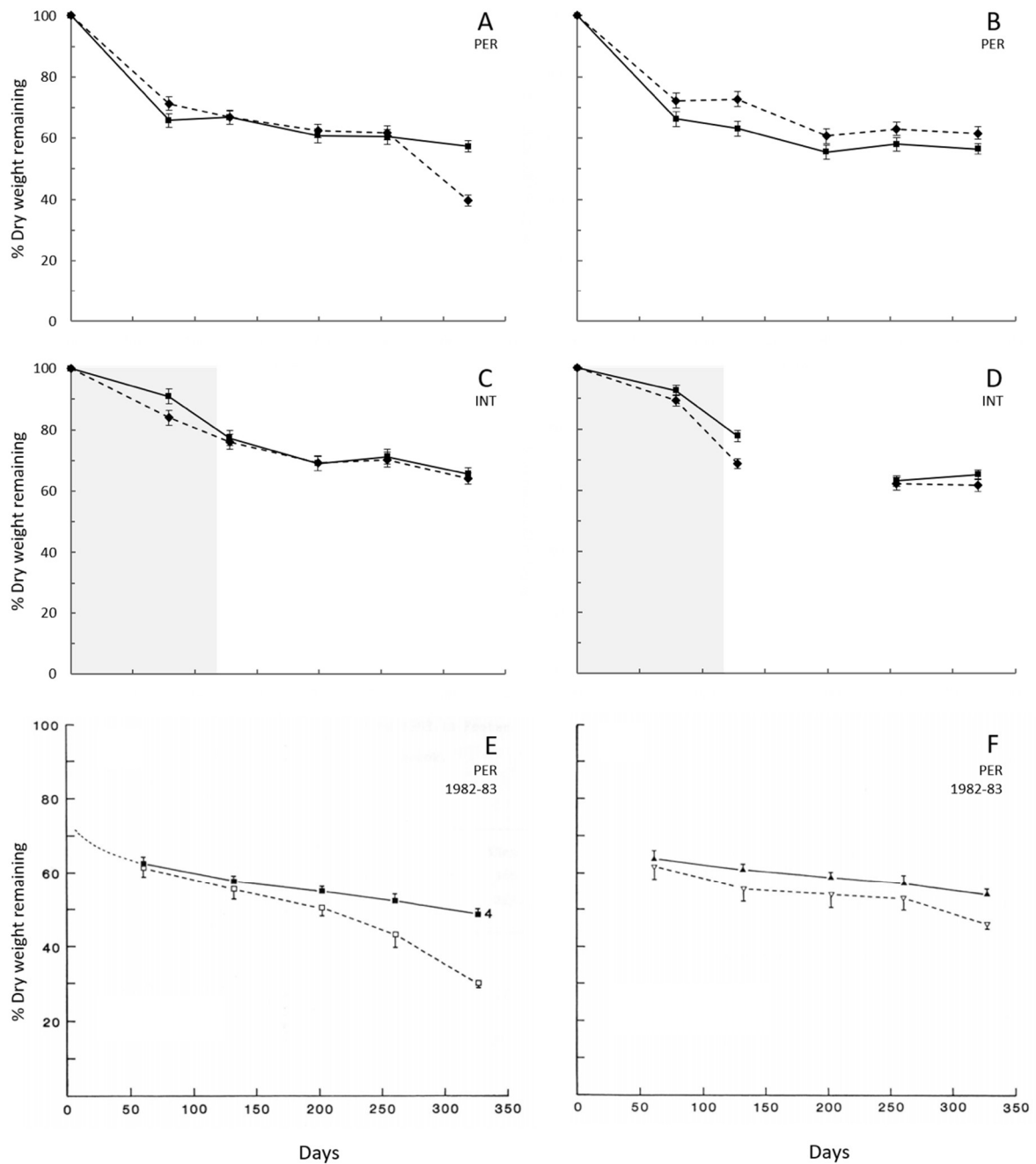


Figure 6.3: Mean decomposition rates of *Eucalyptus marginata* leaves in coarse mesh (dashed line) and fine mesh (solid line) at four experimental reaches: two perennially flowing, PER, (A) WF, Waterfall Gully, and (B) SS, Seldom Seen Brook; and two formerly perennial, now- intermittent reaches, INT, (C) CC, Curtis Creek, and (D) CH, Chandler Brook. Grey shading (C & D) indicates approximate dry period, error bars are ± 1 standard error. Panels E and F are decomposition rates measured at Foster Brook (perennial equivalent to Waterfall Gully) and Seldom Seen Brook, measured in 1982-83 by Bunn (1988c); error bars are 95% CI. Panels E and F reprinted by permission from Springer Nature GmbH: Hydrobiologia (Processing of leaf litter in two northern jarrah forest streams, Western

Unusually, fine mesh bags were processed faster than coarse mesh bags in perennial Seldom Seen Brook ($F_{1,47} = 18.4$, $P < 0.001$) at all sampling times (Figure 5.3b), providing no support for hypothesis 3b. Perennial Waterfall Gully was the only stream with a significant interaction between mesh and time ($F_{4,43} = 10.6$, $P < 0.001$), caused by an accelerated weight loss in coarse mesh bags between 255 and 320 days (Figure 6.3a). So that by the end of the experiment, overall mean weight loss did not differ between coarse and fine mesh bags except in perennial Waterfall Gully (stream x mesh type $F_{4,62} = 7.57$, $P < 0.001$; Table S1, Figure 6.3a). Consequently, hypotheses 3a and 3b were not supported.

Comparison of leaf bag macroinvertebrate assemblages

Flow regime (perennial or intermittent) alone did not explain differences in assemblage composition (pseudo- $F_{1,119} = 1.7$, $P = 0.18$), except during mid-spring (199 d; pseudo- $F_{1,23} = 1.8$, $P = 0.001$, Table 6.2). In contrast, stream nested within reach type was significant overall (pseudo- $F_{2,119} = 7.23$, $P < 0.001$) and at every sampling time (Table 6.2). Therefore, the differences in faunal assemblages between streams were large, dwarfing any influence of flow regime. There was a significant three-way interaction between mesh size, sampling time and reach within stream (pseudo- $F_{5,119} = 1.65$, $P = 0.002$) caused by inconsistent patterns of differences in assemblage composition between mesh size and sampling time in each reach.

Table 6.2: PERMANOVA results testing the effects of reach type (perennial or intermittent) and stream nested within reach type, on macroinvertebrate assemblage composition in experimental leaf packs in Northern Jarrah forest streams (Reach type: 2 levels; stream: 4 levels, Waterfall Gully (WF), Seldom Seen Brook (SS), Curtis Creek (CC) and Chandler Brook (CH)). * at 79 days only perennial streams had flow, therefore only 2 streams were tested (WF, SS); ** dangerous flows at Chandler Brook at 199 days prevented sampling leaf packs, thus only three streams were sampled (WF, SS, CC).

Sample time	Factors	Pseudo-F statistic	df	P
All	Reach type	1.7	1, 2.41	0.18
	Stream (reach type)	7.2	2, 119	0.001
79 days*	Reach type	-	-	-
	Stream (reach type)	5.3	2, 23	0.001
128 days	Reach type	2.1	1, 27	0.07
	Stream (reach type)	3.6	3, 27	0.001
199 days**	Reach type	1.8	1, 23	0.001
	Stream (reach type)	1.9	3, 38	0.01
255 days	Reach type	1.7	1, 38	0.2
	Stream (reach type)	1.9	2, 23	0.001
320 days	Reach type	1.8	1, 51	0.3
	Stream (reach type)	4.5	3, 51	0.001

The ANOSIM results showed that macroinvertebrate assemblages differed significantly between streams (Global $R = 0.391$, $P = 0.001$) and mesh sizes (Global $R = 0.357$, $P = 0.001$). Assemblages differed considerably between mesh sizes at Waterfall Gully (Global $R = 0.587$, $P = 0.001$) and Curtis Creek (Global $R = 0.43$, $P = 0.001$), by a smaller amount in Seldom Seen Brook (Global $R = 0.191$, $P = 0.001$), but not at Chandler Brook (Global $R = 0.007$, $P = 0.42$). Most invertebrates were excluded by the fine mesh except for several Chironomidae (including the leaf miner *Stenochironomus* spp.), Ceratopogonidae and Oligochaeta, and in Waterfall Gully, the predatory trichopteran *Taschorema pallescens* (Table 6.3). These animals would have entered the bags as early instars, smaller than 200 μm , and the bags then

likely acted as a predation refuge (except at Waterfall Gully where *T. pallescens* likely fed on the Diptera).

The assemblages living in coarse mesh bags differed significantly among streams (Global $R = 0.412$, $P = 0.001$; Figure 6.4). At Waterfall Gully, times differed (Global $R = 0.255$, $P = 0.001$) because assemblages differed early in the decomposition process (between 79 and 128 days: $R = 0.442$, $P = 0.02$, and 128 and 255 days: $R = 0.44$, $P = 0.02$). Changes in the relative abundances of common species, such as the amphipod *Uroctena* sp., the chironomid *Riethia* spp. and mayfly *Neboissophlebia occidentalis*, drove dissimilarity between samples. Towards the end of the experiment (between 199, 255 and 320 days) there was no change in assemblage composition ($R = 0.05$, $P = 0.3$) despite the arrival of shredding caddisflies between 255 and 320 days. Pairwise tests showed that larvae of the shredding caddisfly *Notoperata tenax* contributed to dissimilarity between 320 days and each other sample time (79 days = 8%; 128 days = 11%; 199 days = 9%; 255 days = 7%). However, the contribution from *N. tenax* was outweighed by changes in abundance of common species including *Uroctena* spp. and *Riethia* spp. *Notoperata* did not arrive in any other reach in 2018-19.

Table 6.3: Density of taxa with more than one occurrence in coarse mesh leaf bags collected from perennial (Waterfall Gully and Seldom Seen Brook) and formerly perennial, now- intermittent (Curtis creek and Chandler Brook) streams in the Northern Jarrah Forest, Western Australia. Common taxa were assigned to one of three functional feeding groups (collectors, shredders and predators). Number of remaining rare taxa and total number of taxa also given.

Taxon		FFG	Average density no./ bag-1			
			Waterfall Gully	Seldom Seen	Curtis Creek	Chandler Brook
Trichoptera						
Leptoceridae	<i>Notoperata tenax</i>	Sh	1			
Ephemeroptera						
Leptophlebiidae	<i>Neboissophlebia occidentalis</i>	Co	1.7			
	<i>Bibulmena kadjina</i>	Co		2.2		
	<i>Nyungara bunni</i>	Co	0.6			
Odonata						
Telephlebiidae	<i>Austroaeschna anacantha</i>	Pr	0.4			
Austrocorduliidae	<i>Lathrocordulia metallica</i>	Pr	0.4			
Diptera						
Chironomiinae	<i>Stenochironomus</i> sp.	Sh	0.7	0.2		0.5
	<i>Riethia</i> sp.	Co	3.7	0.4		
	<i>Polypedilum</i> sp.	Co	0.5	0.7	0.3	0.2
	<i>Chironomus</i> sp.	Co		0.2	0.3	3
	<i>Tanytarsus</i> sp.	Co		0.3		2.4
Orthoclaadiinae	<i>Cricotopus</i> sp.	Co	0.6	0.4		
Tanypodiinae	<i>Austropelopia</i> sp.	Pr	1	0.2		
Ceratopogonidae	<i>Monohelea</i> sp.	Co/ Pr	0.6	0.1		
	<i>Nilobezzia</i> sp.	Co/ Pr			0.4	
Tipulidae	<i>Tipula</i> sp.	Co/Sh			0.1	
Limoniidae	<i>Molophilus</i> sp.	Co			0.3	
Coleoptera						
Scirtidae	Scirtidae spp. (larval)	Co		1.3	0.1	
Amphipoda						
Perthiidae	<i>Perthia acutitelson</i>	Sh		0.5	0.1	
Paramelitidae	<i>Uroctena</i> sp.	Co/Sh	12.6	0.3	5.3	
Decapoda						
Parastacidae	<i>Cherax quinquecarinatus</i>	Sh		0.1		0.4
Oligochaeta	Oligochaeta	Co	0.9	0.2	0.4	
Total average density			26.4	8.5	8.1	6.9
% of total individuals			91.70%	87%	91%	90%
Remaining rare taxa			27	18	12	5
Total taxa			40	34	21	10

At Seldom Seen Brook, coarse mesh leaf bags exhibited distinct faunal assemblages at all sample times (Global $R = 0.236$, $P = 0.001$) except 320 days (summer, January), which was not significantly different to any other time. This is explained by an increase in species

richness in the warmer months (10 species in autumn and mid-winter, 7 in early spring, *cf.* 22 in spring and 18 in mid-summer). The mayfly *Bibulmena kadjina* was present throughout the year and contributed to 34 - 54% average similarity at 79, 128 and 255 days. Chironomids *Polypedilum* spp., *Riethia* spp., and *Tanytarsus* spp., and the amphipods *Uroctena* sp. and *Perthia acutitelson* also occurred throughout the year; changes to their abundance were the main contributors to dissimilarity between sample times. These amphipods are potential contributors to shredding, however patterns in their abundance were not predictably related to leaf litter decomposition (see below).

After initial leaching of leaves at intermittent reaches in Curtis Creek and Chandler Brook in early winter, faunal assemblages showed little temporal pattern, but differed between streams (Table 6.2). Assemblages at Curtis Creek did not differ across the year: combinations of the amphipod *Uroctena* sp., chironomids *Polypedilum* sp., and *Limnophyes* sp., and larval Tipulidae (*Molophilus* sp. and *Tipula* sp.) were the key contributors to similarity within time groups and changed little in their abundance over time. Assemblages did not differ between mesh sizes at Chandler Brook and only seven coarse bags contained any fauna, which was dominated by Chironomidae (Table 6.3). Assemblages differed between times (Global $R = 0.241$, $P = 0.02$), but with no clear pattern over the experiment. *Chironomus* sp. dominated the fauna, along with *Tanytarsus* sp., the tanypod *Austropelopia* sp., and tipulid *Molophilus* sp (Table 6.3). Changes in the relative abundance of these dipterans drove dissimilarity between times in this depauperate assemblage.

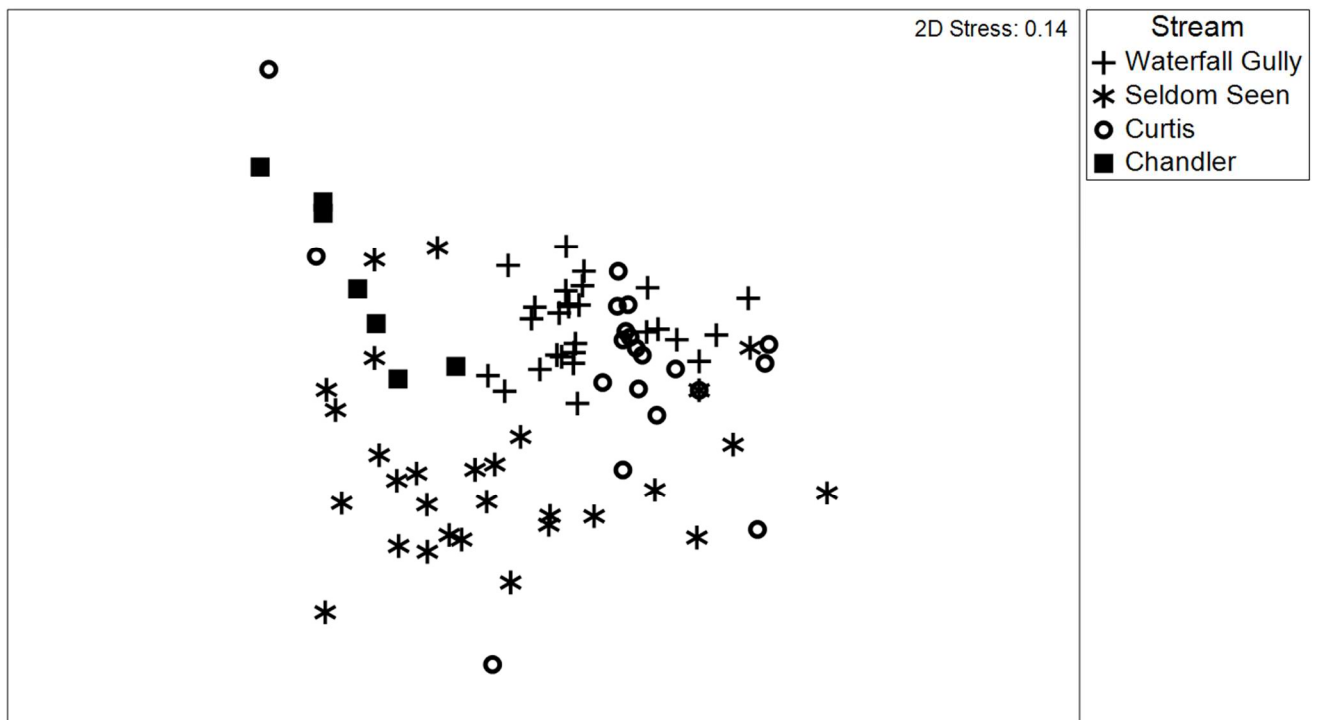


Figure 6.4: nMDS ordination of macroinvertebrate assemblage composition within coarse mesh bags containing *Eucalyptus marginata* leaves, at four experimental reaches: two perennially flowing reaches (Waterfall Gully and Seldom Seen Brook) and two formerly perennial, now- intermittent reaches (Curtis Creek and Chandler Brook), over five sampling times. Stress = 0.14.

Functional feeding groups

Three predator species (caddisfly *T. pallescens*, dragonfly *Austroaeschna anacantha*, and Tanypodinae; Figure 6.5) were found in very low densities in the coarse-mesh leaf bags. Predators were very rare on coarse-mesh leaf bags in intermittent Curtis Creek and Chandler Brook and consisted almost entirely of Tanypodinae chironomids (Table 6.3). Collectors were the most abundant and speciose FFG and their abundance differed between some stream by time combinations ($F_{9,71} = 4.6$, $P < 0.001$; Figure 6.5; Table 6.3) but without clear patterns. Perennial Waterfall Gully had much higher densities of collectors than other stream reaches, and shredders were most abundant and diverse there (Figure 6.5; Table 6.4).

The ANCOVAs showed that density of shredders on leaves in coarse bags was strongly related to remaining leaf weight at perennial Waterfall Gully ($F_{1,21} = 12.3$, $P = 0.002$, $R^2 =$

0.86) and weakly related at intermittent Curtis Creek ($F_{1,17} = 13.6$, $P = 0.002$, $R^2 = 0.41$, but was not associated with leaf weight loss in coarse bags at perennial Seldom Seen Brook ($F_{1,20} = 0.22$, $P = 0.64$) or intermittent Chandler Brook ($F_{1,9} = 0.034$, $P = 0.85$). This provides further evidence that hypothesis 3a was not true. Densities of shredders were very low on leaf bags in all reaches, and only four taxa (caddisflies *N. tenax*, *Lectrides parilis*, amphipod *P. acutitelson*, and leaf mining chironomid *Stenochironomus* sp.) were present (Table 6.1). *Stenochironomus* sp. was an unexpected inclusion in the shredding guild. During the warmer months this species was observed in high numbers inside the leaves, consuming the inner contents and producing FPOM, which was released when the softened leaf cuticle (left behind) was broken. This is a completely different shredding process to that performed by caddisflies (Figure 6.2), nonetheless these chironomids contributed to leaf litter decomposition. Although recorded in-stream, the amphipod *Uroctena* sp. was not recorded on leaf bags by Bunn (1985, 1988c) however it was common on leaf bags at perennial Waterfall Gully and intermittent Curtis Creek in 2018-19 (Table 6.4). Although some members of this genus are known to be shredders that skeletonise leaves, *Uroctena* sp. did not contribute to shredding in this experiment.

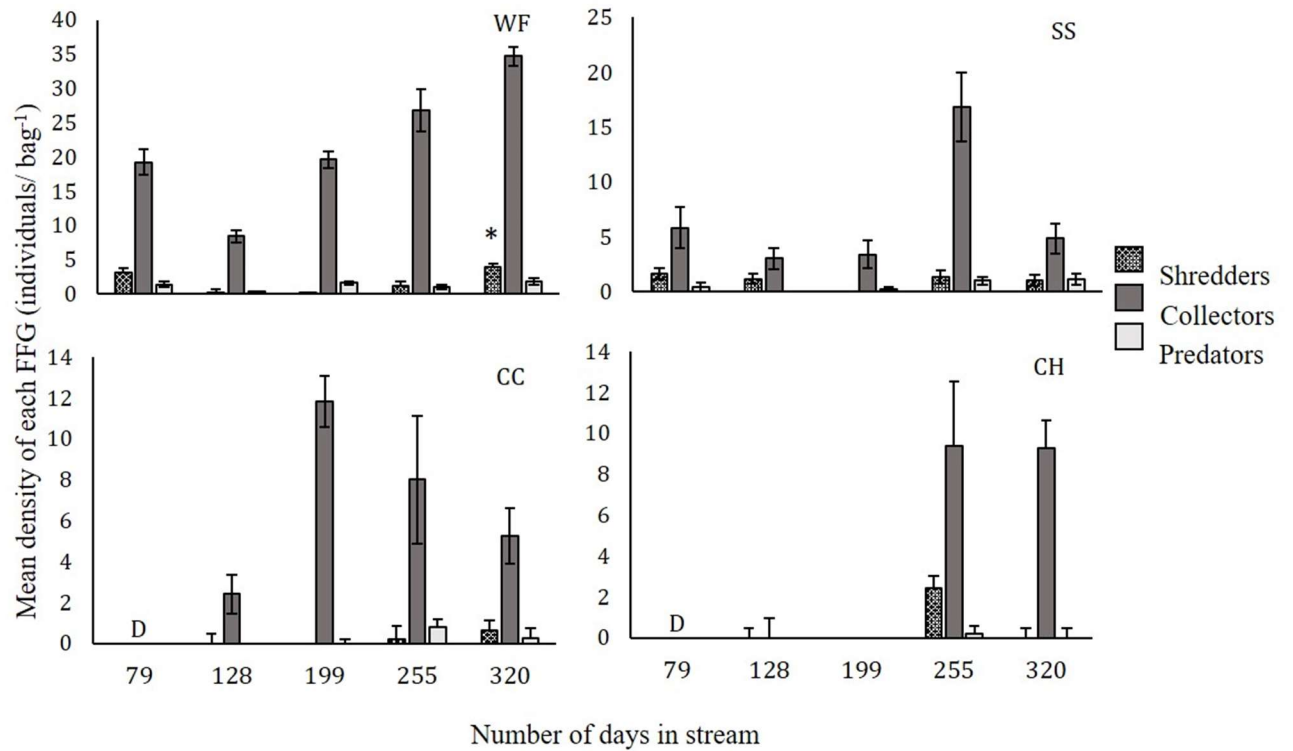


Figure 6.5: Mean densities of individuals of functional feeding groups (per coarse-mesh bag) collected from two perennially flowing (WF, Waterfall Gully; SS, Seldom Seen Brook) and two formerly-perennial, now- intermittent (CC, Curtis creek; CH, Chandler Brook) stream reaches. Intermittent reaches were dry at day 0 (start of experiment) and 79 days and flowing at 128 days. Asterisk indicates significant association (ANCOVA) between shredder density and leaf weight loss. Error bars are ± 1 standard error. Note the differences in the size of the y axis between stream reaches.

Table 6.4: Presence of key shredding invertebrate species in Northern Jarrah Forest streams in the leaf bag experiment from 1982-83 (Bunn 1988c) and present day leaf bag assemblages. Bunn (1988c) data includes species designated as “common”, rare taxa not available (FB Foster Brook, SS Seldom Seen Brook). Present-day data includes all occurrences (WF Waterfall Gully, CC Curtis Creek, Ch Chandler Brook). P = present; R = rare (number of individuals given); * = absent in leaf bags, though located in 2016-17 survey data (Carey unpublished data). *Perthia* identification differences due to updated taxonomy. It is unclear whether *Stenochironomus* sp. occurred in Bunn (1988c), however its activity as a leaf miner was not reported = ?; The amphipod *Uroctena* sp. was not collected from leaf bags by Bunn (1988), but was recorded in Bunn (1986; et al., 1986) = •.

Taxon	Bunn 1988		2018-19 experiment			
	FB	SS	WF	SS	CC	Ch
<i>Perthia</i>						
<i>branchialis/acutitelson</i>	P	P	P	P	-	-
<i>Notoperata tenax</i>	P	P	P	-	-	-
<i>Leptoperla australis</i>	P	P	-	*	*	-
<i>Newmanoperla exigua</i>	P	P	-	-	-	-
<i>Lectrides parilis</i>	P	P	R(4)	R(2)	-	-
<i>Stenochironomus</i> sp.	?	?	P	P	P	P

Procedural control experiment

The mean weight remaining in leaf bags tethered to bricks only differed from flat packs at one sampling time (September: mean brick weight $5.43 \text{ g} \pm 0.22$; flat pack weight $6.16 \text{ g} \pm 0.22$; $P = 0.011$, Figure 6.5). Macroinvertebrate assemblages also did not differ between brick tethered or flat packs ($R = 0.0006$, $P = 0.6$; Figure 6.7a), although as expected mesh sizes showed different assemblages ($R = 0.445$, $P < 0.001$; Figure 6.7b) for both methods across all sample times. Since results from these two methods only differed at one time (and not at the end of the experiment), direct comparisons between the 1982-83 data and the 2018-19 data were valid.

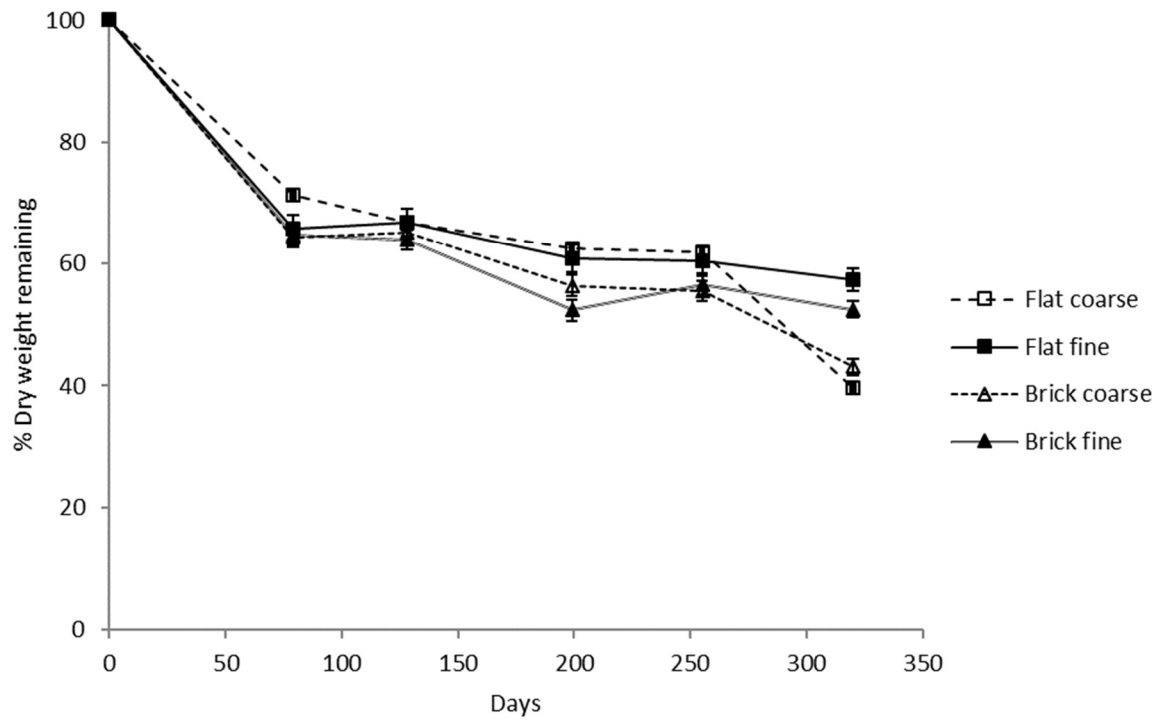


Figure 6.6: By the end of the experiment, breakdown of *Eucalyptus marginata* leaf litter at Waterfall Gully did not differ between leaf bags tethered to house bricks and those attached flat to the substrate on a sheet of wire mesh; but differed between mesh sizes because fine mesh retained more leaf material than coarse mesh bags. Error bars ± 1 SE.

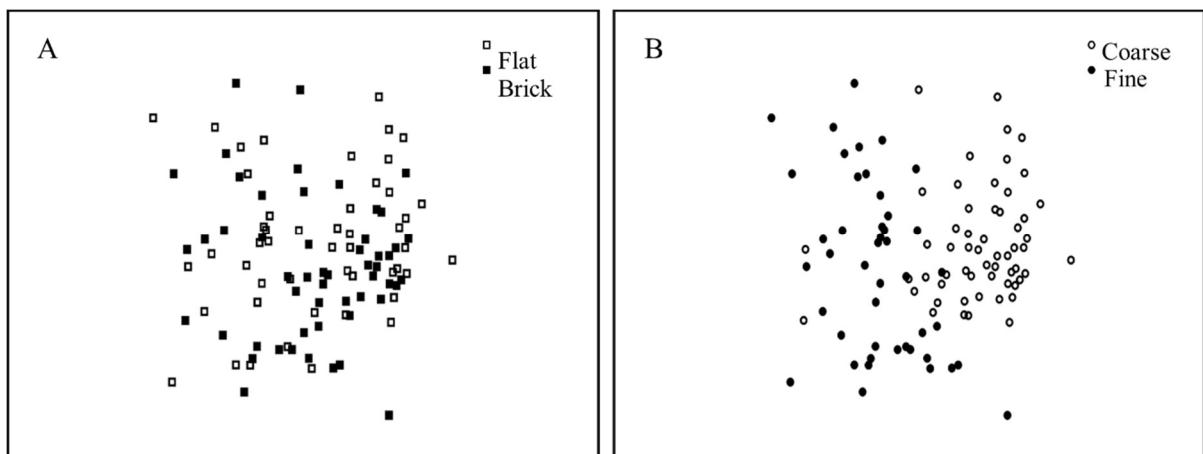


Figure 6.7: Macroinvertebrate assemblages inhabiting leaf bags did not differ between (A) methods (brick tether and attached flat), whereas (B) coarse and fine mesh bags harboured different assemblages. Stress = 0.19.

Discussion

Differences in E. marginata leaf litter decomposition between perennial and formerly-perennial streams

For approximately four months, leaf bags at intermittent Curtis Creek and Chandler Brook were exposed on the dry streambed, whilst bags were immersed in perennially flowing Waterfall Gully and Seldom Seen Brook. At this time, leaf decomposition differed markedly between the flow regimes. We had hypothesised (H_1) that, because of the role of flow in leaching and abrasion (Von Schiller et al., 2017; Datry et al., 2018), decomposition would accelerate when flow resumed, resulting in similar overall losses between intermittent and perennial reaches by the end of the experiment. Leaf decomposition did accelerate upon immersion, but losses in Curtis Creek and Chandler Brook caught up to those in perennial streams at different times (at 320 and 255 days respectively). Nevertheless, the delay in immersion did not change the ultimate outcome for leaf decomposition at the end of the experiment.

Terrestrial processes occurring on dry streambeds precondition leaves to readily lose mass when flows resume in winter (Gessner et al., 1999; synthesis in Datry et al., 2017): in this study leaf bags sitting on a dry streambed lost around 10% of their initial mass within the first 3 months. On terrestrial soils and dry streambed sediments, moisture availability, soil/sediment composition, and ultraviolet irradiation all influence organic matter decomposition (del Campo & Gomez, 2016; Abelho & Descals, 2019; Shumilova et al., 2019). Photodegradation directly mineralises some organic compounds and may play a role in increasing lability of phenolic compounds in the leaf, facilitating their removal (Austin & Vivanco, 2006; King et al., 2012; Qiu et al., 2013). Ground moisture is probably the main determinant of terrestrial decomposition rate, and in SWA stream beds are generally very dry over the summer (when riparian trees shed leaves). *E. marginata* leaves on the forest floor

break down very slowly (29% mass in six months: O'Connell & Menagé, 1983), but rapidly leach soluble compounds when immersed. So, while the delayed onset of wet-season rainfall now exposes leaves on dry streambeds to a longer dry period, a significant proportion of mass (around 27% estimated by O'Connell & Menagé, 1983) disappears very quickly. Hence the ultimate fate of these highly refractory leaves may not be greatly affected by stream intermittency.

The contribution of microbial communities to litter decomposition was not quantified directly in this study. However, other studies have shown that the growth of fungal hyphomycetes and microbes may be negatively impacted by emersion (e.g. Bruder et al., 2011; Abelho & Descals, 2019). Post-drying recovery of microbial decomposers may lag behind that of intermittent streams where perennial streams are exposed to drying (e.g. Mora-Gomez et al., 2018; cf. Timoner et al., 2012; Foulquier et al., 2015). It is possible that a change in the microbial assemblages on leaf litter associated with intermittent flow regimes may have affected other components of the decomposition process, such as the quality of the food source for shredding macroinvertebrates. However, evidence from the fine mesh bags suggests that decomposition unrelated to shredder activity was maintained at similar levels in intermittent and perennial streams.

Perennial Waterfall Gully lost the highest amount of leaf weight due to the activity of shredding caddisflies (*N. tenax*) which arrived towards the end of the experiment; but these were absent from the other streams. Consequently, both Hypothesis 3a - shredding macroinvertebrates would colonise leaves in now-intermittent reaches and consume a similar amount of material; and Hypothesis 3b – coarse mesh bags would lose more material than fine mesh bags, were rejected. In perennial Seldom Seen Brook, fine mesh bags weighed less during the experiment than coarse mesh bags due to higher numbers of leaf-mining Chironomidae (*Stenochironomus* sp.) in fine mesh bags. *Stenochironomus* sp. burrows under

the leaf cuticle, growing inside the leaf until pupation. Fewer than ten larvae can almost completely consume the inner contents of a jarrah leaf, leaving behind a “paper bag” of softened cuticle and FPOM. It appears likely that *Stenochironomus* sp. was more abundant in fine mesh bags because the bags created a predation refuge. While this situation was an artefact of the leaf bag construction, this is the way *Stenochironomus* sp. processes leaves in natural leaf packs. Therefore, it appears likely that leaf-mining chironomids are contributing to leaf decomposition in Seldom Seen Brook and the absence of shredding Trichoptera may have increased the relative importance of this activity for the decomposition process.

Most knowledge of the ecology of leaf mining *Stenochironomus* larvae comes from Brazil, where they are typical of Atlantic forest streams (Serpa et al., 2020). de Mello Cioneck et al. (2021) found that they colonised fine and coarse mesh bags, that they were an important shredder whose densities correlated with leaf decomposition rates, and that they were an indicator of less-degraded streams. They also observed that more ‘typical’ shredder taxa (e.g. caddisflies) were rare in their study streams. The streams used for our experiment in 2018-19 are in good condition aside from flow regime change. Potentially, *Stenochironomus* larvae have replaced some of the shredder taxa reported by Bunn (1988c: caddisflies, stoneflies) at least partially compensating for their loss. Although the absence of a significant correlation between shredder density and leaf weight loss in Seldom Seen Brook raises questions about their overall contribution to leaf decomposition.

Intermittency was not associated with clear differences in leaf bag fauna. Instead, assemblages differed between stream reaches and mesh sizes and the size of these differences was similar. The fine mesh excluded most invertebrate taxa except for some Chironomidae, Ceratopogonidae and Oligochaete species that were able to get through the fine mesh as juveniles. In perennial Waterfall Gully, the predatory trichopteran *T. pallescens* entered fine mesh bags as an early instar and presumably preyed upon the Chironomidae larvae, but in the

other streams, fine mesh bags likely acted as a predation refuge for these worm-bodied invertebrates. Temporal patterns of change in invertebrate assemblages on coarse mesh bags differed among stream reaches. Of the two perennial reaches, Waterfall Gully showed changes in the fauna from early in the decomposition process. In contrast, assemblages in Seldom Seen Brook differed as species richness changed with season, with more species present in the warmer than in the cooler months. Again, mayflies were abundant together with amphipods and chironomids, but the dominant mayfly species differed between the perennial reaches.

In the intermittent reaches, there was little temporal change in assemblage composition. In Curtis Creek, amphipods, chironomids and larval Tipulidae dominated the assemblages and mayflies were less common. The fauna in Chandler Brook was even less diverse and dominated by Chironomidae, and although their numbers fluctuated through time, there was no clear pattern. Furthermore, there was no difference between mesh sizes in this stream (Chironomidae colonised fine mesh bags as first instar larvae).

Analysis of covariance showed a strong relationship between shredder density and leaf weight loss in one perennial (Waterfall Gully) reach and a weaker (but significant) relationship in one intermittent (Curtis Creek) reach but there was no relationship in the other two streams. Rather than being a direct consequence of flow regime, the existence of this relationship appears to depend on the fauna present in a reach. Flow regime determines which species can survive in a stream, so it may correspond with a more or less diverse and abundant shredder guild. However, as perennial Waterfall Gully and intermittent Curtis Creek had more diverse shredder guilds than perennial Seldom Seen Brook or intermittent Chandler Brook, there does not seem to be a direct relationship between flow regime and shredder diversity.

Differences in leaf litter decomposition between perennial streams before and after 35 years of climatic drying

Overall, mean leaf weights remaining after 320 *cf* 327 days and litter decomposition pattern changed little in perennial streams in the NJF over 35 years, despite declines in total annual discharge. Therefore, little had changed in the control streams. Somewhat surprisingly, mean leaf weight remaining in the intermittent streams also did not differ from that in perennial streams in 2018-19. It appears that the outcome of decomposition processes on leaf weight loss was not altered by the transition to intermittency. However, the loss of shredding macroinvertebrate species (notably large larval caddisflies and stoneflies) from all streams has likely reduced the role of shredders in leaf decomposition. The appearance of leaf-mining Chironomidae in some streams may have partially compensated for this loss, however, the lack of difference between all the stream reaches in both time periods (1982-83, 2018-19) suggests that other factors that we did not quantify, such as microbial activity, may have compensated for the loss of shredders.

The invertebrate assemblage composition in perennial streams in 1982-83 was not substantially different to the composition in perennial Waterfall Gully in 2018-19, but in now-intermittent Curtis Creek and Chandler Brook, assemblage composition changed greatly from when they were perennial (Bunn, 1985). In Seldom Seen Brook, although leaf bags lost a similar amount of weight by the end of the experiments in 1983 and 2019, the fauna colonizing the bags had changed because of the loss of trichopteran shredders and their replacement by leaf-mining *Stenochironomus* sp. During the 1980s, all the major shredding taxa active at Foster Brook were also present at Seldom Seen Brook although in lower numbers. Interestingly, *Stenochironomus* was not reported by Bunn (1986; et al., 1986; 1988c), although it may have been listed as a morphospecies. However, even if it was present, it was certainly not as abundant in Bunn's study, as its leaf mining activity would have been observed and discussed. The contribution of *Stenochironomus* sp. to litter

decomposition may increase as a secondary effect of the increasingly drying climate, due to its short life cycle and higher tolerances to temperature and low dissolved oxygen (compared to leptocerid Trichoptera) and potentially to predator release, given the low numbers of predators recorded on the leaf bags in 2018-19.

The replacement of trichopterans with dipterans as leaf processors provides some evidence of redundancy in these stream ecosystems: a drought-tolerant dipteran has replaced intolerant trichopterans and sustained the leaf decomposition. More broadly, although the invertebrates have changed, leaf weight loss in the intermittent streams caught up to loss in the perennial streams, showing that decomposition is still occurring at similar rates in the intermittent streams. This is reassuring from the point of view of maintaining ecosystem processes. However, given that strongly desiccation-tolerant shredding Leptoceridae are known in headwater streams in southern Australia (Wickson et al., 2012), and there are desiccation-tolerant Leptoceridae (*Triplectides australis*) found in wetlands on the nearby coastal plain (Chester et al, 2013), it is puzzling that this important segment of the shredder guild is absent from these intermittent streams. This requires further investigation.

Bunn (1988c) found that shredding macroinvertebrates were responsible for approximately 26% of *E. marginata* litter processing in NJF streams. Additionally, *N. tenax*, and the stoneflies *Leptoperla australica* and *Newmanoperla exigua* were the most abundant shredders and were widespread within the Wungong Catchment at the time (Bunn 1986; Bunn et al., 1986). No stoneflies were collected during this study, however *Leptoperla australica* was patchily distributed in low numbers at Seldom Seen Brook and Curtis Creek in 2016-17; *Newmanoperla exigua* was not found from 2016-2019. Two amphipods, *Uroctena* spp. (Paramelitidae), and *Perthia acutitelson* (Perthiidae) were present on leaf bags in all streams in 2018-19. *Perthia* was listed as a shredder in Bunn (1985, 1988c) but *Uroctena* (although present) was not. Their year-around presence did not affect leaf weight loss, suggesting that

in these streams *Uroctena* were consuming FPOM and using the experimental leaf bags as shelter.

Bunn (1985, 1988c) described leaf breakdown in perennial NJF streams as a three-stage process. The first stage, leaching, has only changed in timing between perennial and intermittent streams, creating no delay in overall processing. In the second stage, in summer-autumn, little leaf weight loss occurred which Bunn attributed to the thick waxy cuticle and high polyphenol content of jarrah leaves slowing microbial colonisation and thus their palatability for shredders. He found that with the onset of winter rains, increased abrasion likely accelerated microbial growth and increased palatability, leading to a third stage of more rapid decomposition. In intermittent streams, the summer-autumn ‘slow’ period occurs while they are dry, and our results show that this does not ultimately inhibit weight loss. So, one reason for the lack of difference in leaf weight loss between Bunn’s study and ours could be that the timing of stream drying coincides with a period in which not much leaf decomposition occurs even in flowing streams. However, Bunn (1985, 1988c) did find that invertebrate shredders did contribute significantly to leaf decomposition, with approximately 25% more weight lost from coarse than fine mesh bags in his study. He found that shredders did not appear on leaf bags until winter, and that the timing of their larval life stage was ideal for using well-conditioned jarrah leaves in winter and spring. These species are now either rare (in perennial reaches) or absent (from intermittent reaches) in NJF streams.

Conclusions

In some studies, loss of shredding caddisflies had more effect on leaf litter decomposition than a temperature increase of 5 °C, even in multispecies assemblages (Boyero et al., 2014; also see Boyero et al., 2007). The loss of climate-sensitive shredders was found to have substantial impacts on litter processing where shredder assemblages had higher diversity and

biomass (Wenisch et al., 2017). Currently, evidence of the importance of shredder species identity to organic matter processing in stream environments, and the impacts of species loss, is equivocal. The circumstances under which loss of shredding species strongly affects litter decomposition are not yet clear. So, they should not be assumed to be minimal nor should it be assumed that they will necessarily be replaced by drought-tolerant species with faster life cycles.

As far as we are aware, this study is the first to measure changes in decomposition of *Eucalyptus* leaves in streams that have undergone flow regime change due to climatic drying. Using a partial mBACI design we have shown that transition to intermittency slowed litter decomposition early in the process, but that leaf loss later caught up to perennial reaches. This “catch up” happened despite the obvious declines in historically occurring shredder fauna. Decomposition processes in these streams appear to be robust, but further studies of a larger number of streams are needed to show whether this is generally the case. We urge researchers in regions where drying has not yet caused a transition to intermittency, to gather baseline data on stream processes (not just patterns) and to do this at regular intervals to provide a better description of how processes are changed through time. In SWA, few streams remain perennial and this presents an enormous challenge for study design. For example, Waterfall Gully is the only stream that remains perennial along its entire length in the Wungong Catchment. This reinforces the need to collect baseline data before the (predicted) widespread transition to intermittency in other regions.

As evident here, there are challenges in using pre-existing, pre-global change data to identify the effects of particular impacts. We were able to use a procedural control experiment to show successfully that the leaf bag tethering method used by Bunn (1988c) did not give different results to the flat pack method we used. Similarly, *E. marginata* forest has been well-studied, so we could show that it was unlikely that leaf composition had changed due to

climate change (before leaves entered streams). Also, our design partitioned this variation into the before-control, after-control comparison, which showed no differences, demonstrating that any changes in leaf composition had little effect on decomposition in-stream. Other challenges may be more difficult to deal with. Destruction of sites potentially useful for “before and after” data, as occurred following Bunn’s study (to Foster Brook), is likely to pose a considerable problem, given the global expansion of urbanisation, agriculture and water-use infrastructure is likely to have destroyed sites for which there is pre-impact data. Furthermore, data collected for pre-climate change studies is likely to have been collected for a specific purpose. In many cases, sites will have been studied because they were in some way impaired making their use as ‘before’ sites challenging. This is especially the case where replication at ecosystem scales was historically low, as it was here, with leaf bags placed in only two stream reaches (Bunn, 1988c).

Nevertheless, an increasing number of studies are making use of pre-existing, pre-global change data to identify the effects of a variety of global changes on species and both terrestrial and aquatic ecosystems (Table 6.5). These studies show a range of changes in species distribution and life histories, but they also show some circumstances where global change has not entailed a corresponding change in ecology, at least for some species (e.g. Matsuda et al. 2018, Lavery et al., 2020, LeMoine et al. 2020, this study, Table 6.5). It may be difficult to accurately predict the outcome of global change impacts on species, as not all species responded as predicted. An increasing number of these types of studies, complemented with experimental studies, should improve the quality of predictions of the outcomes of global change.

Table 6.5: Examples of papers comparing datasets from before and after globally important changes across at least two decades, taken from Global Change Biology in 2018 to 2021.

Response measured	Stressor	Timespan	Summary of results	Reference
Ground crickets in Japan	Increased temperatures (global warming) temperatures for embryonic development present at 31-35 ° latitude in 1970s now present at 36 °latitude.	1969-1976 <i>cf.</i> 2015-2017 40+ years	Life history of ground crickets did not change over time but shifted northwards with temperature change.	Matsuda et al. (2018) Northward expansion of the bivoltine life cycle of the cricket over the last four decades. <i>Global Change Biology</i> , 24, 5622-5628.
Terrestrial vegetation site occupancy (Canada)	Increased temperatures (global warming)	1972 <i>cf.</i> 2015 43 years 1970 <i>cf.</i> 2012 42 years 1973 <i>cf.</i> 2016 43 years	In locations with more warming, there was more movement of plants across an elevational gradient. Increasing magnitude of change in species richness and composition from west to east.	Becker-Scarpitta et al. (2019) Four decades of plant community change along a continental gradient of warming. <i>Global Change Biology</i> , 25, 1629-1641.
Invertebrates in alpine streams in European Alps	Increased temperatures (global warming) and flow regulation	1977-79 <i>cf.</i> 1996-2015 19 – 38 years	Cold-adapted species declined and thermophilic species became more widespread.	Bruno et al. (2019) Structural and functional responses of invertebrate communities to climate change and flow regulation in alpine catchments. <i>Global Change Biology</i> , 25, 1612-1628.
Composition of breeding birds and latitudinal distribution of birds in 2 deserts in North America.	Increased temperatures (global warming)	1908-1948 <i>cf.</i> 2013-16 >60 years	Reduced occupancy, range contractions, change in community structure, limited range expansion of species from the warm, desert into the cooler one. Both deserts now support more species tolerant of drier, warmer conditions.	Iknayan & Beissinger (2020) In transition: Avian biogeographic responses to a century of climate change across desert biomes. <i>Global Change Biology</i> , 26, 3268-3284.
Mammal abundance and diversity in Solomon Islands	Logging of tropical forest	1992 <i>cf.</i> 2014-15 20+ years	Widely distributed generalist species did not decline in response to logging but short-range endemic species did.	Lavery et al. (2020) Ecological generalism and resilience of tropical island mammals to

				logging: A 23-year test. <i>Global Change Biology</i> , 26, 3285-3293.
Habitat occupancy by 4 freshwater fish species across 280 sites	Increased temperatures (global warming)	1993-1995 <i>cf.</i> 2011-2013. ~ 20 years	Small-bodied cool-water fish sp showed 48% decline in site occupancy, small-bodied warm water species expanded range. One large-bodied salmonid showed a small % decline, other sp. showed no change	LeMoine et al. (2020) Landscape resistance mediates native fish species distribution shifts and vulnerability to climate change in riverscapes. <i>Global Change Biology</i> , 26, 5492-5508.
Composition of drifting insects in streams in Hong Kong	Increased temperatures (global warming)	1983-84 <i>cf.</i> 2013-16 30+ years	Warming was associated with the loss of mayfly taxa and an increase in dipterans.	Dudgeon et al. (2021) Shifts in aquatic insect composition in a tropical forest stream after three decades of climatic warming. <i>Global Change Biology</i> , 26, 6399-6412.
Decomposition of leaves and invertebrate assemblages on leaf bags.	Change in flow regime from perennial to seasonal flows	1982-83 <i>cf.</i> 2018-19 36+ years	Decomposition (weight lost from leaf bags) did not change through time, but shredder species changed from Trichoptera, Plecoptera to Diptera.	This study

Supplementary material

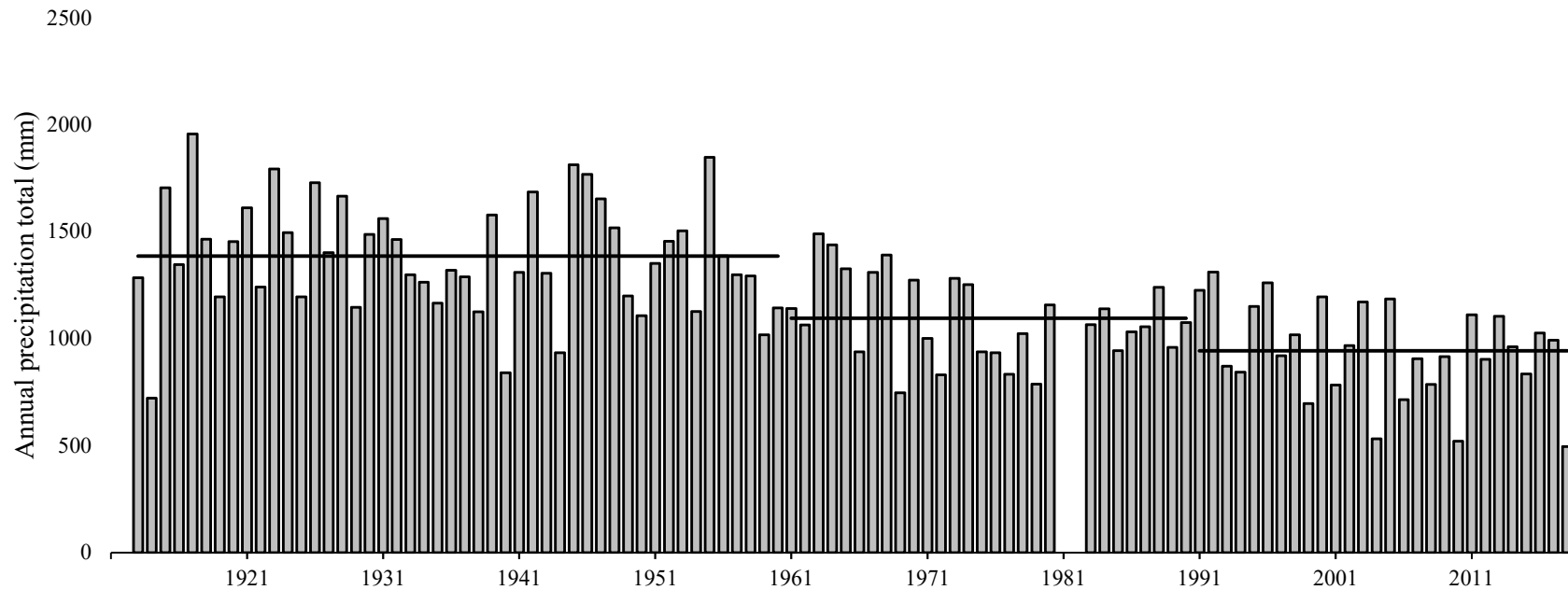


Figure S6.1[2.2]: Annual rainfall total 1913 to 2018, recorded at Wungong Dam. Horizontal bar shows long term average rainfall at the site, 1913 to 1960; 1961 to 1990; and 1991 to 2018. Years 1981 and 1982 are missing data. Chart produced using rainfall data freely available from the Australian Bureau of Meteorology (station number 009044; bom.gov.au/climate/data).

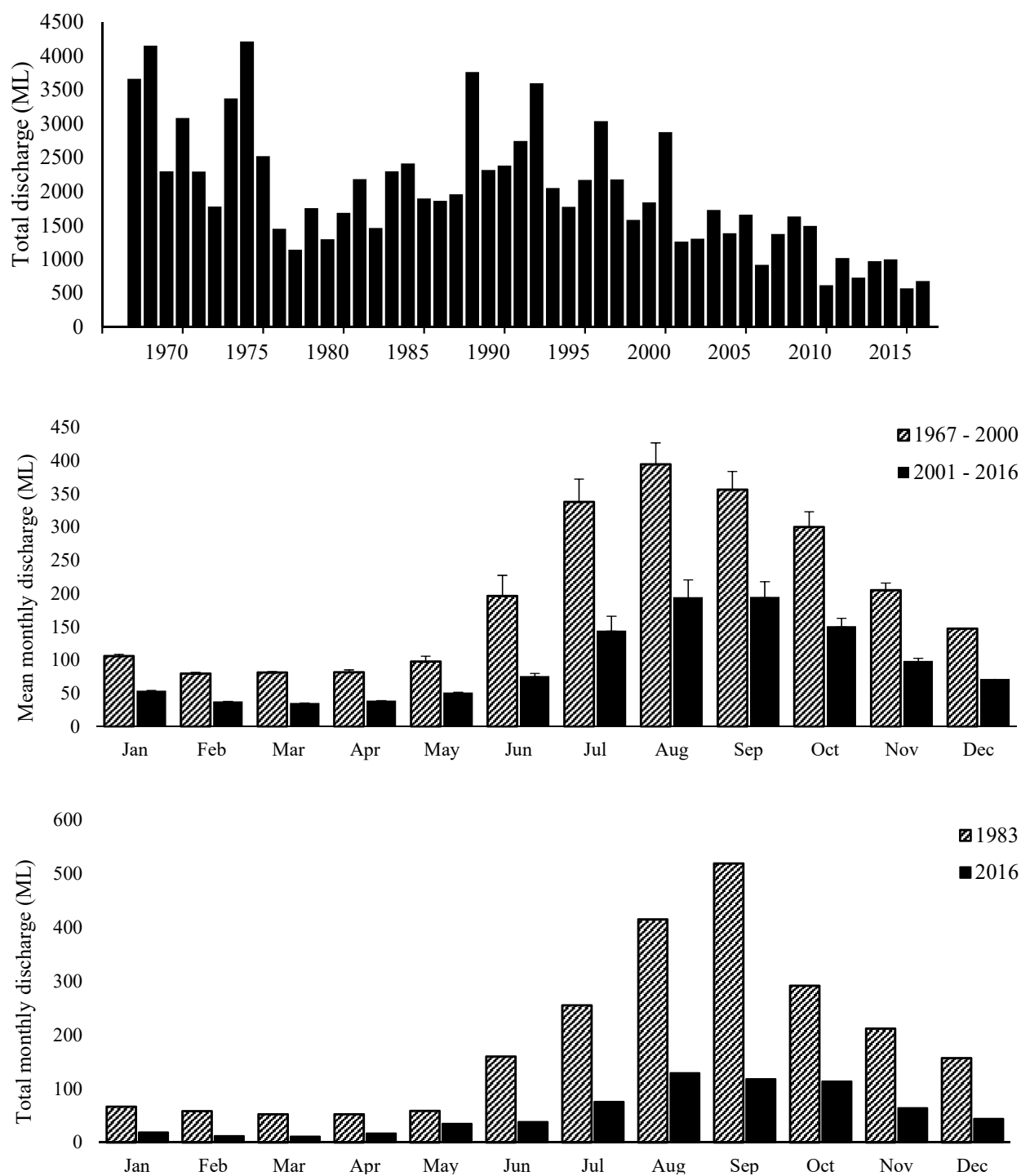


Figure S6.2: Annual total discharge (ML) at Waterfall Gully from 1967 to 2016 (top). Centre: mean ($\pm$ SE) monthly streamflow (ML discharge) at Waterfall Gully, years 1967 to 2000 inclusive, compared to years 2001 to 2016 inclusive. Bottom: total streamflow at Waterfall Gully per month in 1983 (coinciding with Bunn 1988c) and 2016 (most recent year of complete flow data available). All charts produced using daily streamflow measurements, freely available from the Western Australian Department of Water and Environmental Regulation (gauge number 616023; wir.water.wa.gov.au).

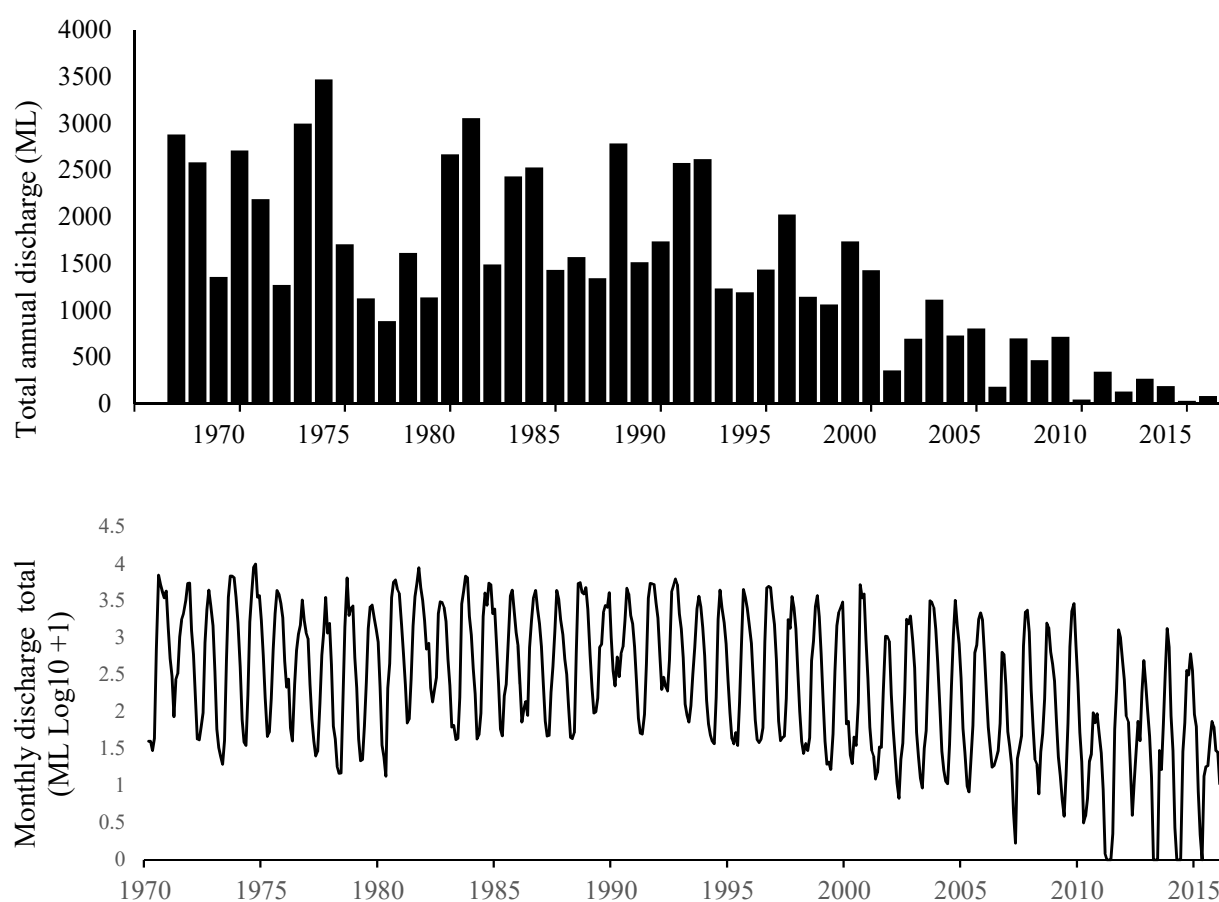


Figure S6.3a: Annual total discharge (ML) at Seldom Seen Brook from 1967 to 2016 (top). Bottom: monthly streamflow totals at Seldom Seen Brook 1970 to 2015 inclusive, log transformed so months of zero flow can be seen (separately from those with very low flows). Seldom Seen Brook became intermittently flowing in 2011, however a spring on one reach maintained a trickle of perennial flow in some years since, including 2018. All charts produced using daily streamflow measurements, freely available from the Western Australian Department of Water and Environmental Regulation (gauge number 616021; wir.water.wa.gov.au).

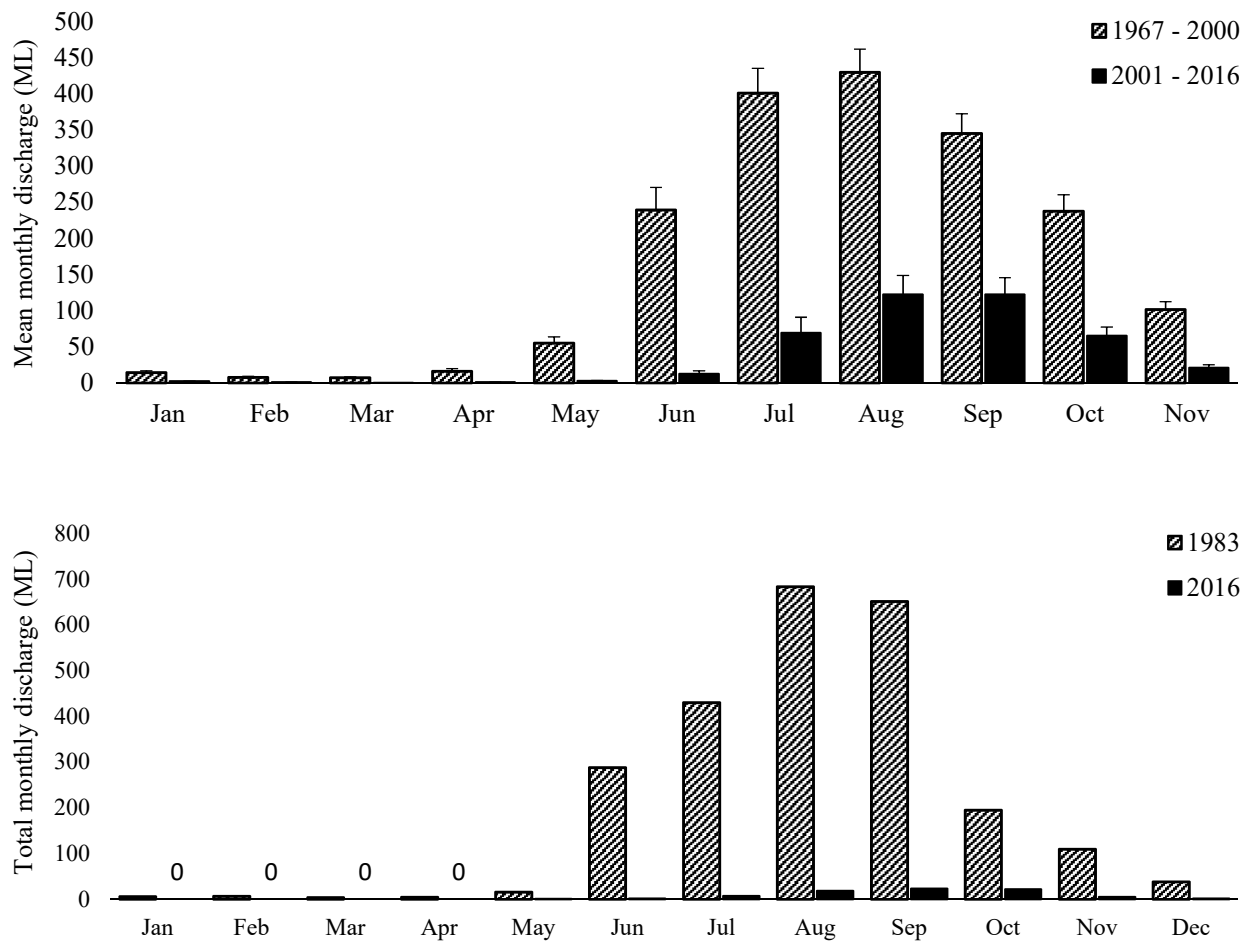


Figure S6.3b: Mean ($\pm$ SE) monthly streamflow (ML discharge) at Seldom Seen Brook, years 1967 to 2000 inclusive, compared to years 2001 to 2016 inclusive (top). Bottom: total streamflow per month at Seldom Seen Brook in 1983 (coinciding with Bunn 1988c) and 2016 (most recent year with complete flow data available). Data labels given for months with zero flow. All charts produced using daily streamflow measurements, freely available from the Western Australian Department of Water and Environmental Regulation (gauge number 616021; wir.water.wa.gov.au).

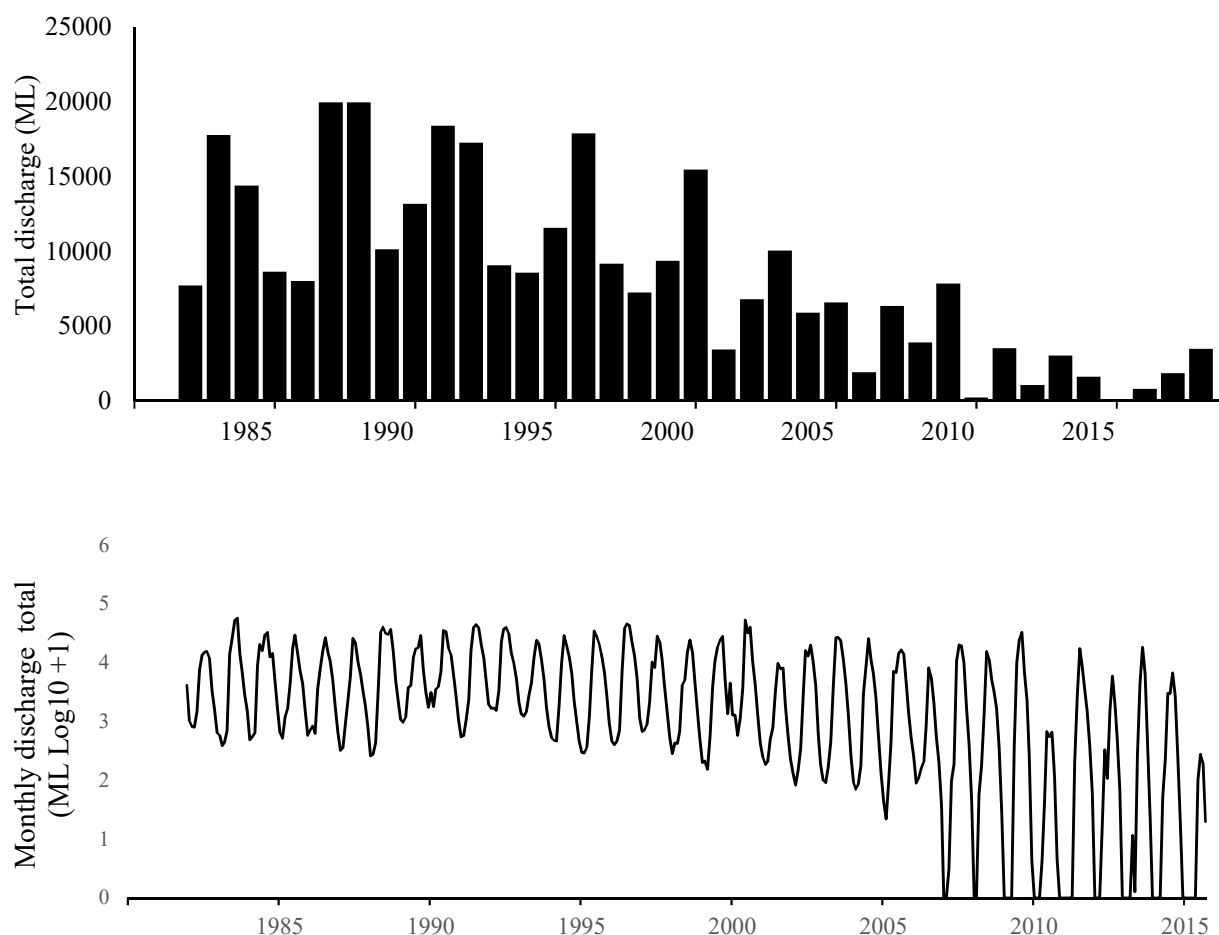


Figure S6.4a: Annual total discharge (ML) at Wungong Brook from 1982 to 2016 (top). Bottom: monthly streamflow totals at Wungong Brook 1982 to 2015 inclusive, log transformed so months of zero flow can be seen (separately from those with very low flows). Wungong Brook became intermittently flowing in 2007 and has flowed intermittently every year since (correct at 2019). All charts produced using daily streamflow measurements, freely available from the Western Australian Department of Water and Environmental Regulation (gauge number 616021; wir.water.wa.gov.au).

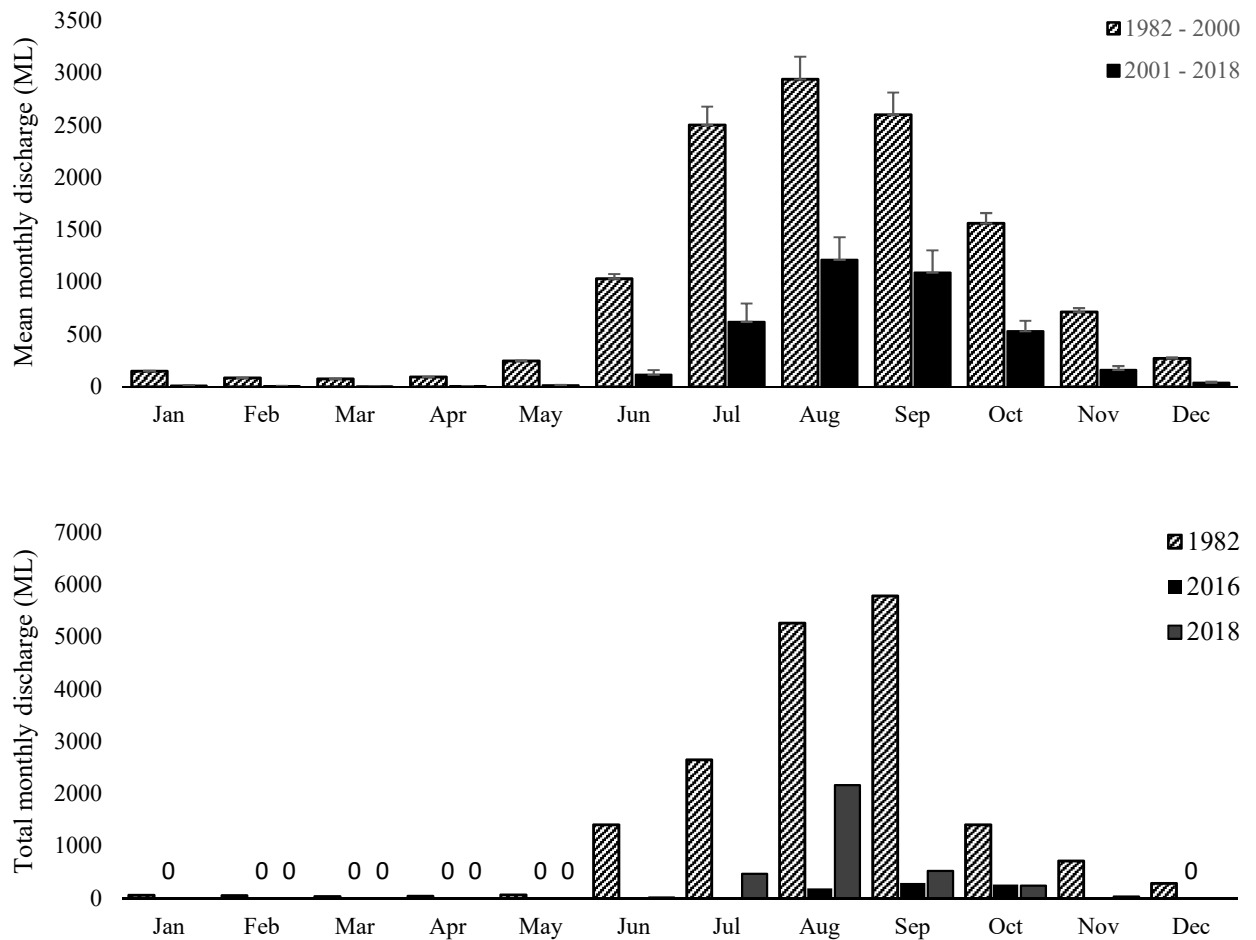


Figure S6.4b: Mean ($\pm$ SE) monthly streamflow (ML) at Wungong Brook, years 1967 to 2000

inclusive, compared to years 2001 to 2016 inclusive (top). Bottom: total streamflow at Wungong per month in 1983 (coinciding with Bunn 1988c), 2016 (aligning with data availability at Waterfall Gully and Seldom Seen Creek) and 2018 (most recent year with complete data). Data labels show months given for months with zero flow. By comparison to post- 2000 stream flows, 2018 was a high flow year, interannual variability that is typical in the region, however this also shows the downward shift in range of streamflow volumes. All charts produced using daily streamflow measurements, freely available from the Western Australian Department of Water and Environmental Regulation (gauge number 616041; wir.water.wa.gov.au).

Chapter 7

Dry season refuges, refugia and flow-regime change in southern Australian streams.

The following chapter is drafted in accordance with the journal *Freshwater Biology*.

The manuscript is currently prepared for submission.

The following authors contributed to this manuscript as outlined below:

	Author order	Concept & study design	Fieldwork	Laboratory work	Data analysis	Writing manuscript	Review/edit paper
Nicole Carey	First	X	X	X	X	X	X
Edwin Chester	Second	X	X				X
Scott Strachan	Third	X	X				X
Belinda Robson	Fourth	X			X		X

By signing this document, the Candidate and Principal Supervisor acknowledge that the above information is accurate and has been agreed to by all other authors.

Candidate

Principal Supervisor

Abstract

1. Climate change is causing stream flow regimes to commence drying in many regions globally. This includes southwestern Australia (SWA) which has already experienced widespread loss of perennial streams. In drier landscapes, ecological refuges and evolutionary refugia are expected to become critical for conserving freshwater biodiversity. This study aimed to characterise dry season refuges in a forested headwater catchment recently impacted by severe permanent drying, to determine the contribution of refuges to species persistence and community recovery in recently intermittent streams.
2. During the dry seasons of 2017 and 2018, dry season refuges within recently intermittent streams were located and their macroinvertebrate assemblages sampled. Refuges included small spring-fed pools, perched receding pools and subterranean refuges associated with granite inselbergs. There were no hyporheic refuges present. Dry sediments (inorganic sediments and organic matter) were collected from each intermittent stream for rehydration, and a subsample of inorganic sediments were aerated to break dormancy of species that require flow.
3. Mantel tests were used to compare the contribution of refuges to species assemblages in the early wet season, when intermittent streams had begun to flow. Analysis of similarities was used to compare patterns between dry and wet season assemblages of intermittent streams and the single remaining perennial stream in the catchment, to infer contribution of the perennial stream to community recovery in intermittent streams.
4. Refuge types had very different species assemblages: spring-fed refuges supported several locally endemic species, but perched pools were dominated by opportunistic

colonists. Several taxa were found aestivating in dry sediments, primarily adult Coleoptera and larval Chironomidae. Inselberg springs supported small populations of endemic Trichoptera and Ephemeroptera and are a potential subterranean evolutionary refugium for an endemic amphipod. However, dry season refuges did not significantly contribute to community recovery. Rather, assemblages in early winter flows were similar to those inhabiting the perennial stream, showing that recolonisation from perennial streams is likely the primary process of community recovery in these newly intermittent streams.

5. Dry season refuges allowed a few locally endemic species to persist and supported a suite of colonizing species formerly absent or rare when all streams were perennial. However, continued loss of perennial streams in SWA will result in landscape wide reductions in local diversity as colonisation sources are lost, because there are no evolutionary refugia in this landscape for relictual species to retreat into.
6. Refuges will likely play a pivotal role in persistence of species able to use them, however the existence of refuges is also under threat from intensifying climatic drying. Thus, identification and protection of future evolutionary refugia is a matter of urgency in regions facing drier climates, as it is ultimately evolutionary refugia that will become critical reservoirs of freshwater biodiversity as perennial streams and refuges are lost.

Keywords

Climate drying, inselbergs, intermittent streams, IRES, refugia, drought refuges

Introduction

There is increasing interest in the roles of both evolutionary refugia and ecological refuges as climate change progresses across most biomes, because of their potential to stem biodiversity

loss and increase the resilience of ecological communities (Davis et al., 2013; Bogan et al., 2017). Evolutionary refugia operate on the scale of millennia, being places that organisms retreat to and persist in under changing climates. Refugia may become perpetually isolated in time and space unless climatic change again produces similar environmental conditions across the wider landscape (Keppel et al., 2012). Ecological refuges are places that offer protection from disturbance or adverse biotic interactions that operate at the scale of minutes to decades (Keppel et al., 2012). In the context of freshwaters in dry landscapes, evolutionary refugia tend to be permanent, groundwater fed water bodies, decoupled from local precipitation (Davis et al., 2013). Ecological refuges may also consist of permanent aquatic habitats however, they are often intrinsically linked with prevailing precipitation and are thus more variable in time and space than genuine refugia (Davis et al., 2013). Freshwater refugia typically harbour relictual short-range endemic species that cannot disperse to other habitats in the broader landscape, making these habitats critically important to the persistence of these species; while ecological refuges are habitats that support aquatic fauna through the dry season and facilitate recolonisation upon resumption of favorable conditions (Sheldon et al., 2010; Robson et al., 2013). Many regions of the globe are predicted to become drier as climate change progresses, causing drier flow regimes and transition of perennial streams to intermittency (e.g. Crabot et al., 2021; Zipper et al., 2021). Thus, habitat that harbours species over both long (refugia) and short (refuges) time scales will become more important to species persistence. However, the ways in which these habitats function are still poorly understood, making the mitigation of threats from anthropogenic stressors difficult.

Dry season refuges are a critical component of species resilience and recovery in intermittent rivers and ephemeral streams (IRES) that undergo regular predictable drying (typical of mediterranean and arid climates; Robson et al., 2013; Bogan et al., 2017). IRES provide spatially and temporally dynamic patchworks of flowing or standing water and dry states

(Datry et al. 2017a). They undergo regular drying along part or all of their length, a process that crosses several biotic thresholds with which resident organisms must contend in order to persist. Initially, flows cease and streams become a series of longitudinally disconnected pools, followed by intensification of environmental and biotic conditions as pools recede. Some pools dry completely and the hyporheic zone retracts, so that vertical connectivity is lost. The final stage is loss of streambed moisture and desiccation of the hyporheic zone (Boulton, 2003). At each stage of drying, the presence of refuges (either within or external to the stream channel) offers respite to organisms that cannot withstand drying and facilitates recolonisation by those species upon resumption of flow (Robson et al., 2013; Bogan et al., 2017). Different types of dry season refuge play differing roles in promoting persistence of organisms and biodiversity in IRES, interacting with the resilience (e.g. ability to disperse) or resistance traits (physiological tolerances, abilities to withstand drying *in situ*) of species (Bogan et al., 2017; Table 7.1). Therefore, species occupancy of a refuge will depend on connectivity and conditions within the refuge, and the species' ability to disperse into the refuge and survive there (Bogan et al., 2017; Robson et al., 2013, Chester et al., 2015).

Understanding the relationship between geology, microclimate and refugia may improve prediction of the occurrence of refuges and refugia under changing climates (*sensu* Keppel et al., 2012) and may assist conservation of critically important habitats in IRES. The relationship between geology and refuges is demonstrated in some cases, for example in the presence of hyporheos, which may be freely connected to underlying superficial aquifers, perched above a confining layer, or absent due to underlying lithology (White, 1993; Malard et al., 2002). Geology is also a fundamental determinant of the presence of surface water refuges and refugia, with features such as rocky ranges and cracks in rock aquifers determining presence of springs and permanent pools, which are known to be highly important habitats in arid zones (Davis et al., 2013; Bogan et al., 2014; Davis et al., 2017;

also see Springer & Stevens, 2008). In granitic landscapes, the degree of exposure of outcropping bedrock among the regolith (weathered material) determines local water balance, with areas of regolith acting as free aquifers and the weathering front acting as an impermeable layer (Centeno et al., 2010). Higher outcrop to regolith ratio increases the amount of runoff and water stored in regolith, creating a water balance equivalent to higher rainfall irrespective of prevailing climate (Centeno et al., 2010). In terrestrial plants, this influence on the water balance has been shown to extend the ranges of mesic species at arid margins (Hopper et al., 1997; and the opposite McGann 2002), thus outcrops are associated with very high floristic diversity (Yates et al., 2019). Whilst the presence of geologically determined aquatic habitats in arid zones (e.g. outcrop springs, rocky pools) have been shown to exert strong influences on aquatic fauna (Davis et al., 2017; Davis et al., 2018), the relationship between geology and aquatic refuges in non-desert landscapes undergoing climatic drying has not yet been explored.

The mediterranean climate region of southwestern Australia (SWA) has been drying for at least five decades (Bates et al., 2008) and as a result, formerly common perennial headwater streams have transitioned to intermittent flow regimes (Petrone et al. 2010; Hughes et al., 2012; Kinal & Stoneman, 2012). The stream fauna in SWA are primarily Gondwanan relicts, having evolved in cooler wetter times (Davies 2010; Davies & Stewart, 2013), and the transition to intermittency has negatively impacted this fauna (Chapters 3 & 5). The presence and use of refuges in this drying landscape may be particularly important in facilitating survival of relictual species under climate change (Davis et al., 2013). Thus, we aimed to locate dry season refuges in formerly perennial, now-intermittent headwater streams, and identify the aquatic macroinvertebrate species using them. We compared the species composition of refuges to the assemblage present during the flowing phase to determine the relative importance of *in situ* dry season refuges compared with recolonisation from perennial

streams for maintaining biodiversity in newly intermittent streams. We expected that the range of drought refuges used by macroinvertebrates in the Wungong Brook catchment would be similar to those recorded in other southern Australian IRES with relictual fauna (e.g. Chester & Robson, 2011; Mackie et al., 2013; Chester et al., 2015). Species persistence in intermittent streams is dependent on their life history adaptations to drying, grouped as either resistance or resilience traits (Bogan et al., 2017). Resistance traits enable a species to persist *in situ* as a desiccation resistant life stage, commonly desiccation resistant reproductive cysts, or aestivation of aquatic obligate life stages. Resilience traits facilitate recolonisation upon return of flows, including the ability to retreat to perennial or subterranean refuges, or a long-lived terrestrial adult stage (Strachan et al., 2015). Southern Australian streams tend to have a resilient fauna, whereby recolonisation from perennial refuges is more prevalent than *in situ* desiccation resistance, however a range of drought survival strategies are known to contribute to assemblage recovery in IRES (Chester & Robson, 2011; Robson et al., 2011; see also Lytle et al., 2008; Sheldon et al., 2010; Bogan et al., 2015; Pařil et al., 2019). Therefore, the hypotheses tested were: H₁ species assemblages will differ between refuge types, with spring-fed reaches and pools supporting a higher proportion of species without drought resistance or resilience traits than drying pools and dry sediments; H₂ taxa richness will be highest in the perennial spring-fed reach, followed by perennial spring-fed refuge pools then by drying pools, and it will be lowest in dry sediment, organic matter and aestivating under stones; H₃ different taxa will emerge from organic matter compared with inorganic sediments, and from aerated compared with non-aerated samples, but species richness will not differ between sediment types or aeration states; H₄ there will be a significant concordance between species present during the flowing phases of the streams and species present in refuges during the dry phase; H₅ assemblages within summer refuges would be significantly different to the assemblage in a perennial stream,

because the latter would predominantly contain species without adaptations to intermittency and drought refuge use. These two assemblages would become more similar upon resumption of flow in intermittent streams, as flying insects emerged from the perennial reach and recolonised intermittent streams.

Table 7.1: Synthesis of studies reporting various types of refuges, refugia in IRES.

Refuge type	Study	Location (Climate)	Roles identified	Organism traits	Also see
Evolutionary refugia	Davis et al., 2013	Central Australia (Arid)	Permanent ground water source decoupled from local precipitation patterns Harbour relictual species that cannot survive in the surrounding landscape and do not disperse Can act as a refuge during the dry season, but not strictly a drought refuge	Sufficient powers of dispersal to reach and occupy refugia, and to disperse from the refugia upon return of wet season	Keppel et al., 2012
Springs	Davis et al., 2017	Arid zones	Permanent ground water source decoupled from local precipitation patterns. Harbour relictual species that cannot survive in the surrounding landscape.	High local population sizes.	Murphy et al., 2009; 2010 Rossini et al., 2018
Perennially flowing reaches	Chester & Robson, 2011	Victoria Range, south-eastern Australia (Mediterranean)	Support organisms not adapted to drying Support rheophiles Promote recolonisation throughout stream networks via drift and aerial flight	Resilience traits Longitudinal and lateral dispersal capability	Paltridge et al., 1997; Fritz & Dodds, 2004; Bonada et al., 2007; Crabot et al., 2020
Permanent pools	Chester & Robson, 2011	Victoria Range, south-eastern Australia (Mediterranean)	Support organisms not adapted to drying Along with perennial flowing reaches, act as dry season refuge across broad landscape Promote community resilience and recolonisation	Resilience traits Longitudinal and lateral dispersal capability Withstand lentic conditions	Sheldon et al., 2010; Bogan et al., 2019; Bonada et al., 2020
	Banega-Medina et al., 2021	Central-southern Chile (Mediterranean)	Provide refuge for fauna with poor powers of dispersal (e.g. Bivalvia, Gastropoda, Decapoda) Provide habitat facilitating dispersal of species suited to lentic conditions (e.g. some Ephemeroptera and Diptera)		
Artificial refuges (including farm dams, fire dams, drainage ditches, canals)	Chester & Robson, 2013	Review (Temperate, mediterranean, arid, tropical)	May partially or fully represent species assemblages within natural waterbodies Often secondary habitats. Presence of macrophytes, natural sediments, terrestrial vegetation, hydroperiod associated with higher diversity May become more important as natural habitats diminish (e.g. climate change, development, invasive species)	Dispersal abilities according to proximity and connectivity to natural waterways Tolerance for conditions within artificial waterbodies	Dodemaide et al., 2018; Chester et al., 2013
Hyporheos	Stubbington et al., 2013	Review (Temperate, mediterranean, arid, tropical)	Sub-surface refuge for aquatic taxa Increases community resilience by supporting both active refuge seekers and passive occupants during dry phases	Morphological, physiological and behavioural adaptations	Boulton et al., 1992; Paltridge et al., 1997; Fenoglio et al., 2006;

Refuge type	Study	Location (Climate)	Roles identified	Organism traits	Also see
			Utility as a refuge variable, dependent on factors including: hydroperiod (length of dry), riparian vegetation and shading, sediment properties and moisture content, organism adaptations	to occupy hyporheic zone	Wood et al., 2010; Young et al., 2011; Vander Vorste et al., 2016; Paril et al., 2019
Pholeteros (crayfish burrows)	Johnston & Robson, 2009	Grampians, south-east Australia (Mediterranean)	Commensal relationship between small, non-burrowing crayfish and large burrowing crayfish in seasonal wetlands	Survive drying by retreating to humid microclimate	Boulton et al., 1992; Reid et al., 2006
	Noro & Buckup, 2007	Guaipa Lake, South-east Brazil (humid sub-tropical)	Macroinvertebrate fauna use burrows of <i>Parastacus defossus</i> during dry season Includes apparently specialist isopods, tardigrades and some Copepoda.	Commensalism Ability to survive in groundwater	
Leaf litter patches	Chester & Robson, 2011	Victoria Range, south-east Australia (Mediterranean)	Desiccation resistant forms of Diptera and Plecoptera. Same assemblage as dry sediments.	Desiccation resistant cysts	Boulton et al., 1992; Robson et al., 2008 Strachan et al., 2015
Beneath stones	Chester & Robson, 2011	Victoria Range, south-east Australia (Mediterranean)	Maintaining moisture for aestivating species (such as isopod <i>Synamphisopus doegi</i> and dragonfly <i>Austroaeschna subapicalis</i>)	Aestivation	Mackie et al., 2013; Chester et al., 2014
Dry sediments	Stubbington & Datry, 2012	Review (Temperate, mediterranean, arid, tropical)	Supports organism propagules for recolonisation upon flow resumption. Increasing environmental harshness reduces invertebrate seedbank richness and similarity to flowing-river assemblages.	Desiccation resistance, aestivation	Robson et al., 2008; Mackie et al., 2013; Strachan, et al., 2014
	Chester & Robson, 2011	Victoria Range, south-east Australia (Mediterranean)	Desiccation resistant forms of Diptera and Plecoptera.		
Aestivation in damp sediments	Wickson et al., 2012	South-east Australia (Mediterranean)	Aestivation for >100 days in larval Trichoptera (<i>Lectrides varians</i>) in desiccation/rewetting experiment	Aestivation	Dietz-Brantley et al., 2002; Mackie et al., 2013; Strachan et al., 2015
	Paltridge et al., 1997	Magela Creek, northern Australia (tropical)	Aestivation reported in adult copepods, late-instar Diptera, early-instar Ephemeroptera		

Methods

Climate change impacts on Southwest Australian streams

Mediterranean climates typically show marked inter-annual variation in rainfall, but in SWA rainfall has declined by 20% of long-term mean since 1970 (Hope et al., 2010; McFarlane et al., 2020). Concurrently, a disproportionately larger decline in streamflow occurred, of around 70% (Petrone et al., 2010; Silberstein et al., 2012). Disconnection of groundwater from streambeds occurred around the years 2000 – 2001 in most catchments, resulting in continued decline in streamflow independent of annual rainfall (Petrone et al. 2010; Hughes, Petrone & Silberstein, 2012; Kinal & Stoneman, 2012). Within the study area, rainfall totals (measured at Wungong Dam, Bureau of Meteorology station 009044) have steadily declined since records began in 1913, and the 1990-2017 mean was 27% below that between 1913-1990. Of the streams that were perennial in the 1980s (Bunn 1985), only one stream (Waterfall Gully) and one spring-fed flowing reach (in Seldom Seen Brook, SS2 in Figure 7.1) remain perennial, all other streams are now intermittent.

Study streams

Dry season refuges were explored and sampled in the forested headwater streams of the Wungong Brook catchment, within the northern jarrah forest (NJF) of SWA (Figure 7.1). This catchment is an entirely forested protected water supply catchment (since 1909), so there has been no change in catchment land-use as climatic drying has progressed. Flow gauging data clearly shows the transition from perennial to seasonal flows in most of these streams (Figure 2.9; Figure 2.11). The remaining perennial stream in the Wungong catchment, Waterfall Gully, is spring-fed. The headwaters of other streams in the Wungong catchment (e.g. Curtis Creek, Coccinea Creek) arise from the bases of large granite inselbergs (Figure 7.1). Wungong Brook contains large pools overlying bedrock (in the upper reaches) or weathered regolith (middle and lower reaches) including kaolitic clays, which support

retention of pools well into the dry season (Figure 7.3). The perennially flowing reach of Seldom Seen Brook is fed by a spring emerging from the base of a granite inselberg (Figure 7.1). Also, two ephemeral creeks arising from granite outcrop springs were explored along their lengths and included as potential refuges for fauna with ability to use subterranean refuges (labelled GRO; Figure 7.1). These streams have likely always had intermittent flow regimes (both are ungauged); flows occur during a short period in winter-spring (August-September), beginning as trickling flows from the top of the inselbergs rather than seeping from the base (Figure 7.2). Altogether, five surface water refuge types were identified in the channels of Wungong catchment streams: two granite inselberg springs (Curtis Creek; Figure 7.2), one spring-fed perennially flowing reach (Seldom Seen Brook), an artificial refuge (a weir pool on Seldom Seen Brook), and pools on Wungong Brook that persisted well into the dry season (Figure 7.3). The pools on Wungong Brook were hydrologically dissimilar to the spring-fed reach and seeps, being perched receding pools (*sensu* Bonada et al., 2020; Bogan et al., 2019).

The following terms are defined for clarity throughout text: “dry season refuges” refers to all refuge types sampled during this study; “surface water refuges” encompass all summer dry season refuges containing surface water; whereas “spring-fed refuges” are all groundwater connected reaches and pools, including inselberg seeps and the weir pool on Seldom Seen Creek, but not perched pools on the Wungong Brook. “Subterranean refuge” refers to inselberg springs, which overflow only in winter (separate to inselberg seeps, a spring-fed refuge). “Dry streambed refuges” encompass all non-surface water refuges, including under stones, organic matter and dry sediments.

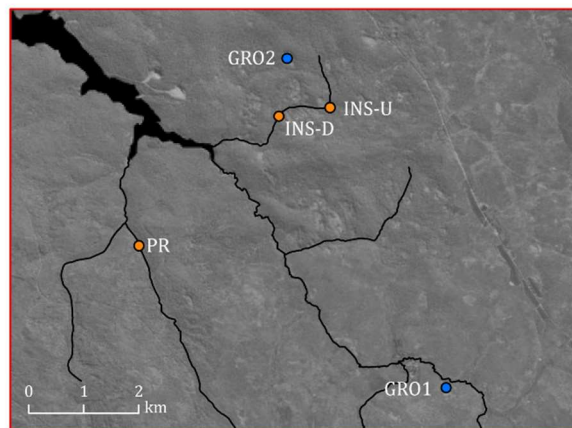
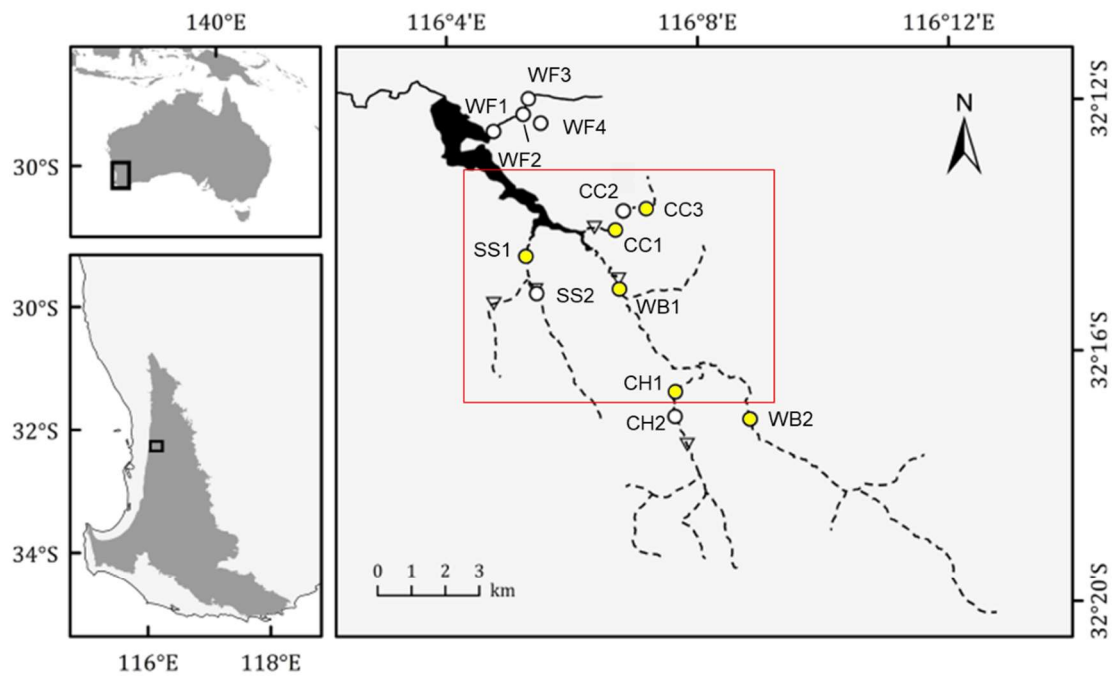


Figure 7.1: Stream study sites in the Wungong Catchment, southwestern Australia - perennial reaches (solid lines), formerly perennial, now intermittent reaches (dashed lines), and flow gauging stations (inverted triangles; flow gauge at WF1 not visible). Yellow indicates collection locations for sediment rehydration experiment. WF = Waterfall Gully; CC = Curtis Creek; SS = Seldom Seen Brook; CH = Chandler Brook; WB = Wungong Brook. Inset = Location of spring-fed refuges, summer refuges orange (PR = spring on Seldom Seen Brook, feeding a perennial reach and pool; INS-U = upstream inselberg spring, INS-D = downstream inselberg spring), winter discharging springs atop granite outcrops (GRO1 & GRO2).

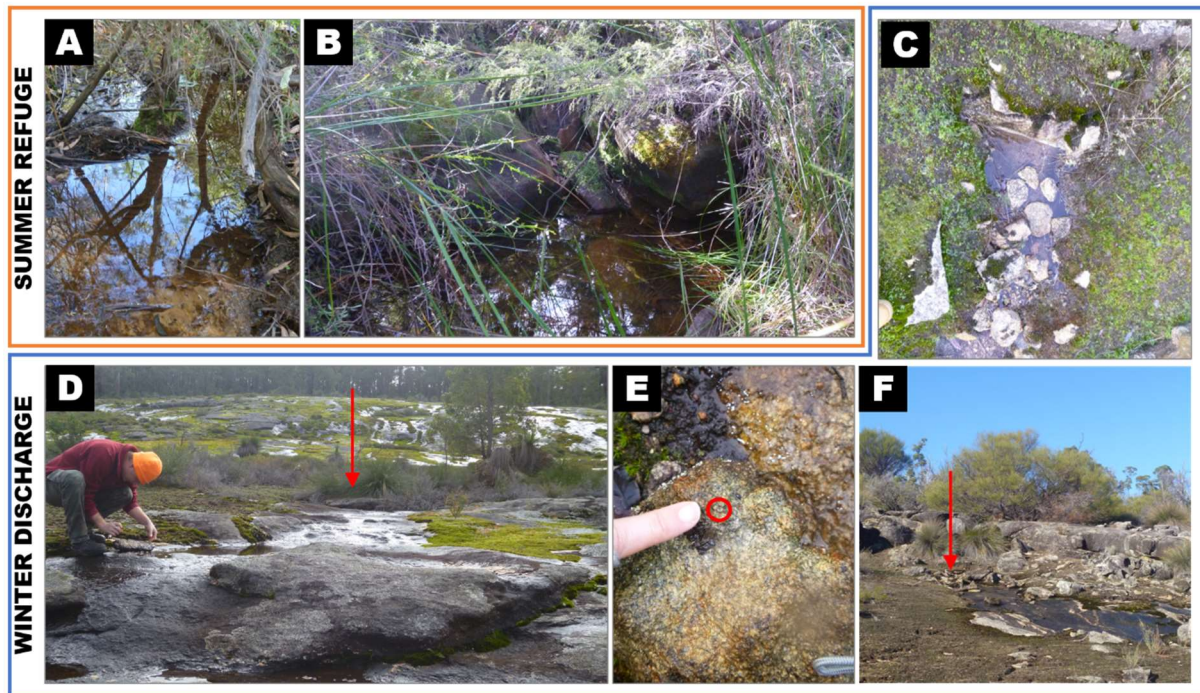


Figure 7.2: Examples of granite inselberg refuges for aquatic macroinvertebrates: summer dry season refuges (A, B, both on Curtis Creek) consist of spring-fed inselberg seeps formed around the base of the inselberg; and subterranean refuges (C – F) which are dry at the surface during summer and autumn, however winter discharge from the top of the inselberg brings *Amphipoda Uroctena* sp. to the surface, revealing cracks in the granite as a subterranean refuge. C – discharge spring atop an inselberg, which eventually flows to Wungong Brook; D – arrow points to fissure in the granite from which *Uroctena* sp. emerge (here on Mount Curtis); E – *Uroctena* sp. living in a film of water discharging from the top of a granite inselberg shown in C; F – arrow points to spring shown in C.

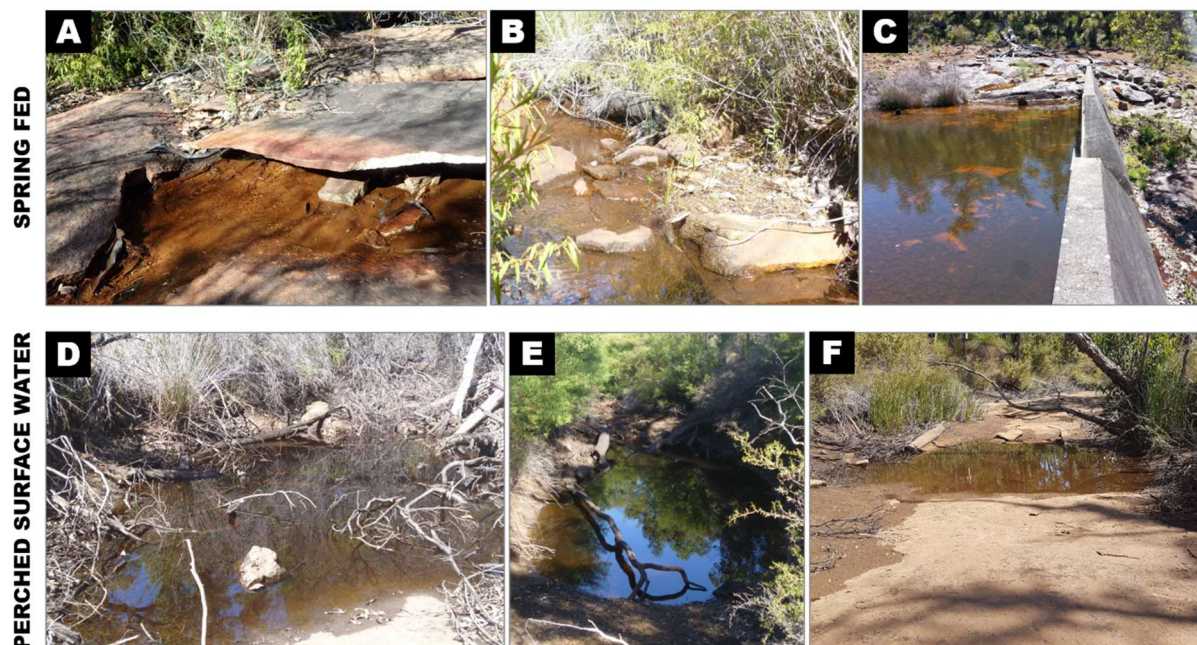


Figure 7.3: Examples of dry season refuges sampled in the autumn of 2017 and 2018, including a spring fed perennial reach (A, B) and weir pool (C) on Seldom Seen Brook, and perched, drying pools on the Wungong Brook (D – F).

Study design

Sampling aimed to capture all refuge types, in both flowing and standing water and on inselbergs; and all possible resilience and resistance traits, including capacities for desiccation resistance or aestivation, or for retreating to subterranean refuges. The study streams primarily overlay bedrock or shallow regolith and are thus perched along most of their lengths; since there was little evidence of a hyporheic zone in most instances this refuge type was not included.

The sampling design comprised two dry seasons (2017, 2018) during which all potential refuge sites were sampled for invertebrates (Table 7.2). In addition, inorganic sediments and organic matter were collected for a pilot rehydration experiment in 2017 and a larger experiment in 2018, to identify species using desiccation-resistant propagules in streambed sediments. Six replicate samples each of inorganic sediment and organic matter were

collected from eight dry reaches distributed on four streams and one granite outcrop (Curtis Creek = 3, Seldom Seen Brook = 1, Chandler Brook = 2, Wungong Brook = 2, GRO1); the 2017 pilot study took six replicate sediment samples from these same sites.

Table 7.2: Distribution of surface water refuge types present during 2017/18. Perched pools retained water into the late dry season, but some likely dried eventually. Outcrop springs were present in winter.

*Two pools were present, but one had been destroyed by feral pigs. N.S. = not sampled.

Season Type	Dry Spring-fed reach		Dry Spring-fed pools		Dry Perched pools		Wet Outcrop springs	
	2017	2018	2017	2018	2017	2018	2017	2018
Seldom Seen Brook	1	1	1	1				
Curtis Creek			2	1*				
Wungong Brook					6	N.S.		
Chandler Brook								
Ephemeral tributaries (2)							2	N.S.

During the early austral dry season of 2017 (February), dry streambeds were explored along most of their length in search of remnant pools or seeps which may have acted as refuges for aquatic fauna. Where accessible, several kilometres of each of Curtis Creek, Seldom Seen Brook, Chandler Brook and Wungong Brook were searched on foot and the location of any surface water refuges were marked using GPS. Searches generally extended upstream and downstream of study sites used in Chapters 3, 4 & 5, and the headwaters of each stream were also located and searched. These refuges were sampled in the early (February and March) and late dry season (April and May) in 2017, and inselberg springs were sampled again in the late dry season of 2018. During exploration for surface water refuges, dry streambeds were also searched for larger stones or woody debris which may have sheltered aestivating invertebrates (e.g. Chester & Robson 2011; Strachan et al., 2014). Spring-fed refuge pools occurred in Curtis Creek ($n = 2$) and Seldom Seen Brook ($n = 1$) but not in Wungong or

Chandler Brooks. Wungong Brook retained at least six perched pools well into the dry season (April) with at least two remaining until the late dry (May).

Invertebrate sampling

Larger refuges (large pools, perennially flowing reach at Seldom Seen Brook) were sampled using kick samples (500 μ m mesh dip net). Sediment was vigorously disturbed by foot and the net swept through the water multiple times to capture dislodged invertebrates. This was performed several times to capture all available habitats within the refuge, including woody debris and emergent vegetation (between two and six sweeps per pool depending on pool size). At the smaller inselberg spring-fed pools on Curtis Creek, a 200 mm aquarium net (250 μ m mesh) was used, gently disturbing the sediment by hand and sweeping the net through dislodged debris to capture invertebrates. The different method was necessary, because the dip net was too large to effectively sample small refuges, and to minimise potentially negative impacts of sampling on these fragile environments.

After each sweep, the contents of the net were then emptied into a white tray filled with stream water. Two observers live-picked invertebrates for preservation in 70% ethanol. Invertebrates were picked until no new morphospecies were found (generally 10 minutes), then the contents was returned to the refuge so that any remaining invertebrates and propagules would survive. Consequently, data collected were presence/absence. Preserved invertebrates were identified to the lowest possible taxonomic level: species level for most groups excluding Diptera (family level except Chironomidae, which were identified to genus), Ostracoda and *Uroctena* amphipods (genus level); Oligochaetes and other worms (Class level), and mites (Order).

In July 2017, inselberg springs atop the headwaters of tributaries to Curtis Creek and Wungong Brook began to flow. The flow from both inselberg springs was rich in *Uroctena* sp. amphipods, which were collected by hand and preserved in 70% ethanol.

To address H₄ and H₅, dry season refuge assemblages were compared with assemblages recorded from intermittent streams upon resumption of flows, and from perennial streams (i.e. data collected for Chapters 3 & 4). Concurrently with refuge sampling in April and May 2017, four sites on Waterfall Gully (the only perennial stream remaining in the catchment) were sampled using a 500 µm mesh dipnet, in much the same way as a Sürber sampler: the opening of the net was positioned on the streambed facing flow, and a 30 x 30 cm area immediately in front of the net gently disturbed by hand, dislodging invertebrates which were then swept into the net by flow. This method was also used to sample intermittent reaches on the onset of flows in June (most reaches) and August (all reaches). At other times, when surface water was present but flow insufficient to wash material into the net, kick samples were taken (method as for larger refuge pools).

Sediment rehydration experiments

Dry sediment and organic matter were collected from six intermittent stream sites, across Wungong Brook (WB1 & WB2), Seldom Seen Brook (SS1), Chandler Brook (CH1), Curtis Creek (CC1 & CC3), and at intervals along the length of the southern granite outcrop spring (GRO1 in 2017 only; Figure 7.1). Flow gauging records show that Wungong Brook ceased flow in November 2016 and 2017 (Table 7.3), and the other intermittent streams had ceased to flow at or prior to this. Drying of perched pools occurred up to three months later and was followed by a dry period of up to eight months, during which sediment was collected. The length of time each stream was dry prior to collection of sediments varied between two and eight months (Table 7.3).

Table 7.3: Approximate length of time sediments were dry prior to collections in May 2017 and 2018.

Wungong Brook ceased flow on the 27th of November 2016 and 28th of November 2017 (DWER gauge 616041). Sediment collections were undertaken on the 17th of May 2017 and the 9th of May 2018

Stream reaches	Month of drying	Length of time dry
Wungong Brook – all	February	2 to 3 months
Curtis Creek – lower	November	6 months
Curtis Creek – middle & upper	February	2 to 3 months
Chandler Brook – all	November	6 months
Seldom Seen Brook – lower	November	6 months
Inselberg springs (2)	September	8 months

In the 2018 experiment, six samples of sediment and organic matter were taken from each stream site (total $n = 72$). Sampling prioritised the deepest points of dried pools, or locations within the thalweg. The top 20-30 mm of sediment over an 0.35 m^2 area were collected using a hand trowel, to a volume of 500 mL. Sediment samples were placed in lidded plastic containers for transport to the laboratory. Immediately upon return to the laboratory, each sediment sample was emptied into a white tray and examined at 10x magnification under a dissecting microscope mounted on a moveable arm, while picking out any animate organisms for preservation in 70% ethanol (primarily terrestrial invertebrates such as springtails). To capture any organisms requiring flowing water as a hatching cue, a subset of the inorganic sediment samples was separated for an aeration treatment (Chester & Robson, 2011). One 100 mL subsample was taken from each sediment sample and pooled into two composite samples per reach for the aeration treatment. The remainder of each sediment sample, and whole organic matter samples, were then placed separately in 1 L clear plastic tubs, filled to three-quarters with dechlorinated tap water, and gauze mesh was secured over the opening to prevent entry or escape of invertebrates. The composite samples were placed in a 1 L tub along with a small air stone and filled to three quarters with dechlorinated water and covered

with mesh. Tubs were placed on a laboratory bench where they received natural light from windows that were closed to prevent flying adult insects from entering the laboratory.

After 24 hours, the contents of each sample were gently emptied into a white tray and examined under the microscope at 10x magnification. Live invertebrates were removed and preserved in 70% ethanol, after which the sediment and water were carefully placed back into the 1L tub and topped up with dechlorinated water where necessary. This process was repeated at 48 hours, and then every 3 to 5 days for a maximum of 35 days. At each examination, all live invertebrates were removed, except in instances where allowing growth would aid identification (e.g. stonefly nymphs). Daily visual checks were performed on each tub and any live organisms (often Coleoptera) were removed and preserved.

The 2017 pilot experiment used six haphazardly collected sediment samples from three intermittent stream study sites (WB1, WB2 and SS1) and one granite outcrop site (GRO1; total $n = 24$). Samples were collected and treated in the same manner as described above, excluding the aeration treatment.

Data analysis

To address H_1 , data from each refuge type were transformed to presence/absence and Bray-Curtis similarity matrices were computed using Primer-e v.7 (Clarke et al., 2014).

Presence/absence was appropriate as hypotheses aimed to characterise species assemblages of drought refuges, rather than measuring abundance or density of individuals. Species assemblages between refuge types were examined using a One-way Analysis of Similarity (ANOSIM) ($n = 9$; six *in situ* refuges and three rehydration treatments). Ordination plots (nMDS) were used to visualise differences between refuge types and dispersion of samples within refuge types, and distances to centroids calculated and ordinated. Uneven sample sizes and differing methods precluded univariate analysis to examine differences in taxa richness between refuge types (H_2), thus total species richness per refuge type was compared.

To examine taxa richness of rehydrated sediments (H_3), a two-way Analysis of Variance (ANOVA) was used to test for significant differences between sediment types ($n = 3$; inorganic sediment, organic matter and aerated sediments) and time periods ($n = 2$; 1 – 3 days, aestivating taxa, vs. 4 – 30 days, aestivating and hatching taxa) (SPSS v. 23).

Assumptions of normality and homogeneity of variance were met. Testing stream as a factor was not possible due to unevenness - most sediment samples from Curtis Creek, Seldom Seen Brook and Chandler Brook failed to produce organisms, whereas most samples from the Wungong Brook were productive (ultimately, only Wungong Brook samples were analysed statistically). Species assemblages were qualitatively compared between sediment types and years.

To assess the contribution of dry season refuges to community recovery in intermittent streams (H_4), Mantel tests were used to correlate the assemblages in refuges (during autumn) with those present in early winter (June, some reaches flowing) and mid-winter (August, all reaches flowing). Species assemblages from each refuge type were pooled according to their respective stream reach (upstream and downstream) corresponding with location of June and August sampling sites, records from granite outcrops were pooled with the receiving stream reach (Figure 1; Table S3). Chandler Brook was excluded, as a single oligochaete comprised the drought refuge data for that stream. This gave six stream reaches with which to compare refuges, early winter and mid-winter assemblages.

Finally, to assess the contribution of the perennial stream to recolonisation of intermittent streams (H_5), the assemblages within autumn refuges, and during early winter and mid-winter flows of intermittent streams, were compared to assemblages concurrently present at the only remaining perennial stream in the catchment (Waterfall Gully). Assemblage composition and abundance data were collected concurrently from Waterfall Gully in 2017 (presented in Chapters 3 & 4), and treated in the same manner as described for intermittent stream data

(above for H₄): transformed to presence/absence data, and pooled according to respective stream reach (upstream, downstream and a tributary; Figure 7.1; Table 7.4). This gave three sets of directly comparable assemblage composition data to compare with autumn refuges (April/May), early winter (June) and mid-winter (August) flows at intermittent streams. A suite of five one-way ANOSIM tests were used. Three tests examined associations between the assemblage present at Waterfall Gully during autumn, and intermittent streams at each hydroperiod stage: perennial (autumn) vs intermittent (autumn); perennial (autumn) vs. intermittent (June); and perennial (autumn) vs. intermittent (August). Then, two tests examined associations in early- and mid-winter: perennial (June) vs. intermittent (June); perennial (August) vs. intermittent (August). Similarity percentage analyses (SIMPER) were used to identify species contributing to dissimilarity between stream types for each comparison.

Table 7.4: Grouping of sites by section per stream (upstream, downstream, or tributary) for comparative analysis (RELATE and ANOSIM). Stream order given.

Stream	Site	Section	Stream Order
Waterfall Gully	WF1	Downstream (below confluence WF4)	2
	WF2	Upstream	1
	WF3	Upstream	1
	WF4	Tributary	1
Curtis Creek	CC1	Downstream	1
	CC2	Upstream	1
	CC3	Upstream	1
Seldom Seen Brook	SS1	Downstream (ephemeral)	2
	SS2	Upstream (spring-fed)	1
Wungong Brook	WB1	Downstream	3
	WB2	Upstream	2
Chandler Brook	CH1	Not included	1
	CH2	Not included	1
Southern outcrop	GRO1	Wungong upstream	1
Northern outcrop	GRO2	Curtis downstream	1

Results

Assemblage composition and taxa richness within dry season refuge types (H_1 & H_2)

A total of 58 taxa were recorded across *in situ* refuges within the Wungong catchment, with different species characteristic of different refuge types (Tables 7.5 & 7.6). Receded perched pools on the Wungong Brook supported the highest number of taxa (29). The spring-fed reach and weir pool supported the next highest number of species (23 and 21, respectively), followed by rehydrated sediments (18), organic matter (17), aerated sediments (12), inselberg springs (9 taxa) and lastly under stones (6 species aestivating) and winter outcrop springs (5 species). Because the highest number of species were recorded from perched receded pools, rather than perennially flowing spring-fed reaches and inselberg springs, H_2 was not

supported. However, presence of surface water was associated with higher numbers of taxa, with dry streambed refuges supporting fewer taxa.

Species assemblages differed between refuge types (Global $R = 0.53$, $P = 0.001$) supporting H_1 , however pairwise tests showed overlaps among some refuge types (notably between dry sediments and most other refuges; Figure 7.4; Table S7.1). Assemblages of receded pools differed from all other refuge types ($R = 0.51$ to 0.95 , $P < 0.01$). The assemblage of drying pools included several species of highly mobile, widespread, predatory Coleoptera and Hemiptera, as well as several species of Chironomidae with continental/cosmopolitan distributions (Table S7.1). Both the cosmopolitan mayfly *Cloeon fluviatile* and SWA endemic mayfly *Bibulmena kadjina* were present in these pools, alongside the SWA endemic amphipod *Perthia acutitelson*, burrowing crayfish *Cherax quinquecarinatus*, and occasionally the widespread dragonfly species *Hemicordulia tau* and *Pantala flavescens*.

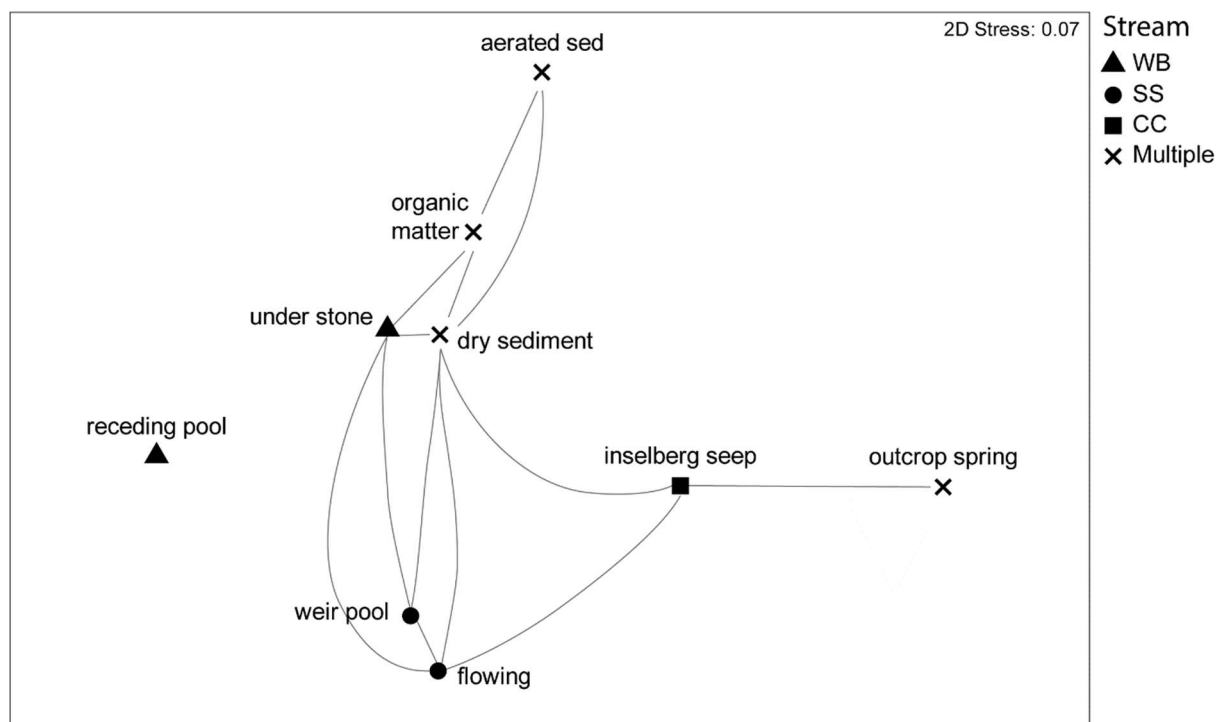


Figure 7.4: Ordination of distance to centroids of macroinvertebrate assemblages recorded from dry season refuges sampled in 2017-18, including *in situ* surface water refuges and under stones, *ex situ* rehydration experiments, and from subterranean refuges flowing in winter (outcrop spring). Streams are Wungong Brook (WB), Seldom Seen Brook (SS) and Curtis Creek (CC), unless the refuge type occurred across multiple streams (sediment rehydration and outcrop springs). Lines connect refuge types that did not differ in one-way ANOSIM pairwise tests ($P > 0.05$). 2D Stress = 0.07.

The macroinvertebrate assemblage of the spring-fed reach on Seldom Seen Brook differed from winter outcrop springs, rehydrated organic matter and aerated sediments (ANOSIM $P < 0.01$; Table S7.1), but did not differ from the weir pool, inselberg seeps, dry sediments and under stones (Figure 7.4). The spring-fed reach supported several Odonata, including *Archeosynthemis leachii*, mayfly *B. kadjina* and the amphipod *Uroctena* sp., as well as several Coleoptera including SWA endemic *Sternopriscus browni*. At the bottom of this reach, the weir pool also supported a population of the IUCN listed mussel *Westralunio carteri*, water scorpion *Ranatra diminuta*, nymphs of endemic *Austroaeschna anacantha* and the increasingly rare, ancient damselfly species *Miniargiolestes minimus*. The Dytiscidae

beetles *Necterosoma darwini*, *Platynectes decempunctatus*, and amphipod *P. acutitelson*, were present in spring-fed refuges, as well as aestivating in dry sediments and organic matter (Table S7.1).

The inselberg seeps on Curtis Creek did not differ from the spring-fed flowing reaches, dry sediments or winter outcrop springs (Figure 7.4) but did differ from the weir pool ($R = 0.29$, $P = 0.05$). Inselberg seeps had fewer taxa than other surface water refuges (9 species), however, they supported species rare/absent at other refuge types, including SWA endemic caddisfly *Triplectides* sp. AV1. Other species included the groundwater-dependent amphipod *Uroctena* sp. and mayfly *B. kadjina*. Opportunistic predators were comparatively rare, with single records of Dytiscidae *Rhantus suturalis* and Veliidae *Nesidovelia paramoena*. When these springs were revisited in April 2018, extensive damage from feral pigs (*Sus scrofa*), a destructive pest in the NJF, had occurred just prior to sampling at the lower spring. No invertebrates were found at that time, as the pigs had turned the previously clear pool into a mud wallow so that the sediment was thoroughly disturbed and the water turbid, extirpating any organisms taking refuge there. The caddisfly *Triplectides* AV1 and *B. kadjina* were collected at the upper spring-fed pool which had not been disturbed by pigs.

Winter outcrop springs were different to all other refuge types, except the seeps at their base flow in summer (Figure 7.4; Table S7.1). These springs were dominated by the amphipod *Uroctena* sp. emerging from cracks in the granite in which they had survived the preceding summer; other taxa included flatworms, and larval Scirtidae beetles. Several species were recovered from under stones or woody debris on the streambed in April 2017, but only from recently dried pools on the Wungong Brook. These included two dragonflies (*Adversaeschna brevistyla* and *Procordulia affinis*), adult Coleoptera and the amphipod *P. acutitelson* (Table 7.5; Table S7.1). No live aquatic invertebrates were located from this refuge type in 2018.

Table 7.5: Abbreviated species list recorded from *in situ* refuges in the dry seasons of 2017-18.

Emphasis is given to taxa of known relictual origin, or taxa that formerly were most-common prior to flow regime change (Bunn 1985; Bunn et al., 1986). Number of taxa per higher grouping given in parentheses. Full species list see Table S7.1.

Refuge type	Phase	<i>n</i> taxa	Species
Spring-fed perennial reach	Dry phase	23	Ephemeroptera (<i>Bibulmena kadjina</i>); Trichoptera (<i>Lectrides parilis</i> , <i>Triplectides</i> AV1); Odonata (<i>Archeosynthemis leachii</i> , <i>A. occidentalis</i>); Coleoptera (2); Amphipoda (<i>Perthia acutitelson</i> ; <i>Uroctena</i> sp.); <i>Ilyodromus</i> sp.; Chironomidae (10)
Weir pool	Dry phase	21	Acarina; Odonata (<i>Austroaeschna anacantha</i> , <i>Miniargiolestes minimus</i> , +1); Ephemeroptera (<i>B. kadjina</i>); Trichoptera (<i>L. parilis</i> , <i>Triplectides</i> sp. AV1); Coleoptera (<i>Sternopriscus browni</i>); Hemiptera (4); Culicidae; Chironomidae (5) Amphipoda (<i>P. acutitelson</i>); Decapoda (<i>Cherax quinquecarinatus</i>); Cyclopoida; Bivalva (<i>Westralunio carteri</i>)
Inselberg seeps	Dry phase	9	Ephemeroptera (<i>B. kadjina</i>); Trichoptera (<i>Triplectides</i> sp. AV1); Coleoptera (2); Hemiptera (2); Amphipoda (<i>Uroctena</i> sp.); Culicidae; Platyhelminthes
Remnant pools	Dry phase	25	Ephemeroptera (2); Odonata (2); Coleoptera (4); Hemiptera (8) Chironomidae (5); Amphipoda (<i>P. acutitelson</i>); Decapoda (<i>C. quinquecarinatus</i>); Cyclopoida; Oligochaeta
Under stones/woody debris	Dry phase	6	Odonata (<i>Adversaeschna brevistyla</i> , <i>Procordulia affinis</i>); Coleoptera (2); Amphipoda (<i>P. acutitelson</i>); Ostracoda (<i>Cypretta</i> sp.)
Inselberg spring	Flowing phase	5	<i>Uroctena</i> sp., Scirtidae sp. (L), Orthoclaadiinae sp., <i>Nousia</i> sp. AV16; Turbellaria
Inorganic sediment	Dry phase		
Organic matter	Dry phase		See Table 7.6 for species composition
Aerated sediment	Dry phase		

Sediment rehydration experiment (H₃)

A total of 29 taxa emerged from rehydrated sediments in the 2018 experiment, across inorganic sediment, organic matter and aeration treatments, and an additional four species were present in the 2017 pilot study (Table 7.6). In 2018, there were no significant differences in taxa richness between sediment types ($F_{2,45} = 0.9$, $P = 0.4$) or days ($F_{1,45} = 0.6$, $P = 0.5$) nor was there an interaction between sediment types and days ($F_{2,45} = 0.4$, $P = 0.7$) (Figure 7.5). H₃ was not supported, but this is perhaps not surprising given that these samples came from the Wungong Brook sites which are all connected, because most samples collected from Curtis Creek, Seldom Seen Brook and Chandler Brook yielded no invertebrates throughout the experiment.

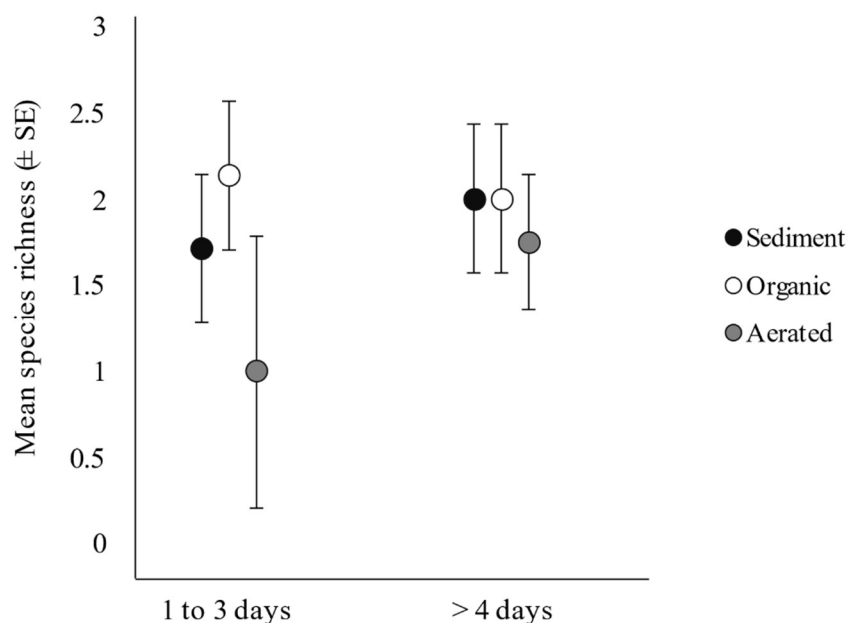


Figure 7.5: Estimated marginal means (± 1 standard error) of species richness emerging from rehydrated sediment samples (inorganic sediments, organic matter and aerated sediments), upon initial rehydration (1 to 3 days) and after 4 days (to a maximum of 35 days). There were no significant differences between sediment types or time periods (ANOVA $P > 0.05$).

Adult Coleoptera were more speciose from organic sediments, whereas more species of Chironomidae were recorded from inorganic sediments (including the aeration treatment). Nymphs of the stonefly *Leptoperla australica* were exclusively found in aerated sediments, indicating their hatching is cued by moving water (with onset of flow). Upon initial rehydration (1 to 3 days) mature specimens of several species emerged, indicating these individuals were aestivating rather than occurring as desiccation resistant propagules (Table 7.6). These included several beetles, the hemipteran *Nesidovelia paramoena*, and two caddisflies: single individuals of *Triplectides* sp. AV1 and *Hudsonema aptus* (both Leptoceridae) from a single sediment sample from the upper reaches of Curtis Creek. Small dipterans (primarily Chironomidae and Tipulidae) were also collected in the initial days of the experiment, indicating hatching. The amphipod *P. acutitelson* also emerged from 2017 sediment samples from the Wungong Brook, showing this species has the capacity to aestivate. Several Chironomidae emerged, primarily cosmopolitan genera found widely across the Australian arid zone including *Chironomus* sp., *Polypedilum* sp. *Tanytarsus* sp. (both mature and early instars observed), but also included SWA endemic chironomid *Paralimnophyes pullulus*. Several species of adult dytiscid beetles were also aestivating, again widespread species common in the arid zone (Table 7.6). Ostracoda also appeared to be aestivating and three species eventually emerged (some hatching from eggs) as the experiment progressed.

Table 7.6: Taxa recorded during 2018 rehydration trials. All Coleoptera are adults unless specified larvae [L], all Diptera are larvae unless specified pupa [P].

Additional 2017 records given. + denotes taxon was abundant (>10) in at least one sample of that substrate type; (imm) = immatures; * denotes taxon recorded in 2017 trial only.

Substrate type	Days	n taxa	EPT	Coleoptera	Chironomidae	Crustacea	Others
Sediment	1 - 3	8	<i>Hudsonema aptus</i> , <i>Triplectides</i> AV1	<i>Enochrus deserticola</i> , <i>Helochaeres percyi</i>	Chironomidae (imm); <i>Cricotopus</i> sp., <i>Chironomus</i> sp., <i>Tanytarsus</i> sp.		Tipulidae (imm). Oribatida
	>4	8		<i>Allodessus bistrigatus</i>	<i>Polypedilum</i> sp., Orthoclaadiinae unident. sp. 1#, <i>Chironomus</i> sp. +, <i>Tanytarsus</i> sp.	<i>Cypretta</i> sp., Ostracoda sp. 3 (mature + imm)	Culicidae, Dolichopodidae
Organic matter	1 - 3	5		Scirtidae sp. [L] (imm)	Orthoclaadiinae unident. sp. 2#		Oligochaeta, Culicidae, <i>Nesidovelia paramoena</i>
	>4	13		<i>Hydrochus</i> sp., <i>H. percyi</i> <i>Limnoxenus zealandicus</i> , <i>Allodessus bistrigatus</i> +, Scirtidae sp. [L]	<i>Chironomus</i> sp., <i>Polypedilum</i> sp., <i>Tanytarsus</i> sp.,	Cyclopoida, <i>Cypretta</i> sp., Ostracoda sp. 3, <i>Ilyodromus</i> sp. +	Dolichopodidae
Aerated sediment	1 - 3	1				<i>Cypretta</i> sp.,	
	>4	9	<i>Leptoperla australica</i> (imm)	<i>H. percyi</i>	<i>Chironomus</i> sp., <i>Paralimnophyes pullulus</i>	<i>Ilyodromus</i> sp., <i>Cypretta</i> sp., Ostracoda sp. 3 (imm),	Oligochaeta, Ceratopogonidae [P]
2017 sediments	1 - 3	9		<i>A. bistrigatus</i> , <i>Limnodessus compactus</i> , <i>H. percyi</i> , <i>Rhantus suturalis</i> *	<i>Chironomus</i> sp., <i>Tanytarsus</i> sp., Orthoclaadiinae unident. sp. 1, <i>Polypedilum</i> sp.	<i>Perthia acutitelson</i> *	
	>4	15		<i>E. deserticola</i> , <i>H. percyi</i> , <i>A. bistrigatus</i> , <i>Paracymus pygmaea</i> <i>Playnectes decempunctatus</i> *, <i>Necterosoma darwini</i>	<i>Polypedilum</i> sp., Orthoclaadiinae unident. sp. 1, <i>Chironomus</i> sp., <i>Tanytarsus</i> sp.,	<i>Perthia acutitelson</i> *, <i>Cypretta</i> sp.	Dolichopodidae, Culicidae, Oligochaeta

Contribution of dry season refuges to stream assemblage composition (H₄)

Altogether, 69 taxa were recorded from *in situ* dry season refuges and resting stages, or approximately 46% of the 151 taxa recorded from intermittent streams during the 2016/17 hydroperiod (noting that organisms using desiccation resistant resting stages were likely underestimated). However, Mantel tests showed no correlation between dry season refuge assemblages and assemblages when intermittent streams commenced flowing in early winter (June $\rho = -0.16$, $P = 0.6$) or in mid-winter (August $\rho = 0.5$, $P = 0.1$), providing no support for H₄.

Contribution of perennial stream assemblages to the composition of intermittent stream assemblages (H₅)

In autumn, assemblages in the perennial stream were distinct from those using refuges in intermittent streams (Global $R = 0.6$, $P = 0.01$), because a large number of species residing in the perennial stream were absent from refuges (Table 7.7). There were also species present in refuges that were never present at Waterfall Gully — Ostracoda, Copepoda, Hemiptera and Coleoptera, some present as desiccation resistant forms (Table 7.7). Assemblages in intermittent streams following flow onset in June (early-winter) were also distinct from those in the perennial stream. However, by August (mid-winter) assemblages did not differ between the flow types (Table 7.7). Species present in autumn in the perennial stream included several groups of winged insects (including a mayfly, caddisflies and several Chironomidae) which were also present in the intermittent streams, suggesting that the perennial stream may have been the recolonisation source for those species. Altogether, it appears likely that species present in the perennial stream in autumn were an integral part of community recovery in intermittent streams following the onset of flows, supporting H₅. However, there were also a suite of primarily endemic relictual species that were exclusively found in the perennial stream (e.g. endemic Odonata *Zephyrogomphus lateralis*, *Austrosynthemis cyanitincta*, *Archeosynthemis spiniger*; caddisflies *L. parilis*, *Hydrobiosella*

sp. AV17, *Smicrophylax australis*, *Taschorema pallescens*), that did not colonise the intermittent streams even after two months of flow.

Table 7.7: ANOSIM and SIMPER results comparing assemblage composition (presence/absence) between perennial and intermittent reaches, in autumn (dry season refuge phase) and early flowing phases (early and mid-winter). Significant pairwise differences in bold (global R = 0.6, P = 0.01). Key taxa contributing to differences as identified by SIMPER are summarised. PER = Perennial stream (Waterfall Gully), INT = recently intermittent streams.

Time sampled PER	Time sampled INT	Pairwise R	P	SIMPER key taxa
Autumn	Autumn	0.6	0.01	Perennial: Dominated by sensitive taxa (e.g. <i>Neobissephlebia occidentalis</i> , <i>Lectrides parilis</i> , <i>Zephyrogomphus lateralis</i>) generalist taxa (e.g. <i>Paramerina</i> sp., <i>Polypedilum</i> sp.) and crustacean species also found in refuges (e.g. <i>Uroctena</i> sp., <i>Cherax quinquecarinatus</i>). Intermittent: Ostracoda (e.g. <i>Cypretta</i> sp., <i>Ilyodromus</i> sp.) refuge-seeking stream taxa (e.g. <i>Bibulmena kadjina</i> , <i>Triplectides</i> AV1, <i>Uroctena</i> sp.) and generalist taxa (e.g. <i>Rhantus suturalis</i> , <i>Chironomus</i> sp., <i>Nesidovelia paramoena</i>)
Autumn	Early winter	0.4	0.05	Sensitive taxa contribute greatest percent dissimilarity between flow regimes. Intermittent: Remaining autumn taxa (e.g. <i>Necterosoma darwini</i> , <i>Bibulmena kadjina</i>) and early colonists (e.g. <i>Archeosynthemis leachii</i> , <i>Leptoperla australica</i>)
Autumn	Mid-winter	0.01	0.4	Species in common include <i>Nyungara bunni</i> , <i>Triplectides</i> AV21, <i>Hydrobiosella michealseni</i> , <i>Stempellina</i> sp., Tipulidae, Empididae. A suite of sensitive taxa do not colonise intermittent streams, including <i>Austrosynthemis cyanitincta</i> and <i>Hydrobiosella</i> sp. AV17
Early winter	Early winter	0.4	0.05	Perennial: Winter assemblages of sensitive species arrive (e.g. <i>Taschorema pallescens</i> , <i>Smicrophylax australis</i> , <i>Archeosynthemis spiniger</i>) contributing to dissimilarity with intermittent reaches
Mid-winter	Mid-winter	0.02	0.4	Intermittent: arrivals of chironomids previously found in perennial stream (<i>Riethia</i> sp., <i>Austropelopia</i> sp., <i>Stempellina</i> sp., <i>Stictocladius</i> sp.). Several Odonata, Trichoptera and Ephemeroptera contribute to dissimilarity and are restricted to the perennial stream

Discussion

A range of dry season refuges were present in the Wungong catchment supporting different macroinvertebrate assemblages, which included species using either resilience or resistance traits to persist. Altogether, the refuges supported nearly half of the species found across all intermittent streams during the flowing phase. Perched pools on Wungong Brook had marginally higher species richness than spring-fed reaches on Seldom Seen Brook, and rehydrated sediments and organic matter had greater richness than inselberg springs, contrary to the expectation that spring-fed reaches would support the most taxa. However, very different species were found in spring-fed refuges compared to receding pools, including several endemic Gondwanan relictual species that would otherwise have faced extirpation from much of the catchment (Chapter 3 & 5) because they have no desiccation-resistant stage.

The distinction between ecological refuges, being locations where components of biodiversity retreat to survive a temporary disturbance (timescales of days to years) and evolutionary refugia, being locations where species retreat and persist under changing climates and environmental conditions (timescales of centuries to millenia; Keppel et al., 2012; Davis et al., 2013), is especially pertinent when considering the roles of spring-fed habitats in conserving species in the NJF. Even though a handful of endemic relictual species were found to survive the dry season in spring-fed refuges, the bulk of SWA relictual species were restricted to the single remaining permanent stream (Waterfall Gully), and were extirpated from streams that had dried, regardless of the presence of refuges. Their continued persistence under a drying climate is going to depend on the existence of evolutionary refugia in the landscape, being surface expressions of ancient groundwater, where local hydrology is disconnected to prevailing climate (*sensu* Davis et al., 2013). In absence of such habitats, it is almost certain these species will be lost from catchments as perennial streams dry.

Surface water refuges

The surface water refuges were either groundwater-fed habitats associated with granite inselberg springs or receding pools in the main channel of Wungong Brook that are perched above granite bedrock or impermeable clay. Inselberg springs constitute an unexplored drought refuge in southern Australia, however the association between topographical features and both refuges and refugia is well demonstrated in arid central Australia (Davis et al., 2013; Davis et al., 2017). During summer, although inselberg seeps harboured fewer species than the larger spring-fed reach and weir pool, they supported an assemblage of endemic insect species and amphipods. In winter, springs at the summit of these inselbergs began to flow, connecting subterranean *Uroctena* sp. amphipods with the surface, and supporting taxa such as platyhelminths and an intermittent stream specialist mayfly (*Nousia* sp. AV16). The number of taxa inhabiting these inselberg springs (in both summer and winter) was likely underestimated in this study because of their cryptic nature, but also because extensive sampling would likely damage populations within these small refuges.

In contrast, the receding pools on Wungong Brook were dominated by highly mobile predators. Some samples from these pools consisted almost entirely of adult predatory Coleoptera and Hemiptera, and partially eaten Notonectidae nymphs. These predatory species were rare or absent in these streams when they were perennially flowing (Bunn 1985; Bunn et al., 1986; Bunn 1986) but have been recognised elsewhere as a characteristic group of opportunist lentic predators that specialise in feeding on aquatic animals trapped in drying pools (e.g. Boulton & Lake, 1992; Boulton 2003; Bogan & Lytle, 2007; Chester & Robson, 2011; Bonada et al., 2020). Intense ecological interactions are typical of the drying-pool phase in IRES (Boulton et al., 1992; Boulton 2003), which might imply inhospitable conditions for vulnerable species. Nevertheless, whilst dominance by coleopteran and hemipteran colonists is often reported from disconnected pools in IRES (Bonada et al., 2020),

this phase in the hydroperiod has been shown to provide refuge for many taxa, including those with poor powers of dispersal or suited to lentic conditions (Hill & Milner, 2018; Banega-Medina et al., 2020; Bonada et al., 2020). Bogan et al. (2019) found that remnant pools were a critical drought refuge for aquatic invertebrates during an exceptional supraseasonal drought in Californian med-region streams. However, very few SWA endemic stream taxa occurred in the receding pools in Wungong Brook. The transition to intermittency caused extirpations of species that could not adapt their life histories (Chapter 5), however for those able to withstand pool conditions this refuge type may be an ecological trap owing to the presence of predatory insects.

Although Mantel tests showed that assemblages of dry season refuges were not correlated with community recovery, there were several locally endemic species that relied on the presence of spring-fed refuges to persist. These included the mayfly *Bibulmena kadjina*, which has adjusted its life history phenology to persist in spring-fed pools (Chapter 5), and caddisfly *Triplectides* sp. AV1, present in inselberg springs. The spring-fed reach and connected artificial weir pool supported several species not found elsewhere, including a species (*Miniargiolestes minimus*) from an ancient group of damselflies only found in SWA; related species (*Archiargiolestes* spp.), once common in the NJF, have not been recorded there since the 1970s (Pennifold, 2018). This shows that spring-fed pools and reaches are important refuges for some key endemic species.

Dry streambed refuges

Generally, aestivation was the most common strategy used by species in dry sediments, including twelve species of adult Coleoptera which were not among the common fauna prior to drying (cf. Bunn, 1985; Bunn et al., 1986). Larvae of two widespread species of dragonfly (*Adversaeschna brevistyla* Aeshnidae and *Procordulia affinis* Corduliidae) were found aestivating under stones, an ability also known in other large Australian dragonflies (e.g.

Hemicordulia tau, Strachan et al., 2014). The dragonfly *Austroaeschna subapicalis* (Telephlebiidae) has also been found to aestivate (Chester & Robson, 2011) however its congeneric *A. anacantha*, formerly the most common dragonfly in the NJF, is now completely restricted to perennial streams (Chapter 5). Chironomidae were likely hatching from desiccation resistant eggs, because larger specimens were present only after several days; but it is also possible that chironomids occurred in dry sediments as aestivating larvae (Strachan et al., 2015; Stubbington et al., 2016). Unexpectedly, two individual leptocerid caddisfly larvae were found aestivating in sediments in the upper reach of Curtis Creek. Aestivation has been previously demonstrated in leptocerid caddisflies; for example, Wickson et al. (2012) showed that *Lectrides varians* are capable of aestivation for over 100 days, and *Triplectides australis* aestivates in wetland sediments (Chester et al. 2013; Strachan et al., 2014). Here, a larva of *Triplectides* sp. AV1 was found aestivating in dry sediments, in addition to being present in inselberg seeps thus expressing both resistance and resilience traits. A single larva of *Hudsonema aptus*, a flow dependant suspension feeder, was also found aestivating, and was otherwise only found in perennial Waterfall Gully (Chapter 3). This relictual species can therefore express a drought resistance trait, but its distribution may still be constrained by its feeding method to streams with continuous flow. The SWA endemic amphipod *Perthia acutitelson* was also found aestivating, an ability not widely known among epigeal Amphipoda, but known in other macrocrustaceans including the isopods *Synamphisopus doegi* and *Heterias* sp. (Chester & Robson, 2011; Mackie et al., 2013; but see Vadher et al., 2018; Fišer 2019). Although these species were found aestivating in sediments that had been dry for two to three months, it is uncertain whether these individuals would have survived a further two months until flow resumed.

The intermittent stream specialist mayfly *Nousia* sp. AV16 (Letophlebiidae) has become the most common leptophlebiid mayfly in NJF intermittent streams (Chapter 3 & 5), and is likely

using desiccation resistant cysts to repopulate intermittent streams, however it did not emerge from rehydrated sediments. Some taxa require flow to break dormancy (Stubbington & Datry, 2013; see also Chester & Robson, 2011), as demonstrated by the stonefly *Leptoperla australica* which emerged only from aerated sediments. For *Nousia* sp. AV16, temperature may be a necessary hatching cue as it has a short, tightly synchronised life cycle completed entirely in mid-spring (Chapter 5). The hatching experiment was conducted indoors in an air-conditioned laboratory, therefore any cue related to winter stream temperatures were not replicated (water temperatures were *circa.* 8°C in intermittent streams in winter; Chapter 5). Alternatively, distribution of flying insect propagules in streams can be highly patchy according to the spatial and temporal distribution of oviposition resources (Lancaster et al., 2010a, b; Reich et al., 2011; Lancaster et al., 2020b). There is no data on the oviposition behaviours, site selectivity, or corresponding aggregation of propagules for SWA stream invertebrate fauna. Whilst a much larger sediment rehydration experiment would likely uncover more species, adequate characterisation of dry sediment refuges in newly intermittent streams would greatly benefit from further research into insect oviposition behaviours and processes governing distribution of eggs in streambeds.

Whilst the SWA endemic fauna have reportedly withstood drier times in the deep past (Bunn & Davies, 1990), they are predominantly of cool-water stenothermic lineages rather than an arid zone fauna, therefore absence of a substantial propagule bank is not surprising. As shown by the extirpation of much of this fauna from intermittent streams, any adaptations among the relictual fauna are to withstand shorter periods of drying than have occurred since the transition to intermittency post-2000/01. Furthermore, the near absence of taxa emerging from sediments in the drier stream reaches (Chandler Brook and lower Seldom Seen Brook), indicates a harshness threshold precluding *in situ* persistence of even widespread colonist species. These streams are completely dry for over six months each year, and flow gauging

records show the transition to this state was instantaneous. Furthermore, because these streams lack hyporheic or damp sediment refuges, they are potentially harsher than those in other sediment rehydration studies (e.g. Chester & Robson, 2011; Stubbington & Datry, 2013). Altogether, the extremity of recent drying in many SWA headwater streams has given inhabitant fauna few opportunities to adapt, that is, to express resistance or resilience traits that may have allowed their survival through previously drier times.

Contribution of refuges to community recovery in intermittent streams

Dry season refuges are understood to interact with species traits to play differing roles in community recovery in IRES (Robson et al., 2011; Bogan et al., 2017), however there was no significant correlation between autumn dry refuge assemblages and those present when intermittent streams were flowing. Rather, community recovery in intermittent streams was related to species present at Waterfall Gully (the only stream in the catchment that remains perennially flowing), both in the dry season and early winter. Whilst organism movement was not directly measured, this result shows that recolonisation from perennial streams is likely more important in community recovery in these streams than *in situ* dry season refuges. Chester & Robson (2011) also found that perennial pools and flowing reaches supplied colonists to the broader catchment in southeastern Australian streams affected by suprasedseasonal drought, playing a greater role than drought refuges within intermittent streams. In this study, it appears most likely that it was the movement of flying insects (including Chironomidae, other Diptera and Ephemeroptera) from Waterfall Gully into intermittent streams that eliminated differences in assemblage composition between perennial and intermittent reaches by August. However, as also observed by Chester & Robson (2011), numerous species found in intermittent reaches when flowing were either present only in refuges or persisted in drought resistant stages when the streams were dry. Thus, whilst the

contribution of refuges was not significant to the assemblage overall, these habitats were important for several endemic species allowing them to persist in the catchment.

Biodiversity conservation and threatening processes in NJF dry season refuges

Very few macroinvertebrates appear on conservation listings for SWA, although several authors have identified many taxa of concern (Sutcliffe 2003; Bush & Theischinger 2004; Pennifold 2017, 2018) and identified climate change as a key threat (Davies 2010; Pennifold 2018). Gondwanan relictual species appear disproportionately negatively impacted by a transition to intermittency (Chapter 3), as streams sustained by runoff do not flow long enough for life cycles to be completed (Chapter 5). Whilst temporary aquatic habitats in newly intermittent streams may provide stepping stones for dispersal of widespread species (a process predicted to become more important as climatic drying progresses; Cañedo-Argüelles et al., 2020), it is likely that surface water refuges maintained by groundwater discharge will be integral to the persistence of several SWA endemic species as perennial streams continue to be lost to drying.

Spring-fed refuges were very rare in headwater streams, and of small spatial extent, and are therefore more vulnerable to severe disturbance. Introduced feral pigs (*Sus scrofa*) completely destroyed one refuge during this study, turning an inselberg spring pool into a turbid mud wallow. Feral pigs preferentially live near surface water, including wetlands, riverine pools and springs, upon which they have devastating impacts through their foraging activity (digging, turning over soil) and wallowing behaviours (Mihailou & Massaro, 2021). Pig damage has been shown to directly and severely impact invertebrate communities on exposed sediments of seasonal wetlands and intermittent rivers (Marshall et al., 2020), and alter microbial and invertebrate assemblages in streams, even where extensive physical damage is not evident (Kaller & Kelso, 2006). Hooved animals are absent from Australian native macrofauna, and since their introduction they have contributed to degradation of land

and waterways across the continent (Bengsen et al., 2014; Mihailou & Massaro, 2021). Although these streams occur in a completely forested catchment protected for water supply, small refuges and the aquatic species reliant upon them are threatened by severe non-natural disturbances in the form of feral pig activity. Furthermore, many regions globally are experiencing increasing fire frequency and severity due to global warming (Abatzoglou et al., 2019; Abram et al., 2021). Fires may impact small drought refuges directly through removal of riparian canopy, sedimentation, severely reduced water quality, and consequent mortality of inhabitant fauna, and stochastic events caused by fires, such as tree falls and landslips can completely destroy headwater springs (Verkaik et al., 2013; Dahm et al., 2015). Whilst stream macroinvertebrate communities in southern Australia have been shown to be resilient to large fires (Robson et al., 2018), mortality of individuals within these small refuges may remove whole populations in landscapes where species are otherwise extirpated by drying. Therefore, direct interventions to protect headwater spring refuges, such as removal of feral ungulates, will become more necessary to reduce the number of threatening processes acting upon them. Some of the methods used to protect larger desert spring ecosystems (such as feral mammal control, Davis et al., 2017) may need to be adapted for headwater springs.

Predicting refuges and future refugia: a conservation priority under drying climates

As anthropogenic climate change progresses, it is essential to identify both climate refuges and future evolutionary refugia across biomes (*sensu* Keppel et al., 2012) to enable species to persist in landscapes. Furthermore, some habitats that serve as ecological refuges may become evolutionary refugia under changing climates (Davis et al., 2013). For example, perennially flowing streams in SWA (such as Waterfall Gully) are becoming “ark type” refuges (*sensu* Robson et al., 2013) in which a suite of endemic species are increasingly isolated from the surrounding environment. Perennial flows will continue to diminish across SWA (Andrys et al., 2017; McFarlane et al., 2020), but if they are maintained at Waterfall

Gully it may become an evolutionary refugium in future as species lose capacity to disperse beyond that stream. Such species would need to sustain sufficient local population size to maintain their genetic diversity and avoid inbreeding depression (*cf.* Murphy et al., 2009; 2010); but this is unlikely to be possible for all the species currently relying on this refuge (e.g. dragonflies).

Anthropogenic refuges for aquatic fauna have become more important as natural habitats are altered by climate change (Chester et al., 2013; Robson et al., 2013). This study included one artificial waterbody, a weir pool fortuitously situated at the downstream edge of a spring-fed reach, which supported many taxa not present elsewhere including species of conservation significance. Recognition of particular types of anthropogenic waterbodies as potential refuges is important given widespread loss of habitat that such refuges might replace, however with weir pools, there is a potential conflict with the objective of restoring natural stream flows by removing weirs—a conflict that may be difficult to resolve in some cases. For example, the mussel *Westralunio cartei* (Hyriidae) is entirely dependent on permanent water and so is now absent from most of the catchment studied here. The weir pool in Seldom Seen Brook is a refuge, but the weir wall is also a barrier to recruitment, since larvae attach to native freshwater fish to disperse and flows are no longer sufficient to permit fish passage (if they ever were). Hence all individuals left in the pool are old and reproductively isolated, and it appears that extirpation is inevitable as the population ages (*cf.* Chester et al. 2014). Removal of the weir pool would lead to immediate loss of *W. carteri* because of flow intermittency, and native fish are no longer present in the stream. Other SWA endemic species would also be extirpated, so on balance more native species are protected and sustained in Seldom Seen Brook by retaining the weir than by removing it — but in other situations, the balance may differ. Such trade-offs must be considered in restoration and

biodiversity management (Chester & Robson, 2011) and optimal decisions will be made on a case-by-case basis.

The dry season refuges in the Wungong catchment have developed only recently: prior to widespread flow regime change perennially flowing streams were the norm, so the aquatic fauna were, until recently, perennial stream assemblages (Bunn, 1985). The species now occupying dry season refuges have either adapted to the loss of perennial flow or have newly arrived as intermittency created habitat favourable to lentic and desiccation resistant species. The interactions between refuge types and organism life history traits in streams with a long history of intermittency are now established as fundamental processes in community organisation within IRES (e.g. Robson et al., 2011; Bogan et al., 2015; Strachan et al., 2015; Stubbington et al., 2017; Table 7.1). The occurrence of dry season refuges (and the ability of species to use them) was therefore expected to be a critical determinant of community trajectory in formerly perennial streams. However, with the exception of a few refuges containing locally endemic species, it was the remnant perennial stream, a potential evolutionary refugium, that influenced assemblage recovery following the onset of flows. Similar patterns might be expected in other landscapes undergoing drying that harbour relictual faunas (e.g. Chester & Robson, 2011), but published research is currently largely lacking.

Understanding the relationship between geology and freshwater refuges would improve predictions of their occurrence across landscapes. Complex topography likely determines the presence of refuges through interactions with the local water balance (*sensu* Centeno et al., 2010). Here, granite inselbergs retained groundwater-fed summer refuges, and also subterranean refuges for fauna including species from an endemic genus of groundwater dependent amphipods. Potentially, these inselbergs will become evolutionary refugia for *Uroctena* sp., as perennial streams are lost and these amphipods become totally dependent on subterranean water during the dry season. Further eastwards, in the Western Australian

Wheatbelt and Great Western Woodlands, rock pools (gnammas) on the top of granite inselbergs support endemic freshwater plant (Tuckett et al., 2010) and animal species (Jocqué et al., 2007; Timms, 2012a,b), as well as having high cultural and other ecological values. The presence of macrophyte species evolved to live only in these rock pools provides evidence that they are freshwater refugia. In this study, the SWA endemic amphipods *Uroctena* sp. were exclusively found in spring-fed pools and reaches associated with granite inselbergs, indicating these are evolutionary refugia for these animals. There is a clear need to further investigate inselbergs as potential refuges and evolutionary refugia under climate change in other landscapes and continents. In arid central Australia, aquatic evolutionary refugia are connected to ancient groundwater discharge and almost always associated with topographical features that have high cultural and other ecological values (Davis et al., 2017; Davis et al. 2021). In Davis et al. (2021), all of the evolutionary refugia within a ~250,000 km² area were protected within National Parks or Indigenous Protected Areas. In contrast, very little is known of the occurrence of drought refugia in southern Australia and there are no defined predictors. One approach would be to investigate the origins of groundwater expressions associated with granite inselbergs by analysing isotope signatures as in Davis et al. (2021). Generally, expressions of older groundwater are decoupled from variations in rainfall, whereas recently formed aquifers are more vulnerable to a drying climate and may require other management actions to protect or augment flows. An improved understanding of landscape factors producing dry season refuges or refugia could lead to greater success in conserving species vulnerable to drying through climate change. Management actions resulting from this knowledge will necessarily be focussed on habitat that is limited in spatial extent, but which may be critical for maintaining species across the broader landscape.

Table S7.1: Fauna and number of individuals* recorded from in situ surface water refuges or from under stones/woody debris during streambed searches in 2017/18 dry seasons, or 2017 wet season from inselberg springs (groundwater refuge). *Note different sampling methods between reach types.

Order/Family	Lowest taxon	Season	Dry	Dry	Dry	Dry	Dry	Wet
			Spring-fed reach	Weir pool	Drying pools	Under stones	Inselberg seeps	Outcrop spring
Acariformes	Acarina		1	1				
Oligochaeta	Oligochaeta				13			
Platyhelminthes	Flatworms							2
Crustacea								
Decapoda	<i>Cherax quinquecarinatus</i>		12	11	6			
Amphipoda	<i>Perthia acutitelson</i>		6	1	7	2	1	
	<i>Uroctena</i> sp.		4				25	13
Ostracoda	<i>Cypretta</i> sp.					1		
	<i>Ilyodromus</i> sp.			2				
Copepoda	Cyclopoida			1	14			
Insecta								
Ephemeroptera	<i>Bibulmena kadjina</i>		13	25			3	
	<i>Cloeon fluviatile</i>				23			
Odonata	<i>Archeosynthemis leachii</i>		3					
	<i>Austroaeschna anacantha</i>			1				
	<i>Adversaeschna brevistyla</i>			2		1		
	<i>Hemicordulia tau</i>				2			
	<i>Miniargiolestes minimus</i>			1				
	<i>Pantala flavescens</i>				1			
	<i>Procordulia affinis</i>					1		
Trichoptera	<i>Lectrides parilis</i>		4	1				
	<i>Triplectides</i> AV1		1	2			1	
Hemiptera								
Corixidae	<i>Agraptocorixa</i> sp.				1			
	<i>Diaprepocoris</i> sp. F			1				
	<i>Micronecta</i> sp.			1				
	<i>Siagara mullaka</i>				1			
Nepidae	<i>Ranatra dispar</i>			2				
Notonectidae	<i>Anisops hackeri</i>				45			
	<i>Notonecta handlischii</i>				1			
	Notonectidae juv			1	1			

	<i>Paranisops endmyion</i>			1			
Mesovelidae	<i>Microvelia australiensis</i>			1			
	Mesovelidae nymph			3			
Veliidae	<i>Nesidovelia paramoena</i>					2	
	Veliidae juv					2	
Coleoptera	<i>Eretes australis</i>			5			
	<i>Necterosoma darwini</i>			47	2		
	<i>Plantynectes decempunctatus</i>	1		7	1		
	<i>Rhantus sutralis</i>			67		1	
	Scirtidae	4				1	
	<i>Sternopriscus browni</i>		3				
Diptera	Culicidae		1			3	
	Dolichopodidae sp.						1
	Psychodidae sp					1	
	Sciomyzidae	1					
(Chironomidae)	<i>Chironomus</i> spp.	46	6	14			
	<i>Cladotanytarsus</i> spp.			4			
	<i>Cricotopus</i> spp.	11					
	<i>Dicrotendipes</i> spp.			14			
	Orthoclaadiinae (unident.)						
	<i>Paralimnophyes</i> spp.	1					
	<i>Paramerina</i> spp.	5	1				
	<i>Polypedilum</i> spp.	11	6	3			
	<i>Riethia</i> spp.	1					
	<i>Stempellina</i> sp.	12	1				
	<i>Stenochironomus</i> spp.	2		5			
	Tanypod 1	2					
	Tanytarsus	44	2	16			
	Total richness	21	23	24	6	9	5

Chapter 8

General Discussion

The transition in flow regime from perennial to intermittency crossed a critical abiotic threshold in Northern Jarrah Forest (NJF) stream ecosystems, altering community composition and biodiversity patterns, and extirpating many formerly common species, including entire functional groups (e.g. large insect shredders). Although adaptations to flow intermittency are often reported among the aquatic fauna of Mediterranean (hereafter med-) and arid regions (Robson et al., 2011; Bogan et al., 2017), clearly the speed of recent drying in southwestern Australia (SWA) has outpaced the ability of many species to adapt.

Alterations of the hydrological cycle are expected to intensify across the globe over the next decade and beyond (IPCC 2021) therefore the threshold shift in flow regime that occurred across SWA will ensue in other regions of the world (and has already been documented in some cases, e.g. Bogan & Lytle, 2011; Piano et al., 2020; Crabot et al., 2021; Zipper et al., 2021). Because these widespread changes to flow regime are imminent (if not already happening; Polade et al., 2017; Zipper et al., 2021) studies that directly examine the biological reality of climate change impacts are vital to successfully predict extinction risk and opportunities for intervention at this precipice of global environmental change.

Below, I summarise the major findings from the five empirical research chapters in this thesis (Table 8.1). Sections are set out according to themes, not the order of chapters in the thesis.

Then, I synthesise this information to create a predictive framework for the future of NJF stream communities under projected climate change, and outline the factors influencing community and functional responses to flow regime drying.

Table 8.1: Summary of key research outcomes. PER = streams that retained perennial flow regimes; INT = streams that transitioned from perennial to intermittent flow regimes. IRES = intermittent rivers and ephemeral streams. NJF = Northern Jarrah Forest. SWA = Southwestern Australia.

Chapter title	Research aims & abbreviated hypotheses	Key outcomes
Ch. 3 “Loss of functionally important and regionally-endemic species from streams forced into intermittency by global warming” Target journal: <i>Global Change Biology</i>	Quantify and describe <i>in situ</i> impacts of flow regime drying on NJF stream communities at two temporal scales: 1) differences between contemporary assemblages and those historically present; and 2) seasonal patterns over one year between PER and INT streams.	Gondwanan relictual insect taxa (particularly Trichoptera, Ephemeroptera and Odonata), formerly the most common species in NJF streams, were severely negatively impacted by drying. New arrivals were predominantly widespread, tolerant species. Biogeography may partly predict species extirpations among vulnerable groups. The relictual SWA stream fauna is existentially threatened by climate change.
Ch. 4 “Reduced alpha diversity and increased turnover (β-diversity) in formerly-perennial headwater streams affected by permanent drying” Target journal: <i>Oecologia</i>	To compare patterns of α (richness) and β diversity (turnover and nestedness) between PER and INT streams at two scales: spatial patterns within contemporary assemblages; and temporal patterns between 1981/82 (pre-intermittency) and contemporary assemblages at the same sites. This is a partial mBACI study design, PER = control reaches, INT = impact reaches.	Spatial and temporal β diversity was dominated by turnover, not nestedness, contrary to patterns commonly reported in med-region IRES.
Ch. 5 “Life-history traits are poor predictors of species responses to flow regime change in headwater streams” Carey, N., Chester, E. T., Robson, B. R. (2021). <i>Global Change Biology</i> 27: 3547-3564 DOI: 10.1111/gcb.15673	To test predictions of life-history responses of 13 common macroinvertebrate species (12 insects and 1 crustacean) to <i>in situ</i> flow regime change, based on documented life histories and published traits (generally congeners or confamilials) as a predictor of future response. Size distributions, phenology, voltinism and synchrony were compared between a historic (1982/83) study and present (2016/17), between PER and INT reaches, thus comprising a partial mBACI study design. Hypotheses were made based previous phenology of 13 species (12 insects and 1 crustacean; Bunn 1988) coupled with likely	Restriction to perennial reaches was the most common response (7 spp.) including those predicted to readily adapt. Two species were absent altogether. Four species adapted, including those predicted not to. Confamilial Leptophlebiidae mayflies showed opposite responses in synchrony and voltinism, a stonefly changed its hatching and development time, and an amphipod utilized subterranean refuges to survive.

Chapter title	Research aims & abbreviated hypotheses	Key outcomes
	desiccation resistance or resilience traits of similar species published in the scientific literature (Table 5.1)	None of the observed responses matched predictions, thus known traits could not predict flexibility (or lack thereof) in life histories, challenging the use of traits as predictors of survival under climate change.
Ch. 6 “Flow regime change alters shredder identity but not leaf litter decomposition in headwater streams affected by severe, permanent drying” Carey, N., Chester, E. T. & Robson, B. J. 2021. <i>Freshwater Biology</i> 66(9): 1813-1830. DOI:10.1111/fwb.13794	Examine changes to the decomposition process of jarrah leaves (key carbon source) and the role of macroinvertebrate shredders in streams impacted by flow regime drying, using a leaf bag experiment and two pre-planned comparisons (between contemporary PER and INT; temporal comparison between historic (1981/82) and contemporary breakdown patterns in PER and INT; an mBACI design).	Leaf litter decomposition appears robust to flow regime change. Large shredding caddisflies are extirpated from INT streams, leaf mining Chironomidae appeared to provide some functional redundancy for loss of shredders in INT streams.
Ch. 7 “Drought refuges, refugia and flow-regime change in southern Australian streams” Target journal: <i>Freshwater Biology</i>	Identify drought refuges in INT streams and characterise the macroinvertebrate species using them. Conduct sediment rehydration experiments to identify taxa using desiccation resistant life stages (e.g. eggs and aestivation). Determine the contribution of drought refuges to community recovery, compared to colonisation from the remaining PER stream (MANTEL tests).	Each refuge type supported distinct assemblages, and springfed refuges were found to support several relictual species throughout the dry season. Resilience to drying was more common than resistance traits among the fauna, and dry sediments supported more aestivating taxa rather than a large propagule bank. Drought refuge assemblages were not related to community recovery, however PER stream likely supplied colonists to INT streams.

Summary of key findings

The existence of Stuart Bunn's work from the early 1980s provided quality 'before' drying information on macroinvertebrate communities and life histories with which to examine the direct impacts of flow regime drying. This was a valuable opportunity for comparative research, without which the critical findings in this thesis would not have been possible. This research is an empirical test of predicted outcomes based on the literature, which in some cases directly contradicted processes understood to underpin invertebrate adaptations to stream intermittency (i.e. known life history traits were not able to predict survival in practically every case, or that the fauna of a med-region with a history of aridity should be robust to climate drying). Where similar opportunities for comparative research exist this should be made a priority, as among its many values is the potential to unearth errors of logic in accepted theory that do not reflect ecological reality. In this case, elucidating the gap between predicted outcomes and actual responses of invertebrate communities revealed that catastrophic biodiversity loss is imminent in SWA streams – there is no longer time for errors in understanding if biodiversity loss is to be stemmed in this present era of rapid global warming.

Altered biodiversity patterns caused by drying

Changes to regional biodiversity under climate change is a topic of great current interest and increasingly, biodiversity partitioning studies are used to examine changes to alpha (α) and beta (β) diversity patterns to predict changes in landscape (gamma, γ) diversity (Anderson et al., 2011; Sarremejane et al., 2017; Soininen et al., 2018; Crabot et al., 2021). Disturbance may alter β diversity patterns, but studies report both increased heterogeneity and increased homogeneity between sites (Baselga, 2010; Socolar et al., 2016; Rolls et al., 2018; Robson et al., 2018). Partitioning β diversity into nestedness and turnover components can further predict trajectories of community response to environmental change, an increase in

nestedness may indicate subtractive homogenization at impacted sites (*sensu* Rolls et al., 2018), whereas high turnover among impacted sites can indicate stochastic extirpations and/or arrivals increasing assemblage heterogeneity, subsequent to local habitat controls and dispersal (Socolar et al., 2016). Examination of β diversity patterns, such as between stream reaches in a drainage basin, can elucidate landscape connectivity and the roles of local environmental factors, which when scaled up may predict vulnerability to extinction under climate change, or identify potential ark-type refuges or refugia (Robson et al., 2013; Socolar et al., 2016).

Within med-region stream networks, perennial streams typically support more species (higher α diversity) than intermittent streams (Soria et al., 2018), which harbour a subset of species with drought resistance or resilience traits, and may include specialist taxa (Chester & Robson, 2011; Datry, 2014; Leigh & Datry 2016; Stubbington et al., 2017). That is, flow impermanence usually causes a nested pattern of macroinvertebrate assemblages in stream networks (Bonada et al., 2007; Datry et al., 2014; Cañedo-Argüelles et al., 2015, Leigh et al., 2016; Ruhí et al., 2017), and thus the imposition of novel intermittent flow regimes was expected to produce a nested pattern across Wungong catchment streams (see also Vander Vorste et al., 2021). However, contrary to this prediction, spatial and temporal β diversity was overwhelmingly comprised of species turnover between reaches, increasing among-reach heterogeneity (Table 8.1). Temporal declines in α diversity at recently intermittent reaches meant that species lost were only partially replaced by newly arriving species (*c.f.* Bunn et al., 1986; Chapter 3). Thus, rather than observing patterns understood to be typical of med-region intermittent stream networks, the disturbance caused by flow regime change has increased reach-scale heterogeneity of macroinvertebrate assemblages (*sensu* Houseman et al., 2008; Rolls et al., 2018), including between reaches that are longitudinally connected by flow during winter and spring.

High turnover may be indicative of stream networks in transition to drier flow regimes; the few studies directly examining the impacts of flow regime change on macroinvertebrate biodiversity patterns also report increased β diversity, primarily comprised of turnover (Aspin et al., 2018; Pařil et al., 2019; Crabot et al., 2021; although Piano et al., 2020 report homogenization but at higher taxonomic resolution). In the Wungong catchment, reach-scale environmental factors (notably increased variation in local hydrology) are assumed to be maintaining the drought-tolerant remnant of the perennial assemblage and providing space for newly arriving species. This assumes that the assemblage of each reach is a non-random product of the past assemblage, modified by extirpation, organism dispersal, drought resilience or resistance traits, interacting with local environmental conditions (e.g. Cañedo-Argüelles et al., 2015; Rolle et al., 2016; Datry et al., 2017b; Sarremejane et al., 2017). Furthermore, analysis of patterns in temporal β diversity between Bunn (1985; Bunn et al., 1986) and contemporary flow regimes found that assemblages at the driest reaches (CC1 and CH2) were partially nested within the previous assemblage. This nesting is almost certainly due to the rapid dry down at these reaches, whereby disconnected pools do not last long enough for highly mobile colonists to establish (e.g. newly arriving Coleoptera, Hemiptera, Odonata, *Cloeon* mayflies; see below), as occurs at other streams where pools last longer. The downstream reach on Seldom Seen Brook (SS1) was also a partially nested assemblage, however drift from the perennial upstream reach (SS2) maintained α diversity there. Whilst dispersal ability is predicted to become more important in determining broadscale community structure as climates continue to dry (Cañedo-Argüelles et al., 2020), this analysis of temporal patterns in α and β diversity instead suggests that the influence of dispersal has limits (e.g. limited by short hydroperiods). It points to another threshold shift as flow regimes become drier: the eventual loss of perennial reaches (such as SS2) and faster recession of disconnected pools. This will cause an increase in temporal nestedness as species are lost but

not replaced, and subtractive homogenization of assemblages as the range of hydroregimes present in a catchment becomes much more narrow and thus more similar.

Interactions between biogeography, seasonality and flow regime

The endemic invertebrates of SWA upland streams are primarily relictual species of Gondwanan lineages (i.e. from cooler, wetter climates, Davies & Stewart, 2013), which have survived the continental aridification of Australia and previously drier times in SWA (Bunn & Davies, 1990). However, species that either persisted, colonized, or were extirpated from recently intermittent streams showed differences in their biogeography. The majority of species restricted to Waterfall Gully had Gondwanan lineages, including most locally endemic Trichoptera, Odonata, and Ephemeroptera (Figure 8.1; Appendix 1). Contrastingly, many of the species that became more common have northern affinities and were generalists found widely across the continent's IRES (e.g. beetle *Allodessus bistrigatus* and *Eretes australis*, mayfly *Cloeon fluviatile*, *Tanytarsus* sp. chironomids). In this instance, biogeographic history may partly inform extinction risk under climate change (see also Whittaker et al., 2005; for freshwater fish Olden et al., 2010), and is of particular relevance in SWA where opportunities for species to retreat to higher latitudes or altitudes are practically nil (Davies, 2010). Because there is little taxonomic redundancy among this ancient but depauperate fauna (many genera and some families are represented by a single species; Bunn & Davies, 1990) climate change is poised to greatly reduce the taxonomic diversity among SWA stream and wetland (Atkinson et al., 2021) fauna.

Formerly, small seasonal differences in macroinvertebrate assemblages were associated with predictable increases and decreases in streamflow, with a core assemblage of species present year-around that either had long life cycles or bred continuously (Bunn et al., 1986; Bunn, 1988a). Following transition to intermittency, divergence in biogeographic affinities was also evident between the lotic and lentic phases. During the lotic phase, species of Gondwanan

affinity were among the most common, for example: dragonfly *Archaeosynthemis leachii* was common during early winter flows; leptophlebiid mayfly *Nousia* sp. AV16 was the most common mayfly of intermittent streams, and the philopotamid caddisfly *Hydrobiosella michaelsoni* was abundant in early winter at Chandler Brook. The mayfly *Nousia* sp. AV16 is apparently a new arrival to these streams post-intermittency, however it is likely this species was always present within the catchment, breeding in the ephemeral streams associated with granite outcrop springs, and colonising newly-intermittent streams from there. The species that characterized the lentic phase at intermittent streams tended to be widespread, tolerant species, including typical ‘pool’ taxa (*sensu* Bonada et al., 2007; Bogan et al., 2019). At Wungong Brook and Curtis Creek, disconnected pools lasted several months giving successive invertebrate communities sufficient time to establish. This included a suite of opportunistic Coleoptera and Hemiptera, cosmopolitan mayfly *Cloeon fluviatile*, and widespread dragonflies *Hemicordulia tau* and *Anax papuensis*. This increase in seasonal turnover aligns with the “time sharing” dynamic observed in many IRES (*sensu* Bogan & Lytle, 2007; Garcia-Roger et al., 2011; see also Bogan et al., 2019; Bonada et al., 2020). This was not observed at Chandler Brook, which had the shortest hydroperiod, lowest richness and was further along the trajectory of drying than the other streams. Whilst the lotic phase provided a short period of high flow habitat for a few specialized endemic taxa (e.g. *Nousia* sp. AV16 and *H. michaelsoni*), rapid dry down meant there was insufficient time for the establishment of a lentic phase fauna.

Contrasts occurred between species biogeography and persistence under rapidly changing flow regimes, and this may be applicable to other med-regions where the endemic fauna is relictual. For example, the med-climate regions of southeastern Australia, Chile, and South Africa are also of Gondwanan lineages. It is probable that species there will show similar extirpations as climatic drying ensues in those regions (Filipe et al., 2013; Figueroa et al.,

2013; DeMoor & Day, 2013; Davies & Stewart, 2013; Bonada & Resh, 2013). Their extinction risk will then depend on opportunities to retreat to higher latitudes or altitudes, and the presence of evolutionary refugia in the landscape (*sensu* Keppel et al., 2012; Davies et al., 2013).

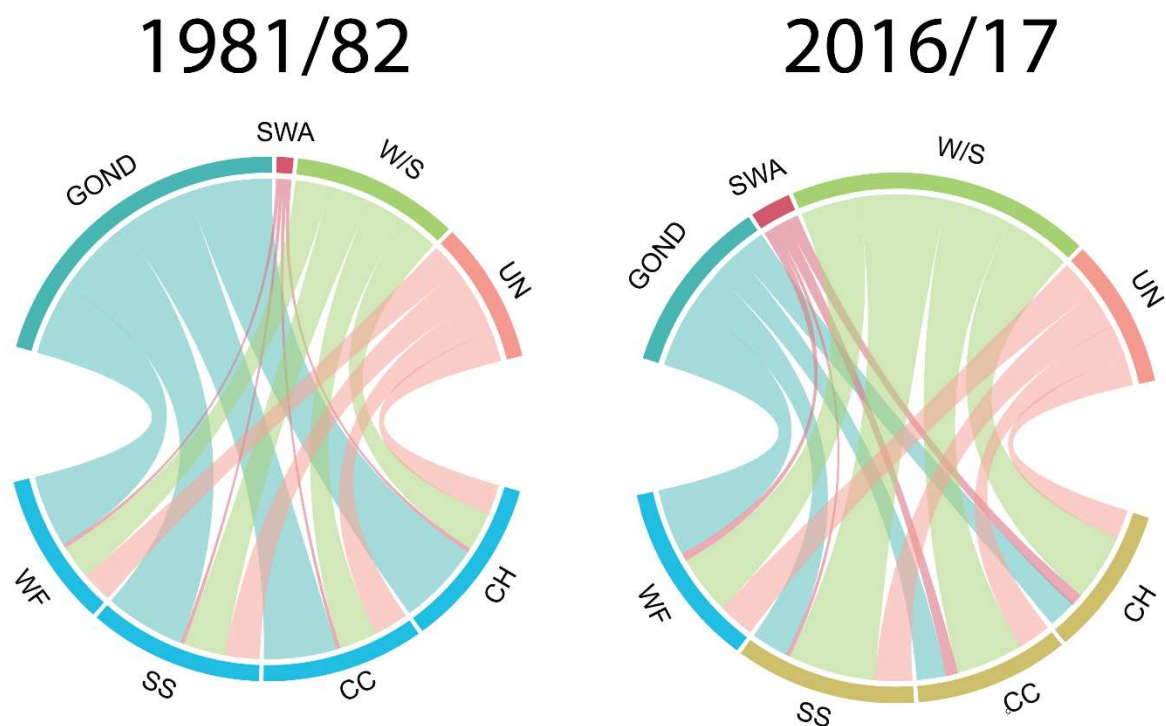


Figure 8.1: Temporal change in biogeography of dominant species present in 1981/82 (Bunn 1985) revisited in 2016/17. Data is the standardized proportional dominance dataset used in for temporal comparisons in Chapters 3 and 4 (all species that comprised at least 5% of individuals at least one site, at least one sample time). GOND = Gondwanan endemic; SWA = southwest Australian endemic, other affinity; W/S = widespread or cosmopolitan species; UN = unknown affinity, usually higher taxonomic identification. Classification of biogeographic affinity as Table 4.2 and Table 4.3, see also Appendix 1.

Maintenance of leaf litter breakdown and an unexpected source of functional redundancy

Flow regime is a fundamental abiotic control of stream ecosystem function (Gasith & Resh, 1999; Bernal et al., 2013) including the breakdown of leaf litter, a key process in the detrital carbon pathway moderated by macroinvertebrates (Bunn 1988b, 1988c; Boyero et al.,

2007; Tank et al., 2010). Flow impermanence typically slows detrital breakdown rates, by interrupting the physical processes of leaching and abrasion, and altering the biotic communities associated with this process (Corti et al., 2011; Datry et al., 2011; Abril et al., 2016; Von Schiller et al., 2017). However, leaf decomposition on dry streambeds is not static, as terrestrial processes including photodegradation, colonization by fungi and microbes continue, preconditioning particulate matter and nutrients to be released as a pulse upon the resumption of flow (Datry et al., 2012). Thus in intermittent streams, breakdown of allochthonous material and transport of particulate carbon and nutrients longitudinally and laterally is variably moderated by both aquatic and terrestrial processes (Datry et al., 2018a).

The transition to intermittency, and associated differences in physical and biotic processes, did initially alter decomposition rates of the key carbon source (*Eucalyptus marginata* leaves; Bunn 1988b; Davies, 1993) as leaves positioned on dry streambeds retained more weight than those immersed at the perennial control reach. However, as expected, weight loss increased with the resumption of flow (likely primed by terrestrial processes and photodegradation; Dieter et al., 2011; Shumilova et al., 2019) so that by the end of the experiment, leaf weight loss was similar across flow regimes (Table S8.1). This was despite the near complete loss of the shredding guild at reaches affected by intermittency, notably large caddisflies which were present at Waterfall Gully (Table 6.3). Shredders were present at intermittent reaches (e.g. caddisfly *Triplectides* sp AV1, and stonefly *Leptoperla australica*; Chapter 3) but their distributions were highly patchy across intermittent streams and no shredder species colonized leaf bags. Some species of *Uroctena* amphipods are known to shred leaves however no evidence of shredding activity was associated with their presence (e.g. skeletonized leaves), nor were they reported as shredders in Bunn's study. Species boundaries within *Uroctena* are uncertain and currently under review, therefore the lack of shredding may be due to species differences in feeding.

The leaf mining chironomid *Stenochironomus* sp. was present across study reaches, and was the most frequent (and often, only) shredder found in the intermittent reaches.

Stenochironomus sp. was not reported as a shredder by Bunn (1988b,c), and would not have been missed because the way it processes leaves is obviously different to that of other shredders such as caddisflies (see Figure 8.2). Thus, the inclusion of this species as a key shredder is likely a recent development and indicates that this species may be providing some functional redundancy for the loss of other large shredding species (i.e. caddisflies, stoneflies) in these streams. *Stenochironomus* has a global distribution, and is one of the few truly leaf mining genera of Chironomidae (Berg, 1995). Most recent ecological research on *Stenochironomus* comes from South America, where it is regarded as an important shredder in tropical streams and indicator of good water quality (Henriques-Oliviera et al., 2003; Serpa et al., 2020; de Mello Cioneck et al., 2021).



Figure 8.2[6.2]: Remnants of *Eucalyptus marginata* leaves, showing shredding activity of *Notoperata tenax* (Trichoptera; A) and *Stenochironomus* sp. (Chironomidae; B). Leaves were collected in January 2019 (320 days) from coarse mesh bags in Waterfall Gully (A) and Seldom Seen Brook (B).

The Chironomidae had as complex a suite of responses to flow regime drying as the macroinvertebrate assemblage as a whole. There were changes in occupancy, extirpations, new arrivals, and increased dominance of cosmopolitan species among the chironomid fauna. Overall, a greater proportion of the assemblage in 2016/17 comprised Chironomidae than in 1981/82. A greater relative proportion of Chironomidae appears to be a common compositional change related to disturbance caused by drying (Ledger et al., 2013; Hille et al., 2014; but see Aspin et al., 2018, 2019), and increased dominance of Chironomidae has

also been reported from streams subject to increasing temperatures (Dudgeon et al., 2020). However, few studies examine finer scale differences in functional and taxonomic organisation of this group, opting to leave identifications at the family or sub-family level (Raunio et al., 2011). These differences may be important because the chironomid species occupy most habitat and functional niches (Berg 1995; Pinder 1995), and their functional importance (including as a source of redundancy) is likely to increase as climate change progresses. Secondly, species losses among the Chironomidae are likely to be unnoticed, thus biodiversity declines and potential reduction in functional redundancy may not be reported. Cañedo-Argüelles et al. (2016) explored the utility of the Chironomidae as indicators of dryness, concluding that chironomid assemblages are sensitive to drying, and good indicators of water scarcity. My results concur with this, as analysis of the Chironomidae alone would have indicated similar responses to flow regime change as the whole assemblage. Unfortunately, this group are often ignored unless they have reached pest status. However, genera known to be pollution tolerant (e.g. *Chironomus* sp., *Polypedilum* sp., *Tanytarsus* sp.) were the most adaptable to flow regime drying in this study and expressed drought resistance traits (likely both aestivation and desiccation resistant eggs) when those of Gondwanan lineages (e.g. *Riethia*, *Stictocladius*) apparently did not. This is testament to the resilience and adaptability of these cosmopolitan genera to a wide range of conditions, which will allow them to fill functional gaps created by the retreat of the many other organisms negatively impacted by drying. This redundancy is likely, as here, to maintain important ecosystem processes (e.g. algal grazing, litter breakdown), so the value of these cosmopolitan genera should be recognised.

Life history flexibility

The flow regime is the most important factor shaping organism life histories in streams, as the presence of absence of water is an inescapable abiotic control with which aquatic

organisms must contend (Gasith & Resh, 1999; Bunn & Arthington, 2002). Certain life history traits have obvious advantages to organisms in temporary waters, including fast development, small body size and desiccation resistance (Williams, 2006; Strachan et al., 2015). However, flexibility in life histories, such as the presence of multiple cohorts per year, ability to split cohorts, and variable larval development rates also suggest adaptability to shorter flow regimes (e.g. Arscott et al., 2010; Storey & Quinn, 2011). In nearly every case, previous life history traits coupled with predicted drought resistance/resilience traits (*sensu* Strachan et al., 2015) did not accurately predict actual response to stream flow drying. Extirpations were higher than predicted using these traits, and three of the four species that did adapt did so in ways that were not predicted (Table 5.1) Indeed, as far back as 1973 Lehmkuhl questioned the validity of fixed life history phenology when considering environmental change, and my results support the notion that these traits provide limited predictive power in the face of substantial environmental change.

Three Leptophlebiidae mayflies illustrate this disparity between predicted and actual response to flow regime change. Two species, *Neboissophlebia occidentalis* and *Nyungara bunni* had very similar phenological traits in 1982/83: two loosely overlapping cohorts, including one winter-spring cohort, and two periods of unsynchronized emergence in spring and autumn. Both were thus predicted to have the same response (increased synchrony and switch to a single winter cohort). However, *N. occidentalis* was absent from streams affected by intermittency, whereas *N. bunni* switched to a single, fast growing cohort and highly synchronous emergence in early summer, well before streams had dried. The third species, *Bibulmena kadjina*, had the most remarkable response of all – previously this species had a single, strongly synchronized cohort, larvae grew slowly throughout the entire year which suggested extirpation was likely in streams that are now dry for several months of the year. Rather, larval synchrony completely broke down, and nymphs were present in a range of

sizes throughout the hydroperiod at intermittent streams. Then, the onset of stream disconnection in early summer triggered another response – one group of nymphs matured and emerged, and a second group persisted in dry season refuges throughout summer and autumn, growing to an enormous size in comparison to the slow growing nymphs of 1982/83 (Figures 5.6 & 5.7). The completely disparate responses by three confamilial mayflies demonstrates very clearly the gap in understanding of the key role of phenological flexibility in determining species survival under drying climates. Furthermore, flexibility (or lack thereof) across multiple life history stages is will play a role in species persistence. For example, the absence of *N. occidentalis* from intermittent streams is unlikely to be a result of phenological mismatch between presence of adults and flowing water at intermittent streams. Instead, another process such as lack of lateral dispersal by adults, or inflexibility in oviposition site selection, may be hindering their recolonisation. Populations of *N. bunni* were much smaller at intermittent streams than at perennial Waterfall Gully and because long lived adults are not known among the Ephemeroptera, perennial streams are the likely population source of this species, unless adults are laying desiccation-resistant eggs in dry streambeds – which is highly improbable. If life history traits for which detailed information exists (as here) could not accurately predict responses by most species, then traits for which we have no information are also likely to be equally important in predicting survivorship.

Life history flexibility in aquatic invertebrates is well demonstrated in response to water temperatures, which may trigger faster development, earlier emergence and larger adult size (e.g. in mayflies, Briers et al., 2004; Everall et al., 2015; damselflies, Block & Stoks, 2005), and differences in development rate have been observed between perennial and intermittent streams in select cases (e.g. Robinson & Buser, 2007; Storey & Quinn, 2011). In a mesocosm study, Lund et al. (2016) demonstrated a degree of flexibility among caddisflies from alpine ponds in response to shorter hydroperiods, including expedited emergence and cannibalism

(postulated to aid quicker maturation of survivors through a high protein food source).

However, complete population mortality was concurrently observed in the actual environment as the hydroperiod was even shorter than the expedited emergence timing. There are few similar studies of phenological responses to novel drying, but there is a clear need for more such studies as freshwater habitats globally undergo threshold changes in hydroperiod. Descriptive life history research, such as presented in Chapter 5, is the only true test of species responses to real climate impacts in their environments. Where possible, multiple before-after control-impact study designs should also be used to gather empirical life history information, which will improve efficacy of predictive models using traits.

Projected impacts of drying on stream ecosystems

The flow regime drying that occurred after 2000/01 in the Wungong catchment was a threshold change (*sensu* Beisner et al., 2003; Suding & Hobbs, 2009), which caused a near complete reorganization of aquatic communities. This included altered macroinvertebrate assemblage composition, a change in biogeographic provenance among the key fauna, a reduction in taxonomic distinctness (*sensu* Marchant, 2007), and increased risk of localized stochastic extinctions due to high turnover (*sensu* Baselga, 2010; Socolar et al., 2016). There was some functional redundancy within the existing fauna, demonstrated among the Chironomidae – species with small body sizes, rapid development, high fecundity, and likely desiccation resistance traits (Williams, 2006; Arscott et al., 2010; Strachan et al., 2015). Increased habitat heterogeneity opened niches for new colonists with more resilient or resistant life histories (for example widespread Coleoptera, and *Nousia* sp. AV16 mayflies). However, despite these substantial changes, ecosystem function has been maintained, patterns of α diversity align to those understood as common among med-region IRES networks (Leigh & Datry, 2016; Soria et al., 2017), loss of large shredders has been partially compensated for by terrestrial breakdown processes and the arrival of larval *Stenochironomus*

sp., and some locally endemic species maintained populations by using new drought refuges within the landscape to survive dry periods.

Unfortunately, climatic drying in this region is expected to intensify in years to come even if greenhouse gas emissions are reined in (Andrys et al., 2017; Hoegh-Guldberg et al., 2018; IPCC 2021). The single remaining perennial stream (Waterfall Gully) supported the bulk of the historic invertebrate assemblage, comprising ancient relictual species, and likely supports community recovery in intermittent streams as a source of dipteran and ephemeropteran colonists (Chapter 6). Thus, a substantial proportion of the biodiversity within this catchment is reliant on Waterfall Gully maintaining perennial flows. Flow at Waterfall Gully has already declined considerably corresponding with a 20% decline in rainfall (Figure 2.5). Inevitably, perennial flows at Waterfall Gully will cease. Drought and drying cross several critical thresholds of physical tolerance among aquatic fauna, which at the reach scale results in several step-wise transitions in community structure (Boulton, 2003; Bogan et al., 2015). The widely used model in Boulton (2003) provides a useful framework with which to project future changes to catchment wide aquatic diversity, as future drying thresholds are crossed. Below I use this model, coupled with evidence from this thesis to predict future changes in gamma (γ) diversity and turnover (β diversity) of SWA forested headwater streams, as climate change progresses (Figure 8.3).

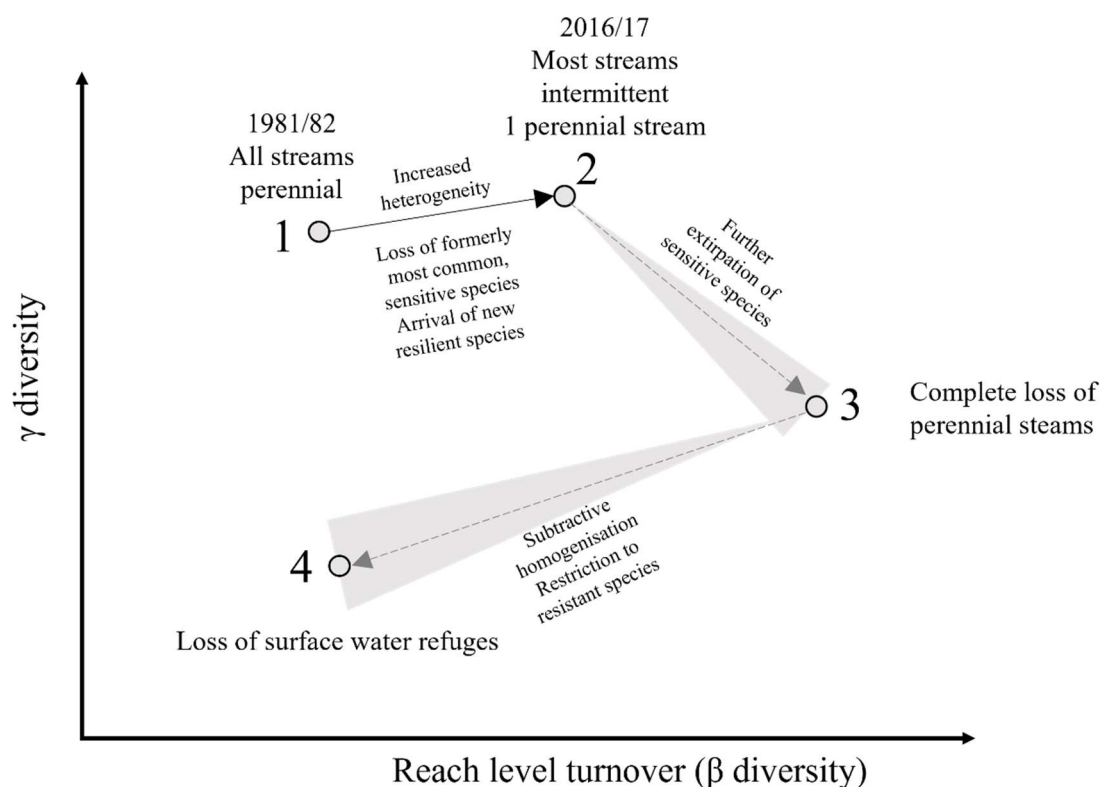


Figure 8.3: Observed changes to regional (γ) diversity and turnover (β diversity) forms the basis of projected changes to γ diversity and species composition with continued climatic drying. Threshold stages in drying based on the model by Boulton (2003): Stage 1 to 2 = initial disconnection and drying (in most, not all streams); stage 2 to 3 = complete seasonal loss of flow; Stage 3 to 4 = seasonal loss of surface water.

Stage 1 to 2: Widespread transition to intermittent flow regimes (1981/82 – 2016/17)

Between the historic and present study, there was a slight increase in gamma (γ diversity) across the Wungong catchment. The transition to intermittency in most streams of the catchment, which spatially vary in hydroperiod and length of the drying phase, has greatly increased habitat heterogeneity on the whole and resulted in an approximate doubling of catchment β diversity (turnover). Typically, macroinvertebrate communities in SWA perennial streams are characterized by high temporal persistence (Bunn et al., 1986; Bunn & Davies, 2000; see also Metzeling et al., 2002), whilst this has likely changed at the

intermittent streams, it is maintenance of stable populations at Waterfall Gully that is sustaining a component of community recovery in intermittent streams. Summer refuges and drying pools provide habitats for widespread colonist species, sustaining their dispersal across this drying landscape (Bogan et al., 2019; Bonada et al., 2020; Cañedo Argüelles et al., 2020) and promoting a “time sharing” dynamic (*sensu* Bogan & Lytle, 2007) between summer cosmopolitan opportunists, and high flow intermittent stream specialists in winter. Perennial streams act as ark-type refuges for the original assemblage of SWA endemic and Gondwana relictual fauna (*sensu* Robson et al., 2013), and retains the highest alpha (α diversity).

Stage 2 to 3: Loss of perennial streams

In SWA, groundwater is continuing to decline independently of interannual variations in precipitation (Petrone et al., 2010; McFarlane et al., 2020), which eventually will result in the next threshold change in water regime in the NJF, being the loss of the few remaining perennial streams. This will cause an almost instantaneous extinction of at least half of the formerly most common species in the catchment (Table 3.2; Appendix 1), including most of the larger relictual insect species (i.e. nearly all endemic species of Odonata, Trichoptera, and Ephemeroptera). Recolonisation processes supported by movement of taxa from Waterfall Gully to the intermittent streams each winter will also cease; thus this decline in γ diversity may be even greater.

Accompanying the sudden drop in γ diversity is a further increase in catchment turnover. Subtractive heterogenization (*sensu* Socolar et al., 2016) caused by transition to intermittency at Waterfall Gully will cause reaches within this stream to become as different to each other as to reaches on the other streams. Small refuges may occur on Waterfall Gully, particularly at the headwater spring. Dry season refuges within this landscape may continue to support small populations of a few endemic species, including mayfly *Bibulmena kadjina*

(Leptophlebiidae) and caddisfly *Triplectides* sp. AV1 (Leptoceridae). Spring fed refuges will be particularly important for endemic species, notably the amphipod *Uroctena* sp., which occurs both in surface water and subterranean refuges associated with granite inselbergs. However, populations of some species within these refuges may become very small and non-viable as perennial habitats dwindle. High turnover is maintained in this stage of drying by the presence of refuges in the landscape, however, these spatially small habitats are vulnerable to other degrading factors (such as feral pigs) and are more susceptible to damage from fires. Therefore, the species reliant on these refuges will be in a precarious position and will likely require human interventions in the form of feral animal management and protection in small, managed reserves to continue to support biodiversity.

Stage 3 to 4: Complete loss of surface water refuges

Loss of all surface water during the dry season is analogous to the final threshold of drying in Boulton (2003), comprising complete annual desiccation. This final threshold of drying in the stream network is accompanied by sharp declines in both γ diversity and turnover (Figure 8.3), as assemblages are restricted to desiccation resistant specialists and the most resilient species (e.g. long lived Coleoptera which are able to survive the dry season as terrestrial adults). Species sorting is thus determined by the presence of these traits among the fauna, along with local habitat factors, likely causing an increase in nestedness among reaches. Because desiccation resistance traits were not common among this stream fauna (Chapter 6), in comparison to faunas of other temporary waterbodies dominated by microcrustaceans (Williams, 2006), α diversity is likely to be quite low.

Under progressively changing climates, evolutionary refugia will become critical for conserving biodiversity (Keppel et al., 2012). Examples of aquatic evolutionary refugia occur across the Australian desert zone, these are permanent surface water habitats fed by ancient groundwater, decoupled from the prevailing climate (Davies et al., 2013). These arid zone

spring fed systems are reservoirs of freshwater biodiversity, as an ark for relictual species, and may also act as ecological refuges, sustaining dispersal of riverine species across the broader landscape. Desert springs have also been demonstrated as critical habitats in US Arizona desert (Bogan et al., 2014); and other regions. In contrast, no ancient groundwater-fed refugia exist in SWA, and with the exception of potential subterranean refugia for *Uroctena* sp. amphipods, there is limited opportunity for SWA stream species to retreat to evolutionary refugia. Furthermore, there is little scope for range contractions southwards and none for movement to higher altitudes (Davies, 2010). In regions undergoing climatic drying, it is the presence of evolutionary refugia that will be critical to sustaining freshwater biodiversity, and future research should focus on identifying refugia as well drought refuges in landscapes.

Conclusion: the impacts of flow regime drying on headwater streams.

Freshwater biodiversity is in decline globally, and climatic drying imposes an abiotic environmental filter which will alter community trajectories in headwater streams (Dudgeon et al., 2006; Poff, 2018). Faced with threshold environmental change, the maintenance of ecosystem function is reliant on the responses of the most common and functionally important taxa (*sensu* Pyne & Poff, 2017), which in this case mostly ranged from negative (reduced occurrence), to catastrophic (complete extirpation) among the formerly most common taxa. Headwater streams contribute disproportionately to biodiversity beyond their own bounds and can exert strong influences on downstream community composition, particularly where headwaters are perennial (Finn et al., 2011; Crabot et al., 2020). Therefore, step wise threshold shifts in flow regime will negatively impact regional biodiversity through losses of sensitive species, and disruption of dispersal and colonization processes across stream networks as well as in reaches directly affected by drying.

Climate change is not the only stressor acting upon stream communities in the Anthropocene – land cover change, pollution, water extraction and river regulation are a few examples of human alterations to stream networks that degrade ecosystems under present day conditions (Ormerod, et al., 2010; Datry et al., 2018b). Whereas historically species may have withstood drier climates, human alteration of the environment has effectively reduced opportunities for species to move and reduced availability of potential refuges and refugia, and thus the likelihood of ‘historical accidents’ which otherwise would have meant survival. If species are unable to use refuges that may have otherwise existed, for example by physical removal of refuges (landuse change, pollution, infilling) or interrupted dispersal pathways (river regulation, including physical barriers and large reservoirs) then opportunities to persist in landscapes become limited.

This research demonstrated that even in a forested catchment protected for water supply, where these other human impacts were largely absent, stream flow drying exceeded the capacity for many species to adapt. Therefore, the assumption that previous survival of drier climates predicts future survival is not supported by this research; even in absence of other major human impacts (see also Davies, 2010). However, there are a few opportunities for direct interventions to assist freshwater fauna in this landscape. Because of dwindling streamflow, large reservoirs such as the Wungong Dam now contribute negligibly to water security in SWA; however only a small amount of water would be required to artificially maintain baseflow at Waterfall Gully and the perennial reach on Seldom Seen Brook, preventing their transition to intermittency. This would have the immediate effect of sustaining populations of many Gondwanan relictual species in the catchment, without which their extinction is imminent. Should there be a return to wetter conditions in future, these species would then be able to disperse from this artificially maintained refugium to repopulate other streams; there is no coming back from the do-nothing scenario.

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APPENDICES

Appendix 1: Flow gauges

Flow gauges in the Wungong Catchment			
Site			
reference	LATITUDE	LONGITUDE	SPHEROID
616021	-32.2509	116.0905	GDA94
616022	-32.2545	116.0794	GDA94
616023	-32.2094	116.0786	GDA94
-			
616041	32.24792129	116.1124386	GDA94
616124	-32.2917	116.1305871	GDA94
616143	-32.234227	116.1059808	GDA94
-			
616144	32.22943291	116.1143907	GDA94

Flow gauging data used in Figure 2.3			
Site			
reference	LATITUDE	LONGITUDE	SPHEROID
609018	-33.9434952	115.6914468	GDA94
-			
609017	33.79755883	115.9536219	GDA94
-			
609016	33.91819459	116.0976479	GDA94
-			
611009	33.52937463	115.9686106	GDA94
-			
613002	33.08708274	116.0389452	GDA94
-			
613020	32.93017747	116.0337154	GDA94
-			
614047	32.75559937	116.0999537	GDA94
614017	-32.5946	116.0287	GDA94
614062	-32.5829	116.0273	GDA94
-			
616041	32.24792129	116.1124386	GDA94
616021	-32.2509	116.0905	GDA94
616022	-32.2545	116.0794	GDA94
616023	-32.2094	116.0786	GDA94
-			
6141030	32.51212041	116.0392579	GDA94

Appendix 2: Selected species summaries of pre-drying and post-drying distribution in Wungong catchment streams (*c.f.* Bunn 1985), biogeographic affinity (where known) and references for further information. Bunn 1985 status: Common; Present = appears in species list, but not numerically dominant; Not recorded; Not recorded (known) = not found in Wungong catchment streams, but known from nearby streams (ARL studies, reported in Bunn 1985). Gondwanan endemic refers to SWA endemic species where Gondwanan lineage is known. References cited are listed at the end of this appendix.

Higher grouping	Species	Bunn 1985 status	Response	Affinity	Further information:
Amphipoda	<i>Uroctena</i> sp.	Common	Still present – groundwater dependent	Gondwanan endemic	Williams & Barnard 1988; Eberhard et al., 2004; Davies & Storey, 2012
	<i>Perthia</i> sp.	Common	Still present, rarer	Gondwanan endemic	Williams & Barnard 1988; Eberhard et al., 2004
Decapoda	<i>Cherax quinquecarinatus</i>	Common	Common	Gondwanan endemic	Beatty et al., 2005
	<i>Cypretta</i> sp.	Not recorded	Common	Genus is widespread; species unknown	
	<i>Ilyodromus</i> sp.	Common	Present	Genus is widespread; species unknown	De Dekker, 1980; Davis & Christidis, 1999; Halse et al., 2002
Mollusca	<i>Glacidorbis occidentalis</i>	Present	Not found	Gondwanan endemic	Bunn, Davies & Edwards, 1989; Ponder & Avern, 2000
Chironomidae	<i>Botryocladus bibulmun</i>	Common	Not found - possibly taxonomy	Gondwanan endemic	Cranston & Edwards, 1999; Krosch et al., 2011
	<i>Diplocladius/Stictocladus</i>	Common	Perennial streams	Gondwanan endemic	Cranston & Sæther, 2010
	<i>Riethia</i> sp.	Common	Perennial streams	Gondwanan, widespread southern (Genus)	Krosch et al. 2020
	<i>Ablabesmyia</i> sp.	Common	Rare	Genus is cosmopolitan	Leung et al., 2011a
	<i>Austropeleopis</i> sp.				
	Orthoclaadiinae "Don Edward V43"	Unknown	Rare, present in intermittent streams	Gondwanan endemic	
	Orthoclaadiinae "Don Edward V44"	Unknown	Rare, present in intermittent streams	Gondwanan endemic	Leung et al., 2011b

Higher grouping	Species	Bunn 1985 status	Response	Affinity	Further information:
	Orthoclaadiinae "Don Edward V15"	Unknown	Rare, present in intermittent streams	Gondwanan endemic	
	Unknown Orthoclad sp.	Unknown	Apparently restricted to inselbergs	Unknown	
	Orthoclad DEC1	Unknown	Common in intermittent streams	Unknown	
	<i>Paramerina levidensis</i>	Common		Genus is cosmopolitan; <i>P. levidensis</i> appears endemic	
	<i>Stenochironomus</i> sp.	Unknown			Berg 1995
	<i>Yarrhpelopia</i> sp. [PC]	Unknown	Rare - Seldom Seen Brook SS2	Gondwanan	Cranston et al., 2021
	Pentanuerini sp. "V20" [PC]	Present	Rare - Seldom Seen Brook SS2	Gondwanan	Leung et al., 2011 a; Cranston et al., 2021 General Chironomidae reference material: Edwards, 1986; Madden et al., 2010; Leung et al., 2011a,b; Cranston 2019
Plecoptera	<i>Leptoperla australica</i>	Common	Patchily distributed in intermittent streams	Gondwanan endemic	
	<i>Newmanoperla exigua</i>	Common	Not found during this study	Gondwanan endemic	
Megaloptera	<i>Apochauiodes cervulus</i>	Common	Perennial streams	Gondwanan endemic	Hynes & Bunn, 1984; Sutcliffe et al., 2002, 2003
			Common perennial streams, rare intermittent streams		
Tricoptera	<i>Notalina</i> AV14, AV16	Genus common - name changes	Not found	Gondwanan endemic	Cartwright et al., 2013; Sutcliffe 2003
	<i>Acritoptila</i> sp.	Common		Gondwanan endemic	Cartwright et al., 2013; Sutcliffe 2003
	<i>Atriplectides dubius</i>	Not recorded (known)	Perennial streams - rare	Gondwanan endemic	Cartwright et al., 2013; Sutcliffe 2003
	<i>Ecnomina</i> AV5	Genus common - name changes	Perennial streams	Gondwanan endemic	Cartwright et al., 2013; Sutcliffe 2003

Higher grouping	Species	Bunn 1985 status	Response	Affinity	Further information:
	<i>Ecnomina</i> AV8	Genus common - name changes	Perennial streams	Gondwanan endemic	Cartwright et al., 2013; Sutcliffe 2003
	<i>Ecnomus</i> sp.	Genus common - name changes	Perennial streams	Gondwanan endemic	Cartwright et al., 2013; Sutcliffe 2003
	<i>Hudsonema aptus</i>	Common	Perennial streams	Gondwanan endemic	Cartwright et al., 2013; Sutcliffe 2003;
	<i>Hydrobiosella</i> AV17	Unknown	Perennial streams	Gondwanan endemic	
	<i>Hydrobiosella michaelsoni</i>	Patchy	Patchily distributed - but locally in high densities	Gondwanan endemic	
	<i>Lectrides parilis</i>	Common	Perennial streams	Gondwanan endemic	
	Leptoc Gen A AV1	Unknown	Perennial streams	Gondwanan endemic	
	<i>Maydenoptila baynesi</i>	Common	Perennial streams	SWA endemic, genus is widespread	Wells 1983
	<i>Oxethira retracta</i>	Common	Perennial streams	SWA endemic, genus is widespread	Wells 1983
	<i>Helyethira</i> sp.	Not recorded		Genus is widespread	
	<i>Notoperata</i> AV4	Unknown		Northern, species unknown	Wells 1983
	<i>Notoperata tenax</i>	Common	Present	Gondwanan endemic	Cartwright et al., 2013; Sutcliffe 2003
	<i>Smicrophylax australis</i>	Common		Gondwanan endemic	Cartwright et al., 2013; Sutcliffe 2003
	<i>Apsilochorema urdalum</i>	Present	Not found	Gondwanan endemic	
	<i>Taschorema pallescens</i>	Common		Gondwanan endemic	Restricted to >1000mm rainfall zone; Sutcliffe 2003
	<i>Triplectides australis</i>	Not recorded	Rare	Widespread Australian endemic	
	<i>Triplectides</i> AV1	Present - name change	Patchily distributed in intermittent streams	Gondwanan endemic	
	<i>Triplectides</i> AV21	Present - name change	Perennial streams	Gondwanan endemic	St. Clair 2002
Odonata	<i>Aeschna brevistyla</i>	Not recorded	New arrival in intermittent streams - resilient	Cosmopolitan	
	<i>Archeosynthemis leachii</i>	Present		Gondwanan endemic	

Higher grouping	Species	Bunn 1985 status	Response	Affinity	Further information:
	<i>Archeosynthemis occidentalis</i>	Unknown	Patchy in nitermittent streams - resistant	Gondwanan endemic	
	<i>Archeosynthemis spiniger</i>	Unknown	Perennial streams - rare	Gondwanan endemic	High concern - Pennifold 2018
	<i>Armagomphus armiger</i>	Present	Perennial streams - rare	Gondwanan endemic	
	<i>Austroaeschna anacantha</i>	Common	Perennial streams	Gondwanan endemic	
	<i>Austroepigomphus gordonii</i>	Not recorded	Patchy in nitermittent streams - resilient	Widespread Australian endemic	
	<i>Austrolestes analis</i>	Not recorded	Patchy in nitermittent streams - resilient	Widespread Australian endemic	
	<i>Austrolestes annulosus</i>	Not recorded	Patchy in nitermittent streams - resilient	Widespread Australian endemic	
	<i>Austrolestes io</i>	Not recorded	Patchy in nitermittent streams - resilient	Widespread Australian endemic	
	<i>Crocothemis nigrifrons</i>	Not recorded	Rare	Widespread Australian endemic	
	<i>Hemianax papuensis</i>	Not recorded	Common intermittent streams	Widespread Australian endemic	
	<i>Hemicordulia tau</i>	Not recorded	Common intermittent streams	Widespread Australian endemic	
	<i>Hespercordulia berthoudi</i>	Present	Perennial streams	Gondwanan endemic	Sutcliffe 2003; high concern Pennifold 2018
	<i>Lathrocordulia metallica</i>	Common	Perennial streams	Gondwanan endemic	Sutcliffe 2003
	<i>Austrosynthemis cyanitincta</i>	Common	Perennial streams	Gondwanan endemic	Sutcliffe 2003
	<i>Miniargiolestes minimus</i>	Present	Perennial streams	Gondwanan endemic	Sutcliffe 2003; congeners appear extinct Pennifold 2018
	<i>Orthetrum caledonicum</i>	Present	Patchy	Widespread	
	<i>Procordulia affinis</i>	Not recorded	Rare	Widespread	
	<i>Zephyrogomphus lateralis</i>	Present	Perennial streams	Gondwanan endemic	Sutcliffe 2003

Higher grouping	Species	Bunn 1985 status	Response	Affinity	Further information:
Hemiptera	<i>Agraptocorixa</i> sp.	Not recorded	Common in intermittent streams	Widespread	Martin & Anderson, 2006
	<i>Anisops hackeri</i>	Not recorded	Common in intermittent streams	Widespread	
	<i>Micronecta gracilis</i>	Not recorded (genus known)	Common in intermittent streams	Widespread	
	<i>Microvelia australiensis</i>	Not recorded	Present in intermittent streams	Widespread	
	<i>Nesidovelia paramoena</i>	Not recorded	Common in intermittent streams	Widespread	
	<i>Paranisops endimyion</i>	Not recorded	Present in intermittent streams	Widespread	
	<i>Siagara mullaka</i>	Not recorded	Common in intermittent streams	Widespread	
Coleoptera	<i>Allodessus bistrigatus</i>	Not recorded	Common in intermittent streams		
	<i>Anacaena littoralis</i>	Not recorded	Present in intermittent streams		
	<i>Coelostoma fabricii</i>	Not present	Present in intermittent streams	Cosmopolitan	
	<i>Eretes australis</i>	Not present	Present in intermittent streams	Cosmopolitan	
	<i>Haliphus gibbus</i>	Not present	Present in intermittent streams	Cosmopolitan	
	<i>Hyphydrus elegans</i>	Not present	Present in intermittent streams	Cosmopolitan	
	<i>Lancetes lanceolatus</i>	Not recorded	Common in intermittent streams		
	<i>Limbodessus compactus</i>	Not recorded	Common in intermittent streams	Cosmopolitan	
	<i>Limnoxenus zealandicus</i>	Not recorded	Common in intermittent streams	Cosmopolitan	
	<i>Necterosoma darwini</i>	Not recorded (known)	Common in intermittent streams	Southern Australia	

Higher grouping	Species	Bunn 1985 status	Response	Affinity	Further information:
	<i>Neohydrocoptus fubfasciatus</i> (larvae)	Not recorded	Present in intermittent streams		
	<i>Paracymus pygmaeus</i>	Not recorded	Present in intermittent streams	Widespread	
	<i>Platynectes decempunctatus</i>	Not recorded (genus known)	Common in intermittent streams	Widespread	
	<i>Rhantus sutralis</i>	Not recorded (known)	Common in intermittent streams	Widespread	
	<i>Sternolophus australis</i>	Not recorded	Present in intermittent streams	Widespread/ Northern Australia	
	<i>Sternopriscus browni</i>	Common	Rare	SWA endemic, genus is widespread	Hendrich & Watts, 2004
	Ephemeroptera <i>Bibulmena kadjina</i>	Common	Patchily distributed in intermittent steams	Gondwanan endemic	
	<i>Nyungara bunni</i>	Common	Patchily distributed nn intermittent steams	Gondwanan endemic	Dean 1987; Dean 1988; Christidis 2006; Gatti et al., 2021
	<i>Neboissophlebia occidentalis</i>	Common	Perennial streams	Gondwanan endemic	
	<i>Offadens soror</i>	Common	Perennial streams		
	<i>Tasmanocoenis tillyardi</i>	Present – Seldom Seen Brook	Not found during this study	Southern Australia	
	<i>Nousia</i> sp. AV16	Not recorded	Common in intermittent streams	Genus southern Australia, Gondwanic affinity	Christidis 2006
	<i>Cloeon fluviatile</i>	Not recorded (known)	Common in intermittent streams	Widespread	Webb & Suter, 2011

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Appendix 3: Species list

PHYLUM/CLASS	Order	Lowest taxon
ANNELIDA		
	<u>Oligochaeta</u>	Oligochaeta
	<u>Gordioidea</u>	Gordioidea
	<u>Arhynchobdellida</u>	
	<u>Hirudinidae</u>	Hirudinidae
NEMATODA		Nematoda
PLATYHELMINTHES		Turbellaria
ARTHROPODA		
COLLEMBOLA		Collembola
ARACHNIDA	<u>Trombidiformes</u>	Acarina Tromidoidea Oribatida
CRUSTACEA		
	<u>Ostracoda</u>	<i>Cypretta</i> sp. <i>Ilyodromus</i> sp. Cyprididae (Ostracod 3) <i>Uroctena</i> sp. <i>Perthia acutitelson</i> <i>Cherax quinquecarinatus</i>
	<u>Amphipoda</u>	
	<u>Decapoda</u>	
INSECTA	<u>Coleoptera</u>	
	Halipidae	<i>Halipus gibbus</i>
	Curculionidae	Curculionidae
	Dytiscidae	<i>Allodessus bistrigatus</i> <i>Eretes australis</i> <i>Lancetes lanceolatus</i> <i>Limbodessus compactus</i> <i>Liodessus</i> sp. <i>Necterosoma darwini</i> <i>Onychohydrus</i> sp. <i>Platynectes decempunctatus</i> <i>Rhantus australis</i> <i>Sternopriscus browni</i>
	Hydrophilidae	<i>Anacaena littoralis</i> <i>Berosus</i> sp. <i>Coelostoma fabricii</i> <i>Enochrus</i> sp. <i>Hydrophilus</i> <i>Hyphydrus elegans</i> <i>Limnoxenus zealandicus</i> <i>Paracymus pygmaeus</i> <i>Sternolophus australis</i> <i>Neohydrocoptus subfasciatus</i>
	Hydraenidae	<i>Limnebius</i> sp.
	Scirtidae	Scirtidae

Diptera

Chironomidae (Aphroteniinae)	<i>Aphrotenia</i> sp. <i>Chironomus</i> sp. <i>Cladotanytarsus</i> sp. <i>Cryptochironomus</i> sp. <i>Dicrotendipes</i> sp. <i>Limnophyes</i> sp. <i>Microchironomus</i> sp. <i>Microtendipes</i> sp. <i>Parakeifferiella</i> sp. <i>Paratanytarsus</i> sp. <i>Paratendipes</i> sp. <i>Polypedilum</i> sp. <i>Riethia</i> sp. <i>Rheotanytarsus</i> sp. <i>Skusella</i> sp. <i>Stempellina</i> sp. <i>Stenochironomus</i> sp. <i>Stictochironomus</i> sp. <i>Tanytarsus</i> sp. <i>Xenochironomus</i> sp.
(Orthocladiinae)	<i>Anzacladius</i> sp. <i>Corynoneura</i> sp. <i>Cricotopus</i> sp. Don Edward 43 Don Edward 44 DonEdwardV15 Orthoclad DEC1 Orthoclad unident <i>Paralimnophyes</i> sp. <i>Rheocricotopus</i> sp. <i>Stictocladius</i> sp. <i>Stictocladius V52</i> <i>Thienemanniella</i> sp.
(Tanypodinae)	<i>Austropelopia</i> sp. <i>Monopelopia</i> sp. <i>Paramerina</i> ? <i>levidensis</i> Pentanuerini sp. "V20" <i>Procladius</i> sp. <i>Thienemannimyia</i> sp. <i>Yarrhpelopia</i> sp.
Athericidae	Athericidae
Ceratopogonidae	<i>Bezzia</i> sp. <i>Culicoides</i> sp. <i>Monohela</i> sp. <i>Nilobezzia</i> sp. Ceratopogoninae sp Dasyheleinae Forcipomyiinae
Dolichopodidae	Dolichopodidae
Empididae	Empididae
Culicidae	<i>Anopheles</i> sp. <i>Culex</i> sp. Culicidae sp.

Psychodidae	Psychodidae
Simuliidae	<i>Austrosimulium</i> sp. <i>Cnephia tonnoiri</i> <i>Simulium</i> sp.
Sciomyzidae	Sciomyzidae
Tabanidae	Tabanidae
Tipulidae	Limoniinae <i>Molophilus</i> sp. <i>Limnophila</i> sp. <i>Tipula</i> sp.
<u>Ephemeroptera</u>	
Baetidae	<i>Cloeon fluviatile</i> <i>Offadens soror</i>
Leptophlebiidae	<i>Bibulmena kadjina</i> <i>Neboissophlebia occidentalis</i> <i>Nousia</i> sp. AV 16 <i>Nyungara bunni</i>
<u>Hemiptera</u>	
Corixidae	<i>Agraptocorixa</i> sp. <i>Micronecta gracilis</i> <i>Siagara mullaka</i>
Mesoveliidae	<i>Microvelia australiensis</i>
Notonectidae	<i>Anisops hackeri</i> <i>Paranisops endimyion</i>
Veliidae	<i>Nesidovelia paramoena</i>
<u>Megaloptera</u>	
Corydalidae	<i>Apochauliodes cervulus</i>
<u>Odonata</u>	
Aeschnidae	<i>Aeschna brevistyla</i> <i>Anax papuensis</i>
Austrocordulidae	<i>Lathrocordulia metallica</i>
Gomphidae	<i>Armagomphus armiger</i> <i>Austroepigomphus gordonii</i> <i>Zephyrogomphus lateralis</i>
Hemicorduliidae	<i>Hemicordulia tau</i> <i>Procordulia affinis</i>
Lestidae	<i>Austrolestes analis</i> <i>Austrolestes annulosus</i> <i>Austrolestes io</i>
Libellulidae	<i>Crocothemis nigrifrons</i> <i>Nannophya occidentalis</i> <i>Orthetrum caledonicum</i> <i>Pantala flavescens</i>

Megapodagrionidae	<i>Miniargiolestes minimus</i>
Oxygastridae	<i>Hespercordulia berthoudi</i>
Synthemistidae	<i>Archeosynthemis leachii</i> <i>Archeosynthemis occidentalis</i> <i>Archeosynthemis spiniger</i> <i>Austrosynthemis cyanitincta</i>
Telephlebiidae	<i>Austroaeschna anacantha</i>
<u>Plecoptera</u>	
Gripopterygidae	<i>Leptoperla australica</i>
<u>Trichoptera</u>	
Ecnomidae	<i>Ecnomina</i> sp. AV5 <i>Ecnomina</i> sp. AV8 <i>Ecnomus</i> sp.
Hydrobiosidae	<i>Taschorema pallescens</i>
Hydrophychidae	<i>Smicrophylax australis</i>
Hydroptilidae	<i>Hellyethira</i> sp. <i>Maydenoptila baynesi</i> <i>Oxyethira brevis</i>
Leptoceridae	<i>Atriplectides dubius</i> <i>Hudsonema aptus</i> <i>Lectrides parilis</i> Leptoc Gen A AV1 <i>Leptoperla australica</i> <i>Notalina</i> sp. AV15 <i>Notalina</i> sp. AV14 <i>Notalina</i> sp. AV16 <i>Notoperata</i> sp. AV4 <i>Notoperata tenax</i> <i>Triplectides australis</i> <i>Triplectides</i> sp. AV1 <i>Triplectides</i> sp. AV21
Philopotamidae	<i>Hydrobiosella</i> sp. AV17 <i>Hydrobiosella michaelsoni</i>

